
The Polyexponential Trace-Tower Program

Moving Levels, Faber and Inverse Monodromy,
Positive Spectra, and Arithmetic Barriers

Complete Research Monograph

Logical-status convention. **U-PASS** denotes an unconditional theorem with a complete proof; **X-PASS** an unconditional finite theorem supported by exact reproducible computation; **R-PASS** a theorem relative to the explicitly cited arithmetic independence input from companion paper [001]; **C-PASS** a proved implication from a displayed open hypothesis; and **OPEN** a statement not proved and never used silently in an unconditional argument.

KEIO

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Abstract

This monograph develops a single trace-tower program around the integer-order poly-exponentials

$$E_q(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n! n^q} \quad (q \geq 1).$$

At finite level, their inverse-power traces are normalized Faber polynomials. The trace field collapses to one level coordinate, and two even traces recover that coordinate exactly; moving through arithmetically independent levels restores a transversal tower of independent coordinates. In the opposite direction, the inverse surfaces have maximal finitary symmetric monodromy and finite packets of inverse branches and first jets are algebraically free. This contrast—maximal local branch freedom versus exact global trace collapse—is the organizing principle of the volume.

The finite Faber polynomials are absolutely indecomposable in every order and their geometric monodromy reduces uniformly to the alternating/symmetric dichotomy. Full symmetric monodromy is proved in even degree, in every prime degree, in the infinite family $m = p + 2$, at fixed degree and sufficiently large polyexponential order, and in the exact computer-assisted rectangles recorded in the reproducibility appendix. A finite-jet Faber transfer gives isolated transpositions in wider polylogarithmic–hypergeometric families. For every integer polylogarithmic order, a strict Kaluza-sign argument proves all-degree absolute indecomposability and yields symmetric monodromy in every prime degree and in the family $m = p + 2$; exact fixed-order window audits add the stated finite odd-degree rectangles. The logarithmic member further gives type-III supernecklace polynomials with explicit arithmetic and collision-geometric consequences.

The same framework supplies positive trace-class carriers, strict total positivity, moment and Hankel towers, finite Jacobi inverse spectra, and Gamma-jet spectral transfers. It retains the exact Ramanujan–Meijer–Gamma transmutations of the preceding manuscript and the Euler factorial Padé/continued-fraction results, including full unit-boundary convergence at $t = -1$ for $p = 2, 3, 5, 13$ and one rational subsequence converging at the real place and at every fixed p -adic place. Finally, the volume states the precise approximation, reflexivity, p -adic height, Stokes-framing, and differential-separability barriers that remain. No unconditional theorem here proves the irrationality or transcendence of Euler’s constant, the Euler–Gompertz constant, or an odd zeta value.

Companion input and scope. Whenever arithmetic independence at algebraic arguments is used, the statement is marked **R-PASS** and cites the companion paper [001]. The finite and inverse monodromy theorems are logically independent of that arithmetic input. The general $f_{\sigma, \tau}$ large- τ Morse assertion, full odd-composite geometric S_m , the global derivative-separability conjecture, and the individual irrationality or transcendence of γ and δ remain open and are not promoted to theorems.

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Part I

Universal Traces and Arithmetic Moving Levels

1. Introduction and exact scope

Write

$$\text{Ein}(z) = \int_0^z \frac{1 - e^{-t}}{t} dt = \sum_{r \geq 1} \frac{(-1)^{r-1} z^r}{r r!},$$

and let

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right), \quad \delta = e E_1(1).$$

Neither γ nor δ is known to be irrational; see Lagarias' survey [10]. The elementary positive-axis identity

$$\text{Ein}(x) = \gamma + \log x + E_1(x) \quad (x > 0) \tag{1}$$

places γ at the boundary between algebraic-point E -values and an asymptotic connection constant.

The companion paper [8] determined the algebraic relations among all nonzero algebraic-point Ein -values over the pure exponential field, constructed several finite Euler witnesses, and computed the first two inverse traces of the divisor of $\text{Ein} - \gamma$. The present paper begins where that finite theory stops. Its organizing question is:

What remains true when the level of the divisor is allowed to move through a countable algebraically independent family converging to γ ?

The results have four layers. In the spectral layer, the two even traces of orders two and four recover the level exactly, so the entire fixed-level inverse-trace ring is one-dimensional; moving the level restores one algebraically independent coordinate per level. The order-two row has a positive telescoping structure and hence a genuine operator-theoretic realization. Algebraic levels of a natural gamma-free inverse map have transcendental preimages. All finite linear witnesses are governed by one normalized-certificate uniqueness lemma. The dyadic tail is the unique minimizer in a natural cumulative-positive augmentation cone, while outside that cone three-point forms are already dense. The resulting coefficient-height transition identifies the exact scale at which the first tail can be cancelled. In the transmutation layer, the four-term Euler recurrence from the 2026 Ramanujan Challenge is connected explicitly to the official Meijer- G construction, the cubic Pilehood family, a finite probability law, a regularized Gamma determinant, and a full rational zeta-jet tower. Its denominator and numerator live in a flat dual-number SL_3 system, equivalently an ordinary SL_6 system. In the obstruction layer, the particular residual-prime, fixed-place, Rodrigues, natural-index P-recursive, and strict-Stokes constructions studied here yield scoped saturation identities, budget inequalities, or sharply stated missing inputs.

The principal results are as follows.

Theorem 1.1 (Universal trace and transversal independence). *For $c \in \mathbb{C}^\times$, let Z_c be the zero multiset of $\text{Ein}(z) - c$. For every $m \geq 2$,*

$$T_m(c) := \sum_{\rho \in Z_c} \rho^{-m} = P_m(c^{-1}),$$

where $P_m \in \mathbb{Q}[X]$ is monic of degree m . If $\{c_j : j \in J\}$ is algebraically independent over a characteristic-zero field F , and one integer $m_j \geq 2$ is selected for each j , then $\{T_{m_j}(c_j) : j \in J\}$ is algebraically independent over F . Moreover

$$c^{-1} = 144T_4(c) - 144T_2(c)^2 - 14T_2(c),$$

so $F[T_m(c) : m \geq 2] = F[c^{-1}]$. The map $c \mapsto (T_2(c), T_3(c))$ is injective except for the single unordered pair

$$\{6 - 2\sqrt{3}, 6 + 2\sqrt{3}\},$$

whose common trace pair is $(-1/24, 1/96)$. For each $m \geq 2$, the nonzero roots of P_m also give a nonempty finite set of algebraic levels at which T_m vanishes and every point of the corresponding divisor is transcendental.

Theorem 1.2 (Finite-rank tails and incomplete-Gamma jets). *Let $\Gamma \subset \overline{\mathbb{Q}}_{>0}^\times$ have multiplicative rank r . For every finite $A \subset \Gamma$,*

$$\text{trdeg}_K K(E_1(\alpha) : \alpha \in A) \geq |A| - r - 1.$$

After deleting at most $r + 1$ values, the remaining full E_1 -tail family is jointly algebraically independent over K . The same bound holds for the combined classical family $\{E_1(\alpha), \text{Ei}(\alpha), \text{Ci}(\alpha), \text{Si}(\alpha)\}$.

For the regularized lower incomplete-Gamma jets

$$R_{\alpha,k} = [s^k] \{s^{-1} - \Gamma_<(s, \alpha)\},$$

every finite $S \subset \Gamma \times \mathbb{N}_0$ satisfies

$$\text{trdeg}_K K(R_{\alpha,k} : (\alpha, k) \in S) \geq |S| - r.$$

Thus at most r exceptions need be removed from the complete two-index lower-jet family. The corresponding upper jets have defect at most r plus the number of distinct jet orders.

Theorem 1.3 (Dyadic moving zero-trace tower). *For $N \geq 1$, put*

$$Y_N = (N + 1) \text{Ein}(2^N) - N \text{Ein}(2^{N+1}).$$

Then $\{Y_N : N \geq 1\}$ is countably algebraically independent over K , $Y_N \downarrow \gamma$, and

$$Y_N - \gamma \sim \frac{N + 1}{2^N} e^{-2^N}.$$

For every choice $m_N \geq 2$, the exact traces

$$\tau_{m_N, N} := \sum_{\text{Ein}(\rho) = Y_N} \rho^{-m_N}$$

are countably algebraically independent over K . For fixed m , $\tau_{m_N, N} \rightarrow S_m$, the m -th inverse trace of $\text{Ein} - \gamma$.

Theorem 1.4 (Inverse levels and critical coefficients). *The function*

$$\mathcal{R}(q) = E_1(1) - 2E_1(q) + E_1(q^2)$$

maps $(1, \infty)$ increasingly onto $(0, E_1(1))$. The inverse image of every algebraic number in this interval is transcendental. In particular, the unique roots $r_m > 1$ of $\mathcal{R}(r_m) = 1/m$, $m \geq 5$, are all transcendental.

For each fixed algebraic $q > 1$, the critical three-point cancellation coefficients $u_N(q)$ defined in (86) satisfy, for every finite I ,

$$\text{trdeg}_K K(u_N(q) : N \in I) \geq |I| - 1.$$

Theorem 1.5 (Positive spectral realizations). *Set*

$$\tau_N = \frac{1}{Y_N^2} - \frac{1}{2Y_N}, \quad d_0 = \tau_1, \quad d_N = \tau_{N+1} - \tau_N \quad (N \geq 1).$$

Then $d_N > 0$, the d_N are countably algebraically independent over K , and

$$S_2 = \sum_{N \geq 0} d_N.$$

The diagonal operator $Ae_N = d_N e_N$ belongs to every Schatten class and has trace S_2 . Moreover

$$\Delta_2(z) = \det \left(I - \frac{z^2}{2} A \right) = \prod_{N \geq 0} \left(1 - \frac{d_N}{2} z^2 \right)$$

is an order-zero Laguerre–Pólya entire function. Its positive zeros are countably algebraically independent over K , and

$$\sum_{\rho \in Z(\Delta_2)} \rho^{-2} = S_2.$$

The analogous order-four increments define a positive diagonal operator A_4 with trace S_4 , and, writing $A_2 = A$,

$$\gamma = \left\{ 144 \text{Tr } A_4 - 144(\text{Tr } A_2)^2 - 14 \text{Tr } A_2 \right\}^{-1}.$$

Theorem 1.6 (Cross-index witness rigidity). *Let $q_* \in (0, 1)$ be the real solution of $\text{Ein}(q_*) = \gamma$. Let $\beta_n \in (0, 1)$ be the positive real zero of*

$$G_n(z) = n \text{Ein}(z) - \text{Ein}(z^n) - (n-1)\gamma,$$

and, for odd $n \geq 3$, let $\lambda_n < -1$ be its negative real zero. Then at most one member of

$$\{q_*\} \cup \{\beta_n : n \geq 2\} \cup \{\lambda_n : n \geq 3, n \text{ odd}\}$$

is algebraic.

Theorem 1.7 (Critical moving-point barrier). *Any theorem of the natural form*

$$|P(F(\xi))| \geq \exp(-C(1 + |\xi|)^c \Phi(\deg P, \log H(P)))$$

intended to hold uniformly for all fixed E -function systems and positive integral sample points, or already for all nonzero polynomial values of the displayed system, must have $c \geq 1$. The obstruction occurs for the fixed system

$$F = (\text{Ein}(z), \text{Ein}(2z), \text{Ein}(4z))$$

and the fixed linear form $P = X_1 - 2X_2 + X_3$. General sufficiency at $c = 1$ is not known; at that scale the same example forces at least the polynomial loss ξ^{-1} . Within the augmentation-one stratum the dyadic coefficients uniquely minimize the tail in the cumulative-positive cone. Outside that cone three points give dense tail values, while cancellation of the first active tail requires at least $\log H \geq (q - 1)q^N + O(1)$. For consecutive augmentation-one triples this transition has a matching construction; at natural point cost a normalized lower bound for that affine template must pay height exponent at least two.

For the dyadic tower, algebraicity of S_2 is equivalent to the existence of one algebraic coefficient s for which

$$-\log |2sY_N^2 + Y_N - 2| \sim 2^N.$$

Theorem 1.8 (Exact Ramanujan transmutation package). *The two distinguished solutions of the Euler recurrence (136) have the finite sums (138)–(139), satisfy the recurrence for every $n \geq 0$, and obey $p_n/q_n \rightarrow \gamma$. Their terminating ${}_2F_2$ state and the official Meijer- G state are joined by the rational coboundary (178); the normalized conserved pairing is the finite identity (183). The denominator system embeds in the flat SL_3 field (156)–(159), and its zero-first jet lift is the flat SL_6 system (164).*

The Euler–Beta–log transform (196)–(197) identifies the same solutions with the cubic Pilehood family. The rational carrier $\mathcal{K}_a$ has denominator and numerator as its zero- and first-jets; $\mathcal{K}_a(s)/\mathcal{K}_a(0)$ converges to $\Gamma(1-s)^3\Gamma(1+s)^2$ and yields the zeta limits (204). Its scaled integer numerator is the monic contact polynomial of theorem 26.3. All recurrence solutions have the limiting form (147). Moreover, with $b = n + 4$ and $D_b = \text{lcm}(1, \dots, b)$,

$$\frac{b!}{D_b} \gcd\left(b, \frac{D_b}{b}\right) \mid \gcd(p_n, q_n),$$

and the factorial-six part of the exact Casoratian (171) comes from the determinant of the denominator-cleared official dual transfer.

Theorem 1.9 (Arithmetic and Stokes barriers). *The following statements hold unconditionally.*

- (i) *The dyadic level (Y_N) and its order-two trace (τ_N) are not P -recursive.*
- (ii) *The two-base form of proposition 36.2 remains exponentially small with*

$$-\log\{Y_N(2) - Y_{M_N}(3)\} = 2^N + N \log 2 - \log(N + 1) + O(1).$$

Thus multiplicative independence of the two bases does not remove this moving-point scale.

- (iii) *Every finite multi-level trace field is exactly the $\mathbb{G}_a$ -invariant field of the gauge shift $G \mapsto G + t, H_i \mapsto H_i - t$. The completed moving tower breaks this formal symmetry through its boundary limit.*
- (iv) *Every nondegenerate rank-one real Hilbert quotient obtained from the fixed residual-direction constraints is exactly saturated under a candidate $\gamma = a/b$. For the optimally saddled uncombined Prévost form, the lcm-only height-one ledger has deficit $2 - \log(3 + 2\sqrt{2})$.*
- (v) *The fixed-prime Rodrigues family (306) places $p^{n-\ell}$ in the endpoint coefficient. Even if this whole factor is credited as fixed-place gain, the resulting ledger and the archimedean saddle satisfy (316).*
- (vi) *The displayed unframed lateral transition of the elementary differential operator annihilating $F = -\text{Ein}$ is gamma-free. The Euler constant occurs in the connection framing and, under $\gamma \in \overline{\mathbb{Q}}$, can be removed by an algebraic shear. A framed Euler–Tate matched-line lifting*

theorem would imply the $\overline{\mathbb{Q}}$ -linear independence of $1, \pi, \gamma$, and hence the transcendence of $\gamma, \gamma/\pi$, and S_2 .

The qualification “transversal” in [theorems 1.1 and 1.3](#) is essential: one chooses one trace order at each level. At a fixed level all trace orders are algebraic over the same one-dimensional level field. Likewise, the infinite sum in [theorem 1.5](#) has no automatic arithmetic consequence. An infinite sum of algebraically independent transcendental numbers may be algebraic.

Organization. [Section 2](#) records the exact input from the companion paper. [Section 3](#) proves finite-rank cofinite independence for both connection coordinates and classical tails, together with the complete cancellation geometry; [section 4](#) exports the engine to sine-integral graphs, generalized Dickman transforms, and regularized incomplete-Gamma jets; the characteristic-zero algebraic-matroid realization is isolated in [appendix A](#) because it uses an external linearization theorem. [Section 5](#) develops the universal trace calculus and identifies its finite multi-level field with the exact shift quotient. [Section 6](#) proves the dyadic and all-order moving towers. [Section 7](#) constructs the two positive operators and the order-two Laguerre–Pólya determinant. [Section 8](#) gives fixed-coefficient power-orbit towers, transcendental inverse levels, and the logarithmic lattice. [Section 9](#) proves normalized certificate uniqueness, cross-index rigidity, and the one-defect cancellation theorem. [Section 10](#) gives the relative five-class exclusion and the exact finite-descent criterion. [Sections 22 and 23](#) solve the four-term Euler recurrence, classify all of its limits, and construct the flat SL_3/SL_6 field. [Sections 24 and 25](#) give the exact official gauge and the Euler–Beta–log offset ladder. [Sections 26 and 27](#) develop the rational Gamma jets, contact polynomial, finite probability law, and zero geometry; [section 28](#) gives the regularized determinant realization. [Section 29](#) proves the real and p -adic Euler–Padé convergence results. [Section 36](#) proves the sharp inside/outside augmentation dichotomy and isolates the critical moving-point obstruction. [Section 37](#) replaces an invalid asymmetric-integral heuristic by an exact evaluation. [Sections 38 and 39](#) establish the residual-prime, Ridout, and Rodrigues barriers. [Sections 40 to 42](#) separate unframed Stokes data from the mixed arithmetic-Gevrey and canonical motivic framings, prove the conditional dilation and Gamma-jet ranks, and isolate the strengthened matched-line target. The final section records what remains open.

Dependency and status ledger. To keep imported input, internal proof, and conditional extrapolation separate, the logical dependencies are summarized below.

Source	Imported statement	Status in this paper
Companion paper [8, Thms. 1.4, 5.1 and Lem. 11.2]	Algebraic independence of algebraic-point resonant Ein_q -values and of the balanced dyadic coordinates	Standing arithmetic input; the tail, incomplete-Gamma, trace, carrier, and rigidity deductions are proved here
Fischler–Rivoal [6]	Definitions of mixed functions; conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$; conditional linear specialization	The dilation and all-order Gamma-jet families are constructed here; every conclusion using the intersection conjecture is explicitly conditional
Official Challenge solution [32, §5.2]	Official Meijer– G normalization and Miller-selected limiting direction	Imported normalization; the rational coboundary, bilinear identity, and transmutations are proved here
Rosen–Sidman–Theran [33, Thm. 1.10]	Linear representability over an algebraically closed characteristic-zero field	Used only in Appendix A
Powers [31, §5]	A concluding multisection algebraic-independence assertion	Used only under an explicit hypothesis in Remark 4.7; no unconditional result depends on it

2. Input from the companion paper

Throughout,

$$K = \mathcal{E}_{\text{exp}} = \overline{\mathbb{Q}}(e^\alpha : \alpha \in \overline{\mathbb{Q}}),$$

and K^{alg} denotes its relative algebraic closure in $\mathbb{C}$. We use the following theorem from [8, Thm. 5.1].

Theorem 2.1 (Algebraic-point Ein-independence). *The countable family*

$$\{\text{Ein}(\alpha) : \alpha \in \overline{\mathbb{Q}}^\times\}$$

is algebraically independent over K , in the finite-subset sense. Consequently it remains algebraically independent over K^{alg} .

For the incomplete-Gamma application below we also use the resonant extension of this input. Put

$$\text{Ein}_q(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n^q n!} \quad (q \geq 1).$$

By [8, Thm. 1.4], the family

$$\{\text{Ein}_q(\alpha) : q \geq 1, \alpha \in \overline{\mathbb{Q}}^\times\} \tag{2}$$

is algebraically independent over K , in the finite-subset sense.

The final assertion is standard: algebraic independence is preserved under algebraic extension of the base field. We shall repeatedly use the following linear consequence.

Lemma 2.2 (Full-rank linear images). *Let $x_1, \dots, x_r$ be algebraically independent over a field F , and let $A \in M_{s \times r}(F)$ have row rank s . Then the s coordinates of $A(x_1, \dots, x_r)^\top$ are algebraically independent over F .*

Proof. Complete the rows of A to an invertible r -by- r matrix over F . The resulting linear change of variables is an automorphism of the rational function field $F(x_1, \dots, x_r)$. $\square$

For later comparison we also recall the Euler-level traces from [8, Thm. 14.5]. If Z_γ is the zero multiset of $\text{Ein}(z) - \gamma$, then

$$S_2 := \sum_{\rho \in Z_\gamma} \rho^{-2} = \frac{1}{\gamma^2} - \frac{1}{2\gamma}, \quad (3)$$

$$S_3 := \sum_{\rho \in Z_\gamma} \rho^{-3} = \frac{1}{\gamma^3} - \frac{3}{4\gamma^2} + \frac{1}{6\gamma}, \quad (4)$$

and

$$\gamma = \frac{24S_2 + 1}{24S_3 + 6S_2}. \quad (5)$$

In particular S_2 is algebraic if and only if γ is algebraic.

3. Finite-rank independence and cancellation geometry

The algebraic-point theorem has a useful export form when the arguments lie in a multiplicative group of finite rank. We restrict here to positive algebraic arguments, so that the real logarithm is single-valued and no additional $2\pi i$ -coordinate occurs. Put

$$c_\alpha = \text{Ein}(\alpha) - \log \alpha = \gamma + E_1(\alpha) \quad (\alpha \in \overline{\mathbb{Q}}_{>0}). \quad (6)$$

Theorem 3.1 (Finite-rank deficit and cofinite independence). *Let $\Gamma \subset \overline{\mathbb{Q}}_{>0}^\times$ have multiplicative rank r . For every finite $A \subset \Gamma$,*

$$\text{trdeg}_K K(c_\alpha : \alpha \in A) \geq |A| - r. \quad (7)$$

Consequently there is a set $E \subset \Gamma$, $|E| \leq r$, such that

$$\{c_\alpha : \alpha \in \Gamma \setminus E\} \text{ is algebraically independent over } K. \quad (8)$$

Fix $\alpha_0 \in \Gamma$. There is likewise a set $E_{\alpha_0} \subset \Gamma \setminus \{\alpha_0\}$, $|E_{\alpha_0}| \leq r$, for which

$$\{E_1(\alpha) - E_1(\alpha_0) : \alpha \in \Gamma \setminus (E_{\alpha_0} \cup \{\alpha_0\})\} \quad (9)$$

is algebraically independent over K .

Proof. Choose $\lambda_1, \dots, \lambda_r$ as a basis of the $\mathbb{Q}$ -vector space spanned by $\{\log \alpha : \alpha \in \Gamma\}$; torsion causes no difficulty because all elements are positive. For every finite A , all $\log \alpha$, $\alpha \in A$, lie in the $\mathbb{Q}$ -span of the λ_j . By (6),

$$K(\text{Ein}(\alpha) : \alpha \in A) \subset K(c_\alpha : \alpha \in A, \lambda_1, \dots, \lambda_r).$$

The left side has transcendence degree $|A|$ by [theorem 2.1](#); this proves (7).

Choose a maximal algebraically independent subfamily of $\{c_\alpha : \alpha \in \Gamma\}$. If its complement contained $r + 1$ elements, the algebraic dependence of each of them on a finite part of the maximal family would produce a finite set with deficit at least $r + 1$, contradicting (7). This proves (8).

For distinct $\alpha_1, \dots, \alpha_m \neq \alpha_0$, the differences $\text{Ein}(\alpha_i) - \text{Ein}(\alpha_0)$ are algebraically independent over K : they are full-rank linear images of the independent coordinates at $\alpha_0, \alpha_1, \dots, \alpha_m$. Subtracting the logarithmic differences costs at most the same r logarithmic coordinates. The first argument and the maximality argument therefore apply verbatim. $\square$

The common Euler term costs at most one further transcendence degree. This turns the shift bookkeeping into an unconditional theorem about the tails themselves.

Theorem 3.2 (Finite-rank tail defect and cofinite independence). *Let $\Gamma \subset \overline{\mathbb{Q}}_{>0}^\times$ have multiplicative rank r . For every finite $A \subset \Gamma$,*

$$\boxed{\text{trdeg}_K K(E_1(\alpha) : \alpha \in A) \geq |A| - r - 1.} \quad (10)$$

There is a set $E \subset \Gamma$, $|E| \leq r + 1$, such that

$$\{E_1(\alpha) : \alpha \in \Gamma \setminus E\} \quad (11)$$

is algebraically independent over K . In particular at most $r + 1$ members of the full tail family are algebraic over K , and at most $r + 1$ members of

$$\{\gamma\} \cup \{E_1(\alpha) : \alpha \in \Gamma\}$$

are algebraic over K . If $\gamma \in K^{\text{alg}}$, every occurrence of $r + 1$ in these tail assertions can be replaced by r .

Proof. By [theorem 3.1](#),

$$|A| - r \leq \text{trdeg}_K K(c_\alpha : \alpha \in A).$$

Since $c_\alpha = \gamma + E_1(\alpha)$, the field on the right is contained in $K(\gamma, E_1(\alpha) : \alpha \in A)$. Adjoining γ costs at most one transcendence degree, which proves (10). Choose a maximal algebraically independent subfamily of the tails. If its complement contained $r + 2$ elements, adjoining the finitely many members of the maximal family needed to algebraize them would contradict (10). This proves (11) and the first algebraicity count.

If γ is algebraic over K , adjoining it costs nothing and the same proof gives defect r . If it is transcendental, it is not counted among the algebraic members. The assertion for the union follows in both cases. $\square$

Corollary 3.3 (The two-prime triangular grid). *For $n \geq 1$, put*

$$A_n = \{2^k 3^\ell : k, \ell \geq 0, 1 \leq k + \ell \leq n\}, \quad |A_n| = \frac{n(n+3)}{2}.$$

Among $\{c_\alpha : \alpha \in A_n\}$, all but at most two can be chosen jointly algebraically independent over K , and at most two are algebraic. Among $\{E_1(\alpha) : \alpha \in A_n\}$, all but at most three can be chosen jointly algebraically independent, and at most three are algebraic. Thus at $n = 20$ the grid has 230 entries: at least 228 of the first family, and at least 227 of the tail family, belong to jointly algebraically independent subfamilies.

Proof. The group generated by 2 and 3 has multiplicative rank two. Apply [theorems 3.1](#) and [3.2](#); the cardinality is $\sum_{j=1}^n (j+1) = n(n+3)/2$. $\square$

Corollary 3.4 (Four classical special-function families). *Let $\Gamma \subset \overline{\mathbb{Q}}_{>0}^\times$ have multiplicative rank r . From the combined family*

$$\{E_1(\alpha), \text{Ei}(\alpha), \text{Ci}(\alpha), \text{Si}(\alpha) : \alpha \in \Gamma\} \quad (12)$$

delete at most $r + 1$ members. The remaining family is algebraically independent over K ; in particular at most $r + 1$ members of the full family are algebraic over K .

For $\alpha = 1, \dots, M$, the $4M$ displayed values have transcendence degree at least

$$4M - \pi(M) - 1 \quad (13)$$

over K , and at most $\pi(M) + 1$ of them are algebraic over K .

Proof. For each positive α , the four independent coordinates

$$\text{Ein}(\alpha), \quad \text{Ein}(-\alpha), \quad \frac{\text{Ein}(i\alpha) + \text{Ein}(-i\alpha)}{2}, \quad \frac{\text{Ein}(i\alpha) - \text{Ein}(-i\alpha)}{2i}$$

are respectively

$$\gamma + \log \alpha + E_1(\alpha), \quad \gamma + \log \alpha - \text{Ei}(\alpha), \quad \gamma + \log \alpha - \text{Ci}(\alpha), \quad \text{Si}(\alpha).$$

Across distinct positive α , the underlying points $\{\pm\alpha, \pm i\alpha\}$ are disjoint; hence all these coordinates are algebraically independent over K by [theorem 2.1](#) and [lemma 2.2](#). For any finite selection from (12), adjoining γ and at most r logarithmic generators recovers the same number of independent coordinates. The resulting defect is at most $r + 1$, and maximality gives the cofinite claim. The integers up to M generate a multiplicative group of rank $\pi(M)$, which proves (13). $\square$

The finite-dimensional form identifies exactly where the deficit occurs. Let $\alpha_1, \dots, \alpha_n$ be positive algebraic numbers contained in a rank- r group. Choose logarithmic generators $\lambda = (\log g_1, \dots, \log g_r)^\top$ and an exponent matrix $L \in M_{n \times r}(\mathbb{Q})$ such that

$$(\log \alpha_1, \dots, \log \alpha_n)^\top = L\lambda. \quad (14)$$

Theorem 3.5 (Linear-image rank deficit). *Let $M \in M_{m \times n}(\overline{\mathbb{Q}})$ have row rank s , and put $t = \text{rank}(ML)$. With $c = (c_{\alpha_1}, \dots, c_{\alpha_n})^\top$, one has*

$$\boxed{\text{trdeg}_K K(Mc) \geq s - t \geq s - r.} \quad (15)$$

Proof. Write $x = (\text{Ein}(\alpha_1), \dots, \text{Ein}(\alpha_n))^\top$. Then $x = c + L\lambda$. Choose a row selector $R \in M_{s \times m}(\overline{\mathbb{Q}})$ for which RM has row rank s . The s coordinates of Rmx are algebraically independent by [lemma 2.2](#). On the other hand,

$$RMx = R(Mc) + (RML)\lambda, \quad \text{rank}(RML) \leq \text{rank}(ML) = t.$$

Thus $\text{trdeg}_K K((RML)\lambda) \leq t$, while $R(Mc)$ is a linear image of Mc . Therefore

$$s \leq \text{trdeg}_K K(Mc) + t,$$

which is the assertion. $\square$

There is also a sharp statement in the directions where both the Euler coefficient and all logarithms cancel.

Corollary 3.6 (Complete cancellation space). *Let*

$$U = \ker \begin{pmatrix} \mathbf{1}^\top \\ L^\top \end{pmatrix} \subset \overline{\mathbb{Q}}^n, \quad d = \dim U = n - \text{rank}[\mathbf{1}, L]. \quad (16)$$

For any basis $u_1, \dots, u_d$ of U , the numbers

$$W_j = \sum_{i=1}^n u_{j,i} \text{Ein}(\alpha_i) = \sum_{i=1}^n u_{j,i} E_1(\alpha_i) \quad (1 \leq j \leq d) \quad (17)$$

are algebraically independent over K . Thus (16) is not only the complete space of gamma-free, log-free coefficient directions: every independent direction produces a new algebraically independent period.

Proof. The two defining equations for U , together with (1), prove the equality in (17). The basis rows have rank d ; apply lemma 2.2 to the algebraically independent Ein-coordinates. $\square$

For a directed graph $G = (V, E)$, let ∂ be its incidence matrix, and attach a positive algebraic number α_v to each vertex. If L is the corresponding exponent matrix, then theorem 3.5 gives the graph form

$$\text{trdeg}_K K(E_1(\alpha_{t(e)}) - E_1(\alpha_{s(e)}) : e \in E) \geq |V| - c(G) - \text{rank}(\partial^T L), \quad (18)$$

where $c(G)$ is the number of connected components. This is a genuine incidence-geometric extension of the one-dimensional prime-defect counts used below.

4. External realizations of the independence engine

We record three consequences that export the algebraic-point input beyond the exponential-integral language. They classify relations rather than asserting new irrationality results for Euler's constant.

Sine-integral graphic matroids

Let $G = (V, E)$ be a finite loopless directed multigraph, and choose nonzero algebraic numbers a_v such that $a_u^2 \neq a_v^2$ for $u \neq v$. Put

$$s_v = \text{Si}(a_v) = \frac{\text{Ein}(ia_v) - \text{Ein}(-ia_v)}{2i}, \quad I_e = s_{t(e)} - s_{s(e)} = \int_{a_{s(e)}}^{a_{t(e)}} \frac{\sin z}{z} dz.$$

Theorem 4.1 (Complete graphic relation ideal). *Let $\partial_G : K^E \rightarrow K^V$ be the incidence map $\partial_G e = e_{t(e)} - e_{s(e)}$. Then*

$$\ker(K[X_e : e \in E] \longrightarrow K[I_e : e \in E]) = \left\langle \sum_{e \in E} c_e X_e : c \in \ker \partial_G \right\rangle. \quad (19)$$

Consequently

$$\text{trdeg}_K K(I_e : e \in E) = |V| - c(G), \quad (20)$$

and a set of edge integrals is algebraically independent exactly when its edges form a forest.

Proof. The pairs $\{ia_v, -ia_v\}$ are disjoint. The full-rank linear-image lemma applied to the corresponding algebraically independent Ein values shows that the s_v are algebraically independent. The edge vector is $I = \partial_G^T s$. Hence its coordinate ring is the symmetric algebra of the incidence row space, whose defining ideal is generated by the linear cycle relations. The rank of the incidence matrix is $|V| - c(G)$, and a set of incidence columns is independent exactly when the corresponding edges form a forest in the underlying multigraph. $\square$

This polynomial-relation statement strengthens the disjoint-endpoint linear-independence results for sine-integral values in [21] by using the stronger input from the companion paper.

Generalized Dickman log-Laplace values

For the generalized Dickman distribution of [23], the Laplace transform is

$$\psi_{\theta,a}(s) = \exp\{-\theta \text{Ein}(as)\}. \quad (21)$$

Take positive algebraic θ_j, a_j, s_j , set $x_j = a_j s_j$, and

$$\Lambda_j = -\log \psi_{\theta_j, a_j}(s_j) = \theta_j \text{Ein}(x_j).$$

Theorem 4.2 (Dickman log-Laplace classification). *For every finite J ,*

$$\boxed{\operatorname{trdeg}_K K(\Lambda_j : j \in J) = \#\{x_j : j \in J\}.} \quad (22)$$

For each distinct x , choose a representative j_x . The complete polynomial relation ideal is

$$\ker(K[X_j : j \in J] \rightarrow K[\Lambda_j : j \in J]) = \langle \theta_{j_x} X_j - \theta_j X_{j_x} : x_j = x, j \neq j_x \rangle. \quad (23)$$

Moreover, for integers m_j ,

$$\boxed{\prod_{j \in J} \psi_{\theta_j, a_j}(s_j)^{m_j} = 1 \iff \sum_{j: x_j = x} m_j \theta_j = 0 \quad \text{for every distinct } x.} \quad (24)$$

Proof. Choose one representative for each distinct x_j . The corresponding $\operatorname{Ein}(x_j)$ are algebraically independent; every Λ_j is a nonzero algebraic multiple of its block representative. This proves the transcendence degree and the linear defining ideal. The Laplace values are positive real numbers, so a multiplicative relation equal to one is equivalent, after the real logarithm, to a linear relation among the distinct $\operatorname{Ein}(x)$; algebraic independence forces the displayed block conditions. $\square$

The theorem classifies the logarithmic values and the multiplicative relations equal to one. It does not claim to classify every polynomial relation among the Laplace values themselves.

Regularized incomplete-Gamma jets

For $\alpha > 0$ algebraic, let

$$\Gamma_{<}(s, \alpha) = \int_0^\alpha e^{-t} t^{s-1} dt, \quad \Gamma_{>}(s, \alpha) = \int_\alpha^\infty e^{-t} t^{s-1} dt,$$

initially in their half-planes of convergence and then by continuation. At $s = 0$, define

$$R_{\alpha, k} = [s^k] \{s^{-1} - \Gamma_{<}(s, \alpha)\}, \quad U_{\alpha, k} = [s^k] \Gamma_{>}(s, \alpha) \quad (k \geq 0). \quad (25)$$

This boundary-Laurent and multipoint regime is distinct from the positive-parameter incomplete-Gamma special values studied, for example, in [26]; no priority claim beyond the precise statements proved below is intended.

Theorem 4.3 (Cofinite independence of regularized lower jets). *Let $\Gamma \subset \overline{\mathbb{Q}}_{>0}^\times$ have multiplicative rank r , and let $S \subset \Gamma \times \mathbb{N}_0$ be finite. Then*

$$\boxed{\operatorname{trdeg}_K K(R_{\alpha, k} : (\alpha, k) \in S) \geq |S| - r.} \quad (26)$$

Consequently, after deleting at most r members, the entire two-index family $\{R_{\alpha, k} : \alpha \in \Gamma, k \geq 0\}$ is algebraically independent over K . In particular at most r of these values are algebraic over K .

More explicitly, with $L_\alpha = \log \alpha$,

$$R_{\alpha, k} = -\frac{L_\alpha^{k+1}}{(k+1)!} + \sum_{j=0}^k \frac{(-1)^j L_\alpha^{k-j}}{(k-j)!} \operatorname{Ein}_{j+1}(\alpha). \quad (27)$$

Thus $R_{\alpha, 0} = c_\alpha$, while

$$R_{1, k} = (-1)^k \operatorname{Ein}_{k+1}(1) \quad (28)$$

and the latter all-order family is algebraically independent over K without exceptions.

Proof. Splitting $e^{-t} = 1 + (e^{-t} - 1)$, expanding α^s , and then expanding $(n + s)^{-1}$ gives (27). For the finite set S , collect the finitely many coordinates $\text{Ein}_j(\alpha)$ that occur on the right. The rows defining the $R_{\alpha,k}$ have rank $|S|$: within each fixed α , the row of order k has the new highest coordinate $\text{Ein}_{k+1}(\alpha)$ with nonzero coefficient, and different α -blocks are disjoint.

Let X be the resulting vector of M Ein-coordinates and let $L = K(\lambda_1, \dots, \lambda_r)$, where the λ_i span the real logarithms of Γ . Over L , complete the $|S|$ displayed rows to an invertible linear coordinate system (R, Z) on X , with $|Z| = M - |S|$. By (2),

$$M = \text{trdeg}_K K(X) \leq \text{trdeg}_K L(R, Z) \leq r + \text{trdeg}_K K(R) + M - |S|.$$

This proves (26). The maximality argument of [theorem 3.1](#) then gives the cofinite assertion. Setting $L_1 = 0$ proves (28). $\square$

Write $\Gamma(s) = s^{-1} + \sum_{k \geq 0} g_k s^k$. Since $\Gamma = \Gamma_{<} + \Gamma_{>}$, one has

$$U_{\alpha,k} = g_k + R_{\alpha,k}. \quad (29)$$

Corollary 4.4 (Upper-jet defect). *For finite $S \subset \Gamma \times \mathbb{N}_0$, let $d(S) = \#\{k : (\alpha, k) \in S\}$. Then*

$$\boxed{\text{trdeg}_K K(U_{\alpha,k} : (\alpha, k) \in S) \geq |S| - r - d(S).} \quad (30)$$

For every fixed k , all but at most $r + 1$ members of $\{U_{\alpha,k} : \alpha \in \Gamma\}$ form an algebraically independent family over K . At $k = 0$, $U_{\alpha,0} = E_1(\alpha)$, so this recovers [theorem 3.2](#).

Proof. By (29), adjoining the $d(S)$ Gamma coefficients g_k to the upper jets generates all lower jets in S . Combine (26) with the transcendence-degree inequality. For fixed k , apply the same maximality argument as before. $\square$

The finite-order shift ledger can now be replaced by a complete invariant ring. This is a structural classification, not an irrationality theorem.

Proposition 4.5 (Complete finite Gamma-jet quotient). *Fix $N \geq 0$, let*

$$A_N(s) = 1 + \sum_{k=0}^N g_k s^{k+1} \pmod{s^{N+2}}, \quad T_k = U_{1,k}, \quad d_k = g_k - T_k,$$

and put $\kappa_j = [s^j] \log A_N(s)$. On the polynomial ring in the g_k, T_k , define a $\mathbb{G}_a$ -action by

$$A_N(s) \mapsto e^{-ts} A_N(s) \pmod{s^{N+2}}, \quad d_k \mapsto d_k. \quad (31)$$

Then

$$\boxed{\mathbb{Q}[g_0, \dots, g_N, T_0, \dots, T_N]^{\mathbb{G}_a} = \mathbb{Q}[\kappa_2, \dots, \kappa_{N+1}, d_0, \dots, d_N].} \quad (32)$$

The analogous equality holds for fraction fields. At the actual Gamma split,

$$\kappa_1 = -\gamma, \text{quad}\kappa_j = \frac{(-1)^j}{j} \zeta(j) \ (j \geq 2), \text{quadd}_k = (-1)^{k+1} \text{Ein}_{k+1}(1). \quad (33)$$

Thus γ is the unique orbit coordinate at every finite order; the zeta cumulants and resonant Ein-coordinates generate all polynomial and rational invariants.

Proof. The change of variables

$$(g_0, \dots, g_N, T_0, \dots, T_N) \longleftrightarrow (\kappa_1, \dots, \kappa_{N+1}, d_0, \dots, d_N)$$

is triangular and polynomial in both directions, by the truncated formal identities $\log A_N$ and $A_N = \exp(\log A_N)$. Under (31), $\kappa_1 \mapsto \kappa_1 - t$ and every other displayed coordinate is fixed. The invariant ring of translation in one variable is generated by the remaining variables, proving (32). Finally $A_N(s)$ is the truncation of $s\Gamma(s) = \Gamma(1+s)$, so

$$\log \Gamma(1+s) = -\gamma s + \sum_{j \geq 2} \frac{(-1)^j}{j} \zeta(j) s^j.$$

Equation (29) at $\alpha = 1$, together with (28), gives the formula for d_k . $\square$

Remark 4.6 (Correct reading of the all-order ledger). The gauge orbit has dimension exactly one at every order. Any additional unknown excess in a truncated equation count is not another copy of the Euler shift: the new odd-zeta cumulants are themselves invariant arithmetic coordinates. Consequently a growing unknown-minus-equation count is useful bookkeeping, but is not by itself a no-go theorem.

The corresponding export to arbitrary characteristic-zero algebraic matroids uses an external linearization theorem and is therefore separated from these direct applications; see [appendix A](#).

Remark 4.7 (A conditional density consequence of Powers). In the notation of Powers' later multisection manuscript [31, §5], assume that

$$G_1(1), H_1(1), \dots, G_N(1), H_N(1)$$

are algebraically independent over $\overline{\mathbb{Q}}(e)$ for every N . This assertion appears in the concluding multisection discussion rather than as a proved numbered theorem in the cited version. There

$$\begin{aligned} G_n(1) &= \frac{\tilde{\delta}^{(n)} - \delta^{(n)}}{2e}, \\ H_n(1) &= \gamma^{(n)} + \frac{\tilde{\delta}^{(n)} + \delta^{(n)}}{2e}. \end{aligned}$$

Conditional on it, combine the two generalized Euler–Gompertz rows

$$\delta^{(1)}, \dots, \delta^{(N)}, \quad \tilde{\delta}^{(1)}, \dots, \tilde{\delta}^{(N)}.$$

Their field satisfies

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\delta^{(1)}, \dots, \delta^{(N)}, \tilde{\delta}^{(1)}, \dots, \tilde{\delta}^{(N)}) \geq \left\lfloor \frac{3N}{2} \right\rfloor - 1.$$

Indeed the Gamma-moment identities quoted by Powers give

$$\text{trdeg}_{\overline{\mathbb{Q}}} \overline{\mathbb{Q}}(\gamma^{(1)}, \dots, \gamma^{(N)}) \leq N - \left\lfloor \frac{N}{2} \right\rfloor + 1.$$

The assumed $2N$ independent coordinates, together with the transcendence of e , lie in the field generated by e , the two Euler–Gompertz rows, and these Gamma moments. Subtracting transcendence degrees yields the displayed bound. Greedy basis selection in the associated algebraic matroid, ordered as $\delta^{(1)}, \tilde{\delta}^{(1)}, \delta^{(2)}, \dots$, then gives a jointly algebraically independent subfamily of lower density at least $3/4$. This entire paragraph is conditional, and no unconditional result elsewhere in the paper depends on it.

5. Universal level-trace polynomials

For $c \in \mathbb{C}^\times$, set

$$F_c(z) = \text{Ein}(z) - c,$$

and denote by Z_c its zero multiset, with multiplicities. Since F_c has order one, its inverse-power sums converge absolutely from order two onward.

Theorem 5.1 (Universal level-trace polynomial). *For $m \geq 2$, define*

$$P_m(X) := -m[z^m] \log(1 - X \text{Ein}(z)). \quad (34)$$

Then $P_m \in \mathbb{Q}[X]$, its constant term is zero, and it is monic of degree m . Moreover

$$T_m(c) := \sum_{\rho \in Z_c} \rho^{-m} = P_m(c^{-1}). \quad (35)$$

In particular

$$P_2(X) = X^2 - \frac{X}{2}, \quad (36)$$

$$P_3(X) = X^3 - \frac{3}{4}X^2 + \frac{1}{6}X, \quad (37)$$

$$P_4(X) = X^4 - X^3 + \frac{25}{72}X^2 - \frac{1}{24}X. \quad (38)$$

Proof. A genus-one Hadamard factorization has the form

$$F_c(z) = -c e^{a_c z} \prod_{\rho \in Z_c} \left(1 - \frac{z}{\rho}\right) e^{z/\rho}.$$

Taking the logarithm at the origin gives, for every $m \geq 2$,

$$[z^m] \log \frac{F_c(z)}{F_c(0)} = -\frac{1}{m} \sum_{\rho \in Z_c} \rho^{-m}.$$

On the other hand,

$$\frac{F_c(z)}{F_c(0)} = 1 - \frac{\text{Ein}(z)}{c}.$$

Comparison proves (35). Formula (34) is a polynomial in X of degree at most m . Since $\text{Ein}(z) = z + O(z^2)$, the contribution of $X^m \text{Ein}(z)^m/m$ to (34) is X^m . Thus P_m is monic of degree m . The displayed formulas follow by expanding through degree four. $\square$

Remark 5.2 (Multiplicity and genus). No simplicity assertion is required: all zeros are counted with their multiplicities. The first inverse sum is not used. At order one the canonical genus-one exponential factor absorbs the non-absolutely convergent sum, whereas (35) is absolute for $m \geq 2$.

Corollary 5.3 (Transversal trace independence). *Let $F \subset \mathbb{C}$ be a field of characteristic zero, and let $\{c_j : j \in J\} \subset \mathbb{C}^\times$ be algebraically independent over F . Choose one integer $m_j \geq 2$ for each j . Then*

$$\{T_{m_j}(c_j) : j \in J\}$$

is algebraically independent over F , in the finite-subset sense.

Proof. Fix a finite set $I \subset J$, and write $x_j = c_j^{-1}$ and $t_j = P_{m_j}(x_j)$. Since

$$P_{m_j}(x_j) - t_j = 0,$$

the field $F(x_j : j \in I)$ is algebraic over $F(t_j : j \in I)$. Hence

$$\text{trdeg}_F F(t_j : j \in I) = \text{trdeg}_F F(x_j : j \in I) = |I|.$$

□

Theorem 5.4 (Exact even-trace recovery and ring closure). *For every $c \neq 0$,*

$$\boxed{c^{-1} = 144T_4(c) - 144T_2(c)^2 - 14T_2(c).} \quad (39)$$

Consequently, for every characteristic-zero field $F \subset \mathbb{C}$,

$$\boxed{F[T_m(c) : m \geq 2] = F[T_2(c), T_4(c)] = F[c^{-1}]}$$

and

$$\boxed{F(T_2(c), T_4(c)) = F(c).}$$

Proof. With $x = c^{-1}$, (36) and (38) give

$$144P_4(x) - 144P_2(x)^2 - 14P_2(x) = x.$$

This proves (39). Every trace is a polynomial in x by theorem 5.1, while the displayed identity recovers x from T_2, T_4 . Taking fraction fields and using $F(x) = F(c)$ finishes the proof. □

The preceding recovery identity also gives the exact, rather than merely heuristic, form of the finite-stage shift obstruction. It is important to distinguish this gauge action, which fixes every connection coordinate, from common translation of the connection coordinates themselves; augmentation zero pertains to the latter action, not to the former.

Theorem 5.5 (Exact finite-stage shift quotient). *Let B be a characteristic-zero field, put*

$$R = B[G, H_1, \dots, H_n], \quad c_i = G + H_i,$$

and let $\mathbb{G}_a$ act by

$$G \mapsto G + t, \quad H_i \mapsto H_i - t. \quad (40)$$

Then

$$\boxed{R^{\mathbb{G}_a} = B[c_1, \dots, c_n], \quad \text{Frac}(R)^{\mathbb{G}_a} = B(c_1, \dots, c_n).} \quad (41)$$

If the universal traces are formed at these levels, then

$$B(T_m(c_i) : 1 \leq i \leq n, m \geq 2) = B(c_1, \dots, c_n), \quad (42)$$

$$B[T_m(c_i) : 1 \leq i \leq n, m \geq 2] = B[c_1^{-1}, \dots, c_n^{-1}]. \quad (43)$$

Thus the complete finite multi-level trace field is exactly the rational quotient by the gauge direction. It loses one orbit coordinate and no others.

Proof. The polynomial change of variables $H_i = c_i - G$ identifies $R = B[c_1, \dots, c_n, G]$, and (40) becomes translation of G with every c_i fixed. In characteristic zero the polynomial and rational invariants of translation in one variable are respectively $B[c_1, \dots, c_n]$ and $B(c_1, \dots, c_n)$, proving (41). Every trace is a polynomial in c_i^{-1} , while (39) recovers c_i^{-1} from $T_2(c_i), T_4(c_i)$. This proves (42)–(43). □

Remark 5.6 (What the formal quotient does and does not prove). The elimination ideal of the equations $c_i = G + H_i$ in $B[G]$ is zero: the same finite invariant data admit a model with $G = 0$ and another with G transcendental. Hence invariant algebraic identities alone cannot decide the arithmetic of the orbit coordinate. This is a statement about the formal axiom system, not about the actual complex number γ . Indeed the analytic completion supplies extra boundary data. If $y_N \rightarrow g$ in a Hausdorff topological field and a continuous action fixes every y_N , it fixes g . Applied to $Y_N \rightarrow \gamma$, and to the two carrier limits $\tau_N \rightarrow S_2$, $\sigma_N \rightarrow S_4$, this is precisely where the finite-stage gauge symmetry breaks; S_2, S_4 recover γ by (44).

Corollary 5.7 (Arithmetic classification of every Euler trace). *For $m \geq 2$, put $S_m = T_m(\gamma)$. Then*

$$S_m \in \overline{\mathbb{Q}} \iff \gamma \in \overline{\mathbb{Q}}, \quad S_m \text{ is transcendental} \iff \gamma \text{ is transcendental}.$$

Moreover the two even traces recover Euler's constant without exceptions:

$$\boxed{\gamma = \frac{1}{144S_4 - 144S_2^2 - 14S_2}}. \quad (44)$$

In particular

$$\gamma \in \mathbb{Q} \iff S_2, S_4 \in \mathbb{Q}.$$

Proof. If $S_m = P_m(\gamma^{-1})$ is algebraic, then γ^{-1} is a root of the nonconstant polynomial $P_m(X) - S_m$ over $\overline{\mathbb{Q}}$, and hence is algebraic. The converse is immediate. Formula (44) is (39) at $c = \gamma$; it also proves the rationality equivalence. $\square$

Theorem 5.8 (The unique two-trace level collision). *For $c, d \in \mathbb{C}^\times$,*

$$(T_2(c), T_3(c)) = (T_2(d), T_3(d))$$

holds if and only if either $c = d$, or

$$\boxed{\{c, d\} = \{6 - 2\sqrt{3}, 6 + 2\sqrt{3}\}}. \quad (45)$$

In the exceptional case the common pair is

$$\boxed{(T_2, T_3) = \left(-\frac{1}{24}, \frac{1}{96}\right)}. \quad (46)$$

Proof. Put $x = c^{-1}$ and $y = d^{-1}$. From (36), equality of the second traces gives

$$(x - y) \left(x + y - \frac{1}{2} \right) = 0.$$

If $x = y$, then $c = d$. Otherwise $x + y = 1/2$, and equality of the third traces, using (37), reduces to

$$x^2 + xy + y^2 - \frac{3}{4}(x + y) + \frac{1}{6} = 0,$$

or $xy = 1/24$. Thus x, y are the roots of

$$X^2 - \frac{1}{2}X + \frac{1}{24},$$

and taking reciprocals gives (45). Direct substitution in (36) and (37) gives (46). $\square$

Corollary 5.9 (Exact vanishing traces with transcendental divisors). *For every $m \geq 2$, define*

$$\mathcal{V}_m = \{c \in \mathbb{C}^\times : T_m(c) = 0\}.$$

Then

$$\boxed{\mathcal{V}_m = \{x^{-1} : x \neq 0, P_m(x) = 0\} \subset \overline{\mathbb{Q}}^\times.} \quad (47)$$

This set is nonempty and finite; its elements arise from exactly $m - 1$ nonzero roots of P_m , counted with multiplicity. For every $c \in \mathcal{V}_m$, every zero ρ of $\text{Ein}(z) - c$ is transcendental, although the absolutely convergent inverse trace satisfies

$$\sum_{\text{Ein}(\rho)=c} \rho^{-m} = 0.$$

For example, $\mathcal{V}_2 = \{2\}$.

Proof. The trace identity (35) gives (47). The polynomial P_m has degree m , constant term zero, and linear coefficient

$$[X]P_m(X) = m[z^m] \text{Ein}(z) = \frac{(-1)^{m-1}}{m!} \neq 0.$$

Hence zero is a simple root and the remaining $m - 1$ roots are nonzero, counted with multiplicity. They are algebraic because $P_m \in \mathbb{Q}[X]$.

Now fix $c \in \mathcal{V}_m$. If a zero ρ of $\text{Ein}(z) - c$ were algebraic, then $\rho \neq 0$ and $\text{Ein}(\rho) = c \in \overline{\mathbb{Q}}$, contradicting theorem 2.1. Thus every point of the divisor is transcendental. Finally $P_2(X) = X(X - 1/2)$ gives the example. $\square$

Thus even all inverse traces at one fixed level still contain only one coordinate. Within the universal inverse-trace construction, varying the trace order at a fixed level produces only the one-dimensional field generated by that level. A transversal algebraically independent inverse-trace family therefore requires varying the level; other constructions may instead vary a coefficient, a power orbit, or an external parameter.

6. Balanced dyadic levels and all-order moving traces

For $N \geq 1$, define

$$Y_N = (N + 1) \text{Ein}(2^N) - N \text{Ein}(2^{N+1}). \quad (48)$$

The companion paper proved the first assertion below. The remaining assertions are the analytic bridge to the Euler level.

Proposition 6.1 (Balanced dyadic approach). *The family $\{Y_N : N \geq 1\}$ is countably algebraically independent over K . Moreover*

$$Y_N = \gamma + \varepsilon_N, \quad \varepsilon_N = (N + 1)E_1(2^N) - NE_1(2^{N+1}), \quad (49)$$

and

$$\varepsilon_N > 0, \quad \varepsilon_N \downarrow 0, \quad \varepsilon_N \sim \frac{N + 1}{2^N} e^{-2^N}. \quad (50)$$

In particular $Y_N \downarrow \gamma$.

Proof. The algebraic independence is [8, Lem. 11.2]; it also follows directly from [theorem 2.1](#) and [lemma 2.2](#), because the finite row family $((N+1)e_N - Ne_{N+1})$ has full row rank.

Substitution of (1) in (48) cancels both the logarithmic and the $N \log 2$ terms and gives (49). Positivity follows from

$$\varepsilon_N = E_1(2^N) + N\{E_1(2^N) - E_1(2^{N+1})\} > 0.$$

For strict decrease put $g(t) = E_1(2^t)$. Since

$$g'(t) = -(\log 2)e^{-2^t}, \quad g''(t) = (\log 2)^2 2^t e^{-2^t} > 0,$$

strict convexity gives

$$\varepsilon_N - \varepsilon_{N+1} = (N+1)\{g(N) - 2g(N+1) + g(N+2)\} > 0.$$

Finally, the standard positive-axis expansion

$$E_1(x) = \frac{e^{-x}}{x} \left(1 + O(x^{-1})\right) \quad (x \rightarrow +\infty) \quad (51)$$

gives the asymptotic in (50); see [2, §6.12i]. $\square$

For $m \geq 2$, let

$$\tau_{m,N} := \sum_{\text{Ein}(\rho)=Y_N} \rho^{-m}, \quad (52)$$

where the solutions are counted with multiplicity.

Theorem 6.2 (All-order transversal tower). *For every $m \geq 2$ and $N \geq 1$,*

$$\tau_{m,N} = P_m(Y_N^{-1}). \quad (53)$$

For any selection $m_N \geq 2$, the family

$$\{\tau_{m_N,N} : N \geq 1\}$$

is countably algebraically independent over K . For each fixed $m \geq 2$,

$$\tau_{m,N} \longrightarrow S_m := P_m(\gamma^{-1}). \quad (54)$$

Proof. Equation (53) is [theorem 5.1](#). Apply [corollary 5.3](#) to the independent levels Y_N . The limit follows from [proposition 6.1](#) and continuity of $P_m(1/x)$ at $x = \gamma$. $\square$

Remark 6.3 (Why transversal is sharp). For a fixed N , the even traces already recover the level exactly:

$$\boxed{Y_N^{-1} = 144\tau_{4,N} - 144\tau_{2,N}^2 - 14\tau_{2,N}.} \quad (55)$$

Thus

$$K[\tau_{m,N} : m \geq 2] = K[\tau_{2,N}, \tau_{4,N}] = K[Y_N^{-1}].$$

The full double family is therefore not algebraically independent: the theorem gives exactly one independent spectral coordinate per moving level.

The order-two row has additional positivity. Put

$$f(x) = \frac{1}{x^2} - \frac{1}{2x} = \frac{2-x}{2x^2}, \quad \tau_N := \tau_{2,N} = f(Y_N). \quad (56)$$

Theorem 6.4 (Exact positive convergence at order two). *One has*

$$0 < \tau_1 < \tau_2 < \cdots < S_2, \quad \tau_N \longrightarrow S_2.$$

More precisely,

$$S_2 - \tau_N = \frac{\varepsilon_N \{\gamma(4 - \gamma) + (2 - \gamma)\varepsilon_N\}}{2\gamma^2(\gamma + \varepsilon_N)^2}. \quad (57)$$

Consequently

$$S_2 - \tau_N \sim \frac{4 - \gamma}{2\gamma^3} \frac{N + 1}{2^N} e^{-2^N}. \quad (58)$$

Proof. Since $E_1(2) < e^{-2}/2$ and $\gamma < 1$, $Y_1 = \gamma + 2E_1(2) - E_1(4) < \gamma + e^{-2} < 2$, so all Y_N lie in $(0, 2)$. But

$$f'(x) = \frac{x - 4}{2x^3} < 0 \quad (0 < x < 2).$$

Thus $Y_N \downarrow \gamma$ implies $\tau_N \uparrow f(\gamma) = S_2$, and positivity follows from $f(x) > 0$ on $(0, 2)$. Direct subtraction with $Y_N = \gamma + \varepsilon_N$ gives (57). Combine it with (50) to obtain (58). $\square$

Theorem 6.5 (Non-P-recursiveness of the dyadic moving sequences). *Neither $(Y_N)_{N \geq 1}$ nor $(\tau_N)_{N \geq 1}$ is P-recursive over $\mathbb{C}$. Equivalently, neither sequence satisfies a nonzero recurrence*

$$\sum_{j=0}^r p_j(N) u_{N+j} = 0 \quad (p_j \in \mathbb{C}[N])$$

for all sufficiently large N .

Proof. Suppose first that such a recurrence holds for $Y_N = \gamma + \varepsilon_N$. Since every polynomial multiple of ε_{N+j} tends to zero, we obtain

$$\gamma \sum_{j=0}^r p_j(N) \longrightarrow 0.$$

The sum is a polynomial, hence it vanishes identically. Let j_0 be the least index for which $p_{j_0} \neq 0$. Dividing the remaining relation by $p_{j_0}(N)\varepsilon_{N+j_0}$ gives

$$1 + \sum_{j>j_0} \frac{p_j(N)}{p_{j_0}(N)} \frac{\varepsilon_{N+j}}{\varepsilon_{N+j_0}} = 0.$$

But (50) implies, for every fixed $j > j_0$,

$$\frac{\varepsilon_{N+j}}{\varepsilon_{N+j_0}} = \exp\{-(2^j - 2^{j_0})2^N + O(N)\},$$

which dominates every rational function of N . The displayed equality therefore tends to $1 = 0$, a contradiction.

For $\tau_N = f(Y_N)$, theorem 6.4 gives

$$\tau_N = S_2 + f'(\gamma)\varepsilon_N + O(\varepsilon_N^2), \quad f'(\gamma) \neq 0.$$

The same least-index argument applies verbatim. $\square$

Remark 6.6 (Scope of the non-P-recursiveness result). The theorem does not say that no holonomic construction can involve the finite-stage values. It says that the natural moving index itself cannot be packaged as a fixed P-recursive sequence. Any application of arithmetic holonomy must therefore replace the moving tower by a genuinely different, fixed differential or recurrence system rather than merely reindexing Y_N .

For orientation, the first values are shown in [table 1](#). The table is illustrative only; no numerical value is used in a proof.

Table 1: The dyadic level and its order-two zero trace.

N	Y_N	τ_N
1	0.671237333907806	1.474569389580200
2	0.588478390885392	2.037963940792663
3	0.577366307471447	2.133831985200994
4	0.577215698103968	2.135171685377284
5	0.577215664901535	2.135171980841625
limit	0.577215664901533	2.135171980841646

7. Positive spectral carriers for S_2 and S_4

Define

$$d_0 = \tau_1, \quad d_N = \tau_{N+1} - \tau_N \quad (N \geq 1). \quad (59)$$

Theorem 7.1 (Positive algebraically independent increments). *The sequence $\{d_N : N \geq 0\}$ is positive and countably algebraically independent over K . It satisfies*

$$\boxed{S_2 = \sum_{N=0}^{\infty} d_N} \quad (60)$$

and, for $N \geq 1$,

$$d_N \sim \frac{4 - \gamma}{2\gamma^3} \frac{N + 1}{2^N} e^{-2^N}. \quad (61)$$

Proof. Positivity follows from [theorem 6.4](#). The finite transformation

$$(\tau_1, \dots, \tau_{M+1}) \longleftrightarrow (d_0, \dots, d_M)$$

is an invertible triangular $\mathbb{Q}$ -linear transformation. The τ_N are algebraically independent by [theorem 6.2](#); hence so are the d_N . The telescoping identity

$$\sum_{N=0}^M d_N = \tau_{M+1}$$

and [theorem 6.4](#) give (60). Finally the ratio of two successive tails in (58) tends to zero, so

$$d_N = (S_2 - \tau_N) - (S_2 - \tau_{N+1}) \sim S_2 - \tau_N,$$

which proves (61). □

Let $\{e_N : N \geq 0\}$ be the standard basis of $\ell^2(\mathbb{N}_0)$.

Corollary 7.2 (A positive operator carrier). *The diagonal operator*

$$\mathbf{A}e_N = d_N e_N$$

is positive, self-adjoint and trace class, with

$$\mathrm{Tr} \mathbf{A} = S_2.$$

More strongly,

$$\sum_{N \geq 0} d_N^p < \infty \quad (p > 0).$$

Thus $\mathbf{A} \in \mathcal{S}_p$ for every $p \geq 1$, and satisfies the corresponding quasi-Schatten condition for $0 < p < 1$. Its nonzero eigenvalues are simple and countably algebraically independent over K .

Proof. All assertions except simplicity follow immediately from [theorem 7.1](#). The double-exponential asymptotic [\(61\)](#) gives summability of every positive power. If $d_j = d_k$ for $j \neq k$, the algebraically independent family would satisfy a nonzero linear relation; hence the eigenvalues are distinct. $\square$

The Fredholm determinant of the operator gives a scalar entire realization.

Theorem 7.3 (Order-zero Laguerre–Pólya realization). *The product*

$$\Delta_2(z) = \det \left(I - \frac{z^2}{2} \mathbf{A} \right) = \prod_{N=0}^{\infty} \left(1 - \frac{d_N}{2} z^2 \right) \quad (62)$$

converges locally uniformly and defines an order-zero entire function in the Laguerre–Pólya class. Its zeros are simple and equal to

$$\pm r_N, \quad r_N = \sqrt{\frac{2}{d_N}}.$$

The positive half $\{r_N : N \geq 0\}$ is countably algebraically independent over K , and

$$\boxed{\sum_{\rho \in Z(\Delta_2)} \rho^{-2} = S_2.} \quad (63)$$

Proof. Since $\sum d_N < \infty$, the product in [\(62\)](#) converges normally on compact sets. Every finite partial product has only real zeros, so their locally uniform limit belongs to the Laguerre–Pólya class; see [\[11, Chapter VIII\]](#). For every $\eta > 0$,

$$\sum_{N \geq 0} r_N^{-\eta} = 2^{-\eta/2} \sum_{N \geq 0} d_N^{\eta/2} < \infty.$$

Thus the zero exponent of convergence, and hence the order of the canonical product, is zero. Simplicity follows from the distinctness of the d_N .

For a finite index set I , the relations $d_N = 2/r_N^2$ imply

$$\mathrm{trdeg}_K K(r_N : N \in I) \geq \mathrm{trdeg}_K K(d_N : N \in I) = |I|,$$

which proves algebraic independence of the positive zeros. Finally, absolute convergence and [\(60\)](#) give

$$\sum_{\rho \in Z(\Delta_2)} \rho^{-2} = \sum_{N \geq 0} \left(r_N^{-2} + (-r_N)^{-2} \right) = \sum_{N \geq 0} d_N = S_2.$$

$\square$

Theorem 7.4 (A second positive carrier and even-trace recovery). *Put*

$$\sigma_N = T_4(Y_N) = P_4(Y_N^{-1}) \quad (N \geq 1),$$

and define

$$d_0^{(4)} = \sigma_1, \quad d_N^{(4)} = \sigma_{N+1} - \sigma_N \quad (N \geq 1).$$

Then every $d_N^{(4)}$ is positive, the family $\{d_N^{(4)} : N \geq 0\}$ is algebraically independent over K , and

$$S_4 = \sum_{N \geq 0} d_N^{(4)}. \quad (64)$$

Consequently the diagonal operator $A_4 = \text{diag}(d_N^{(4)} : N \geq 0)$ belongs to every Schatten class and $\text{Tr } A_4 = S_4$. Together with the order-two carrier $A_2 := A$ of [corollary 7.2](#), it gives

$$\gamma = \left\{ 144 \text{Tr } A_4 - 144(\text{Tr } A_2)^2 - 14 \text{Tr } A_2 \right\}^{-1}. \quad (65)$$

Proof. The earlier bound $Y_N < 2$ and the monotonicity $Y_N \downarrow \gamma$ show that $x_N = Y_N^{-1}$ increases and satisfies $x_N > 1/2$. From [\(38\)](#),

$$P_4(1/2) = \frac{1}{288}, \quad P_4'(1/2) = \frac{1}{18},$$

and

$$P_4''(x) = 12x^2 - 6x + \frac{25}{36} > 0 \quad (x \geq 1/2).$$

Thus P_4 is positive and strictly increasing on this range, so all increments are positive. The σ_N are algebraically independent by [theorem 6.2](#); the triangular change from $(\sigma_1, \dots, \sigma_{M+1})$ to $(d_0^{(4)}, \dots, d_M^{(4)})$ preserves this property. Telescoping and $T_4(Y_N) \rightarrow T_4(\gamma) = S_4$ prove [\(64\)](#). The doubly exponential convergence of Y_N gives summability of every positive power of $d_N^{(4)}$. Finally apply [\(44\)](#). $\square$

Remark 7.5 (Three distinct meanings of “gamma-free”). Each finite datum Y_N, τ_N, d_N is defined by algebraic-point Ein-values without displaying γ . The limit and sum

$$S_2 = \lim_N \tau_N = \sum_N d_N$$

are countable constructions. They do not prove that S_2 belongs to the finite field generated by algebraic-point Ein-values. Indeed $[z^2]\Delta_2(z) = -S_2/2$, so its Taylor coefficients are already beyond what the finite-stage algebraic-independence theorem controls. No claim is made that Δ_2 is an E -function, a D -finite function, or an arithmetic entire function.

Remark 7.6 (No arithmetic conclusion from the infinite sum). Algebraic independence controls finite polynomial relations. It says nothing by itself about the arithmetic nature of an infinite sum. Thus [\(60\)](#) and [\(63\)](#) do not prove that S_2 , or equivalently γ , is transcendental.

8. Further moving-level towers

The dyadic tower is not isolated. This section gives a fixed-coefficient power-orbit family and a logarithmic lattice. They have different decay rates and different independence defects.

8.1. Fixed-coefficient power orbits

Fix an integer $n \geq 2$ and a positive algebraic number $q > 1$. For $k \geq 0$, put

$$B_{n,k}(q) = \frac{n \operatorname{Ein}(q^k) - \operatorname{Ein}(q^{nk})}{n-1}. \quad (66)$$

Theorem 8.1 (Power-orbit moving tower). *For fixed n and q , the family $\{B_{n,k}(q) : k \geq 0\}$ is countably algebraically independent over K . Moreover*

$$B_{n,k}(q) = \gamma + \eta_{n,k}(q), \quad \eta_{n,k}(q) = \frac{nE_1(q^k) - E_1(q^{nk})}{n-1}, \quad (67)$$

and

$$\eta_{n,k} > 0, \quad \eta_{n,k} \downarrow 0, \quad \eta_{n,k} \sim \frac{n}{n-1} \frac{e^{-q^k}}{q^k}. \quad (68)$$

For every choice $m_k \geq 2$, the exact traces

$$T_{m_k}(B_{n,k}(q))$$

are countably algebraically independent over K . For fixed m , they converge to S_m . At $m = 2$ they increase to S_2 and yield, by successive differences, another positive algebraically independent spectral realization of S_2 .

Proof. Let $X_j = \operatorname{Ein}(q^j)$. The distinct algebraic points q^j give an algebraically independent coordinate family by [theorem 2.1](#). The row vectors

$$v_k = ne_k - e_{nk} \quad (k \geq 0)$$

are linearly independent: in a finite relation, the coordinate nk_0 associated with the largest occurring k_0 appears as the terminal coordinate of that row and as the initial coordinate of no row in the finite set. Thus its coefficient vanishes, and descending induction finishes the proof. If the finite set consists only of $k_0 = 0$, then $v_0 = (n-1)e_0 \neq 0$, which is the same conclusion. Apply [lemma 2.2](#).

The connection formula (1) cancels the logarithmic terms and gives (67). For $x > 1$,

$$\frac{d}{dx} \left(\frac{nE_1(x) - E_1(x^n)}{n-1} \right) = \frac{n}{n-1} \frac{e^{-x^n} - e^{-x}}{x} < 0.$$

This proves strict decrease; positivity follows from $E_1(x) > E_1(x^n) > 0$, and the asymptotic follows from (51). The trace assertions follow from [corollary 5.3](#) and [theorem 5.1](#). The order-two construction is the proof of [theorems 7.1](#) and [7.3](#) applied to this decreasing level sequence: indeed $B_{n,0}(q) = \operatorname{Ein}(1) < 1 < 2$, since $(1 - e^{-t})/t < 1$ on $(0, 1]$. $\square$

This tower samples the rank-one witness of the companion paper exactly:

$$B_{n,k}(q) - \gamma = \frac{G_n(q^k)}{n-1}. \quad (69)$$

Remark 8.2 (Fixing n is essential). The full two-parameter family is not algebraically independent. For $r \geq 1$,

$$B_{n^r,k}(q) = \frac{n-1}{n^r-1} \sum_{j=0}^{r-1} n^{r-1-j} B_{n,n^j k}(q). \quad (70)$$

For example $B_{4,k} = (2B_{2,k} + B_{2,2k})/3$.

8.2. Algebraic-level inverse transcendence

The first difference between two members of the $n = 2$ power orbit gives the gamma-free function

$$\mathcal{R}(q) = E_1(1) - 2E_1(q) + E_1(q^2) \quad (q > 1). \quad (71)$$

It has an unexpectedly rigid inverse on algebraic levels.

Theorem 8.3 (Transcendental inverse levels). *The map $\mathcal{R} : (1, \infty) \rightarrow (0, E_1(1))$ is a strictly increasing bijection. If $a \in \overline{\mathbb{Q}} \cap (0, E_1(1))$, then the unique number $r_a > 1$ satisfying $\mathcal{R}(r_a) = a$ is transcendental.*

In particular, for every integer $m \geq 5$, the unique root $r_m > 1$ of

$$E_1(1) - 2E_1(r_m) + E_1(r_m^2) = \frac{1}{m} \quad (72)$$

is transcendental, and

$$r_m = 1 + \sqrt{\frac{e}{m}} + \frac{e}{2m} + O(m^{-3/2}). \quad (73)$$

Proof. Direct differentiation gives

$$\mathcal{R}'(q) = \frac{2}{q}(e^{-q} - e^{-q^2}) > 0.$$

Moreover $\mathcal{R}(1) = 0$ and $\mathcal{R}(q) \rightarrow E_1(1)$ as $q \rightarrow \infty$, proving the bijection. If r_a were algebraic, the connection formula would turn $\mathcal{R}(r_a) = a$ into

$$\text{Ein}(1) - 2\text{Ein}(r_a) + \text{Ein}(r_a^2) = a.$$

The three arguments are distinct nonzero algebraic numbers, so this relation contradicts [theorem 2.1](#).

Splitting the defining integral at (2) and (3), and using $(1/t > 1/2)$ on $(1 < t < 2)$ and $(1/t > 1/3)$ on $(2 < t < 3)$, gives

$$E_1(1) > \frac{1}{2e} + \frac{e^{-2} - e^{-3}}{3} > \frac{1}{5}.$$

Thus $a = 1/m$ is admissible for $m \geq 5$. Finally, writing $q = 1 + h$, direct differentiation at one gives

$$\mathcal{R}(1 + h) = e^{-1}(h^2 - h^3 + O(h^4)).$$

Series inversion at $\mathcal{R} = 1/m$ proves (73). □

8.3. A logarithmic lattice and density-one transcendence

For $r \geq 1$, put

$$c_r = \text{Cin}(r) - \log r = \gamma - \text{Ci}(r). \quad (74)$$

These levels approach γ only polynomially, but the loss caused by the logarithms can be counted by primes.

Lemma 8.4. *One has $c_r > 0$ for every integer $r \geq 1$.*

Proof. For $x > 0$, set $c(x) = \gamma - \text{Ci}(x)$. Since $c'(x) = -\cos x/x$, its local minima occur at $x_k = \pi/2 + 2\pi k$. At the first minimum,

$$\text{Cin}(\pi/2) \geq \int_0^{\pi/2} \left(\frac{t}{2} - \frac{t^3}{24} \right) dt = \frac{\pi^2}{16} - \frac{\pi^4}{1536} > \log(\pi/2),$$

using $1 - \cos t \geq t^2/2 - t^4/24$ on this interval. The last strict inequality follows, for example, from $333/106 < \pi < 355/113$ and $\log(1+u) \leq u - u^2/2 + u^3/3$ for $0 < u < 1$. Thus $c(x_0) > 0$. For $k \geq 1$, integration by parts gives $\text{Ci}(x_k) \leq 2/x_k < 1/2 < \gamma$. Here we used $x_k \geq 5\pi/2$ and the elementary bound $\gamma > 1/2$. Hence all local minima are positive, and so is $c(x)$. In particular $c_r > 0$. $\square$

For each r , let

$$U_r = T_2(c_r) = \frac{1}{c_r^2} - \frac{1}{2c_r}. \quad (75)$$

Theorem 8.5 (Prime-defect lattice theorem). *Let $\pi(M)$ be the number of primes not exceeding M . Then*

$$\text{trdeg}_K K(c_1, \dots, c_M) = \text{trdeg}_K K(U_1, \dots, U_M) \geq M - \pi(M). \quad (76)$$

Consequently at most $\pi(M)$ of the first M members of each family $\{c_r\}$ and $\{U_r\}$ are algebraic over K . Both families are K -transcendental on a set of natural density one.

Proof. The companion result [8, Cor. 5.5] makes $\text{Cin}(1), \dots, \text{Cin}(M)$ algebraically independent over K . Equivalently, this follows from $\text{Cin}(r) = \{\text{Ein}(ir) + \text{Ein}(-ir)\}/2$: the pairs $\{ir, -ir\}$ are disjoint, so these averaging rows have full rank. Since

$$\log r = \sum_{p \leq M} v_p(r) \log p,$$

we have

$$K(\text{Cin}(1), \dots, \text{Cin}(M)) \subset K(c_1, \dots, c_M, \log p : p \leq M).$$

Therefore

$$M \leq \text{trdeg}_K K(c_1, \dots, c_M) + \pi(M),$$

which gives the lower bound for the c_r . Equation

$$2U_r c_r^2 + c_r - 2 = 0$$

shows that $K(c_1, \dots, c_M)$ is algebraic over $K(U_1, \dots, U_M)$, while the reverse field inclusion is rational. Thus the two transcendence degrees agree. If a of the first M coordinates were algebraic over K , the transcendence degree would be at most $M - a$; hence $a \leq \pi(M)$. The prime number theorem completes the density statement. $\square$

The trace U_r also converges to S_2 . Direct subtraction gives

$$U_r - S_2 = \frac{\text{Ci}(r)\{\gamma(4 - \gamma) - (2 - \gamma)\text{Ci}(r)\}}{2\gamma^2(\gamma - \text{Ci}(r))^2}, \quad (77)$$

and the standard expansion of Ci yields

$$\begin{aligned} U_r - S_2 &= \frac{4 - \gamma}{2\gamma^3} \frac{\sin r}{r} - \frac{4 - \gamma \cos r}{2\gamma^3} \frac{1}{r^2} \\ &\quad + \frac{6 - \gamma \sin^2 r}{2\gamma^4} \frac{1}{r^2} + O(r^{-3}). \end{aligned} \quad (78)$$

Thus this lattice is generally only $O(r^{-1})$ from the Euler trace. The theorem does not decide any specified c_r, U_r with $r \geq 2$; $c_1 = \text{Cin}(1)$ is already K -transcendental.

9. Normalized certificates and cross-index rigidity

The moving towers concern limits. We now return to finite witnesses and show that all normalized linear certificates are unique.

Let

$$V = \bigoplus_{\alpha \in \overline{\mathbb{Q}}^\times} K^{\text{alg}}[\alpha]$$

be the free K^{alg} -vector space with finite support on the nonzero algebraic points, and define

$$\Phi : V \longrightarrow \mathbb{C}, \quad \Phi([\alpha]) = \text{Ein}(\alpha).$$

Lemma 9.1 (The evaluation intersection). *The map Φ is injective and*

$$\Phi(V) \cap K^{\text{alg}} = \{0\}. \quad (79)$$

Proof. By [theorem 2.1](#), every finite collection of the displayed Ein-values is algebraically independent over K^{alg} , and hence linearly independent. Moreover any nonzero linear form in algebraically independent variables is transcendental over the base field: extend it to an invertible linear coordinate system. This proves both claims. $\square$

Definition 9.2. A finite K^{alg} -certificate for γ is a triple $(v, c, a) \in V \times (K^{\text{alg}})^\times \times K^{\text{alg}}$ satisfying

$$\Phi(v) = c\gamma + a. \quad (80)$$

Its normalized data are $(v/c, a/c)$.

Theorem 9.3 (Normalized certificate uniqueness). *If (v_1, c_1, a_1) and (v_2, c_2, a_2) are two certificates, then*

$$\boxed{\frac{v_1}{c_1} = \frac{v_2}{c_2}, \quad \frac{a_1}{c_1} = \frac{a_2}{c_2}}. \quad (81)$$

Proof. Subtracting the two normalized identities gives

$$\Phi\left(\frac{v_1}{c_1} - \frac{v_2}{c_2}\right) = \frac{a_1}{c_1} - \frac{a_2}{c_2} \in K^{\text{alg}}.$$

Apply (79). $\square$

This is an exclusion theorem, not an existence theorem. The certificate set may be empty. Its force is that whenever two apparently unrelated algebraicity events would produce certificates, their normalized supports must coincide exactly.

We apply it to the real Euler witnesses recalled in the introduction. The existence, location and within-index algebraic-zero rigidity of q_* , β_n and λ_n were established in [8, Thm. 14.2, Prop. 14.3, and Thm. 14.4].

Theorem 9.4 (Cross-index witness rigidity). *At most one member of*

$$\mathcal{W} = \{q_*\} \cup \{\beta_n : n \geq 2\} \cup \{\lambda_n : n \geq 3, n \text{ odd}\} \quad (82)$$

is algebraic. Thus precisely one of the following alternatives holds:

- (i) *every member of $\mathcal{W}$ is transcendental;*

(ii) *exactly one member is algebraic, and in that event γ and δ are algebraically independent over K .*

Proof. If q_* is algebraic, its normalized certificate vector is

$$v_* = [q_*], \quad \Phi(v_*) = \gamma.$$

If an algebraic number α is a zero of G_n , its normalized certificate vector is

$$v_{n,\alpha} = \frac{n}{n-1}[\alpha] - \frac{1}{n-1}[\alpha^n], \quad \Phi(v_{n,\alpha}) = \gamma. \quad (83)$$

The real-zero classification places α in $(0, 1)$ or, for odd n , in $(-\infty, -1)$. Hence $\alpha \neq \alpha^n$, and $v_{n,\alpha}$ has exactly one positive and one negative support coefficient. It cannot equal the one-point vector v_* .

If $v_{n,\alpha} = v_{m,\beta}$, equality of the positive and negative support coordinates gives

$$\frac{n}{n-1} = \frac{m}{m-1},$$

so $n = m$, and then $\alpha = \beta$. Certificate uniqueness therefore excludes two distinct algebraic members of (82). The algebraic independence of γ, δ in the exceptional case is [8, Thm. 14.2 and Thm. 14.4]. $\square$

9.1. A defect-one family of critical cancellation coefficients

Fix an algebraic number $q > 1$. For $N \geq 1$, put

$$D_N(q) = E_1(q^N) - 2E_1(q^{N+1}) + E_1(q^{N+2}), \quad (84)$$

$$A_N(q) = (N+2)E_1(q^{N+1}) - (N+1)E_1(q^{N+2}), \quad (85)$$

$$u_N(q) = \frac{A_N(q)}{D_N(q)}. \quad (86)$$

Both numerator and denominator are positive: $D_N(q) > 0$ is the strict second-difference inequality for the strictly convex function $t \mapsto E_1(q^t)$, while monotonicity gives $A_N(q) > E_1(q^{N+1}) > 0$. The number $u_N(q)$ is the unique positive coefficient for which the three-point augmentation-one tail vanishes:

$$-u_N E_1(q^N) + (N+2+2u_N)E_1(q^{N+1}) - (N+1+u_N)E_1(q^{N+2}) = 0. \quad (87)$$

Theorem 9.5 (One-defect cancellation family). *For every finite set $I \subset \mathbb{N}_{\geq 1}$,*

$$\boxed{\text{trdeg}_K K(u_N(q) : N \in I) \geq |I| - 1.} \quad (88)$$

Consequently at most one member of $\{u_N(q) : N \geq 1\}$ is algebraic over K . If one member is algebraic over K , all the remaining members are countably algebraically independent over K , and γ, δ are algebraically independent over K . If γ is algebraic over K , then the whole family $\{u_N(q) : N \geq 1\}$ is countably algebraically independent over K .

Finally,

$$u_N(q) \sim \frac{N+2}{q} \exp(-(q-1)q^N). \quad (89)$$

Proof. Let $X_j = \text{Ein}(q^j)$. The connection formula and (87) give

$$-u_N X_N + (N + 2 + 2u_N)X_{N+1} - (N + 1 + u_N)X_{N+2} = \gamma. \quad (90)$$

For finite I , let $U = \bigcup_{N \in I} \{N, N+1, N+2\}$. Since $u_N > 0$, the relations (90), used in decreasing order of N , solve every X_N with $N \in I$ in terms of u_N , γ , and the coordinates X_j with $j \in U \setminus I$. Hence

$$K(X_j : j \in U) \subset K(u_N : N \in I, \gamma, X_j : j \in U \setminus I).$$

The left-hand coordinates are algebraically independent over K , so

$$|U| \leq \text{trdeg}_K K(u_N : N \in I) + 1 + |U| - |I|,$$

which proves (88). If one u_{N_0} is algebraic over K , apply the bound to $I \cup \{N_0\}$ to obtain full independence of every finite set with N_0 omitted.

In that exceptional case (90) expresses γ as a nonconstant K^{alg} -linear form in three algebraically independent Ein-coordinates. Together with $\delta/e = \text{Ein}(1) - \gamma$, where 1 is a fourth distinct algebraic point, two linearly independent coordinate forms are obtained. Thus γ, δ are algebraically independent over K . If instead $\gamma \in K^{\text{alg}}$, the extra “+1” in the preceding transcendence-degree estimate disappears, proving full independence of the u_N .

The asymptotic follows from (51): the first terms in (84) and (85) dominate and

$$\frac{E_1(q^{N+1})}{E_1(q^N)} \sim q^{-1} \exp(-(q-1)q^N).$$

□

10. Relative exclusion and finite trace descent

Put

$$A = \text{Ein}(1), \quad B = \text{Ein}(-1), \quad C = \text{Cin}(1) = \frac{\text{Ein}(i) + \text{Ein}(-i)}{2}.$$

The numbers (A,B,C) are algebraically independent over K , and

$$\begin{aligned} E_1(1) &= A - \gamma, & \text{Ei}(1) &= \gamma - B, \\ \text{Ci}(1) &= \gamma - C, & \text{Chi}(1) &= \gamma - \frac{A+B}{2}, & \delta &= e(A - \gamma). \end{aligned} \quad (91)$$

Theorem 10.1 (Five relative certificate classes). *Among the following five assertions, at most one can hold:*

$$\begin{aligned} \mathcal{C}_0 : & \gamma \in K^{\text{alg}}, \\ \mathcal{C}_1 : & E_1(1) \in K^{\text{alg}} \iff \delta \in K^{\text{alg}}, \\ \mathcal{C}_2 : & \text{Ei}(1) \in K^{\text{alg}}, \\ \mathcal{C}_3 : & \text{Ci}(1) \in K^{\text{alg}}, \\ \mathcal{C}_4 : & \text{Chi}(1) \in K^{\text{alg}}. \end{aligned}$$

Consequently at least four members of

$$\gamma, \quad \delta, \quad E_1(1), \quad \text{Ei}(1), \quad \text{Ci}(1), \quad \text{Chi}(1)$$

are transcendental over K . If the possible exceptional class is not $\mathcal{C}_1$, then at least five are transcendental over K .

Proof. The equivalence in $\mathcal{C}_1$ follows from $\delta = eE_1(1)$ and $e \in K^\times$. The five assertions produce the following normalized certificate vectors, respectively:

$$0, \quad [1], \quad [-1], \quad \frac{[i] + [-i]}{2}, \quad \frac{[1] + [-1]}{2}.$$

For example, if $E_1(1) \in K^{\text{alg}}$, then $\Phi([1]) = \gamma + E_1(1)$; if $\text{Ci}(1) \in K^{\text{alg}}$, then $\Phi(([i] + [-i])/2) = \gamma - \text{Ci}(1)$. These five vectors are distinct, so [theorem 9.3](#) permits at most one assertion. Counting the sizes (1,2,1,1,1) of the five classes gives the final statement. $\square$

Remark 10.2 (The necessary correction). The theorem does *not* say that at most one of the six constants is algebraic over K . The pair $E_1(1), \delta$ forms one relative class: they are simultaneously algebraic or transcendental over K . Over $\overline{\mathbb{Q}}$, the companion paper excludes their simultaneous algebraicity by the additional identity $e = \delta/E_1(1)$ and the transcendence of e .

Corollary 10.3 (Witnesses exclude every relative class). *If one member of the witness set $\mathcal{W}$ in (82) is algebraic, then all six constants displayed in [theorem 10.1](#) are transcendental over K .*

Proof. The normalized vector of an algebraic witness is either $[q_*]$ or a two-point vector with one positive and one negative coefficient as in (83). Its support and sign pattern differ from each of the five vectors in the proof of [theorem 10.1](#). Certificate uniqueness therefore excludes all five relative assertions. $\square$

We next separate finite field membership from countable analytic limits. Let

$$\mathcal{G} = K(\text{Ein}(\alpha) : \alpha \in \overline{\mathbb{Q}}^\times). \quad (92)$$

Every element of $\mathcal{G}$ uses only finitely many algebraic-point values, although the generating family is countable.

Proposition 10.4 (Relative trace descent). *For $m \geq 2$,*

$$S_m = P_m(\gamma^{-1}), \quad \mathcal{G}(S_m) \subset \mathcal{G}(\gamma), \quad [\mathcal{G}(\gamma) : \mathcal{G}(S_m)] \leq m.$$

Consequently

$$\text{trdeg}_{\mathcal{G}} \mathcal{G}(S_m) = \text{trdeg}_{\mathcal{G}} \mathcal{G}(\gamma). \quad (93)$$

Moreover

$$\boxed{\mathcal{G}(S_2, S_3) = \mathcal{G}(S_2, S_4) = \mathcal{G}(\gamma)}, \quad (94)$$

and hence

$$S_2, S_3 \in \mathcal{G} \iff S_2, S_4 \in \mathcal{G} \iff \gamma \in \mathcal{G}. \quad (95)$$

Individually,

$$S_2 \in \mathcal{G} \implies [\mathcal{G}(\gamma) : \mathcal{G}] \leq 2, \quad S_3 \in \mathcal{G} \implies [\mathcal{G}(\gamma) : \mathcal{G}] \leq 3, \quad S_4 \in \mathcal{G} \implies [\mathcal{G}(\gamma) : \mathcal{G}] \leq 4.$$

Proof. The polynomial identity comes from [theorem 5.1](#) at the Euler level. Thus γ^{-1} is algebraic of degree at most m over $\mathcal{G}(S_m)$, proving the first assertions and (93). Equations (5) and (44) prove (94); the remaining statements follow. $\square$

The reverse of the first individual implication for S_2 is false without an additional norm–trace condition.

Theorem 10.5 (Exact quadratic descent criterion). *One has $S_2 \in \mathcal{G}$ if and only if either $\gamma \in \mathcal{G}$, or*

$$[\mathcal{G}(\gamma) : \mathcal{G}] = 2 \quad \text{and} \quad N_{\mathcal{G}(\gamma)/\mathcal{G}}(\gamma) = 2 \operatorname{Tr}_{\mathcal{G}(\gamma)/\mathcal{G}}(\gamma).$$

In the quadratic case the conjugate is

$$\gamma' = -\frac{2\gamma}{2 - \gamma}. \quad (96)$$

Proof. Let $f(x) = x^{-2} - (2x)^{-1}$. If $\gamma \notin \mathcal{G}$ and $f(\gamma) = S_2 \in \mathcal{G}$, then [proposition 10.4](#) gives a quadratic extension. If γ' is its nontrivial conjugate, then

$$f(\gamma) - f(\gamma') = \frac{(\gamma' - \gamma)\{2(\gamma + \gamma') - \gamma\gamma'\}}{2\gamma^2\gamma'^2}.$$

Thus invariance of $f(\gamma)$ is equivalent to $\gamma\gamma' = 2(\gamma + \gamma')$, which is exactly the displayed norm–trace condition and rearranges to (96). Conversely that condition makes $f(\gamma)$ fixed by the quadratic Galois involution and hence an element of $\mathcal{G}$. $\square$

For instance, degree two alone is insufficient: with base field $\mathbb{Q}$ and $x = \sqrt{2}$, one has $f(x) = 1/2 - \sqrt{2}/4 \notin \mathbb{Q}$.

Part II

Finite Faber Monodromy and Barrier Escape

11. Uniform monodromy of polyexponential trace polynomials

11.1. Scope and status

This note proves an all-order structural theorem for the trace polynomials attached to every integer-order polyexponential. The proof has four parts:

1. normalized Faber rigidity turns a polynomial decomposition into a vanishing coefficient of a reciprocal series;
2. a uniform 2-adic argument proves that none of those reciprocal coefficients vanishes;
3. two centered affine invariants exclude the cyclic, dihedral, and three exceptional rational polynomial-monodromy cases in Müller’s classification;
4. the discriminant separates the geometric and arithmetic conclusions.

The result does **not** prove that every trace polynomial is Morse. It proves indecomposability for every pair of indices and reduces its monodromy to the alternating/symmetric dichotomy, with the precise conclusions stated below.

Throughout, $q \geq 1$ and $m \geq 2$ are integers, and

$$E_q(z) := \operatorname{Ein}_q(z) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n! n^q} z^n,$$

$$P_{q,m}(X) := -m[z^m] \log(1 - X E_q(z)).$$

The coefficient formula

$$[X^k] P_{q,m}(X) = \frac{m}{k} [z^m] E_q(z)^k$$

shows that $P_{q,m} \in \mathbf{Q}[X]$, that its constant term is zero, and that it is monic of degree m .

11.2. 1. A general normalized-Faber rigidity lemma

We first work with an arbitrary normalized formal series

$$f(z) = z + \sum_{n \geq 2} a_n z^n$$

over a characteristic-zero field and define

$$P_m^{(f)}(X) := -m[z^m] \log(1 - X f(z)).$$

Put

$$M(w) := \frac{1}{f(1/w)} = w + d_0 + d_1 w^{-1} + d_2 w^{-2} + \dots$$

Let Φ_m be the standard Faber polynomial of M , normalized by

$$\log \frac{M(w) - X}{w} = - \sum_{m \geq 1} \frac{\Phi_m(X)}{m} w^{-m}.$$

11.2.1 Lemma 1 (normalized Faber identity)

For every $m \geq 1$,

$$P_m^{(f)}(X) = \Phi_m(X) - \Phi_m(0).$$

Proof Since $f(1/w) = M(w)^{-1}$,

$$\log(1 - X f(1/w)) = \log \frac{M(w) - X}{w} - \log \frac{M(w)}{w}.$$

Comparing the coefficient of w^{-m} gives the formula. $\square$

11.2.2 Lemma 2 (inner-factor rigidity)

Suppose $m = ab$ with $a, b > 1$. If

$$P_m^{(f)} = G \circ H$$

over a characteristic-zero extension field, then the decomposition may be normalized so that G, H are monic, $G(0) = H(0) = 0$, and under this normalization

$$H = P_b^{(f)}.$$

Proof The Faber identity gives

$$P_m^{(f)}(M(w)) = w^m - \Phi_m(0) + O(w^{-1}).$$

Let Ψ be the formal inverse branch of the monic degree- a polynomial G at infinity. It has a Puiseux expansion

$$\Psi(U) = U^{1/a} + c_0 + c_1 U^{-1/a} + c_2 U^{-2/a} + \dots.$$

Substitution of $U = w^{ab} + O(1)$ shows

$$H(M(w)) = w^b + \text{constant} + O(w^{-1});$$

in particular, there are no powers $w^{b-1}, \dots, w$. The defining uniqueness of the degree- b Faber polynomial implies that H differs from Φ_b by a constant. The normalization $H(0) = 0$ then gives $H = \Phi_b - \Phi_b(0) = P_b^{(f)}$. $\square$

11.2.3 Lemma 3 (reciprocal-coefficient obstruction)

With

$$\frac{1}{f(z)} = z^{-1} + d_0 + d_1 z + d_2 z^2 + \dots,$$

a decomposition of $P_m^{(f)}$ having inner degree b forces

$$d_b = [z^b] \frac{1}{f(z)} = 0.$$

Proof The first negative Faber coefficient is

$$\Phi_b(M(w)) = w^b + b d_b w^{-1} + O(w^{-2}).$$

Here is a formal-residue verification. If L is the compositional inverse of M at infinity, then $\Phi_b(X)$ is the polynomial part of $L(X)^b$. Formal change of variables gives

$$[X^{-1}] L(X)^b = \text{res}_X L(X)^b dX = \text{res}_w w^b M'(w) dw = -b d_b.$$

Substituting $X = M(w)$ into the polynomial/negative-power decomposition of $L(X)^b$ therefore gives the displayed coefficient $+bd_b$. Hence

$$P_b^{(f)}(M(w)) = w^b - \Phi_b(0) + bd_bw^{-1} + O(w^{-2}).$$

In $G(P_b^{(f)}(M(w)))$, the coefficient at

$$w^{m-b-1} = w^{b(a-1)-1}$$

has the unique contribution $abd_b = md_b$ from the leading power of G . Indeed, a constant selection creates deficit b , a negative term w^{-j} creates deficit $b+j$, and there are no terms of degrees $b-1, \dots, 1$; neither another selection in the leading outer power nor a lower outer power can reach deficit $b+1$. On the other hand, $P_m^{(f)}(M(w))$ has no positive power below w^m . Therefore $md_b = 0$, and characteristic zero gives $d_b = 0$. $\square$

11.2.4 Remark (generic jet coordinates)

If $c_k = [X^k]P_m^{(f)}$, then

$$c_k = ma_{m-k+1} + Q_k(a_2, \dots, a_{m-k}).$$

Thus

$$(a_2, \dots, a_m) \longmapsto (c_{m-1}, \dots, c_1)$$

is a triangular polynomial automorphism with diagonal entries m . The specialized polyexponential problem is therefore not caused by a defect in the universal trace construction: the general $(m-1)$ -jet produces the generic monic degree- m , zero-constant polynomial.

11.3. 2. Uniform 2-adic nonvanishing

Write

$$A_q(z) := \frac{E_q(z)}{z} = \sum_{n \geq 0} \frac{(-1)^n}{n!(n+1)^{q+1}} z^n, \quad B_q(z) := \frac{1}{A_q(z)} = \sum_{n \geq 0} B_{q,n} z^n.$$

11.3.1 Theorem 4 (exact reciprocal valuations)

For all $q \geq 1$ and $n \geq 0$,

$$v_2(B_{q,n}) = -(q+1)n.$$

In particular, every $B_{q,n}$ is nonzero.

Proof Let $s_2(n)$ be the sum of the binary digits of n , and put $r = v_2(n+1)$. Legendre's formula $v_2(n!) = n - s_2(n)$ gives

$$v_2\left(\frac{2^{(q+1)n}}{n!(n+1)^{q+1}}\right) = qn + s_2(n) - (q+1)r.$$

This value is zero for $n = 0, 1$. If $n \geq 2$ and $r = 0$, it is positive. If $r \geq 1$, the final r binary digits of n are all one, so $s_2(n) \geq r$, and

$$qn + s_2(n) - (q+1)r = q(n-r) + (s_2(n) - r) > 0.$$

It follows that

$$A_q(2^{q+1}z) \equiv 1 + z \pmod{2}.$$

Taking formal reciprocals in $\mathbf{Z}_2[[z]]$,

$$B_q(2^{q+1}z) \equiv (1+z)^{-1} = \sum_{n \geq 0} z^n \pmod{2}.$$

Thus every $2^{(q+1)n}B_{q,n}$ is a 2-adic unit, which is exactly the claimed valuation. $\square$

Since

$$\frac{1}{E_q(z)} = z^{-1}B_q(z), \quad [z^b]\frac{1}{E_q(z)} = B_{q,b+1},$$

Lemmas 2–3 immediately give the central structural result.

11.3.2 Theorem 5 (all-order indecomposability)

For every $q \geq 1$ and $m \geq 2$, the polynomial $P_{q,m}$ is indecomposable over $\overline{\mathbf{Q}}$, and hence over every characteristic-zero subfield.

11.4. 3. Centered invariants and the cyclic/dihedral cases

For a general series

$$f(z) = z + a_2z^2 + a_3z^3 + a_4z^4 + \cdots,$$

the first upper coefficients of $P_m^{(f)}$ are

$$\begin{aligned} P_m^{(f)}(X) = & X^m + ma_2X^{m-1} + m\left(a_3 + \frac{m-3}{2}a_2^2\right)X^{m-2} \\ & + m\left(a_4 + (m-4)a_2a_3 + \frac{(m-4)(m-5)}{6}a_2^3\right)X^{m-3} + \cdots. \end{aligned}$$

The last displayed term and the corresponding formula below are understood for $m \geq 4$; the Y^{m-2} formula is valid for $m \geq 3$.

Center it by

$$\hat{P}_m(Y) := P_m^{(f)}(Y - a_2) - P_m^{(f)}(-a_2).$$

A direct expansion yields

$$[Y^{m-2}]\hat{P}_m = m(a_3 - a_2^2),$$

$$[Y^{m-3}]\hat{P}_m = m(a_2^3 - 2a_2a_3 + a_4).$$

For $f = E_q$,

$$a_2 = -\frac{1}{2^{q+1}}, \quad a_3 = \frac{1}{2 \cdot 3^{q+1}}, \quad a_4 = -\frac{1}{3 \cdot 2^{2q+3}}.$$

Define

$$N_A(q) := 2^{2q+1} - 3^{q+1}, \quad N_B(q) := 2^{2q+2} - 3^{q+1} - 6^q.$$

Then

$$a_3 - a_2^2 = \frac{N_A(q)}{2^{2q+2}3^{q+1}},$$

$$a_2^3 - 2a_2a_3 + a_4 = \frac{N_B(q)}{2^{3q+3}3^{q+1}}.$$

The first numerator is nonzero by unique factorization, and

$$N_B(q) \equiv 1 \pmod{3}.$$

Thus both centered coefficients are nonzero. A polynomial with cyclic monodromy is linearly equivalent to a power map, whose centered coefficient of degree $m - 2$ vanishes, so the cyclic case is excluded for $m \geq 3$. A polynomial with dihedral monodromy is linearly equivalent to a Chebyshev polynomial, whose centered coefficient of degree $m - 3$ vanishes, so the proper dihedral case is excluded for $m \geq 4$. In degree 3, the dihedral group is $D_3 \cong S_3$, while the cyclic case has already been excluded. Degree 2 is immediate.

11.5. 4. Excluding Müller's exceptional degrees 6, 9, and 10

For a monic centered polynomial

$$F(Y) = Y^m + C_2Y^{m-2} + C_3Y^{m-3} + \cdots, \quad C_2 \neq 0,$$

the quantity

$$\mathcal{I}(F) := \frac{C_3^2}{C_2^3}$$

is invariant under linear equivalence: after monic normalization, input scaling has weights 2 and 3 on C_2, C_3 (with reciprocal weights under the opposite input-scaling convention), and output scaling is fixed by making the polynomial monic. In either convention the displayed ratio is unchanged.

For $P_{q,m}$, the preceding formulas give

$$\mathcal{I}(P_{q,m}) = \frac{R_q}{m}, \quad R_q := 3^{q+1} \frac{N_B(q)^2}{N_A(q)^3}.$$

Müller's three exceptional rational polynomials have the following centered invariants:

degree	representative	$\mathcal{I}$
6	$X^4(X^2 + 6X + 25)$	$18/5$
9	$9X^9 + 108X^7 + 72X^6 + \dots$	$1/27$
10	$(X^2 - 405)^4(X^2 + 50X + 945)$	$-8/81$

For completeness, the centered pairs (C_2, C_3) are respectively

$$(10, -60), \quad (12, 8), \quad (-1800, -24000).$$

Both $N_A(q)$ and $N_B(q)$ are 3-adic units. Moreover $R_1 = -9$, while $R_q > 0$ for $q \geq 2$: indeed $N_A(2) = 5 > 0$, and the ratio of the two exponential terms increases by a factor $4/3$ with q . We can now exclude each row:

- Degree 10 would require $R_q = -80/81$. The sign excludes $q \geq 2$, and $R_1 = -9 \neq -80/81$.
- Degree 9 would require $R_q = 1/3$, impossible because $v_3(R_q) = q + 1$.
- Degree 6 would require $R_q = 108/5$. Comparing 3-adic valuations forces $q = 2$; but $N_A(2) = 5$, $N_B(2) = 1$, and $R_2 = 27/125 \neq 108/5$.

Therefore none of Müller's exceptional rational monodromy polynomials is linearly equivalent to any $P_{q,m}$.

11.6. 5. Monodromy theorem

Let

$$G_{q,m}^{\text{geom}} := \text{Gal}(P_{q,m}(X) - T / \overline{\mathbf{Q}}(T)),$$

$$G_{q,m}^{\text{arith}} := \text{Gal}(P_{q,m}(X) - T / \mathbf{Q}(T)),$$

where Gal denotes the Galois group of the splitting field.

We use the rational-coefficient form of Müller's classification: the geometric monodromy of an indecomposable polynomial in $\mathbf{Q}[X]$ is alternating or symmetric, cyclic, dihedral, or belongs to one of the three explicit rational exceptional cases above. Linear equivalence is taken over $\mathbf{C}$, equivalently here over $\overline{\mathbf{Q}}$. Theorems 5 and Sections 3–4 remove every case except alternating and symmetric.

11.6.1 Theorem 6 (uniform alternating/symmetric reduction)

For all $q \geq 1$ and $m \geq 3$,

$$G_{q,m}^{\text{geom}} \in \{A_m, S_m\}.$$

In fact $G_{q,3}^{\text{geom}} = S_3$ by the degree-3 observation in Section 3. For $m = 2$, both geometric and arithmetic groups are S_2 .

The discriminant is

$$D_{q,m}(T) := \text{Disc}_X(P_{q,m}(X) - T),$$

and, because $P_{q,m}$ is monic of degree m ,

$$\deg_T D_{q,m} = m - 1,$$

$$\text{LC}_T D_{q,m} = (-1)^{m(m-1)/2+m-1} m^m.$$

11.6.2 Theorem 7 (precise unconditional conclusions)

For every $q \geq 1$:

1. If m is even, then

$$G_{q,m}^{\text{geom}} = G_{q,m}^{\text{arith}} = S_m.$$

2. If m is odd and not a perfect square, then

$$G_{q,m}^{\text{arith}} = S_m,$$

while for odd $m \geq 5$ geometrically the present argument leaves $G_{q,m}^{\text{geom}} = A_m$ or S_m . Degree 3 is already S_3 .

3. If m is an odd perfect square, both the arithmetic square-class test and the geometric test require additional information.

Proof For even m , the inertia generator at infinity is an m -cycle and is odd, so the geometric group cannot be A_m . Equivalently, $D_{q,m}$ has odd degree $m - 1$ and cannot be a square. Thus both groups are S_m .

Now let m be odd. The square class of the leading coefficient is

$$(-1)^{(m-1)/2} m.$$

If $m \equiv 3 \pmod{4}$, this is negative and hence not a rational square. If $m \equiv 1 \pmod{4}$ but m is not a square, it is again not a rational square. Therefore $D_{q,m}(T)$ is not a square in $\mathbf{Q}(T)$, so the arithmetic group is not contained in A_m . Since the geometric group is A_m or S_m , the arithmetic group is S_m .

Finally, every odd square is $1 \pmod{8}$, so its displayed leading coefficient is itself a square and $m - 1$ is even. The leading-term test then gives no conclusion. $\square$

11.7. 6. The remaining boundary

The exact unresolved boundary of the uniform proof is therefore:

- **arithmetic:** odd perfect-square degrees $m \geq 9$;
- **geometric:** odd degrees $m \geq 5$, where one must distinguish A_m from S_m .

These are different questions. A nonsquare rational constant multiplying a square polynomial can make the arithmetic group S_m while the geometric group remains A_m . To prove geometric S_m , one must show that $D_{q,m}(T)$ is not of the form

$$cR(T)^2, \quad c \in \mathbf{Q}^\times, R \in \mathbf{Q}[T].$$

For the original order $q = 1$, exact finite-field certificates currently prove

$$G_{1,m}^{\text{geom}} = G_{1,m}^{\text{arith}} = S_m \quad (2 \leq m \leq 401).$$

For odd $57 \leq m \leq 401$, the certificate uses two regular specializations modulo 409 at which the discriminant has opposite quadratic characters; this rules out $cR(T)^2$. The reproducible code and the lower-order exact-Morse certificates are in

- `computation/trace_galois_401.py`,
- `computation/trace_morse_certificates.py`,
- `computation/TRACE_MORSE_EXACT_CERTIFICATES.md`.

This finite verification does not replace an all- m proof and should be reported as a certified range.

11.8. Reference

P. Müller, “Primitive monodromy groups of polynomials,” in *Recent Developments in the Inverse Galois Problem*, Contemporary Mathematics **186** (1995), 385–401.

12. Prime degree: a complete Morse theorem

Write

$$A_q(z) = \frac{E_q(z)}{z} = \sum_{n \geq 0} a_n z^n, \quad a_n = \frac{(-1)^n}{(n+1)!(n+1)^q},$$

and

$$P'_{q,m}(X) = \sum_{j=0}^{m-1} d_j X^j.$$

Direct coefficient extraction gives

$$\boxed{d_j = m[z^{m-j-1}]A_q(z)^{j+1}.} \tag{97}$$

Theorem 12.1 (Prime-degree Morse theorem; **U-PASS**). *Let $m \geq 3$ be prime. For every $q \geq 1$, $P_{q,m}$ is Morse and*

$$\boxed{G_{q,m}^{\text{geom}} = G_{q,m}^{\text{arith}} = S_m.} \tag{98}$$

Proof. Work over $\mathbb{Q}_m$. Formula (97) gives

$$v_m(d_0) = -q, \quad v_m(d_j) \geq 1 \quad (1 \leq j \leq m-2), \quad v_m(d_{m-1}) = 1.$$

Thus the Newton polygon of $P'_{q,m}$ is the single edge

$$(0, -q) \longrightarrow (m-1, 1).$$

Every critical point α_i has valuation

$$v_m(\alpha_i) = -\frac{q+1}{m-1}.$$

After a ramified base extension the edge residual is a binomial $u + vY^{m-1}$. Its derivative is nonzero because $m \nmid m-1$, so its $m-1$ roots are simple and have distinct leading residues y_i .

Let $c_k = [X^k]P_{q,m}$. Then $c_1 = d_0$, $v_m(c_k) \geq 1$ for $2 \leq k \leq m-1$, and $c_m = 1$. At every critical point, α_i^m is the unique term of smallest valuation in $P_{q,m}(\alpha_i)$. Indeed,

$$v_m(c_1\alpha_i) - v_m(\alpha_i^m) = 1,$$

and, for $2 \leq k \leq m-1$,

$$v_m(c_k\alpha_i^k) - v_m(\alpha_i^m) \geq 1 + \frac{(m-k)(q+1)}{m-1} > 0.$$

The leading critical-value residues are therefore y_i^m . Since all y_i^{m-1} equal the same nonzero constant, $y_i^m = Cy_i$, and these residues are pairwise distinct. All critical points are simple and all critical values are distinct, so $P_{q,m}$ is Morse. Its finite branch cycles are transpositions and generate S_m . The arithmetic group contains the geometric one. $\square$

This is an infinite, computation-free theorem: it settles every prime degree simultaneously for every polyexponential order.

13. A moving-prime isolated transposition

Let $m \geq 7$ be odd and choose a prime

$$\frac{m+1}{2} < p < m, \quad r = m - p. \quad (99)$$

Then r is even and $2 \leq r \leq p-2$. The relevant valuations of the coefficients of A_q are

$$v_p(a_n) = \begin{cases} 0, & 0 \leq n \leq p-2, \\ -(q+1), & n = p-1, \\ -1, & p \leq n \leq m-1. \end{cases}$$

Because $m \leq 2p-2$, at most one exceptional factor can occur in any composition contributing to (97). The endpoints $j=1, r$ attain valuation $-q-1$, while every intermediate point lies on or above the horizontal segment. Hence the lower Newton polygon has the three edges

$$(0, -1) \longrightarrow (1, -q-1) \longrightarrow (r, -q-1) \longrightarrow (m-1, 0). \quad (100)$$

The critical points split into one point of valuation q , $r-1$ unit points, and $p-1$ points of valuation $-(q+1)/(p-1)$. The first point α is simple, and its critical value $b = P_{q,m}(\alpha)$ satisfies

$$\boxed{v_p(b) = q-1.} \quad (101)$$

For completeness, the exact valuation requires excluding cancellation between the first two terms. Write $P_{q,m}(X) = \sum_{k \geq 1} c_k X^k$ and $P'_{q,m}(X) = \sum_{j \geq 0} d_j X^j$, so that $c_1 = d_0$ and $c_2 = d_1/2$. The first Newton edge gives

$$d_1 \alpha = -d_0(1 + \varepsilon), \quad v_p(\varepsilon) \geq q.$$

Therefore

$$c_1 \alpha + c_2 \alpha^2 = d_0 \alpha + \frac{d_1}{2} \alpha^2 = \frac{d_0 \alpha}{2} (1 - \varepsilon),$$

which has valuation exactly $q - 1$, since p is odd. For $k \geq 3$,

$$v_p(c_k \alpha^k) \geq (k - 1)q - 1 \geq 2q - 1 > q - 1,$$

so no later term can cancel the leading contribution.

Define the residual lower-row derivative

$$R_{q,r}(X) = P'_{q,r}(X) \bmod p.$$

Keeping the horizontal-edge principal terms gives, up to a nonzero scalar,

$$p^{q+1} P_{q,m}(X) \equiv \frac{X^2}{r} R_{q,r}(X) \pmod{p}, \quad (102)$$

and differentiation gives

$$p^{q+1} P'_{q,m}(X) \equiv \frac{X}{r} (2R_{q,r}(X) + X R'_{q,r}(X)) \pmod{p}. \quad (103)$$

The only potentially dangerous denominator when recovering P from P' is removed by Frobenius:

$$d_{p-1} = m[z^r] A_q(z)^p \equiv m[z^r] A_q(z^p) = 0 \pmod{p} \quad (0 < r < p). \quad (104)$$

Theorem 13.1 (Window-prime criterion; U-PASS). *If*

$$\gcd(P'_{q,r}, P''_{q,r}) = 1 \quad \text{in } \mathbb{F}_p[X], \quad (105)$$

then

$$\boxed{G_{q,m}^{\text{geom}} = G_{q,m}^{\text{arith}} = S_m.} \quad (106)$$

Proof. If a unit critical point β had a critical value of elevated valuation, (102)–(103) would force

$$R_{q,r}(\bar{\beta}) = R'_{q,r}(\bar{\beta}) = 0,$$

contradicting (105). Thus all unit critical values have valuation $-(q + 1)$.

On the final edge the principal derivative is $d_r X^r + m X^{m-1}$. At any of its critical points the principal critical value is

$$d_r X^{r+1} \left(\frac{1}{r+1} - \frac{1}{m} \right) = d_r X^{r+1} \frac{p-1}{m(r+1)},$$

so its valuation is

$$-\frac{m(q+1)}{p-1} < 0.$$

Hence b is the unique critical value of valuation $q - 1$, and it is supported by the single simple point α . Its inertia is a transposition. The pre-existing indecomposability theorem gives primitivity, and a primitive group containing a transposition is S_m . The arithmetic conclusion follows. $\square$

The old degree- m discriminant problem has therefore descended to a gcd for a derivative of degree $r - 1 = m - p - 1$.

Corollary 13.2 (An all-order infinite family; U-PASS). *For every prime $p \geq 5$, every $q \geq 1$, and $m = p + 2$,*

$$\boxed{G_{q,p+2}^{\text{geom}} = G_{q,p+2}^{\text{arith}} = S_{p+2}.} \quad (107)$$

Indeed $r = 2$, so $P'_{q,2} \bmod p$ is linear and square-free. In particular, the former first odd geometric gap $m = 403 = 401 + 2$ and the former first odd-square arithmetic gap $m = 441 = 439 + 2$ are closed for every q , without a full discriminant calculation.

14. Exact finite rectangles

For $p > r$, the reduction of $P_{q,r}$ depends only on $q \bmod (p - 1)$. The criterion can therefore certify infinitely many orders by a finite residue-class computation.

Theorem 14.1 (All orders through degree 1001; X-PASS). *For every $q \geq 1$ and $2 \leq m \leq 1001$,*

$$\boxed{G_{q,m}^{\text{geom}} = G_{q,m}^{\text{arith}} = S_m.} \quad (108)$$

The exact audit finds one window prime working for every residue of q in 492 odd degrees. Prime-degree [theorem 12.1](#) handles $m = 3, 5, 11, 17, 167, 577$. Only two composite degrees require combined certificates:

The finite search policy is itself part of the certificate. The audit is run with `-bound 1001 -candidate-count 10` and examines at most the ten largest eligible strict-window primes in descending order. The displayed prime-degree fallback list is therefore tied to this fixed cutoff and search order, rather than being an intrinsic classification of the degrees.

- At $m = 27$, the primes 17, 19, 23 leave the single progression $q \equiv 1075 \pmod{1584}$. Modulo 28, this progression occupies the seven classes 3, 7, 11, 15, 19, 23, 27. For each class, reduction modulo 29 gives two regular specializations of $\text{Disc}_X(P_{q,27}(X) - T)$ with opposite quadratic character:

$q \bmod 28$	square $(T, D(T))$	nonsquare $(T, D(T))$
3	(3, 7)	(0, 17)
7	(0, 16)	(4, 11)
11	(0, 22)	(5, 27)
15	(2, 16)	(1, 12)
19	(1, 16)	(0, 18)
23	(0, 22)	(1, 15)
27	(0, 22)	(1, 3)

Hence none of the seven discriminant polynomials is a constant times a square, supplying the required odd geometric branch packet.

- At $m = 989$, the bad order residues for $p = 919$ are odd, while those for $p = 929$ are even. Their simultaneous CRT locus is empty.

The script and captured stdout are:

- `trace_allq_modp_audit.py`
SHA-256: 36ba3fd004df1e5a56a7bb6c61ed49cf7542d6f7c934ba458f88c1ebe105c31c;
- `trace_allq_modp_audit_1001.txt`
SHA-256: 1d4867aa086b4315d45d70c52ef27090ad4d1c256cd956cb6b18d72a8be374be.

This theorem asserts full symmetric monodromy. It does *not* assert that every polynomial in this rectangle is Morse.

Theorem 14.2 (Original tower through degree one million; X-PASS). *For every $2 \leq m \leq 10^6$,*

$$\boxed{G_{1,m}^{\text{geom}} = G_{1,m}^{\text{arith}} = S_m.} \quad (109)$$

For each odd $7 \leq m \leq 10^6$, an exact finite-field search tests the largest strict-window prime and, only when necessary, the second largest. The largest residual degree is 114. The largest-prime choice fails only at $(m, p, r) = (215, 211, 4)$ and $(69649, 69623, 26)$; the second prime works in both cases. Even degrees are structural, and prime degrees also follow from [theorem 12.1](#) independently of the audit.

The script and captured stdout are:

- `window_prime_selection_audit.py`
SHA-256: e2eb3b8a14f758d84c0d80a4ed3c0ed11a3cf7eb38e3935960be3122d8fdbc0;
- `window_prime_selection_audit_1000000.txt`
SHA-256: 51d3a027e9f0ce42f4918b39058288cdfdf808e25ded994ed009a701bc7c44f5.

Again, this is a monodromy theorem, not an all-degree Morse theorem.

15. Large order: complex regular polygons replace real critical points

Put

$$G_q(z) = \sum_{n \geq 1} \frac{z^n}{n!n^q}, \quad h_{q;m,k} = [z^m]G_q(z)^k > 0.$$

Then

$$P_{q,m}(X) = \sum_{k=1}^m (-1)^{m-k} \frac{m}{k} h_{q;m,k} X^k. \quad (110)$$

If $n_1 + \cdots + n_k = m$ with every $n_i \geq 1$, the product of the parts satisfies

$$n_1 \cdots n_k \geq m - k + 1. \quad (111)$$

For $2 \leq r \leq m-1$, define

$$\lambda_{m,r} = \frac{m^{(r-1)/(m-1)}}{r}, \quad \theta_m = \max_{2 \leq r \leq m-1} \lambda_{m,r} < 1. \quad (112)$$

The strict inequality follows from the strict decrease of $x \mapsto \log x/(x-1)$ on $(1, \infty)$.

Let

$$s_q = m^{-q/(m-1)}, \quad \tilde{P}_{q,m}(y) = m^{qm/(m-1)} P_{q,m}(s_q y).$$

The endpoint terms $k = 1, m$ remain, while (111)–(112) suppress every intermediate term exponentially.

Theorem 15.1 (Effective large-order Morse theorem; **U-PASS**). *For every fixed $m \geq 2$,*

$$\boxed{\tilde{P}_{q,m}(y) \longrightarrow S_m(y) := y^m + \frac{(-1)^{m-1}}{(m-1)!}y \quad (q \rightarrow \infty)} \quad (113)$$

coefficientwise and locally uniformly, with error $O_{m,R}(\theta_m^q)$ on $|y| \leq R$. Consequently there is an effectively computable $Q(m)$ such that, for every $q \geq Q(m)$,

$$\boxed{P_{q,m} \text{ is Morse,} \quad G_{q,m}^{\text{geom}} = G_{q,m}^{\text{arith}} = S_m.} \quad (114)$$

Proof. The derivative of the limit is

$$S'_m(y) = my^{m-1} + \frac{(-1)^{m-1}}{(m-1)!}.$$

Its roots satisfy

$$y^{m-1} = \frac{(-1)^m}{m!},$$

so they are the distinct vertices of a regular $(m-1)$ -gon. At a critical point y_j ,

$$S_m(y_j) = (-1)^{m-1} \frac{m-1}{m!} y_j,$$

and these critical values are another set of pairwise distinct regular polygon vertices. Thus S_m is Morse. The Morse locus is Zariski open in coefficient space; alternatively, disjoint Rouché circles around these explicit roots and their separated values give an effective bound. The exponential coefficient estimate therefore proves (114). $\square$

For fixed odd m , the limiting critical equation has no real root. Hence sufficiently large q has no real critical point at all. The smallest exact example is $(q, m) = (2, 3)$. A uniform real-transposition strategy is therefore false; complex critical-value separation is the correct replacement.

16. Local Faber transfer beyond the polyexponential tower

The moving-prime mechanism is a finite-jet theorem, not a peculiarity of the differential equation for E_q . Let L be a complete discretely valued characteristic-zero field with residue characteristic $p \geq 5$, uniformizer π , and residue field k . For

$$f(z) = zA(z), \quad A(z) = \sum_{n \geq 0} a_n z^n, \quad a_0 = 1,$$

put $P_m^f = -m[z^m] \log(1 - Xf(z))$. Suppose $m = p + r$, $2 \leq r \leq p - 2$, and, for integers $h > s \geq 0$,

$$\begin{aligned} v(a_n) &\geq 0 & (0 \leq n \leq p-2), \\ v(a_{r-1}) &= 0, \quad v(a_{p-1}) = -h, \\ v(a_n) &\geq -s & (p \leq n \leq m-2), \\ v(a_{m-1}) &= -s. \end{aligned} \quad (115)$$

Let $\bar{A}_{<p}$ be the reduction of the integral truncation below degree $p-1$, and define

$$R_r(X) = \overline{(P_r^f)'(X)} = r \sum_{j=1}^r [z^{r-j}] \bar{A}_{<p}(z)^j X^{j-1}. \quad (116)$$

Theorem 16.1 (Single-spike Faber transfer; **U-PASS**). *Under (115), the derivative Newton polygon is exactly*

$$(0, -s) \longrightarrow (1, -h) \longrightarrow (r, -h) \longrightarrow (m-1, 0). \quad (117)$$

If $\gcd(R_r, R'_r) = 1$ in $k[X]$, then P_m^f has a critical value supported by exactly one simple critical point. The three critical-point clusters and their critical-value valuations are

number	$v(\text{critical point})$	$v(\text{critical value})$
1	$h - s$	$h - 2s$
$r - 1$	0	$-h$
$p - 1$	$-h/(p-1)$	$-mh/(p-1)$.

(118)

Thus its geometric inertia contains a transposition. If the coefficients lie in a number field K , the local hypotheses hold in a completion K_v or a valued extension, and P_m^f is indecomposable over $\overline{K}$, then both its geometric and arithmetic monodromy groups are S_m .

Proof. The universal identities

$$[X^k]P_m^f = \frac{m}{k}[z^{m-k}]A^k, \quad [X^j](P_m^f)' = m[z^{m-j-1}]A^{j+1} \quad (119)$$

show that at most one copy of the unique deep coefficient a_{p-1} can enter. Exact endpoint terms give (117). If $\eta = \overline{\pi^h a_{p-1}}$, the horizontal initial forms are

$$\text{in}_{-h}(P_m^f) = \frac{\bar{m}\eta}{r} X^2 R_r(X), \quad \text{in}_{-h}((P_m^f)') = \frac{\bar{m}\eta}{r} X(2R_r + X R'_r). \quad (120)$$

The apparent coefficient $[X^p]P_m^f = d_{p-1}/p$ is integral because $[z^r]A^p \in p\mathcal{O}_L$ for $0 < r < p$. Consequently the gcd condition prevents cancellation of every unit-cluster critical value. The length-one edge gives one simple point α , and its derivative equation yields $v(P_m^f(\alpha)) = h - 2s$. The other two values in (118) follow from (120) and the final-edge binomial. Since

$$-\frac{mh}{p-1} < -h < h - 2s,$$

the value at α is globally isolated. Primitive polynomial monodromy containing a transposition is S_m . $\square$

For $r = 2$, R_2 is linear, so the gcd condition is automatic. This immediately transfers the local branch construction, for integers $\tau \geq 1$ and $\sigma, \tau \geq 1$, respectively, to

$$\text{Li}_\tau(z) = \sum_{N \geq 1} \frac{z^N}{N^\tau} \quad \text{and} \quad f_{\sigma, \tau}(z) = \sum_{N \geq 1} \frac{z^N}{(N!)^\sigma N^\tau}, \quad (121)$$

with depths $(h, s) = (\tau, 0)$ and $(\sigma + \tau, \sigma)$ over $\mathbb{Q}_p$, respectively. Both families therefore acquire a transposition. The next subsection supplies the missing indecomposability for every pure polylogarithmic member; for $f_{\sigma, \tau}$ it remains a separate gate.

16.1. Global polylogarithmic closure

For an integer $\tau \geq 1$, write

$$L_{\tau, m}(X) := P_m^{\text{Li}_\tau}(X) = -m[z^m] \log(1 - X \text{Li}_\tau(z)), \quad A_\tau(z) := \frac{\text{Li}_\tau(z)}{z}. \quad (122)$$

The missing global input for this family is supplied by a strict form of Kaluza's reciprocal-sign mechanism.

Lemma 16.2 (Strict Kaluza sign for polylogarithms; **U-PASS**). *If*

$$A_\tau(z) = \sum_{n \geq 0} a_n z^n, \quad a_n = (n+1)^{-\tau}, \quad \frac{1}{A_\tau(z)} = \sum_{n \geq 0} b_n z^n,$$

then $b_0 = 1$ and $b_n < 0$ for every $n \geq 1$.

Proof. The positive sequence (a_n) is strictly log-convex, since

$$\rho_n := \frac{a_n}{a_{n-1}} = \left(\frac{n}{n+1} \right)^\tau$$

is strictly increasing. Kaluza's sign criterion [9] gives $b_n \leq 0$ for $n \geq 1$. Strictness can also be read directly from its recurrence. Put

$$\frac{1}{A_\tau(z)} = 1 - \sum_{n \geq 1} c_n z^n.$$

The reciprocal identity gives

$$a_n = \sum_{k=1}^n c_k a_{n-k} \quad (n \geq 1).$$

Thus $c_1 = a_1 > 0$, and comparison with the preceding identity multiplied by ρ_n yields, for $n \geq 2$,

$$c_n = \sum_{k=1}^{n-1} c_k a_{n-1-k} (\rho_n - \rho_{n-k}) > 0. \quad (123)$$

Induction proves $b_n = -c_n < 0$ in every positive degree. $\square$

Theorem 16.3 (Global polylogarithmic Faber theorem; **U-PASS**). *For every integer $\tau \geq 1$ the following hold.*

1. $L_{\tau,m}$ is absolutely indecomposable for every $m \geq 2$.
2. If $\ell \geq 3$ is prime, then $L_{\tau,\ell}$ is Morse and

$$\text{Gal}(L_{\tau,\ell}(X) - T/\overline{\mathbb{Q}}(T)) = \text{Gal}(L_{\tau,\ell}(X) - T/\mathbb{Q}(T)) = S_\ell.$$

3. If $p \geq 5$ is prime, then

$$\text{Gal}(L_{\tau,p+2}(X) - T/\overline{\mathbb{Q}}(T)) = \text{Gal}(L_{\tau,p+2}(X) - T/\mathbb{Q}(T)) = S_{p+2}.$$

4. In the degrees covered by (2) or (3), the off-diagonal collision polynomial

$$\frac{L_{\tau,m}(X) - L_{\tau,m}(Y)}{X - Y}$$

is absolutely irreducible over $\mathbb{Q}$.

Proof. Since

$$\frac{1}{\text{Li}_\tau(z)} = z^{-1} \frac{1}{A_\tau(z)},$$

Lemma 16.2 gives

$$[z^b] \frac{1}{\text{Li}_\tau(z)} = b_{b+1} < 0 \quad (b \geq 0).$$

The reciprocal-coefficient obstruction in Section 11.2.3 says that a decomposition with inner degree $b > 1$ would force this coefficient to vanish. This proves (1).

For (2), work over $\mathbb{Q}_\ell$ and write

$$L'_{\tau,\ell}(X) = \sum_{j=0}^{\ell-1} u_j X^j, \quad u_j = \ell[z^{\ell-j-1}]A_\tau(z)^{j+1}.$$

Then

$$v_\ell(u_0) = 1 - \tau, \quad v_\ell(u_j) \geq 1 \quad (1 \leq j \leq \ell - 2), \quad v_\ell(u_{\ell-1}) = 1.$$

Indeed, the intermediate coefficient extractions involve only denominators $1, \dots, \ell - 1$, which are ℓ -adic units. Hence the derivative Newton polygon is the single edge

$$(0, 1 - \tau) \longrightarrow (\ell - 1, 1).$$

Its residual binomial has $\ell - 1$ distinct roots. The critical points α_i therefore have valuation $-\tau/(\ell - 1)$ and distinct leading residues y_i . If $c_k = [X^k]L_{\tau,\ell}$, then

$$c_1 = u_0, \quad c_k = u_{k-1}/k \quad (2 \leq k \leq \ell - 1), \quad c_\ell = 1.$$

Consequently

$$v_\ell(c_1 \alpha_i) - v_\ell(\alpha_i^\ell) = 1,$$

and, for $2 \leq k \leq \ell - 1$,

$$v_\ell(c_k \alpha_i^k) - v_\ell(\alpha_i^\ell) \geq 1 + \frac{(\ell - k)\tau}{\ell - 1} > 0.$$

Thus α_i^ℓ is the unique leading term of the critical value. Because the edge equation gives $y_i^{\ell-1} = C \neq 0$, the leading critical value residues $y_i^\ell = C y_i$ are pairwise distinct. Hence $L_{\tau,\ell}$ is Morse, its finite branch cycles are transpositions, and its geometric and arithmetic monodromy groups are S_ℓ .

For (3), apply Theorem 16.1 with $r = 2$ and $(h, s) = (\tau, 0)$. The residual polynomial R_2 is linear, and (1) supplies absolute indecomposability, so both monodromy groups are S_{p+2} . Finally, an S_m polynomial cover is two-transitive. The point stabilizer is transitive on the other sheets, which is equivalent to absolute irreducibility of the displayed off-diagonal fiber product. $\square$

Corollary 16.4 (Fixed-order polylogarithmic window; X-PASS). *For every $\tau \in \{1, 2, 3, 4\}$ and every odd $7 \leq m \leq 1001$,*

$$\boxed{\text{Gal}(L_{\tau,m}(X) - T/\overline{\mathbb{Q}}(T)) = \text{Gal}(L_{\tau,m}(X) - T/\mathbb{Q}(T)) = S_m.} \quad (124)$$

Proof. The exact finite-field audit described in Appendix E finds a strict-window prime satisfying $\gcd(R_r, R'_r) = 1$ for every displayed row except $(\tau, m) = (4, 17)$. In each certified row, Theorem 16.1 supplies a transposition and Theorem 16.3(1) supplies primitivity. The sole window exception has prime degree and is therefore covered unconditionally by Theorem 16.3(2). Thus it is a window-method exception, not a monodromy exception. $\square$

The case $\tau = 1$ is $\text{Li}_1(z) = -\log(1 - z)$ and has additional enumerative structure. The independent identity

$$\frac{z}{-\log(1 - z)} = \int_0^1 (1 - z)^t dt$$

also shows directly that every positive-degree reciprocal coefficient is nonzero. For $f(z) = -\log(1-z)$,

$$P_m^f(X) = \sum_{k=1}^m \frac{(k-1)! |s(m, k)|}{(m-1)!} X^k. \quad (125)$$

Thus all the polynomials in (125) are absolutely indecomposable, and Theorem 16.3 gives

$$\boxed{\text{Gal}(P_{p+2}^f(X) - T/\overline{\mathbb{Q}}(T)) = \text{Gal}(P_{p+2}^f(X) - T/\mathbb{Q}(T)) = S_{p+2} \quad (p \geq 5 \text{ prime})} \quad (126)$$

for the logarithmic–Stirling trace tower. No literature-priority claim is made without a dedicated search.

17. Shifted discriminants, partial separability, and conditional density one

For the original order $q = 1$, put

$$F_r(X) = (r!)^2 P'_{1,r}(X) \in \mathbb{Z}[X], \quad \Delta_r = \text{Disc}(F_r). \quad (127)$$

The integrality follows termwise from $\prod n_i! \mid r!$ and $\prod n_i \mid r!$ for every composition $n_1 + \dots + n_k = r$. If $p > r$, the leading coefficient $r(r!)^2$ is a unit modulo p , so

$$\boxed{P'_{1,r} \bmod p \text{ is not square-free} \iff p \mid \Delta_r.} \quad (128)$$

Coefficient majorization by $(e^z - 1)^k$, followed by Mahler's discriminant bound, gives whenever $\Delta_r \neq 0$

$$\log |\Delta_r| = O(r^2 \log r), \quad \omega(\Delta_r) = O(r^2 \log r). \quad (129)$$

Hypothesis DS (OPEN). Every even residual derivative $P'_{1,r}$, $r \geq 2$, is square-free over $\mathbb{Q}$, equivalently $\Delta_r \neq 0$.

Theorem 17.1 (Unconditional partial separability; **U-PASS**). *Let $r = 2s$. For every $q \geq 1$, the derivative $P'_{q,r}$ has at least s simple roots. After the dyadic normalization*

$$c_q = 2^{q+1}, \quad \widehat{P}_{q,r}(X) = c_q^r P_{q,r}(X/c_q),$$

these are exactly the unit roots, while every possible repeated root lies in the strict positive-valuation disk. Moreover, for every odd prime p and every $q \geq 1$,

$$\boxed{P'_{q,p+1}(X) \text{ is square-free over } \mathbb{Q}.} \quad (130)$$

Proof. Put $f_q(z) = E_q(c_q z)/c_q$. Exact coefficient valuations give the order-independent reduction

$$f_q(z) \equiv z + z^2 \pmod{2}.$$

If $a = v_2(r)$, coefficient extraction then yields

$$2^{-a} \widehat{P}'_{q,2s}(X) \equiv X^{s-1} \Lambda_{2s+1}(X) \pmod{2}, \quad (131)$$

$$\Lambda_{2s+1}(X) = \sum_{j=0}^s \binom{s+j}{s-j} X^j. \quad (132)$$

The Lucas recurrence $L_0 = 0, L_1 = X, L_n = X(L_{n-1} + L_{n-2})$ gives $L_{2s+1} = X^{s+1}\Lambda_{2s+1}$. Its nonzero roots are $X = \zeta + \zeta^{-1}$ for nontrivial $(2s+1)$ -st roots of unity, modulo $\zeta \leftrightarrow \zeta^{-1}$. Hence Λ_{2s+1} has s distinct nonzero roots and simple Hensel lifts, proving the first assertion.

For $m = p + 1$, the p -adic Newton polygon of $P'_{q,m}$ is

$$(0, -1) \longrightarrow (1, -q - 1) \longrightarrow (p, 0). \quad (133)$$

The first edge is linear; after adjoining a uniformizer for the possibly fractional second slope, its nonzero-root initial binomial has exponent gap $p - 1$, prime to p . Thus all roots are simple. $\square$

Corollary 17.2 (Exact residual rectangle; X-PASS). *For the original order, exact finite-field computation proves*

$$\boxed{P'_{1,r} \text{ is square-free for every even } 2 \leq r \leq 5000.} \quad (134)$$

At the single good prime 998244353, exact convolution and polynomial gcd arithmetic certify (134). Leading coefficients stay nonzero modulo this prime; for odd $3 \leq s \leq 4999$, those of the polar and its derivative are $s + 1$ and $(s - 1)(s + 1)$, respectively. Thus the relevant degrees are preserved. The same run, together with the trivial constant case $s = 1$, also certifies every odd polar $2H_s + XH'_s$ for $1 \leq s \leq 4999$, where $H_s = \sum_k [z^s](-E_1(-z))^k X^{k-1}$.

The exact audit has SHA-256 hashes

- `even_derivative_separability_audit.cpp`
SHA-256: d9c6f7c6438a9e7bf5d0aed50bbce727f460efcacc720f520c97b444b3f83b4d;
- `even_derivative_separability_audit_5000.txt`
SHA-256: 1e4b4073527d7d2543c8a087c37f0888fdbd99c7ca2b7880a418e7c533e6617b.

This does not prove DS: in row $r = 10$, a later inner Newton face already has inseparable residual polynomial $1 + Y^2$ over $\mathbb{F}_2$.

Theorem 17.3 (Conditional density-one selection; C-PASS). *Assume DS. Let $\mathcal{E}(X)$ be the odd $m \leq X$ for which no prime $(m + 1)/2 < p < m$ passes the residual gcd criterion (105). Then, for every $\varepsilon > 0$,*

$$\boxed{\#\mathcal{E}(X) \ll_\varepsilon X^{15/16+\varepsilon}.} \quad (135)$$

In particular, under DS the original trace tower has symmetric geometric and arithmetic monodromy for a density-one set of odd degrees.

Proof. In a dyadic interval $[X, 2X]$, let $p < m$ be the largest prime and write $r = m - p$. Set $H = X^{5/16}$. If $r \leq H$ and m is exceptional, then (128)–(129) give

$$\#\{m : r \leq H, p \mid \Delta_r\} \ll \sum_{r \leq H} r^2 \log r \ll H^3 \log H.$$

The remaining integers lie more than H places into a prime gap. Maynard's mean-square prime-gap theorem [14, Theorem 3.1]

$$\sum_{p_n \leq x} (p_{n+1} - p_n)^2 \ll_\varepsilon x^{5/4+\varepsilon}$$

implies that their number is $O_\varepsilon(X^{5/4+\varepsilon}/H)$. Balancing the two estimates gives (135); dyadic summation completes the proof. $\square$

The exact equivalence (128) also explains why prime-gap bounds alone do not prove selection in every degree: one must exclude the correlated divisibilities $m - r \mid \Delta_r$ for several offsets. Already at $m = 215$, the largest window prime 211 is bad in row $r = 4$, although the next prime works. Thus a canonical nearest-prime proof is false even though the selectable-prime criterion survives.

18. Exact negative results that prevent a false proof

The failed routes are useful because they make the remaining gap precise.

1. A simple real critical point does not suffice: another critical point can share its critical value. For $q = 1$, every odd degree has a simple positive critical point, but global value separation remains necessary.
2. The natural coefficient triangle is not totally positive. At $q = 1$ it already has an exact negative 9×9 minor. A proof cannot cite global TP_∞ .
3. The dyadic derivative analysis isolates a simple outer critical cluster, but coefficient Newton polygons alone do not exclude an exact collision with a shallower cluster.
4. The moving-prime criterion is sufficient, not automatic. For odd composite m , Bertrand applied to $(m + 1)/2$ supplies a prime $(m + 1)/2 < p < m$; compositeness excludes the only possible upper endpoint $p = m$. Bertrand says nothing, however, about residual square-freeness for an eligible prime.

19. The remaining theorem gate

The cleanest unresolved statement is now finite-field rather than transcendental.

Problem 19.1 (OPEN). Prove that for every $q \geq 1$ and every sufficiently large odd composite m , at least one prime $p = p(q, m)$

$$\frac{m+1}{2} < p < m$$

makes

$$\gcd(P'_{q,m-p}, P''_{q,m-p}) = 1 \pmod{p}.$$

For the original order $q = 1$, exact computation verifies this through $m = 10^6$, always using one of the two largest eligible primes. Under DS, theorem 17.3 proves it for a density-one set, but no argument currently promotes that finite fact to all m . A proof would close the remaining odd composite degree problem by theorem 13.1.

An alternative is to continue the polar expansion at a multiple residual root and prove that its lifted critical value cannot meet the isolated value of (101). This is the precise local collision problem left by the present work.

20. Logarithmic–Stirling and supernecklace applications

The single-spike theorem above transfers an isolated local transposition to the polylogarithmic and factorially weighted families in (121). The two scopes now diverge. The strict Kaluza argument in Theorem 16.3 proves absolute indecomposability for every pure polylogarithmic member and yields global symmetric families. For the factorially weighted $f_{\sigma,\tau}$ family the transfer remains local: without a separate indecomposability theorem it does not by itself

imply full symmetric monodromy. The logarithmic member Li_1 has the additional enumerative refinement developed below.

Put

$$C_m(X) := (m-1)!P_m^{(-\log(1-z))}(X) = \sum_{k=1}^m (k-1)! \begin{bmatrix} m \\ k \end{bmatrix} X^k, \quad (\text{SN-1})$$

where $\begin{bmatrix} m \\ k \end{bmatrix}$ is the unsigned Stirling number of the first kind.

Proposition 20.1 (Type-III supernecklace enumerator; **U-PASS**). *The exponential generating function of the polynomials C_m is*

$$\sum_{m \geq 1} C_m(X) \frac{z^m}{m!} = \log \left(\frac{1}{1 - X \log \frac{1}{1-z}} \right). \quad (\text{SN-2})$$

Consequently C_m is the cycle-number enumerator for a labelled cycle of labelled cycles (the type-III supernecklace construction).

Proof. Expand the outer logarithm and use the classical Stirling exponential generating function:

$$\log \left(\frac{1}{1 - XL(z)} \right) = \sum_{k \geq 1} \frac{X^k}{k} L(z)^k, \quad L(z)^k = k! \sum_{m \geq k} \begin{bmatrix} m \\ k \end{bmatrix} \frac{z^m}{m!},$$

where $L(z) = \log(1/(1-z))$. Coefficient comparison gives (SN-1). The labelled product/cycle interpretation is precisely the standard exponential-formula reading of the same identity. $\square$

Theorem 20.2 (Symmetric supernecklace family; **U-PASS**). *For every prime $p \geq 5$, with $m = p + 2$,*

$$\text{Gal}(C_m(X) - T/\overline{\mathbb{Q}}(T)) = \text{Gal}(C_m(X) - T/\mathbb{Q}(T)) = S_m. \quad (\text{SN-3})$$

Moreover the off-diagonal collision polynomial

$$\boxed{\frac{C_m(X) - C_m(Y)}{X - Y}} \quad (\text{SN-4})$$

is absolutely irreducible over $\mathbb{Q}$.

Proof. Scaling a polynomial by the nonzero constant $(m-1)!$ does not change its geometric or arithmetic monodromy. Equation (126) therefore gives (SN-3). For a polynomial cover, the irreducible factors of the off-diagonal fiber product correspond to the orbits of a point stabilizer on the remaining sheets. The natural action of S_m is two-transitive, so that stabilizer has one such orbit. This proves absolute irreducibility. $\square$

Remark 20.3 (Boundary of the application). Qualitative Hilbert irreducibility supplies many specializations with group S_m , but no quantitative exceptional-set exponent is asserted here. Nor is a genus formula claimed for the collision curve: symmetric monodromy gives absolute irreducibility, while smoothness and the precise ramification divisor require a separate Morse analysis.

Part III

Maximal Inverse Monodromy and Global Trace Collapse

21. All-order inverse monodromy for the polyexponential tower

Uniform incomplete-gamma remainders, contraction boxes, and the covering-space argument are included so that the proof is logically self-contained apart from the quoted standard large-variable remainder theorem. In particular, **global pairwise noncollision of all critical values is neither assumed nor claimed**.

Put

$$F_q(z) = \text{Ein}_q(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n! n^q}, \quad q \geq 1,$$

and put $F_0(z) = 1 - e^{-z}$. Then

$$zF_q'(z) = F_{q-1}(z). \quad (\text{Inv-0.1})$$

The principal result is

Theorem A (all-order maximal inverse monodromy). For every fixed integer $q \geq 1$ and every regular value c_0 (that is, c_0 is not a critical value),

$$\text{Mon}_{c_0}(F_q^{-1}) = \text{FSym}(F_q^{-1}(c_0)) \cong \text{FSym}(\mathbb{N}). \quad (\text{Inv-0.2})$$

For $q = 1$, all critical points are simple and all critical values are pairwise distinct. For $q \geq 2$, those two global statements are not needed: the proof shows that degeneracies and critical-value collisions, if any, form only a finite exceptional set. A finite-exception permutation lemma then upgrades the tail transpositions to the full finitary symmetric group.

21.1. 1. Polynomial plus exponential decomposition

For small u ,

$$\begin{aligned} \Phi(u, z) &:= \sum_{q \geq 1} F_q(z) u^{q-1} \\ &= \frac{z^u \Gamma(1-u) - 1}{u} + z^u \Gamma(-u, z). \end{aligned} \quad (\text{Inv-1.1})$$

Write $w = \text{Log } z + \gamma$ on the principal sheet and define

$$Q_q(w) = [u^q] \exp \left(wu + \sum_{m \geq 2} \frac{\zeta(m)}{m} u^m \right). \quad (\text{Inv-1.2})$$

Thus

$$Q_0 = 1, \quad Q'_q = Q_{q-1}, \quad Q_q(w) = \frac{w^q}{q!} + O_q(|w|^{q-2}), \quad (\text{Inv-1.3})$$

with the obvious interpretation for $q = 1$, and

$$F_q(z) = Q_q(\text{Log } z + \gamma) + R_q(z), \quad R_q(z) = [u^{q-1}]z^u \Gamma(-u, z). \quad (\text{Inv-1.4})$$

The upper-incomplete-gamma expansion, uniformly for $|\arg z| \leq \pi$ with upper and lower boundary values on the cut, gives

$$z^u \Gamma(-u, z) = \frac{e^{-z}}{z} \left(\sum_{k=0}^{N-1} \frac{(-1)^k (1+u)_k}{z^k} + O_N(|z|^{-N}) \right). \quad (\text{Inv-1.5})$$

Here and below the required uniformity has the following precise meaning. Fix $0 < \eta < 1$. The standard large-variable remainder theorem for the upper incomplete gamma function, applied with $a = -u$ in the compact set $|u| \leq \eta$, says that for every $N \geq 1$

$$z^u \Gamma(-u, z) = \frac{e^{-z}}{z} \left(\sum_{k=0}^{N-1} \frac{(-1)^k (1+u)_k}{z^k} + E_N(u, z) \right), \quad \sup_{|u| \leq \eta} |E_N(u, z)| \leq C_{N,\eta} |z|^{-N}. \quad (\text{Inv-1.5a})$$

This holds on $|\arg z| \leq \pi$, with the corresponding boundary branch on either side of the negative axis. It is the compact-parameter form of the usual fixed-parameter expansion (for example, NIST DLMF, §8.11(i)); it follows from the same contour-integral remainder after the parameter is restricted to a compact set. The remainder is holomorphic in u . Consequently Cauchy's formula on $|u| = \eta$ gives, for every fixed j ,

$$|[u^j]E_N(u, z)| \leq C_{N,\eta} \eta^{-j} |z|^{-N}. \quad (\text{Inv-1.5b})$$

On each closed subsector strictly inside $|\arg z| < 3\pi/2$, the same remainder theorem may be differentiated in z , and, after enlarging the constant,

$$\sup_{|u| \leq \eta} |\partial_z E_N(u, z)| \leq C_{N,\eta} |z|^{-N-1}. \quad (\text{Inv-1.5c})$$

For completeness, (1.5c) also follows from (1.5a) without a separate differentiated theorem: put the chosen closed subsector inside a slightly larger one and apply Cauchy's derivative estimate on a circle of radius $\varepsilon|z|$. In particular it holds in the upper and lower sectors about the imaginary axes used in Section 3. Cauchy's formula in u applies to (1.5c) as well. These uniform statements justify both coefficient extraction and the derivative estimate for the contraction maps below.

Coefficient extraction yields

$$R_q(z) = (-1)^{q-1} \frac{e^{-z}}{z^q} \left(1 - \frac{q(q+1)}{2z} + O_q(z^{-2}) \right). \quad (\text{Inv-1.6})$$

Indeed, $(1+u)_k$ has degree k ; its coefficient of u^{q-1} first occurs at $k = q-1$, where it equals 1, and at $k = q$ that coefficient is $1 + \dots + q = q(q+1)/2$. Taking $N \geq q+1$ in (1.5a)–(1.5b) proves (1.6), including its uniform error. For later uniform notation set

$$Q_0 = 1, \quad R_0(z) = -e^{-z}, \quad F_0 = Q_0 + R_0. \quad (\text{Inv-1.7})$$

Formula (1.6), together with (1.7), is the analytic engine for everything that follows.

21.2. 2. No finite asymptotic values in any order

Lemma 2.1. For every $q \geq 1$ and $M > 0$, every connected component of

$$\{z : |F_q(z)| < M\}$$

is bounded. Consequently F_q has no finite asymptotic value.

Let $z = x + iy$, $r = |z|$, $L = \log r$, $\theta = \arg z \in [-\pi, \pi]$, and put $w = \text{Log } z + \gamma$. Uniformly as $r \rightarrow \infty$,

$$Q_q(w) = \frac{w^q}{q!} \left(1 + O_q(L^{-2})\right), \quad R_q(z) = (-1)^{q-1} e^{-z} z^{-q} \left(1 + O_q(r^{-1})\right). \quad (\text{Inv-2.0})$$

If $|F_q(z)| \leq M$, then $|Q_q(w)| \asymp_q L^q$, and (1.4) gives

$$\frac{R_q(z)}{Q_q(w)} = -1 + O_{q,M}(L^{-q}). \quad (\text{Inv-2.0a})$$

Taking logarithmic moduli in (2.0)–(2.0a), and using $\log |w| = \log L + O(L^{-1})$, gives the following estimate uniformly on the bounded sublevel, not merely along a selected sequence:

$$x = -qL - q \log L + \log(q!) + O_{q,M}(L^{-1}). \quad (\text{Inv-2.1})$$

Hence $|y| \sim r$ and $\arg z = \text{sgn}(y)\pi/2 + O_q(L/r)$. Taking arguments in the same cancellation gives

$$(q-1)\pi - y - q\theta \equiv \pi + \arg Q_q(w) + O_{q,M}(L^{-q}) \pmod{2\pi}.$$

Since $\arg Q_q(w) = q \arg w + O_q(L^{-2})$ and $\arg w = \theta/L + O(L^{-2})$, it follows, separately in the upper and lower half-planes, that

$$\text{dist}\left(y, 2\pi\mathbb{Z} + \text{sgn}(y)\frac{q\pi}{2}\right) \leq \frac{C_{q,M}}{L}. \quad (\text{Inv-2.2})$$

To justify explicitly the word “uniformly”, if either (2.1) or (2.2) failed outside every disk, a sequence of counterexamples tending to infinity would contradict (2.0)–(2.0a). Choose $R = R(q, M)$ so that outside $\overline{D(0, R)}$ the sublevel is contained in the horizontal strips of half-width $\pi/4$ centered at the lattices in (2.2). These strips are pairwise disjoint after coincident strips are identified (the upper and lower lattices either agree or differ by π).

The intersection of the sublevel with any one fixed strip is bounded. Indeed, otherwise it would contain a sequence with bounded y and $|x| \rightarrow \infty$, whereas (2.1) says that x has only logarithmic size in $|x|$, an impossibility. Components of an open subset of $\mathbb{C}$ are path-connected. A component avoiding $\overline{D(0, R)}$ is therefore contained in one strip and is bounded. If a component meets the disk, then every remote strip used by it meets $\partial D(0, R)$: follow a path from the disk to a point in that strip and take its last exit from the disk. Only finitely

many of the strips meet that circle, and the sublevel part in each of them is bounded. The component is again bounded.

If $F_q(\gamma(t)) \rightarrow a$ along a curve $\gamma(t) \rightarrow \infty$, then for some $M > |a|$ a tail of γ is a connected unbounded subset of $\{|F_q| < M\}$, hence lies in an unbounded component. This contradiction proves the last assertion of the lemma.

Also, every F_q is surjective. Since $F_0(x)$ has the sign of x , (0.1) and induction show that $F'_q(x) > 0$ for real x . The positive and negative real-axis versions of (1.4)–(1.6) give respectively $F_q(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $F_q(x) \rightarrow -\infty$ as $x \rightarrow -\infty$. Thus no real value is omitted. The function is transcendental entire and has real Taylor coefficients, so a nonreal omitted value would force its distinct conjugate to be omitted as well, contrary to Picard's theorem.

21.3. 3. Large zeros and eventual separation of critical values

The following is the required Hardy-type zero lemma. Recall that $F_p(z) = ze(-z, 1 \mid p+1)$ in Hardy's polyexponential notation, so its leading zero distribution goes back to Hardy. The derivative and critical-value consequences below are what are needed for monodromy.

Lemma 3.1 (zero boxes). Fix $p \geq 1$. Apart from finitely many zeros, the nonzero zeros of F_p are conjugate pairs $\zeta_{p,n}, \overline{\zeta_{p,n}}$, $n \geq n_0(p)$, with exactly one upper zero in every corresponding large phase box. There is a C^1 (indeed real-analytic) parametrized fixed point $\zeta_p(t)$, $t \geq t_0$, whose integer samples are those zeros and for which

$$\begin{aligned}\zeta_p(t) &= 2\pi it - p \log(2\pi t) - p \log \log(2\pi t) + O_p(1), \\ \zeta'_p(t) &= 2\pi i + O_p(t^{-1}), \\ \text{Log } \zeta_{p,n+1} - \text{Log } \zeta_{p,n} &= \frac{1}{n} + O_p\left(\frac{\log n}{n^2}\right).\end{aligned}\tag{Inv-3.1}$$

Every sufficiently large zero is simple.

Here is a direct proof with explicit boxes. Put

$$\mathcal{B}_p(z) = (-1)^{p-1} e^z z^p R_p(z) = 1 - \frac{p(p+1)}{2z} + O_p(z^{-2}).$$

Fix a large positive integer n , put $z_n^0 = 2\pi i n$ and $L_n = \log(2\pi n)$, and initially work on a rough disk

$$E_{p,n} = \{z : |z - z_n^0| \leq C_p \log n\},\tag{Inv-3.2a}$$

where C_p is a sufficiently large fixed constant. On this disk the principal $\text{Log } z$ is defined and

$$w = \text{Log } z + \gamma = L_n + \gamma + \frac{\pi i}{2} + O_p(\log n/n).$$

Thus $Q_p(w) \neq 0$, and we choose its logarithm by

$$\log Q_p(w) = p \text{Log } w - \log(p!) + O_p(w^{-2}).$$

Also $\mathcal{B}_p(z) \neq 0$, and we choose $\log \mathcal{B}_p(z) \rightarrow 0$. The differentiated uniform remainder in Section 1 gives

$$\mathcal{B}'_p(z) = O_p(n^{-2}) \quad (z \in E_{p,n}). \quad (\text{Inv-3.2b})$$

At a zero of F_p , exponentiation gives $e^z z^p Q_p(w) = (-1)^p \mathcal{B}_p(z)$. With $\log((-1)^p) = i\pi p$, the corresponding fixed-point equation is

$$z = T_{p,n}(z) := 2\pi i n + i\pi p - p \operatorname{Log} z - \log Q_p(\operatorname{Log} z + \gamma) + \log \mathcal{B}_p(z). \quad (\text{Inv-3.2c})$$

Using $-i\pi p$ instead would merely replace n by $n + p$; (3.2c) fixes the phase index once and for all. Define

$$c_{p,n} = T_{p,n}(z_n^0), \quad D_{p,n} = \{z : |z - c_{p,n}| \leq 1\}. \quad (\text{Inv-3.2d})$$

The selected logarithms give

$$\begin{aligned} \Re c_{p,n} &= -pL_n - p \log L_n + \log(p!) + O_p(L_n^{-1}), \\ \Im c_{p,n} &= 2\pi n + \frac{p\pi}{2} - \frac{p\pi}{2L_n} + O_p(L_n^{-2}). \end{aligned} \quad (\text{Inv-3.2e})$$

After increasing C_p , the fixed-radius disk $D_{p,n}$ lies in $E_{p,n}$ for every sufficiently large n . On the rough disk,

$$T'_{p,n}(z) = -\frac{p}{z} - \frac{Q_{p-1}(w)}{zQ_p(w)} + \frac{\mathcal{B}'_p(z)}{\mathcal{B}_p(z)} = O_p(n^{-1}). \quad (\text{Inv-3.2f})$$

Consequently, for $z \in D_{p,n}$, the line segment from z_n^0 to z lies in $E_{p,n}$ and

$$|T_{p,n}(z) - c_{p,n}| \leq \frac{C_p}{n} |z - z_n^0| \leq \frac{C_p(1 + \log n)}{n} < 1,$$

while $\sup_{D_{p,n}} |T'_{p,n}| < 1/2$. Thus $T_{p,n}$ has exactly one fixed point in $D_{p,n}$, and exponentiating (3.2c) shows that it is a zero of F_p . The centers of consecutive disks differ by $2\pi i + o(1)$, so these disks are pairwise disjoint for large n .

Conversely, (2.1)–(2.2), applied to the zeros with $M = 1$ and $q = p$, show that every sufficiently large upper zero has a unique integer n for which

$$z = z_n^0 + O_p(\log n), \quad \Im z = 2\pi n + \frac{p\pi}{2} + O_p(L_n^{-1}).$$

With the logarithms selected above it satisfies precisely (3.2c), so (3.2f) gives

$$|z - c_{p,n}| = |T_{p,n}(z) - T_{p,n}(z_n^0)| = O_p(\log n/n) < 1.$$

It is therefore the already constructed fixed point. This proves both existence in every large phase box and exhaustion of all large upper zeros; conjugation gives the lower zeros.

For real t sufficiently large, replace n by t throughout the same construction. The estimates are locally uniform in t , and $1 - T'_{p,t}(\zeta_p(t)) \neq 0$. The real implicit-function theorem gives a local C^1 , indeed real-analytic, fixed point; uniqueness patches these local functions into a single $\zeta_p(t)$. Differentiating (3.2c) yields

$$\zeta_p'(t) = \frac{2\pi i}{1 - T_{p,t}'(\zeta_p(t))} = 2\pi i + O_p(t^{-1}).$$

Equation (3.2e) gives the first line of (3.1), and hence

$$\begin{aligned} \text{Log } \zeta_p(n+1) - \text{Log } \zeta_p(n) &= \int_n^{n+1} \frac{\zeta_p'(t)}{\zeta_p(t)} dt \\ &= \int_n^{n+1} \left(\frac{1}{t} + O_p\left(\frac{\log t}{t^2}\right) \right) dt = \frac{1}{n} + O_p\left(\frac{\log n}{n^2}\right), \end{aligned}$$

which completes (3.1).

Finally, consecutive-remainder division, with (1.7) when $p = 1$, gives

$$\frac{R_{p-1}(z)}{R_p(z)} = -z \left(1 + \frac{p}{z} + O_p(z^{-2}) \right). \quad (\text{Inv-3.2g})$$

At a large zero of F_p , $R_p = -Q_p$, and therefore

$$\begin{aligned} F_{p-1}(z) &= Q_{p-1}(w) + R_{p-1}(z) \\ &= zQ_p(w) \left(1 + O_p(n^{-1}) + O_p((n \log n)^{-1}) \right) = zQ_p(w)(1 + o(1)). \end{aligned} \quad (\text{Inv-3.3})$$

Thus $F_p'(z) = F_{p-1}(z)/z \neq 0$ for every sufficiently large zero, including $p = 1$. Lemma 3.1 follows.

Now let $q \geq 2$, put $p = q - 1$, and define the upper critical values

$$b_{q,n} = F_q(\zeta_{p,n}).$$

At a zero of F_p , the quotient of two consecutive remainders in (1.6) is

$$\frac{R_{p+1}(z)}{R_p(z)} = -\frac{1}{z} \left(1 - \frac{p+1}{z} + O_p(z^{-2}) \right).$$

Since $R_p = -Q_p$ there,

$$b_{q,n} = Q_{p+1}(w_n) + \frac{Q_p(w_n)}{\zeta_{p,n}} + O_p\left(\frac{(\log n)^p}{n^2}\right), \quad w_n = \text{Log } \zeta_{p,n} + \gamma. \quad (\text{Inv-3.4})$$

Using $Q_{p+1}' = Q_p$ and (3.1),

$$\delta w_n := w_{n+1} - w_n = \frac{1}{n} + O_p\left(\frac{\log n}{n^2}\right).$$

Taylor's formula, uniformly for w on the segment between w_n and w_{n+1} , gives

$$Q_{p+1}(w_{n+1}) - Q_{p+1}(w_n) = Q_p(w_n)\delta w_n + O_p((\log n)^{p-1}n^{-2}).$$

Moreover

$$\frac{Q_p(w_{n+1})}{\zeta_{p,n+1}} - \frac{Q_p(w_n)}{\zeta_{p,n}} = O_p((\log n)^p n^{-2}).$$

The difference of the two remainders in (3.4) is bounded by the sum of their absolute bounds. Consequently

$$b_{q,n+1} - b_{q,n} = \frac{Q_p(w_n)}{n} + O_p\left(\frac{(\log n)^{p+1}}{n^2}\right). \quad (\text{Inv-3.5})$$

Since

$$w_n = \log(2\pi n) + \gamma + \frac{\pi i}{2} + O_p\left(\frac{\log n}{n}\right),$$

equation (3.5) gives

$$\begin{aligned} \Re(b_{q,n+1} - b_{q,n}) &= \frac{(\log(2\pi n))^p}{p! n} (1 + o(1)) > 0, \\ \Im(b_{q,n+1} - b_{q,n}) &= \frac{\pi(\log(2\pi n))^{p-1}}{2(p-1)! n} (1 + o(1)) > 0. \end{aligned} \quad (\text{Inv-3.6})$$

The same formulas also give the individual-value asymptotics

$$\begin{aligned} \Re b_{q,n} &= \frac{(\log(2\pi n))^{p+1}}{(p+1)!} (1 + o(1)), \\ \Im b_{q,n} &= \frac{\pi(\log(2\pi n))^p}{2p!} (1 + o(1)) > 0. \end{aligned} \quad (\text{Inv-3.7})$$

In particular, $|b_{q,n}| \rightarrow \infty$, and the lower critical values are their conjugates. We now separate all possible tail collisions. For two upper indices $m > n$, summing the eventually positive real increments in (3.6) gives $\Re b_{q,m} > \Re b_{q,n}$. Upper and lower tail values cannot coincide because (3.7) puts them in opposite half-planes. Finally, the critical points not belonging to the two tails form a finite set, whereas the tail values tend to infinity; after increasing the tail threshold, no tail value equals a value coming from that finite set. We have proved:

Corollary 3.2. For every fixed $q \geq 2$, all but finitely many critical points of F_q are simple, all but finitely many critical values arise from exactly one critical point, and distinct such tail critical points have distinct critical values. The complete critical value set is closed and discrete. Every critical value has only finitely many critical points above it.

Indeed, the critical points of F_q are precisely the nonzero zeros of F_{q-1} . Lemma 3.1 leaves only finitely many of them outside the two tails. Thus the full critical-value set is a finite set together with two sequences escaping to infinity, which proves closedness and discreteness. For any fixed critical value, (3.7) permits only finitely many tail preimages, and there are only finitely many exceptional critical points.

For completeness, the order $q = 1$ calculation used below is global. The critical points are

$$\text{Crit}(F_1) = \{2\pi i k : k \in \mathbb{Z} \setminus \{0\}\},$$

and they are simple because $F_1''(2\pi ik) = F_1'(2\pi ik)/(2\pi ik) = 1/(2\pi ik) \neq 0$. For $k > 0$,

$$F_1(2\pi ik) = \text{Cin}(2\pi k) + i \text{Si}(2\pi k). \quad (\text{Inv-3.8})$$

The function $\text{Cin}(x) = \int_0^x (1 - \cos t)t^{-1} dt$ is strictly increasing at the points $2\pi k$. Also $\text{Si}(2\pi k) > 0$: pair the positive half-wave $[2j\pi, (2j+1)\pi]$ with the following negative half-wave and use the strict decrease of $1/t$. Hence the upper values in (3.8) are pairwise distinct and nonreal, and conjugation separates them from all lower values. Finally (1.4)–(1.6) on the positive imaginary axis give $\text{Cin}(2\pi k) = \log(2\pi k) + \gamma + O(k^{-1})$, so this critical-value set is closed and discrete, and every value has exactly one critical point above it.

For $q = 2$, in fact every critical point is simple: a common nonzero zero of F_1 and F_1' would have to be $2\pi ik$, but $F_1(2\pi ik) = \text{Cin}(2\pi k) + i \text{Si}(2\pi k) \neq 0$. Global noncollision for F_2 remains a finite, numerically verified but unproved problem. It is not required in Theorem A.

21.4. 4. The finite-exception permutation lemma

Lemma 4.1. Let Ω be countably infinite and let $G \leq \text{FSym}(\Omega)$ be transitive. Suppose G is generated by a family of transpositions together with finitely many additional finitary permutations. Then $G = \text{FSym}(\Omega)$.

Let H be the group generated by the transpositions and let its orbits be the connected components of the associated edge graph. On each component, H contains the full finitary symmetric group. If H is transitive, then already $H = \text{FSym}(\Omega)$ and there is nothing to prove. Assume henceforth that it is not transitive. Every H -orbit must meet the finite union of the supports of the exceptional generators; otherwise that proper orbit is G -invariant. Thus there are finitely many H -orbits, and at least one is infinite.

Start from an infinite orbit U , on which G contains $\text{FSym}(U)$. If an exceptional generator g sends $a \in U$ to $b \notin U$, choose $x \in U$ outside the finite support of g . Then

$$g(a x)g^{-1} = (b x) \in G.$$

This cross-transposition adjoins the whole H -orbit of b to U . More explicitly, form a finite incidence graph whose vertices are the H -orbits and in which two vertices are joined when some exceptional generator sends a point of one to a point of the other. This graph is connected: a union of its connected components would otherwise give a nontrivial G -invariant union of H -orbits, contrary to transitivity. Following a spanning tree from the infinite orbit and repeating the above cross-transposition construction therefore absorbs every orbit. Hence every finitary transposition belongs to G , proving the lemma.

21.5. 5. Proof of Theorem A

Let B_q be the critical-value set. Corollary 3.2 (and the explicit $q = 1$ calculation) makes B_q closed and discrete. Set

$$X_q = \mathbb{C} \setminus F_q^{-1}(B_q), \quad Y_q = \mathbb{C} \setminus B_q.$$

The preimage $F_q^{-1}(B_q)$ is closed and discrete. Indeed, near a point whose image is not in B_q this follows from closedness of B_q ; near a point mapping to $b \in B_q$, choose a target disk containing no other point of B_q , so locally the preimage is the discrete zero set of $F_q - b$. The

complement of a closed discrete subset of the plane is path-connected (perturb a polygonal path around the finitely many points it meets), hence X_q is path-connected.

Let $\Delta = D(a, r)$ be a disk with $\overline{\Delta} \subset Y_q$, and let U be a component of $F_q^{-1}(\Delta)$. For $M > |a| + r$, the connected set U lies in one component of $\{|F_q| < M\}$; Lemma 2.1 therefore makes U bounded. The restriction $F_q : U \rightarrow \Delta$ is proper. To see this without any boundary assumption, let $K \Subset \Delta$ and take a sequence $z_j \in U \cap F_q^{-1}(K)$. Boundedness gives a convergent subsequence $z_j \rightarrow z$. Then $F_q(z) \in K \subset \Delta$; since $F_q^{-1}(\Delta)$ is open, a small connected neighborhood of z lies in the same component U , so $z \in U$.

The map on U is unramified. A proper local biholomorphism has finite fibers and is a covering onto its image; that image is nonempty, open, and closed in Δ , hence is all of Δ . Since U is connected and Δ is simply connected, this covering has one sheet. We have therefore proved that

$$F_q : X_q \longrightarrow Y_q$$

is a connected covering and its monodromy is transitive.

The regular fiber is countably infinite. Indeed, if $F_q - c$ had only finitely many zeros, the order-one Hadamard factorization theorem (the order is immediate from the Taylor coefficients) would give

$$F_q(z) - c = P(z)e^{az+b}$$

with a polynomial P . If $a \neq 0$, a ray on which $\Re(az) \rightarrow -\infty$ makes the right side tend to zero and gives the finite asymptotic value c , contrary to Lemma 2.1. If $a = 0$, then F_q would be a polynomial. Thus every regular fiber is infinite; being a discrete subset of $\mathbb{C}$, it is countably infinite.

The finitary nature of a meridian requires the bounded-component conclusion of Lemma 2.1, not merely finiteness of the critical fiber. We record the precise local statement.

Lemma 5.1 (finitary meridians). Let f be entire and suppose every component of the inverse image of every bounded disk is bounded. Let b be an isolated critical value with only finitely many critical points above it. Then a meridian around b has finite support on a regular fiber. If exactly one critical point lies above b and it is simple, the meridian is a single transposition.

Choose a disk Δ_b whose closure contains no critical value of f other than b . The compactness argument above, which did not use absence of critical points, shows that the restriction of f to every component U of $f^{-1}(\Delta_b)$ is a proper finite branched cover onto Δ_b . If U contains no critical point, the map is a connected unramified cover of a disk, hence has degree one, and its sheet is fixed by the meridian. Every critical point in such a component maps to b ; hence only finitely many components contain critical points. Each of those components has finite degree, so the meridian is supported on the finite union of their finite fibers. Its nontrivial cycles are exactly the local ramification cycles. A unique simple critical point therefore gives one transposition. Transporting the local meridian to the chosen base point only conjugates this permutation and preserves its finite support and cycle type. This proves the lemma.

Lemma 5.1 applies to F_q by Lemma 2.1 and Corollary 3.2 (or (3.8) when $q = 1$). Thus every meridian is finitary and all but finitely many meridians are single transpositions. Finally, every loop in Y_q is a finite word in meridians: its compact image lies in a sufficiently large disk whose

boundary avoids B_q , and that disk contains only finitely many points of the closed discrete set B_q . It follows that the entire monodromy group lies in FSym and is generated by the tail transpositions together with finitely many exceptional finitary permutations. It is transitive and acts on a countably infinite fiber, so Lemma 4.1 gives (0.2).

21.6. 6. Algebraic freedom of branches and first jets

Theorem B (all branches are ordinarily algebraically free). On an evenly covered simply connected disk, every finite family of distinct inverse branches of F_q is algebraically independent over $\mathbb{C}(c)$.

Suppose a relation exists and clear its coefficients' denominators to obtain a nonzero polynomial

$$P(c; X_1, \dots, X_n) \in \mathbb{C}[c, X_1, \dots, X_n]$$

which vanishes on the given branches. Choose c_* in the original disk so that $P(c_*; \mathbf{X})$ is not the zero polynomial; only finitely many values are excluded by its nonzero coefficient polynomials. Clearing the denominators has removed all coefficient poles; equivalently, meridian representatives could be perturbed away from their finite pole set. Analytic continuation of the identity around loops based at c_* , together with the n -transitivity of FSym , makes $P(c_*; \mathbf{X})$ vanish on every ordered distinct n -tuple in the infinite fiber. Those tuples are Zariski dense in $\mathbb{C}^n$: fix all but one coordinate, omit the finitely many previously used fiber points, and induct. This contradicts the choice of c_* and proves Theorem B.

There is a sharper jet statement.

Theorem C (all-order first-jet freedom). For every fixed $q \geq 1$, distinct inverse branches $\rho_1, \dots, \rho_n$ satisfy

$$\{\rho_i, \rho_i' : 1 \leq i \leq n\} \text{ algebraically independent over } \mathbb{C}(c). \quad (\text{Inv-6.1})$$

Fix an arbitrary regular value c_* . We first write out the fiber version of the contraction, rather than appeal to it implicitly. On the rough disk $E_{q,n}$ from (3.2a), $Q_q(w) - c_* \neq 0$ for large n , and we choose the logarithm asymptotic to $q \log w - \log(q!)$. The equation $F_q(z) = c_*$ is equivalent to

$$z = T_{q,n}^{c_*}(z) := 2\pi in + i\pi q - q \log z - \log(Q_q(\log z + \gamma) - c_*) + \log \mathcal{B}_q(z). \quad (\text{Inv-6.2})$$

Define $c_{q,n}^{c_*} = T_{q,n}^{c_*}(2\pi in)$ and

$$D_{q,n}^{c_*} = \{z : |z - c_{q,n}^{c_*}| \leq 1\}.$$

The same direct expansion used in (3.2e), now with a fixed lower-order term $-c_*$, gives $|c_{q,n}^{c_*} - 2\pi in| = O_{q,c_*}(\log n)$; hence this disk lies in $E_{q,n}$. Moreover,

$$(T_{q,n}^{c_*})'(z) = -\frac{q}{z} - \frac{Q_{q-1}(w)}{z(Q_q(w) - c_*)} + \frac{\mathcal{B}'_q(z)}{\mathcal{B}_q(z)} = O_{q,c_*}(n^{-1}). \quad (\text{Inv-6.3})$$

Thus, for all sufficiently large n ,

$$|T_{q,n}^{c_*}(z) - c_{q,n}^{c_*}| \leq \frac{C_{q,c_*}}{n} |z - 2\pi in| \leq \frac{C_{q,c_*}(1 + \log n)}{n} < 1 \quad (z \in D_{q,n}^{c_*}),$$

and $\sup_{D_{q,n}^{c_*}} |(T_{q,n}^{c_*})'| < 1/2$. Thus $T_{q,n}^{c_*}(D_{q,n}^{c_*}) \subset D_{q,n}^{c_*}$. Exponentiating (6.2) identifies its unique fixed point $z_n(c_*)$ with a point of $F_q^{-1}(c_*)$. This supplies an infinite sequence $|z_n(c_*)| \rightarrow \infty$, since the consecutive centers again differ by $2\pi i + o(1)$.

At such a point $R_q = c_* - Q_q$. Consecutive-remainder division, including $q = 1$ by (1.7), gives

$$\frac{R_{q-1}}{R_q} = -z \left(1 + \frac{q}{z} + O_q(z^{-2}) \right),$$

and hence

$$F_{q-1}(z) = Q_{q-1}(w) + z(Q_q(w) - c_*) \left(1 + \frac{q}{z} + O_q(z^{-2}) \right). \quad (\text{Inv-6.4})$$

Using (0.1) and $Q_{q-1}(w)/(z(Q_q(w) - c_*)) = O_{q,c_*}((z \log z)^{-1})$, we obtain

$$p(z) := \frac{1}{F_q'(z)} = \frac{1}{Q_q(\log z + \gamma) - c_*} \left(1 + O_{q,c_*}(z^{-1}) \right) \sim \frac{q!}{(\log z)^q}. \quad (\text{Inv-6.5})$$

The jet set

$$J_{q,c_*} = \{(z, p(z)) : F_q(z) = c_*\}$$

is Zariski dense in $\mathbb{C}^2$. Indeed, for a nonzero polynomial $A(z, p)$, choose its largest z -degree d , and within that degree choose the smallest occurring p -degree j . Along the sequence above, the term $a_{d,j} z^d p^j$ dominates terms of the same z -degree and larger p -degree because $p \rightarrow 0$. If N is the largest difference of p -degrees that occurs, it dominates every lower z -degree term because

$$|z|^{-1} |p|^{-N} = O((\log |z|)^{qN} / |z|) \rightarrow 0.$$

Thus A cannot vanish on all of J_{q,c_*} .

We use the following elementary product fact. If an infinite subset $S \subset \mathbb{C}^2$ is Zariski dense, then S with finitely many points removed is still Zariski dense: otherwise the irreducible variety $\mathbb{C}^2$ would be the union of a proper closed set and finitely many points. It follows by induction that

$$S^{[n]} = \{(s_1, \dots, s_n) \in S^n : s_i \neq s_j \ (i \neq j)\}$$

is Zariski dense in $(\mathbb{C}^2)^n$. Indeed, after fixing $s_1, \dots, s_{n-1}$, a polynomial vanishing on $S^{[n]}$ vanishes as a polynomial in its last pair of variables on $S \setminus \{s_1, \dots, s_{n-1}\}$; its coefficients then vanish on the ordered distinct $(n-1)$ -fold product.

Since the construction and density proof apply to every regular c_* , we may now specialize a hypothetical relation at a convenient value. Suppose that the $2n$ first-jet functions satisfy an algebraic relation. Clearing denominators gives a nonzero

$$P(c; Z_1, W_1, \dots, Z_n, W_n) \in \mathbb{C}[c, Z_1, W_1, \dots, Z_n, W_n].$$

Choose c_* in the original regular disk so that its specialization in the jet variables is nonzero. Clearing denominators removes coefficient poles (or, equivalently, the finitely many poles can

be avoided by the loop representatives). Under continuation around a loop, a sheet z and its derivative move together as $(z, 1/F'_q(z))$. The n -transitivity from Theorem A therefore makes the specialized polynomial vanish on the ordered distinct n -fold product of $J_{q,c*}$, which is Zariski dense by the preceding paragraph. This contradicts nonzero specialization and proves (6.1).

21.7. 7. Ordinary freedom versus differential algebra

If $\rho(c)$ is an inverse branch and

$$\mathcal{L}_\rho = \frac{\rho}{\rho'} \frac{d}{dc},$$

then (0.1) gives

$$H_q := \mathcal{L}_\rho^q(c) = 1 - e^{-\rho}, \quad H'_q = \rho'(1 - H_q). \quad (\text{Inv-7.1})$$

After clearing powers of ρ' , (7.1) is a differential-polynomial equation of order at most $q + 1$. Hence every inverse branch is differential-algebraic, while Theorems B–C show extreme ordinary algebraic freedom.

For $q = 1$, (7.1) is exactly

$$\rho\rho'' + (\rho')^3 - (\rho + 1)(\rho')^2 = 0. \quad (\text{Inv-7.2})$$

Theorem C shows that no order-zero or order-one differential equation can hold, so the differential order is exactly two. Differentiating (7.2) expresses every higher derivative rationally in ρ, ρ' . Therefore, for any n distinct branches,

$$\text{trdeg}_{\mathbb{C}(c)} \mathbb{C}(c)\langle \rho_1, \dots, \rho_n \rangle = 2n, \quad \text{dtrdeg}_{\mathbb{C}(c)} = 0. \quad (\text{Inv-7.3})$$

This is the sharpest ordinary/differential separation presently available.

21.8. 8. Coupling to exact two-trace recovery

For a nonzero level c , and the zero divisor of $F_q(z) - c$, let

$$T_m^{(q)}(c) = \sum_{F_q(\rho)=c} \rho^{-m}, \quad m \geq 2.$$

The series is absolutely convergent because $F_q - c$ has order one. We recall the trace calculation, with the same normalization as the finite Faber-monodromy section above. Hadamard factorization gives

$$F_q(z) - c = -c e^{\alpha z} \prod_{F_q(\rho)=c} \left(1 - \frac{z}{\rho}\right) e^{z/\rho},$$

with zeros repeated according to multiplicity. Comparing the coefficient of z^m , $m \geq 2$, in the logarithm at $z = 0$ gives

$$T_m^{(q)}(c) = P_{q,m}(c^{-1}), \quad P_{q,m}(X) := -m[z^m] \log(1 - X F_q(z)). \quad (\text{Inv-8.0})$$

Indeed,

$$P_{q,m}(X) = \sum_{k=1}^m \frac{m}{k} [z^m] F_q(z)^k X^k,$$

so $P_{q,m} \in \mathbb{Q}[X]$ is monic of degree m . Writing $X = c^{-1}$ and using the first four Taylor coefficients of F_q gives

$$\begin{aligned} T_2^{(q)} &= X^2 - \frac{X}{2^q}, \\ T_4^{(q)} &= X^4 - \frac{2X^3}{2^q} + \left(\frac{2}{3^{q+1}} + \frac{1}{2^{2q+1}} \right) X^2 - \frac{X}{6 \cdot 4^q}. \end{aligned} \tag{Inv-8.0a}$$

In particular, if

$$A_q = \frac{2}{3^{q+1}} - \frac{1}{2^{2q+1}},$$

and

$$C_q = -\frac{1}{6 \cdot 4^q} + \frac{2}{3 \cdot 6^q} - \frac{1}{2^{3q+1}} \neq 0,$$

(vanishing would give $4^{q+1} = 6^q + 3^{q+1}$, impossible modulo 3),

then (8.0a) gives

$$T_4^{(q)} - (T_2^{(q)})^2 - A_q T_2^{(q)} = C_q X,$$

and hence

$$c^{-1} = \frac{T_4^{(q)} - (T_2^{(q)})^2 - A_q T_2^{(q)}}{C_q}. \tag{Inv-8.1}$$

Consequently

$$\mathbb{C}(c) = \mathbb{C}(T_2^{(q)}, T_4^{(q)}). \tag{Inv-8.2}$$

Combining (8.2) with Theorems A–C produces the all-order **local-freedom/global-collapse dichotomy**:

- every finite collection of local inverse branches, and even their first jets, is algebraically free;
- the monodromy of all local sheets is the maximal finitary symmetric group;
- nevertheless, all global reciprocal Newton traces of the infinite divisor are functions of two traces, and those two traces recover the level exactly.

This synthesis is substantially stronger than either the monodromy theorem or the trace calculation in isolation.

21.9. 9. Literature boundary and attribution

- NIST Digital Library of Mathematical Functions, §8.11(i), equations 8.11.2–8.11.3 (with the parameter restricted to a compact set), supplies the large-variable incomplete-gamma remainder used in (1.5a). The contour-remainder proof gives compact-parameter uniformity, and Cauchy’s estimates give (1.5b)–(1.5c).
- G. H. Hardy, *On the Zeroes of Certain Classes of Integral Taylor Series, Part II*, Proc. London Math. Soc. (2) **2** (1905), 401–431, DOI: 10.1112/plms/s2-2.1.401, studied the zeros and asymptotics of $e(x, a \mid s)$. Since $F_p(z) = ze(-z, 1 \mid p+1)$, the leading zero-box asymptotic must be credited to this line.
- K. N. Boyadzhiev, *Polyexponentials*, arXiv:0710.1332, surveys the defining identities and Hardy asymptotics.
- L. Zelenko, *Generic monodromy group of Riemann surfaces for inverses to entire functions of finite order*, arXiv:2105.14015, proves $\text{FSym}(\mathbb{N})$ for a generic/“typical” class with globally simple critical points and globally pairwise-distinct critical values.

Thus FSym as an abstract phenomenon is not new. The potentially new content here is the **explicit all-order polyexponential theorem**, the proof that only finitely many exceptional critical events can occur, the finite-exception permutation upgrade (which avoids proving global critical-value noncollision), and the all-order branch/first-jet algebraic freedom. A full submission still requires a systematic MathSciNet/zbMATH priority search.

21.10. 10. What remains open

1. Prove or disprove global pairwise noncollision of the critical values of F_q , already open in the finite remainder for $q = 2$.
2. Determine whether the inverse differential equation in (7.1) has minimal order $q + 1$ for every $q \geq 2$. Theorem C gives only the lower bound two.
3. Determine algebraic independence of higher inverse jets and the exact ordinary transcendence degree of the differential field generated by finitely many branches for $q \geq 2$.
4. Determine whether the local-freedom/global-collapse dichotomy has an analogue for other iterated-integral towers beyond the polyexponentials.

Part IV

Ramanujan–Gamma Transmutations and Euler Continued Fractions

22. An explicit solution of a 2026 Euler recurrence

The Ramanujan Challenge for AI [16] asks, in its Euler-constant problem, for a proof that the quotient of two distinguished solutions of the following four-term recurrence tends to γ . We give a self-contained solution. The finite sums below belong to the multiple-Laguerre framework developed by Wolfs–Van Assche [19]. What is proved here is the adjacent specialization, explicit all-index recurrence certificates, an elementary positive concentration proof, and the flat matrix field.

Put

$$\begin{aligned} A_n &= 8n^3 + 51n^2 + 105n + 68, \\ B_n &= 24n^5 + 337n^4 + 1833n^3 + 4818n^2 + 6092n + 2928, \\ C_n &= (n+2)(n+3)(24n^5 + 273n^4 + 1150n^3 + 2154n^2 + 1635n + 268), \\ D_n &= (n+1)(n+2)^4(n+3)(8n^3 + 75n^2 + 231n + 232), \end{aligned}$$

and

$$A_n u_n = B_n u_{n-1} - C_n u_{n-2} + D_n u_{n-3} \quad (n \geq 0). \quad (136)$$

Theorem 22.1 (Closed solution of the Euler recurrence). *For $n \geq -3$, set $a = n + 3$, $b = n + 4 = a + 1$, and*

$$W_{a,b}(k) = \frac{(a!b!)^2}{(k!)^3(a-k)!(b-k)!} \quad (0 \leq k \leq a). \quad (137)$$

Then the two sequences determined by

$$(p_{-3}, p_{-2}, p_{-1}) = (0, 7, 179), \quad (q_{-3}, q_{-2}, q_{-1}) = (1, 12, 306)$$

and (136) have the exact formulas

$$q_n = b \sum_{k=0}^a W_{a,b}(k), \quad (138)$$

$$p_n = b \sum_{k=0}^a W_{a,b}(k)(3H_k - H_{a-k} - H_{b-k}) + a!. \quad (139)$$

In particular,

$$\boxed{\lim_{n \rightarrow \infty} \frac{p_n}{q_n} = \gamma.} \quad (140)$$

Proof. Substitution of $(a, b) = (0, 1), (1, 2), (2, 3)$ gives the six initial values. The two symbolic telescoping identities in [appendix B](#) prove (136) for both closed forms at every $n \geq 0$. We prove the limit directly.

After cancelling factors independent of k , define a probability law by

$$\Pr(K_n = k) = \frac{\binom{a}{k} \binom{b}{k} / k!}{\sum_{j=0}^a \binom{a}{j} \binom{b}{j} / j!}. \quad (141)$$

Then

$$\frac{p_n}{q_n} = \mathbb{E}(3H_{K_n} - H_{a-K_n} - H_{b-K_n}) + \frac{a!}{q_n}. \quad (142)$$

The ratio of consecutive unnormalized masses is

$$r_k = \frac{(a-k)(b-k)}{(k+1)^3}, \quad (143)$$

which decreases strictly in k . Put $x_n = (ab)^{1/3}$. For each fixed $\eta > 0$, the ratios outside $[(1-\eta)x_n, (1+\eta)x_n]$ are uniformly separated from one once n is large. Multiplying the ratios on either side of the mode gives

$$\Pr(|K_n/x_n - 1| > \eta) = O(e^{-c_\eta x_n}). \quad (144)$$

Hence $K_n/x_n \rightarrow 1$ in probability.

On the event $|K_n/x_n - 1| \leq \eta$, the estimate $H_m = \log m + \gamma + O(m^{-1})$ yields

$$\begin{aligned} 3H_k - H_{a-k} - H_{b-k} &= \gamma + \log \frac{k^3}{(a-k)(b-k)} + o(1) \\ &= \gamma + O(\eta) + o(1). \end{aligned}$$

The expression is $O(\log n)$ on the full support, while the exceptional probability in (144) is exponentially small in x_n . First let $n \rightarrow \infty$, and then let $\eta \downarrow 0$; its expectation therefore tends to γ . Finally the $k = 0$ summand in (138) gives

$$0 \leq \frac{a!}{q_n} \leq \frac{1}{b b!} \rightarrow 0.$$

Equation (142) proves (140). $\square$

Remark 22.2 (What has and has not been solved). The theorem gives an independent proof of the publicly stated limit problem. The official proof and code are now public [32]; the sentence in Version 2.0 saying otherwise is superseded. No priority claim is made for the official Meijer- G trajectory, Miller selection, or the broad multiple-Laguerre family. The comparison with the terminating state is established in section 24. The convergence $p_n/q_n \rightarrow \gamma$ does not by itself imply the irrationality of γ .

The same recurrence has a second nontrivial Apéry limit. Let w_n be the solution with

$$(w_{-3}, w_{-2}, w_{-1}) = (1, 18, 440), \quad (145)$$

and let $\mathfrak{h}$ be the positive constant in (232) below.

Theorem 22.3 (Complete initial-value limit classification). *One has*

$$\boxed{\frac{w_n}{q_n} \rightarrow \mathfrak{h} = 1.4322057346532244148 \dots} \quad (146)$$

For arbitrary rational initial data $(x, y, z) = (u_{-3}, u_{-2}, u_{-1})$, the corresponding solution satisfies

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{u_n}{q_n} &= \frac{1}{136} [(-6x + 179y - 7z)\mathfrak{h} \\ &\quad + (-228x - 134y + 6z)\gamma + (142x - 179y + 7z)]. \end{aligned} \quad (147)$$

Consequently the set of all rational-initial-data limits is exactly

$$\boxed{\mathbb{Q} + \mathbb{Q}\gamma + \mathbb{Q}\mathfrak{h}}. \quad (148)$$

No assertion that this span has dimension three is made.

Proof. The official cyclic gauge at zero has rows

$$U(0) = \begin{pmatrix} 1 & 18 & 440 \\ 0 & 7 & 179 \\ 0 & 1 & 45 \end{pmatrix}. \quad (149)$$

They generate solutions w, p, s , respectively, and $q = w - p + s$. The official Miller-selection theorem, transported through the exact gauge of theorem 24.1, gives

$$\frac{1}{q_n}(w_n, p_n, s_n) \rightarrow (\mathfrak{h}, \gamma, 1 + \gamma - \mathfrak{h}); \quad (150)$$

see [32, §5.2] and [theorem 27.2](#). The identity $q = w - p + s$ normalizes the right side to one. Solving the initial vector (x, y, z) in the basis (q, p, w) gives the three coefficients in (147). Conversely q, p, w realize $1, \gamma, \mathfrak{h}$, proving (148). $\square$

Theorem 22.4 (Complete integer initial lattice). *For a solution of (136),*

$$u_n \in \mathbb{Z} \text{ for every } n \geq -3 \iff x, y, z \in \mathbb{Z} \text{ and } 13x + 6y + z \equiv 0 \pmod{17}. \quad (151)$$

Proof. The first new term is

$$u_0 = \frac{2784x - 402y + 732z}{17}, \quad (152)$$

so integrality forces the stated congruence. Conversely its kernel in $\mathbb{Z}^3$ has basis

$$(1, 0, 4), \quad (0, 1, 11), \quad (0, 0, 17).$$

The finite companion-gauge certificate in [appendix C](#) proves that the three corresponding solutions are integral at every index. Every integral vector satisfying the congruence is their integral combination. $\square$

23. A flat conservative matrix field and exterior arithmetic

For integers $a, b \geq 0$, define

$$F_{a,b}(z) = {}_2F_2\left(\begin{matrix} -a, -b \\ 1, 1 \end{matrix}; z\right), \quad \theta = z \frac{d}{dz}, \quad \mathbf{Y}_{a,b} = \begin{pmatrix} F_{a,b} \\ \theta F_{a,b} \\ \theta^2 F_{a,b} \end{pmatrix}. \quad (153)$$

It satisfies

$$\theta^3 F_{a,b} = z(\theta - a)(\theta - b)F_{a,b}. \quad (154)$$

At $z = 1$, its terminating series gives the exact link

$$F_{a,b}(1) = \sum_{k=0}^a \binom{a}{k} \binom{b}{k} \frac{1}{k!}, \quad q_n = b a! b! F_{a,b}(1) \quad (a = n + 3, b = n + 4). \quad (155)$$

Theorem 23.1 (Flat SL_3 conservative matrix field). *For $a, b > 0$, put*

$$M_a(a, b; z) = \frac{1}{a} \begin{pmatrix} a & -1 & 0 \\ 0 & a & -1 \\ -zab & z(a+b) & a-z \end{pmatrix}, \quad (156)$$

$$M_b(a, b; z) = \frac{1}{b} \begin{pmatrix} b & -1 & 0 \\ 0 & b & -1 \\ -zab & z(a+b) & b-z \end{pmatrix}. \quad (157)$$

Then

$$\mathbf{Y}_{a-1,b} = M_a(a, b; z) \mathbf{Y}_{a,b}, \quad \mathbf{Y}_{a,b-1} = M_b(a, b; z) \mathbf{Y}_{a,b},$$

and

$$\det M_a = \det M_b = 1, \quad (158)$$

$$M_b(a-1, b; z) M_a(a, b; z) = M_a(a, b-1; z) M_b(a, b; z). \quad (159)$$

Thus the denominator system of [theorem 22.1](#) is the adjacent-diagonal path $(a, b) = (n+3, n+4)$ in an explicit flat SL_3 conservative matrix field.

Proof. The two contiguous relations follow by comparing the terminating hypergeometric coefficients and using (154) for the third row. Expansion of the determinants gives 1. Direct matrix multiplication gives the polynomial identity (159); an entrywise symbolic certificate is also included with the source package. $\square$

Remark 23.2 (Path rigidity). Flatness says that changing a lattice path while keeping the same endpoint (a, b) leaves $F_{a,b}$, and hence the endpoint-defined denominator, unchanged. The numerator and quotient in theorem 22.1 are also endpoint-defined by their closed formulas. Rerouting a path therefore does not change these particular values. Changing the endpoint curve is a different problem and is not excluded by this observation.

The numerator is the first Frobenius jet of the same flat connection. Let

$$\Phi_{a,b}(\epsilon, z) = \frac{a!b!}{(1+\epsilon)_a(1+\epsilon)_b} {}_3F_3 \left(\begin{matrix} -a-\epsilon, -b-\epsilon, 1 \\ 1-\epsilon, 1-\epsilon, 1-\epsilon \end{matrix}; z \right). \quad (160)$$

Theorem 23.3 (Flat first-jet SL_6 lift). *Fix positive integers a, b , expand the analytic ϵ -germ at zero modulo ϵ^2 , and put $\mathbf{Y}_{a,b}(\epsilon) = (\Phi_{a,b}, \theta\Phi_{a,b}, \theta^2\Phi_{a,b})^\top$. Then*

$$\mathbf{Y}_{a-1,b}(\epsilon) = \mathcal{M}_a(\epsilon)\mathbf{Y}_{a,b}(\epsilon), \quad \mathbf{Y}_{a,b-1}(\epsilon) = \mathcal{M}_b(\epsilon)\mathbf{Y}_{a,b}(\epsilon), \quad (161)$$

where

$$\mathcal{M}_a(\epsilon) = \frac{1}{a} \begin{pmatrix} a+\epsilon & -1 & 0 \\ 0 & a+\epsilon & -1 \\ -zab - \epsilon z(a+b) & z(a+b) + 2\epsilon z & a - z - 2\epsilon \end{pmatrix}, \quad (162)$$

and

$$\mathcal{M}_b(\epsilon) = \frac{1}{b} \begin{pmatrix} b+\epsilon & -1 & 0 \\ 0 & b+\epsilon & -1 \\ -zab - \epsilon z(a+b) & z(a+b) + 2\epsilon z & b - z - 2\epsilon \end{pmatrix}. \quad (163)$$

In $\mathbb{Q}(a, b, z)[\epsilon]/(\epsilon^2)$, these matrices have determinant one and satisfy the same zero-curvature identity as (159).

Writing $\mathcal{M}_a = \mathcal{M}_{a,0} + \epsilon\mathcal{M}_{a,1}$, the block matrix

$$\begin{pmatrix} \mathcal{M}_{a,0} & 0 \\ \mathcal{M}_{a,1} & \mathcal{M}_{a,0} \end{pmatrix} \in SL_6 \quad (164)$$

transports $(\mathbf{Y}_{a,b(0)})_{\partial_\epsilon \mathbf{Y}_{a,b(0)}}$, and similarly in the b -direction. Thus denominator and numerator path-independence are the zero- and first-jet parts of one flat connection.

At $b = a + 1, z = 1$,

$$b a!b!\Phi_{a,b}(0, 1) = q_{a-3}, \quad b a!b!\partial_\epsilon \Phi_{a,b}(0, 1) = p_{a-3}. \quad (165)$$

The boundary $a = 0, b = 1$ in (165) is verified directly.

Proof. Modulo ϵ^2 , the Frobenius deformation satisfies

$$(\theta - \epsilon)^3 \Phi_{a,b} = z(\theta - a - \epsilon)(\theta - b - \epsilon)\Phi_{a,b}. \quad (166)$$

Coefficient comparison gives the first two rows of (161), while (166) gives the third. Direct determinant expansion and multiplication give determinant one and flatness in the dual-number ring. Expanding a dual-number matrix action gives the block form (164). At $\epsilon = 0$, $\Phi_{a,b} = F_{a,b}$. The derivative of the terminating part gives the harmonic weight in (139); the regularized term at $k = b = a + 1$ is $1/(bb!)$ before multiplication by $b a!b!$, and hence gives exactly the endpoint $+a!$. $\square$

Proposition 23.4 (First-jet exactness barrier). *Let $\mathcal{K}_a$ be the rational carrier defined in (200). For $b = a + 1, z = 1$,*

$$\Phi_{a,b}(\epsilon, 1) - \mathcal{K}_a(\epsilon) = O(\epsilon^2). \quad (167)$$

Return here to the full analytic ϵ -germ. For every $r \geq 1$, the term with $k = b + r$ contributes

$$[\epsilon^2] \Phi_{a,b}(\epsilon, 1)|_{k=b+r} = -\frac{a!b! r!(r-1)!}{(b+r)!^3} \neq 0. \quad (168)$$

The total second-order difference cannot cancel:

$$[\epsilon^2] \{\Phi_{a,b}(\epsilon, 1) - \mathcal{K}_a(\epsilon)\} = \frac{1}{b^2 a!} \left\{ 3H_b - \sum_{r \geq 1} \frac{r!(r-1)!}{(b+1)_r^3} \right\} > 0. \quad (169)$$

Indeed $(b+1)_r \geq (r+1)!$, so the sum is less than one, while $3H_b \geq 3$. Thus the first derivative is exactly terminating after its resonant endpoint, whereas the second derivative develops a genuine infinite hypergeometric tail. This is a barrier for this Frobenius deformation, not a universal no-go for all higher-jet constructions.

Proof. For $0 \leq k \leq a$, the hypergeometric summand is identically the corresponding summand of $\mathcal{J}_a(\epsilon)$. At the resonant index $k = b$, its analytic continuation is

$$\frac{\epsilon}{b^2 a!} \prod_{j=1}^b \left(1 - \frac{\epsilon}{j}\right)^{-3};$$

its linear coefficient is $1/(bb!)$, exactly the correction in $\mathcal{K}_a$, and its quadratic coefficient is $3H_b/(b^2 a!)$. For $k = b + r, r \geq 1$, direct expansion gives (168). Summing those coefficients gives (169); the displayed comparison proves its strict positivity. Hence the zero and first jets agree and the second jets do not. $\square$

Proposition 23.5 (Forced divisor and exact Casoratian). *Let $d_b = \text{lcm}(1, \dots, b)$. For every $n \geq -3$,*

$$\boxed{\frac{b!}{d_b} \gcd\left(b, \frac{d_b}{b}\right) \mid \gcd(p_n, q_n)}. \quad (170)$$

If $u^{(1)}, u^{(2)}, u^{(3)}$ are three solutions of (136), put, for $n \geq -1$,

$$\mathcal{W}_n = \det(u_{n-j}^{(i)})_{1 \leq i \leq 3, 0 \leq j \leq 2},$$

so in particular $\mathcal{W}_{-1} = \det(u_{-1-j}^{(i)})_{1 \leq i \leq 3, 0 \leq j \leq 2}$. Then, for every $n \geq 0$,

$$\boxed{\mathcal{W}_n = \mathcal{W}_{-1} \frac{A_{n+1}}{136} (n+1)!(n+2)!^4 (n+3)!}. \quad (171)$$

Proof. Write

$$W_{a,b}(k) = a!b! \binom{a}{k} \binom{b}{k} \frac{1}{k!}.$$

Put $g_b = \gcd(b, d_b/b)$. After division by $b!/d_b$, each summand of q_n is b times an integer, because $a!/k! \in \mathbb{Z}$. The harmonic part of p_n is likewise b times an integer, since $d_b(3H_k - H_{a-k} - H_{b-k}) \in \mathbb{Z}$. Its endpoint becomes d_b/b . Hence every term of both quotients is divisible by g_b , proving (170).

The companion determinant of (136) is D_n/A_n . Since

$$D_n = (n+1)(n+2)^4(n+3)A_{n+1},$$

iteration from -1 to n telescopes and gives (171). $\square$

Remark 23.6 (Exterior-volume barrier). The flat base field has determinant one, whereas scalar elimination together with its factorial gauge produces the factorial-six Casoratian (171). The forced divisor (170) is a lower bound for the common divisor, not an upper bound. These formulas record the factorial cost but do not rule out further common-divisor cancellation; no small primitive integer linear form is obtained here.

24. Exact Meijer- G /hypergeometric duality

The official solution of the Challenge is now public [32]. It selects a recessive Meijer- G state by Miller's algorithm, whereas sections 22 and 23 uses a terminating ${}_2F_2$ state and a positive saddle. The terminating module is rationally gauge-equivalent to the contragredient of the normalized official rank-three module.

Write

$$F_a(z) = F_{a,a+1}(z), \quad \mathbf{Y}_a = \begin{pmatrix} F_a(1) \\ \theta F_a(1) \\ \theta^2 F_a(1) \end{pmatrix}.$$

The adjacent transition obtained by composing the two contiguous matrices of theorem 23.1 is

$$\mathbf{Y}_a = \mathbf{D}_a \mathbf{Y}_{a+1}, \quad (172)$$

where

$$\mathbf{D}_a = \begin{pmatrix} 1 & -\frac{2a+3}{(a+1)(a+2)} & \frac{1}{(a+1)(a+2)} \\ 1 & \frac{a^2+a-1}{(a+1)(a+2)} & -\frac{2}{a+2} \\ -(2a+1) & \frac{5a^2+11a+5}{(a+1)(a+2)} & \frac{a-2}{a+2} \end{pmatrix}. \quad (173)$$

For $a \geq 1$ this is the composite contiguous transfer. The boundary identity $\mathbf{Y}_0 = \mathbf{D}_0 \mathbf{Y}_1$ is checked directly from the terminating series.

Define

$$\mathcal{H}_a^G(z) = G_{2,3}^{3,2} \left(z \left| \begin{matrix} 1, 1 \\ a+1, a+1, a+1 \end{matrix} \right. \right) \quad (174)$$

with the Mellin-Barnes normalization of [32, §5.2], and let

$$\mathbf{S}_a^G = (\mathcal{H}_a^G(1), \theta \mathcal{H}_a^G(1), \theta^2 \mathcal{H}_a^G(1))$$

be the official row state, with the normalization of [32, §5.2]. With $b = a + 1$, its transfer matrix is

$$\mathbf{S}_{a+1}^G = \mathbf{S}_a^G \mathbf{M}_a^G, \quad \mathbf{M}_a^G = \begin{pmatrix} 0 & b^3 & b^3(3a+4) \\ 0 & -3b^2 & -b^2(8a+11) \\ 1 & 3b & 6a^2+12a+5 \end{pmatrix}. \quad (175)$$

Put

$$\mathbf{P}_a^G = b^{-2} \mathbf{M}_a^G, \quad \mathbf{B}_a = \begin{pmatrix} b^2 & -(2a+1) & 1 \\ 0 & a/b & -1/b \\ 0 & 1/b & 0 \end{pmatrix}. \quad (176)$$

Theorem 24.1 (Exact rational SL_3 coboundary). *For every $a \geq 0$,*

$$\det \mathbf{D}_a = \det \mathbf{B}_a = \det \mathbf{P}_a^G = 1, \quad \det \mathbf{M}_a^G = (a+1)^6, \quad (177)$$

and the exact gauge identity is

$$\boxed{D_a^{-\top} B_{a+1} = B_a (P_a^G)^{-\top}}. \quad (178)$$

If $V_a = Y_a^\top B_a$ and $\Pi_a^G = M_0^G \cdots M_{a-1}^G$, then

$$V_{a+1} = V_a (P_a^G)^{-\top}, \quad (179)$$

$$\boxed{Y_a^\top B_a = (a!)^2 (1, -1, 1) (\Pi_a^G)^{-\top}}. \quad (180)$$

The denominator coordinate has the precise factorial gauge

$$\boxed{q_{a-3} = (a!)^2 V_a e_1}. \quad (181)$$

Equivalently, $\hat{V}_a = (a!)^2 V_a$ has first coordinate q_{a-3} and evolves by the denominator-cleared dual matrix

$$\hat{V}_{a+1} = \hat{V}_a \hat{M}_a^G, \quad \hat{M}_a^G = b^4 (M_a^G)^{-\top}, \quad \det \hat{M}_a^G = b^6. \quad (182)$$

Proof. For $a \geq 1$, equation (172) is the product $M_b(a, a+2; 1) M_a(a+1, a+2; 1)$ from [theorem 23.1](#); multiplication gives (173). At $a = 0$, direct evaluation of the two terminating polynomials gives the same identity. The determinants in (177) are direct expansions. Substitution of [equations \(173\) and \(176\)](#) into (178) reduces its nine entries to rational identities in a .

Rearranging (178) gives $D_a^\top B_a = B_{a+1} (P_a^G)^\top$, which proves (179). Since $Y_0 = (1, 0, 0)^\top$ and $Y_0^\top B_0 = (1, -1, 1)$, iteration gives (180). Finally, $V_a e_1 = b^2 F_a(1)$, whereas (155) gives $q_{a-3} = (a!)^2 b^2 F_a(1)$. This proves (181); the last assertions follow by rescaling. $\square$

The gauge fixes the finite connection pairing, not only its projective direction.

Theorem 24.2 (Finite Meijer- $G/{}_2F_2$ bilinear identity). *For every $a \geq 0$,*

$$\boxed{S_a^G B_a^\top Y_a = (a!)^2}. \quad (183)$$

Writing $b = a + 1$, $F = F_a(1)$, and $H = \mathcal{H}_a^G(1)$, this is

$$\begin{aligned} b^2 H F + (\theta H) \left(-(2a+1)F + \frac{a}{b} \theta F + \frac{1}{b} \theta^2 F \right) \\ + (\theta^2 H) \left(F - \frac{1}{b} \theta F \right) = (a!)^2. \end{aligned} \quad (184)$$

Thus the normalized conserved primal-dual pairing is exactly one.

Proof. The official normalization gives $S_0^G(1, -1, 1)^\top = 1$. Use $S_a^G = S_0^G \Pi_a^G$ and (180); the factors Π_a^G cancel. Expanding $B_a^\top Y_a$ gives (184). $\square$

The determinant-six phenomenon in [proposition 23.5](#) can now be located exactly. If $U(a)$ is the cyclic gauge of the official solution, then

$$\det U(a) = (a+1)^5 (a+2) A_a, \quad \det U(0) = 136. \quad (185)$$

The coboundary transformation

$$C^{\text{comp}}(a) = U(a)^{-1} \hat{M}_a^G U(a+1) \quad (186)$$

gives the scalar companion transfer. Consequently

$$\begin{aligned} \det(\mathbb{C}^{\text{comp}}(0) \cdots \mathbb{C}^{\text{comp}}(n)) &= ((n+1)!)^6 \frac{\det \mathbf{U}(n+1)}{136} \\ &= \frac{A_{n+1}}{136} (n+1)!(n+2)!^4(n+3)!, \end{aligned} \quad (187)$$

which is the multiplier of $\mathcal{W}_{-1}$ in (171). The sixth factorial power comes from the denominator-cleared dual determinant; the shifted factorials and $A_{n+1}/136$ are the endpoint cyclic-gauge determinant.

25. Euler–Beta–log transmutation and the offset ladder

The adjacent sums are an exact integral transform of the cubic Pilehood polynomials, not merely an analogous family. Define

$$\mathcal{Q}_m(t) = \sum_{j=0}^m \binom{m}{j}^3 j! t^j, \quad (188)$$

$$\mathcal{P}_m(t) = \sum_{j=0}^m \binom{m}{j}^3 j! (3H_{m-j} - 2H_j) t^j. \quad (189)$$

These are the cubic polynomials occurring in the Euler-constant construction of Hessami Pilehood–Hessami Pilehood [27, 28]. For $d \geq 1$, put

$$\mathcal{B}_d[f] = \frac{1}{(d-1)!} \int_0^1 (1-t)^{d-1} f(t) dt \quad (190)$$

and

$$\mathcal{T}_{m,d}[f](z) = \frac{(m+d)!}{m!^2} z^m \mathcal{B}_d[f(t/z)]. \quad (191)$$

Theorem 25.1 (Exact Beta–log transmutation). *For every $m \geq 0$ and $d \geq 1$, the first identity below holds for $z \neq 0$ and extends polynomially to $z = 0$; at $z = 1$, the second identity also holds:*

$$F_{m,m+d}(z) = \mathcal{T}_{m,d}[\mathcal{Q}_m](z), \quad (192)$$

$$\mathcal{G}_{m,d}(1) = \mathcal{T}_{m,d}[\mathcal{P}_m + \mathcal{Q}_m \log t](1), \quad (193)$$

where

$$\mathcal{G}_{m,d}(z) = \sum_{k=0}^m \binom{m}{k} \binom{m+d}{k} \frac{z^k}{k!} (3H_k - H_{m-k} - H_{m+d-k}). \quad (194)$$

Moreover the transform intertwines Euler derivations:

$$\theta_z \mathcal{T}_{m,d} = \mathcal{T}_{m,d}(m - \theta_t). \quad (195)$$

For the Challenge specialization $m = a$, $d = 1$, $z = 1$,

$$q_{a-3} = (a+1)^3 a! \int_0^1 \mathcal{Q}_a(t) dt, \quad (196)$$

$$p_{a-3} = (a+1)^3 a! \int_0^1 (\mathcal{P}_a(t) + \mathcal{Q}_a(t) \log t) dt + a!. \quad (197)$$

Proof. Put $j = m - k$. The beta integrals

$$\begin{aligned}\mathcal{B}_d[t^j] &= \frac{j!}{(j+d)!}, \\ \mathcal{B}_d[t^j \log t] &= \frac{j!}{(j+d)!} (H_j - H_{j+d})\end{aligned}$$

turn the coefficient of $z^{m-j} = z^k$ on the right of (192) into

$$\frac{m!(m+d)!}{k!^3(m-k)!(m+d-k)!},$$

which is the coefficient of $F_{m,m+d}$. Adding $3H_{m-j} - 2H_j$ and $H_j - H_{j+d}$ gives exactly $3H_k - H_{m-k} - H_{m+d-k}$, proving (193) at $z = 1$. Differentiating $z^m f(t/z)$ gives (195). Finally equations (196) and (197) follow from equations (138) and (139). $\square$

Corollary 25.2 (Fixed-offset Euler ladder). *For every fixed $d \geq 1$,*

$$\boxed{\frac{\mathcal{G}_{m,d}(1)}{F_{m,m+d}(1)} \longrightarrow \gamma.} \quad (198)$$

Proof. Use the probability weights

$$\mathbb{P}(K = k) \propto \binom{m}{k} \binom{m+d}{k} \frac{1}{k!}.$$

Their consecutive ratio is $(m-k)(m+d-k)/(k+1)^3$, so, with $s_m = (m(m+d))^{1/3}$, the same ratio argument as in theorem 22.1 gives $K/s_m \rightarrow 1$ exponentially in probability. On the window $|K/s_m - 1| \leq \eta$,

$$3H_K - H_{m-K} - H_{m+d-K} = \gamma + \log \frac{K^3}{(m-K)(m+d-K)} + o(1) = \gamma + O(\eta) + o(1).$$

The logarithmic growth on the complement is absorbed by the exponential tail. First let $m \rightarrow \infty$, and then let $\eta \downarrow 0$. Taking expectations proves (198). $\square$

The multiple-Laguerre background and general approximation families are already present in the cited literature and in [19]. The assertion here is the exact transmutation between that cubic system and the 2026 Challenge normalization, together with its full fixed-offset form.

26. Rational Gamma jets and contact polynomials

The numerator and denominator are the first two jets of one rational deformation. Continue to write $b = a + 1$, and define

$$\mathcal{J}_a(s) = a!b! \sum_{k=0}^a \frac{1}{(1-s)_k^3 (1+s)_{a-k} (1+s)_{b-k}}, \quad (199)$$

$$\mathcal{K}_a(s) = \mathcal{J}_a(s) + \frac{s}{b!}. \quad (200)$$

The small final term is essential: it records the resonant endpoint correction $+a!$ in (139).

Theorem 26.1 (Master Gamma-jet transform). *For every $a \geq 0$,*

$$\boxed{q_{a-3} = b a! b! \mathcal{K}_a(0), \quad p_{a-3} = b a! b! \mathcal{K}'_a(0),} \quad (201)$$

and hence

$$\frac{p_{a-3}}{q_{a-3}} = (\log \mathcal{K}_a)'(0). \quad (202)$$

As $a \rightarrow \infty$, locally uniformly on the strip $|\Re s| < 1$,

$$\boxed{\frac{\mathcal{J}_a(s)}{\mathcal{J}_a(0)}, \quad \frac{\mathcal{K}_a(s)}{\mathcal{K}_a(0)} \longrightarrow \Gamma(1-s)^3 \Gamma(1+s)^2.} \quad (203)$$

Consequently, for every $m \geq 2$,

$$\boxed{\frac{\frac{d^m}{ds^m} \log \mathcal{K}_a(s)}{(3 + 2(-1)^m)(m-1)!} \Big|_{s=0} \longrightarrow \zeta(m),} \quad (204)$$

while the first logarithmic jet tends to γ . Every finite-stage logarithmic jet in (204) is rational.

Proof. Differentiating a reciprocal Pochhammer symbol at zero gives

$$\begin{aligned} \frac{d}{ds} \log(1-s)_k^{-3} \Big|_{s=0} &= 3H_k, \\ \frac{d}{ds} \log(1+s)_r^{-1} \Big|_{s=0} &= -H_r. \end{aligned}$$

Equations (138) and (139) now give (201); the derivative of $s/(bb!)$ produces exactly the endpoint $a!$.

Normalize the summands of $\mathcal{J}_a(0)$ to the probability law (141). Uniformly for s in a compact subset of the strip $|\Re s| < 1$, the ratio of a deformed summand to its value at zero is

$$\begin{aligned} &\left(\frac{k!}{(1-s)_k} \right)^3 \frac{(a-k)!}{(1+s)_{a-k}} \frac{(b-k)!}{(1+s)_{b-k}} \\ &= \Gamma(1-s)^3 \Gamma(1+s)^2 k^{3s} (a-k)^{-s} (b-k)^{-s} (1+o(1)). \end{aligned} \quad (205)$$

Choose a shrinking central window $|k/(ab)^{1/3} - 1| \leq \eta_a$, where $\eta_a \downarrow 0$ and $\eta_a^2(ab)^{1/3} \rightarrow \infty$. The concentration estimate (144) makes its complement exponentially small. On the window, $k = (ab)^{1/3}(1+o(1))$, $a-k = a(1+o(1))$, and $b-k = b(1+o(1))$; hence the power factor in (205) tends uniformly to one. Uniform Gamma-ratio bounds on each compact substrip give at most a fixed polynomial in $k+1, a-k+1, b-k+1$, which the exponential tail absorbs. This proves the locally uniform limit for $\mathcal{J}_a$. The extra term in $\mathcal{K}_a$ is $O((bb!)^{-1})$, whereas $\mathcal{J}_a(0) \geq 1$, so the same limit holds for $\mathcal{K}_a$.

Finally,

$$\log\{\Gamma(1-s)^3 \Gamma(1+s)^2\} = \gamma s + \sum_{m \geq 2} \frac{3 + 2(-1)^m}{m} \zeta(m) s^m. \quad (206)$$

Local uniformity permits termwise differentiation of logarithms near zero and proves (204). Rationality at finite a follows because $\mathcal{K}_a \in \mathbb{Q}(s)$ and $\mathcal{K}_a(0) \neq 0$. $\square$

For example, as $a \rightarrow \infty$, the correctly normalized first cases are

$$\frac{(\log \mathcal{K}_a)''(0)}{5} \rightarrow \zeta(2), \quad \frac{(\log \mathcal{K}_a)^{(3)}(0)}{2} \rightarrow \zeta(3), \quad \frac{(\log \mathcal{K}_a)^{(4)}(0)}{30} \rightarrow \zeta(4). \quad (207)$$

The raw derivative $\mathcal{K}_a''(0)$ cannot replace the second logarithmic derivative.

Corollary 26.2 (Exact parity separation). *As $a \rightarrow \infty$, locally uniformly on $|\Re s| < 1$,*

$$\frac{\mathcal{K}_a(s)\mathcal{K}_a(-s)}{\mathcal{K}_a(0)^2} \rightarrow \left(\frac{\pi s}{\sin \pi s} \right)^5, \quad (208)$$

$$\frac{\mathcal{K}_a(s)}{\mathcal{K}_a(-s)} \rightarrow \frac{\Gamma(1-s)}{\Gamma(1+s)}. \quad (209)$$

Thus the logarithm of the first quotient generates only even zeta values, while the logarithm of the second generates γ and the odd zeta values.

Proof. Multiply and divide the two limits in (203), and use $\Gamma(1-s)\Gamma(1+s) = \pi s / \sin \pi s$. $\square$

The Bell-polynomial zeta tower itself is classical [28]. The point of [theorem 26.1](#) and [corollary 26.2](#) is its exact finite connection to the individual 2026 Challenge approximants, including their endpoint term.

The same deformation has a finite integer spectral avatar. Put

$$\mathcal{D}_a(s) = \prod_{j=1}^a (s-j)^3 (s+j)^2 (s+a+1). \quad (210)$$

Theorem 26.3 (Monic integer contact polynomial). *There is a unique polynomial $\mathcal{N}_a \in \mathbb{Z}[s]$ such that*

$$\mathcal{K}_a(s) = \frac{\mathcal{N}_a(s)}{a!(a+1)^2 \mathcal{D}_a(s)}. \quad (211)$$

It is monic of degree $5a+2$, and

$$\deg\{\mathcal{N}_a - s\mathcal{D}_a\} = 3a. \quad (212)$$

Consequently $\mathcal{N}_a$ and $s\mathcal{D}_a$ have the same first $2a+1$ Newton power sums. Explicitly, if the zeros of $\mathcal{N}_a$ are counted with multiplicity, then for $1 \leq m \leq 2a+1$,

$$\sum_{\mathcal{N}_a(\nu)=0} \nu^m = 3 \sum_{j=1}^a j^m + 2 \sum_{j=1}^a (-j)^m + (-a-1)^m. \quad (213)$$

At the opposite end of the polynomial,

$$\mathcal{N}_a(0) = (-1)^a (a+1)(a!)^4 q_{a-3}, \quad (214)$$

$$\mathcal{N}'_a(0) = (-1)^a (a+1)(a!)^4 \left[p_{a-3} - q_{a-3} \left(H_a - \frac{1}{a+1} \right) \right]. \quad (215)$$

Proof. Multiplication of (200) by $a!(a+1)^2 \mathcal{D}_a(s)$ clears every Pochhammer denominator. Each resulting quotient is in $\mathbb{Z}[s]$; the endpoint term becomes exactly $s\mathcal{D}_a(s)$. The latter is monic of degree $5a+2$, while the k -th summand of the remaining part has degree $3a-k$. The $k=0$ term has nonzero leading coefficient, proving monicity and (212).

Two monic degree- $5a+2$ polynomials whose difference has degree $3a$ share their first $2a+1$ coefficients below the leading term. Newton's identities give (213); the roots of $s\mathcal{D}_a$ are 0, three copies of $1, \dots, a$, two copies of $-1, \dots, -a$, and $-a-1$. Finally evaluate (211) and its logarithmic derivative at zero, use (201), and compute

$$\frac{\mathcal{D}'_a(0)}{\mathcal{D}_a(0)} = -H_a + \frac{1}{a+1}.$$

This gives [equations \(214\) and \(215\)](#). $\square$

27. Probabilistic realizations and zero geometry

The positive law in (141) has two complementary realizations: exponential maxima at finite index and independent Bernoulli variables through the zeros of its generating polynomial.

For $m \geq 1$, let M_m be the maximum of m independent $\text{Exp}(1)$ variables, put $M_0 = 0$, and use independent copies whenever superscripts are present. Then

$$\mathbb{E}e^{tM_m} = \prod_{j=1}^m \frac{j}{j-t} = \frac{m!}{(1-t)_m} \quad (\Re t < 1). \quad (216)$$

Theorem 27.1 (Exact finite maxima model). *For $a \geq 0$, let J_a have the law*

$$\mathbb{P}(J_a = k) = \frac{\binom{a}{k} \binom{a+1}{k} / k!}{F_{a,a+1}(1)} \quad (0 \leq k \leq a), \quad (217)$$

and, conditionally on J_a , define

$$Z_a^{(0)} = M_{J_a}^{(1)} + M_{J_a}^{(2)} + M_{J_a}^{(3)} - M_{a-J_a}^{(4)} - M_{a+1-J_a}^{(5)}. \quad (218)$$

Then

$$\boxed{\mathbb{E}e^{tZ_a^{(0)}} = \frac{\mathcal{J}_a(t)}{\mathcal{J}_a(0)}} \quad (|\Re t| < 1). \quad (219)$$

If

$$c_a = \frac{a!}{q_{a-3}}, \quad Z_a = Z_a^{(0)} + c_a, \quad (220)$$

then

$$\boxed{\mathbb{E}Z_a = \frac{p_{a-3}}{q_{a-3}}}. \quad (221)$$

Moreover, as $a \rightarrow \infty$, with independent standard Gumbel variables $G_1, \dots, G_5$,

$$Z_a \longrightarrow X_0 := G_1 + G_2 + G_3 - G_4 - G_5 \quad (222)$$

in distribution and with convergence of every moment. Equivalently, locally uniformly for $|\Re t| < 1$,

$$\mathbb{E}e^{tZ_a} \longrightarrow \Gamma(1-t)^3 \Gamma(1+t)^2. \quad (223)$$

Thus

$$\begin{aligned} \kappa_1(Z_a) &\longrightarrow \gamma, \\ \kappa_m(Z_a) &\longrightarrow (m-1)! \{3 + 2(-1)^m\} \zeta(m) \quad (m \geq 2). \end{aligned} \quad (224)$$

Proof. Multiply the weight in (217) by the five conditional moment generating functions from (216). All ordinary factorials cancel, leaving exactly the summand of $\mathcal{J}_a(t)$; normalization at zero proves (219). Taking the first derivative and using (201) gives (221). Since $c_a \rightarrow 0$, theorem 26.1 gives (223) on a neighborhood of the origin. Uniform exponential moments on every smaller strip imply distributional and all-moment convergence. Finally differentiate the logarithm and use (206). $\square$

The distinction between the two deformations is important:

$$\mathbb{E}e^{tZ_a} = e^{c_a t} \frac{\mathcal{J}_a(t)}{\mathcal{J}_a(0)} \neq \frac{\mathcal{K}_a(t)}{\mathcal{K}_a(0)} \quad (225)$$

at finite a . The function $\mathcal{K}_a$ is the exact rational zero/first-jet carrier; $\mathcal{J}_a$ is the exact probability carrier. They have the same Gamma limit.

We next identify the full official Meijer state. Let

$$A_{n,i} \sim \Gamma(n+1, 1) \quad (1 \leq i \leq 3), \quad E_4, E_5 \sim \text{Exp}(1) \quad (226)$$

be independent, and put

$$X_n = \log \frac{E_4 E_5}{A_{n,1} A_{n,2} A_{n,3}}. \quad (227)$$

Theorem 27.2 (Stop-loss interpretation of the Meijer state). *For the official normalization in section 24,*

$$\boxed{\mathcal{H}_n^G(e^{-x}) = (n!)^3 \mathbb{E}(X_n - x)_+} \quad (x \in \mathbb{R}). \quad (228)$$

Consequently

$$\boxed{\frac{(\mathcal{H}_n^G(1), \theta \mathcal{H}_n^G(1), \theta^2 \mathcal{H}_n^G(1))}{(n!)^3} = (\mathbb{E}(X_n)_+, \mathbb{P}(X_n > 0), f_{X_n}(0))}. \quad (229)$$

The functional shift is

$$\mathcal{H}_{n+1}^G(z) = z \theta^2 \mathcal{H}_n^G(z), \quad (230)$$

and hence

$$f_{X_n}(x) = e^x (n+1)^3 \mathbb{E}(X_{n+1} - x)_+. \quad (231)$$

At $n = 0$,

$$\mathfrak{h} := \mathbb{E}(X_0)_+ = \int_0^\infty e^{-x} \frac{\log x}{x-1} dx = 1.4322057346532244148\dots, \quad (232)$$

whereas

$$\mathbb{E}X_0 = \mathbb{P}(X_0 > 0) = \gamma. \quad (233)$$

In particular $\mathfrak{h} \neq \gamma$.

Proof. The Mellin transform of X_n is

$$\mathbb{E}e^{sX_n} = \frac{\Gamma(1+s)^2 \Gamma(n+1-s)^3}{(n!)^3}, \quad -1 < \Re s < n+1. \quad (234)$$

Perron's inverse Mellin formula on a line $0 < c < n+1$ gives (228); differentiation in x , using $\theta = -d/dx$ at $z = e^{-x}$, gives (229). Shifting the three gamma factors in the Mellin–Barnes integral proves (230), and comparison with (228) gives (231). At $n = 0$, substitute the five exponential densities into $\mathbb{E}(X_0)_+$, integrate four variables first, and apply Frullani's identity; the remaining integral is

$$\int_0^\infty e^{-x} \frac{\log x}{x-1} dx,$$

which proves (232). The probability identity in (233) is due to Powers [30]; the expectation identity is immediate from $\mathbb{E}G_j = \gamma$. $\square$

The Bernoulli realization comes from real-rootedness. For integers $b \geq a \geq 1$, write the zeros of $F_{a,b}$, with multiplicities, as $-\rho_{a,b,1}, \dots, -\rho_{a,b,a}$.

Theorem 27.3 (Poisson–binomial law and zero-density limit). *All zeros of $F_{a,b}$ are negative real numbers, and*

$$\frac{F_{a,b}(z)}{F_{a,b}(1)} = \prod_{j=1}^a (1 - p_j + p_j z), \quad p_j = \frac{1}{1 + \rho_{a,b,j}}. \quad (235)$$

Thus the law with weights proportional to $\binom{a}{k}\binom{b}{k}/k!$ is exactly a sum of independent Bernoulli variables. Its mass sequence is strictly log-concave and ultra-log-concave, and it is monotone in each of a, b in the monotone-likelihood-ratio order.

Fix $d \geq 0$, put $b = a + d$ and $s_a = (ab)^{1/3}$. As $a \rightarrow \infty$, if J_a has these weights, then

$$\mathbb{E}J_a \sim s_a, \quad \text{Var}(J_a) \sim \frac{s_a}{3}, \quad (236)$$

$$\frac{J_a - \mathbb{E}J_a}{\sqrt{\text{Var}(J_a)}} \Rightarrow N(0, 1). \quad (237)$$

The corresponding local central limit theorem holds, and J_a/s_a satisfies a large-deviation principle with speed s_a and rate

$$I(x) = \begin{cases} 3(x \log x - x + 1), & x \geq 0, \\ +\infty, & x < 0, \end{cases} \quad (0 \log 0 := 0). \quad (238)$$

Finally, the locally finite zero measures converge vaguely on $[0, \infty)$:

$$\boxed{\frac{1}{s_a} \sum_{j=1}^a \delta_{\rho_{a,b,j}} \Rightarrow \frac{\sqrt{3}}{2\pi} \rho^{-2/3} d\rho.} \quad (239)$$

Proof. The polynomial $L_a(-z) = \sum_k \binom{a}{k} z^k / k!$ has only negative real zeros. The coefficient sequence $\{\binom{b}{k}\}_{k \geq 0}$ is a classical Pólya–Schur multiplier sequence because its exponential generating function is $L_b(-z)$; see [29]. Applying the multiplier theorem to $L_a(-z)$ proves real-rootedness, and positive coefficients make every root negative. Factorization and normalization at $z = 1$ give (235).

The consecutive mass ratio

$$\frac{w_{k+1}}{w_k} = \frac{(a-k)(b-k)}{(k+1)^3}$$

is strictly decreasing, proving strict log-concavity. Ultra-log-concavity follows because $w_k / \binom{a}{k} = \binom{b}{k} / k!$ has decreasing consecutive ratio. Finally

$$\frac{w_{a+1,b;k}}{w_{a,b;k}} = \frac{a+1}{a+1-k}$$

is increasing in k , and similarly for b , proving the likelihood ratio assertion.

A Taylor expansion about the unique real saddle κ_a , determined by

$$(a - \kappa_a)(b - \kappa_a) = (\kappa_a + 1)^3,$$

shows that the discrete law has one mode or two adjacent modes neighboring κ_a . Its logarithmic curvature is

$$\frac{1}{a - \kappa_a} + \frac{1}{b - \kappa_a} + \frac{3}{\kappa_a + 1} \sim \frac{3}{s_a},$$

which gives (236). The Bernoulli representation, $\text{Var}(J_a) \rightarrow \infty$, and the standard Poisson–binomial local limit theorem [25, Chapter VII] give the central and local limit assertions.

Stirling's formula at $k = xs_a$, uniformly on compact x -intervals, gives (238) and exponential tightness gives the full large-deviation principle.

For $z > 0$, tilt the weights by z^k . Their ratio has the extra factor z , so $J_{a,z}/s_a \rightarrow z^{1/3}$. Therefore

$$\frac{1}{s_a} \frac{F'_{a,b}(z)}{F_{a,b}(z)} = \frac{\mathbb{E}_z J_{a,z}}{s_a z} \rightarrow z^{-2/3}. \quad (240)$$

The left side is the Stieltjes transform of the measure on the left of (239), whereas

$$\int_0^\infty \frac{1}{\rho + z} \frac{\sqrt{3}}{2\pi} \rho^{-2/3} d\rho = z^{-2/3}.$$

Uniqueness and the standard continuity theorem for Stieltjes transforms [20, Chapter I] give the vague limit. $\square$

No simplicity or strict-interlacing assertion is needed above. Those stronger statements require a separate strict multiplier or multiple-orthogonality input and are not claimed here.

Remark 27.4 (Priority boundary for the Ramanujan package). The official solution already proves Question 2 through a rank-three Meijer- G trajectory, a Mellin-Barnes recessive estimate, a rational companion gauge, and inverse-transpose Miller selection [32]. The cubic sums and their Bell-zeta approximants occur in [28, 27], while the five-exponential ratio law, its signed-Gumbel logarithm, and the associated Gamma-product transform occur in [30]. The general multiple-orthogonal and multiple-Laguerre background is developed in [35, 19]. The contribution of the present sections is the explicit finite-level transmutation among the Challenge recurrence, its terminating ${}_2F_2$ dual, the official Meijer- G trajectory, and the cubic Bell family, together with the exact finite probability carrier and the all-order logarithmic-jet limits in the Challenge normalization.

28. The regularized Gamma determinant

The positive trace-class operator of section 7 encodes the single Euler trace S_2 . A different, signed Hilbert-Schmidt operator encodes the complete limiting Gamma jet. On $\ell^2(\mathbb{N})^{\oplus 5}$, define

$$\mathbf{A}_\Gamma = \text{diag} \left(\frac{1}{n}, \frac{1}{n}, \frac{1}{n}, -\frac{1}{n}, -\frac{1}{n} : n \geq 1 \right). \quad (241)$$

Let $\mathbf{A}_{\Gamma,N}$ be its compression to the first N coordinates in each of the five summands. We use the standard Hilbert-Schmidt determinant

$$\det {}_2(I - T) = \det((I - T)e^T).$$

Theorem 28.1 (Regularized determinant and zeta traces). *One has $\mathbf{A}_\Gamma \in \mathcal{S}_2 \setminus \mathcal{S}_1$, and*

$$\boxed{\text{FPTr } \mathbf{A}_\Gamma := \lim_{N \rightarrow \infty} (\text{Tr } \mathbf{A}_{\Gamma,N} - \log N) = \gamma.} \quad (242)$$

For every $m \geq 2$,

$$\boxed{\text{Tr } \mathbf{A}_\Gamma^m = \{3 + 2(-1)^m\} \zeta(m).} \quad (243)$$

Equivalently,

$$\boxed{\Gamma(1-t)^3 \Gamma(1+t)^2 = e^{\gamma t} \det {}_2(I - t\mathbf{A}_\Gamma)^{-1}.} \quad (244)$$

In particular

$$\text{Tr } \mathbf{A}_\Gamma^{2r} = 5\zeta(2r), \quad \text{Tr } \mathbf{A}_\Gamma^{2r+1} = \zeta(2r+1) \quad (r \geq 1). \quad (245)$$

Proof. Square summability and failure of absolute summability are immediate from the spectrum. The truncated trace is H_N , so (242) is Euler's definition. Taking powers of the five eigenvalue strings gives (243). Finally, for a Hilbert–Schmidt operator,

$$\log \det_2(I - tA_\Gamma) = - \sum_{m \geq 2} \frac{t^m}{m} \operatorname{Tr} A_\Gamma^m.$$

Compare with (206); this first proves (244) for $|t| < 1$, and meromorphic continuation gives the identity on $\mathbb{C}$. $\square$

This regularized determinant is an explicit diagonal spectral realization of the Gamma product, but it does not furnish a new irrationality result for odd zeta values. It must also not be confused with the ordinary Fredholm determinant Δ_2 of the positive trace-class operator in theorem 7.3.

29. Cross-place Euler–Padé and continued-fraction convergence

The Padé system for Euler's factorial series gives a separate arithmetic transfer. Put

$$Q_n(T) = \sum_{h=0}^n h! \binom{n}{h}^2 (-T)^h, \quad (246)$$

and let $P_n(T)$, of degree at most $n-1$, be the Padé numerator characterized by

$$Q_n(T) \sum_{m \geq 0} m! T^m - P_n(T) = O(T^{2n}) \quad (247)$$

as a formal series. For a prime p , the series

$$E_p(t) = \sum_{m \geq 0} m! t^m \quad (248)$$

converges for $|t|_p \leq 1$. At a real negative argument put

$$E_\infty(t) = \int_0^\infty \frac{e^{-x}}{1 - tx} dx. \quad (249)$$

The standard Padé remainder gives, for $t \in \mathbb{Z}_p$,

$$v_p \left(E_p(t) - \frac{P_n(t)}{Q_n(t)} \right) \geq 2v_p(n!) - v_p(Q_n(t)), \quad (250)$$

whenever $Q_n(t) \neq 0$; see [22].

Theorem 29.1 (Unit-boundary residue criterion). *Let p be prime and $t \in \mathbb{Z}_p^\times$. For $0 \leq r < p$, define*

$$q_{r,p}(X) = \sum_{h=0}^r h! \binom{r}{h}^2 (-X)^h \in \mathbb{F}_p[X]. \quad (251)$$

If

$$q_{r,p}(\bar{t}) \neq 0 \quad (0 \leq r < p), \quad (252)$$

then every $Q_n(t)$ is a p -adic unit and

$$\boxed{\frac{P_n(t)}{Q_n(t)} \longrightarrow E_p(t)}. \quad (253)$$

The following are sample good nonzero residue classes:

p	$t \pmod{p}$	
3	2	
5	2, 3, 4	
7	4, 5	
11	2, 3, 5, 7	
13	6, 9, 11, 12	

(254)

In particular, at $t = -1$, the full convergent sequence tends to $E_p(-1)$ for $p = 2, 3, 5, 13$.

Proof. Modulo p , all terms with $h \geq p$ vanish and, for $h < p$, $\binom{n}{h} \pmod{p}$ depends only on $n \pmod{p}$. Hence

$$Q_n(t) \equiv q_{r,p}(\bar{t}) \pmod{p} \quad (n \equiv r \pmod{p}). \quad (255)$$

Condition (252) makes every denominator a unit; then (250) tends to infinity. The table is a finite calculation in the indicated residue fields.

For $p = 2, t = -1$, an elementary calculation modulo four gives

$$v_2(Q_n(-1)) = \begin{cases} 0, & n \text{ even,} \\ 1, & n \text{ odd.} \end{cases} \quad (256)$$

The same remainder estimate therefore proves full convergence also at two. For $p = 3, 5, 13$, the assertion is contained in the table. $\square$

The congruence behind a simultaneous subsequence is stronger than periodicity.

Theorem 29.2 (A common real and all-prime subsequence). *For all integers $n \geq 0$ and $k \geq 1$,*

$$Q_{n+k}(T) \equiv Q_n(T) \pmod{k\mathbb{Z}[T]}. \quad (257)$$

Let $N_j = \text{lcm}(1, \dots, j)$. For every fixed negative integer t , the single rational sequence

$$x_j(t) = \frac{P_{N_j}(t)}{Q_{N_j}(t)} \quad (258)$$

satisfies

$$\boxed{x_j(t) \longrightarrow E_\infty(t) \text{ in } \mathbb{R}, \quad x_j(t) \longrightarrow E_p(t) \text{ in } \mathbb{Q}_p \text{ for every fixed } p.} \quad (259)$$

At $t = -1$, the real limit is the Euler–Gompertz constant $\delta = eE_1(1)$, while the p -adic limit is $E_p(-1)$.

Proof. For each h , the polynomial $h! \binom{X}{h} = X(X-1)\cdots(X-h+1)$ belongs to $\mathbb{Z}[X]$. Hence

$$h! \left\{ \binom{n+k}{h}^2 - \binom{n}{h}^2 \right\} = \left\{ h! \binom{n+k}{h} - h! \binom{n}{h} \right\} \left\{ \binom{n+k}{h} + \binom{n}{h} \right\}$$

is divisible by k , coefficient by coefficient in (246). This proves (257). Taking $n = 0, k = N_j$ gives

$$Q_{N_j}(T) \equiv 1 \pmod{N_j\mathbb{Z}[T]}. \quad (260)$$

Thus for every fixed p and all $j \geq p$, $Q_{N_j}(t)$ is a p -adic unit, and (250) proves the p -adic limit. The real Stieltjes–Padé convergence to (249) holds for every $t < 0$, and therefore along the same subsequence. $\square$

Proposition 29.3 (*p*-adic denominator interpolation). *For every prime p, the series*

$$q_p(x) = \sum_{h \geq 0} h! \binom{x}{h}^2 \quad (x \in \mathbb{Z}_p) \quad (261)$$

converges uniformly and defines a 1-Lipschitz function satisfying $q_p(n) = Q_n(-1)$ for $n \in \mathbb{N}_0$. Moreover

$$\sup_{n \geq 0} v_p(Q_n(-1)) = \infty \iff q_p \text{ has a zero in } \mathbb{Z}_p. \quad (262)$$

In particular, if q_p is rootless then the full Euler convergent sequence at $t = -1$ tends to $E_p(-1)$.

Proof. Uniform convergence follows from $v_p(h!) \rightarrow \infty$ and $\binom{x}{h} \in \mathbb{Z}_p$. For each summand,

$$h! \left\{ \binom{x}{h}^2 - \binom{y}{h}^2 \right\} = (x^h - y^h) \left\{ \binom{x}{h} + \binom{y}{h} \right\},$$

which is divisible by $x - y$; passing to the uniform limit gives the Lipschitz bound. The interpolation identity follows because $\binom{n}{h} = 0$ for $h > n$. If the valuations are unbounded, compactness of $\mathbb{Z}_p$ supplies a convergent subsequence of the corresponding integers, and continuity gives a zero. Conversely, approximate a zero by nonnegative integers and use continuity. If there is no zero, compactness bounds $v_p(q_p(x))$, and (250) proves the last assertion. $\square$

The word “simultaneous” in theorem 29.2 is placewise: no convergence in the restricted adelic product topology is asserted. Nor does the theorem prove irrationality of any $E_p(-1)$. At the residue $n = p - 1$,

$$Q_{p-1}(-1) \equiv \sum_{h=0}^{p-1} h! \pmod{p}, \quad (263)$$

so Kurepa’s left-factorial problem appears as one boundary congruence, but is not solved here. Full convergence at $t = -1$ for every prime remains open; the observed growth bound $v_p(Q_n(-1)) = O(\log n)$ is not presently a theorem. The existence of a zero of q_p means only that denominator valuations are unbounded; it is not equivalent to failure of convergence, which would require comparison with $v_p(n!)$.

Part V

Positive Spectra, Jacobi Towers, and Gamma Closures

30. Order-axis polyexponential total positivity and the Jacobi tower

30.1. Standing input

Let

$$K = \overline{\mathbf{Q}}(e^\beta : \beta \in \overline{\mathbf{Q}}), \quad \text{Ein}_q(\alpha) = \sum_{n \geq 1} \frac{(-1)^{n-1} \alpha^n}{n^q n!}.$$

The only arithmetic input is [001, Corollary 8.4]: the values $\text{Ein}_q(\alpha)$, jointly for integers $q \geq 1$ and nonzero algebraic α , are algebraically independent over K .

30.2. 1. Master strict-total-positivity theorem

For real $\alpha, q > 0$, set

$$\mathcal{E}(\alpha, q) = \Gamma(q) \text{Ein}_q(\alpha).$$

Then

$$\mathcal{E}(\alpha, q) = \int_0^\infty y^{q-1} (1 - e^{-\alpha e^{-y}}) dy. \quad (\text{TP-1})$$

30.2.1 Theorem

The kernel $(\alpha, q) \mapsto \mathcal{E}(\alpha, q)$ is strictly totally positive of infinite order on $(0, \infty)^2$. Thus, for every

$$0 < \alpha_1 < \cdots < \alpha_r, \quad 0 < q_1 < \cdots < q_r,$$

one has

$$\det[\mathcal{E}(\alpha_i, q_j)]_{i,j=1}^r > 0. \quad (\text{TP-2})$$

30.2.2 Composition proof, including orientation and strictness

Put

$$B(\alpha, y) = 1 - e^{-\alpha e^{-y}} = e^{-y} \int_0^\alpha e^{-ue^{-y}} du = \int_0^\infty \mathbf{1}_{u < \alpha} e^{-y} e^{-ue^{-y}} du. \quad (\text{TP-3})$$

For increasing α_i and increasing u_j , the Volterra kernel $\mathbf{1}_{u < \alpha}$ is totally nonnegative. For increasing u_i and increasing y_j , the numbers $x_j = e^{-y_j}$ are decreasing, and $[e^{-u_i x_j}]$ has positive determinant: this is the exponential kernel e^{uv} with both u_i and $v_j = -x_j$ increasing. Positive column factors e^{-y_j} do not change signs. Continuous Cauchy–Binet applied to (3) gives total nonnegativity of B . It is strict because the cell

$$0 < u_1 < \alpha_1 < u_2 < \alpha_2 < \cdots < u_r < \alpha_r$$

has positive measure; on it the Volterra determinant equals one and the exponential determinant is positive. Hence $B(\alpha, y)$ is STP_∞ for α, y in increasing order.

The kernel

$$C(q, y) = y^{q-1} = e^{(q-1) \log y}$$

is STP_∞ for increasing $q, y > 0$. A second continuous Cauchy–Binet/Andreief composition applied to (1) proves (2). All integrals are finite because the integrand is bounded near zero and is $O(y^{q-1}e^{-y})$ at infinity.

In particular, every finite ordered family $\alpha \mapsto \mathcal{E}(\alpha, q_j)$ is a strict Chebyshev system: every nonzero linear combination of r such functions has at most $r - 1$ distinct zeros. The transform is also variation diminishing in the standard strict-total-positivity sense. Separately, for every fixed $q > 0$, $\alpha \mapsto \mathcal{E}(\alpha, q)$ is a Bernstein function, since

$$(-1)^{m-1} \partial_\alpha^m \mathcal{E}(\alpha, q) > 0 \quad (m \geq 1).$$

30.3. 2. Arithmetic-generic totally positive matrices and positive geometry

Take algebraic $0 < \alpha_1 < \cdots < \alpha_r$ and positive integers $q_1 < \cdots < q_s$, and put

$$M_{ij} = \Gamma(q_j) \text{Ein}_{q_j}(\alpha_i).$$

Then every minor of M is positive, while its rs entries are jointly algebraically independent over K . Hence M is an explicit K -generic point of the open totally positive matrix cone.

Consequences:

1. Every minor is a positive number transcendental over K .
2. For $r < s$, the maximal minors have exactly the classical Pluecker relations and no accidental relation; their field has transcendence degree $r(s - r) + 1$. After projectivization, the row span is a generic point of the positive Grassmannian and has transcendence degree $r(s - r)$.
3. For a square n -by- n sample, the raw nested Pluecker coordinates first form the corresponding multicone. After separate projectivization, the nested first-row spans give a generic point of the totally positive full flag variety. After projectivizing separately in every rank, its transcendence degree is $n(n - 1)/2$, and its relation ideal is the usual flag Pluecker/incidence ideal.
4. For any infinite increasing algebraic sequence α_i and increasing integer sequence q_i , the leading minors $D_n = \det[M_{ij}]_{1 \leq i, j \leq n}$ are jointly algebraically independent over K . Expand D_n in its private variable M_{nn} ; its coefficient is $D_{n-1} \neq 0$, while earlier D_j do not contain M_{nn} .

30.3.1 Oscillation and Galois corollary

For square M , strict total positivity makes M oscillatory. Its eigenvalues are positive and simple, and their eigenvectors have the usual exact sign-variation pattern. The characteristic coefficients are algebraically independent over K : a universal relation would remain after specializing a generic matrix to a diagonal matrix and would give a relation among elementary symmetric functions. Therefore the n eigenvalues are jointly algebraically independent over K . Over the field generated by its characteristic coefficients, the characteristic polynomial is a generic monic polynomial and has Galois group S_n .

30.4. 3. The order-axis Stieltjes moment problem

For algebraic $\alpha > 0$, define

$$\mu_{\alpha,k} = k! \operatorname{Ein}_{k+1}(\alpha) = \int_0^\infty y^k w_\alpha(y) dy, \quad w_\alpha(y) = 1 - e^{-\alpha e^{-y}}. \quad (\text{TP-4})$$

All $\mu_{\alpha,k}$, jointly in positive algebraic α and $k \geq 0$, are algebraically independent over K .

After normalizing the measure, the change of coordinates $m_{\alpha,k} = \mu_{\alpha,k}/\mu_{\alpha,0}$ is birational. Hence $\mu_{\alpha,0}$ together with all nonconstant probability moments $m_{\alpha,k}$, jointly in α , is algebraically independent. The triangular moment–cumulant transform gives the same assertion for all probability cumulants. In particular, the explicit mean and variance

$$\frac{\operatorname{Ein}_2(\alpha)}{\operatorname{Ein}_1(\alpha)}, \quad \frac{2 \operatorname{Ein}_3(\alpha)}{\operatorname{Ein}_1(\alpha)} - \left(\frac{\operatorname{Ein}_2(\alpha)}{\operatorname{Ein}_1(\alpha)} \right)^2$$

belong to a joint algebraically independent family over K .

The Stieltjes problem is determinate. Indeed,

$$\mu_{\alpha,k} = \alpha k! \left(1 - \frac{\alpha}{2^{k+2}} + O_\alpha(3^{-k}) \right), \quad (\text{TP-5})$$

so $\sum_k \mu_{\alpha,2k}^{-1/(2k)} = \infty$.

The infinite Hankel matrix $[\mu_{\alpha,i+j}]_{i,j \geq 0}$ is strictly totally positive. Every one of its finite minors is positive and transcendental over K . If

$$H_{\alpha,d} = \det[\mu_{\alpha,i+j}]_{0 \leq i,j < d},$$

then $\{H_{\alpha,d} : \alpha \in \overline{\mathbf{Q}}_{>0}, d \geq 1\}$ is jointly algebraically independent over K , since

$$\partial H_{\alpha,d} / \partial \mu_{\alpha,2d-2} = H_{\alpha,d-1}.$$

30.5. 4. Jacobi operator, finite inverse spectra, and Galois groups

Let the monic orthogonal polynomials satisfy

$$p_{n+1}(y) = (y - a_{\alpha,n})p_n(y) - b_{\alpha,n}p_{n-1}(y) \quad (n \geq 1), \quad b_{\alpha,n} > 0,$$

with $p_0 = 1$ and $p_1 = y - a_{\alpha,0}$.

Finite Stieltjes–Favard elimination is birational. Consequently the full family

$$\{\mu_{\alpha,0}, a_{\alpha,n}, b_{\alpha,n+1} : \alpha \in \overline{\mathbf{Q}}_{>0}, n \geq 0\}$$

is jointly algebraically independent over K .

Carleman determinacy makes the minimal Jacobi matrix essentially self-adjoint. Its closure is unitarily equivalent to multiplication by y on

$$L^2\left([0, \infty), \frac{w_\alpha(y)}{\operatorname{Ein}(\alpha)} dy\right).$$

It therefore has simple purely absolutely continuous spectrum $[0, \infty)$ of multiplicity one, with the displayed measure as the spectral measure of the first basis vector.

For the leading N -by- N Jacobi truncation $J_{\alpha,N}$:

- its N eigenvalues are positive, simple, and jointly algebraically independent over K ;
- the spectra of $J_{\alpha,N-1}$ and $J_{\alpha,N}$ strictly interlace and their combined $2N - 1$ points are jointly algebraically independent;
- the complete spectral tower through size N has transcendence degree exactly $2N - 1$, with the top consecutive pair a transcendence basis up to algebraic extension;
- the characteristic polynomial of $J_{\alpha,N}$, over its coefficient field, has generic Galois group S_N ; the pair of consecutive characteristic polynomials has group $S_N \times S_{N-1}$ over their joint coefficient field.

All assertions remain joint for finitely many distinct algebraic α , with one chosen consecutive pair at each parameter.

If $\lambda_{N,j}$ and $\omega_{N,j} > 0$ are the Gaussian nodes and unnormalized quadrature masses, then

$$\mu_{\alpha,k} = \sum_{j=1}^N \omega_{N,j} \lambda_{N,j}^k \quad (0 \leq k \leq 2N - 1).$$

The $2N$ numbers $\lambda_{N,1}, \dots, \lambda_{N,N}, \omega_{N,1}, \dots, \omega_{N,N}$ are jointly algebraically independent over K . This follows because Gaussian quadrature is birational up to the finite splitting extension between these data and $\mu_0, \dots, \mu_{2N-1}$.

30.6. 5. Exact probability law and spectral flow in alpha

The normalized law Y_α has density

$$f_\alpha(y) = \frac{1 - e^{-\alpha e^{-y}}}{\text{Ein}(\alpha)}, \quad y \geq 0,$$

and exact survival function

$$\Pr(Y_\alpha > y) = \frac{\text{Ein}(\alpha e^{-y})}{\text{Ein}(\alpha)}. \quad (\text{TP-6})$$

Its moment generating function for $\Re s < 1$ is

$$\mathbf{E}e^{sY_\alpha} = \frac{\alpha^s \gamma(1-s, \alpha) - (1 - e^{-\alpha})}{s \text{Ein}(\alpha)} = \frac{1}{\text{Ein}(\alpha)} \sum_{n \geq 1} \frac{(-1)^{n-1} \alpha^n}{n!(n-s)}. \quad (\text{TP-7})$$

The meromorphic continuation has a nonzero simple pole at every positive integer; hence this EGF is not D-finite and $(\mu_{\alpha,k})$ is not P-recursive (this analytic statement holds for every real $\alpha > 0$).

The density ratio for $\alpha_2 > \alpha_1$ is strictly increasing in y , so the family is increasing in monotone-likelihood-ratio order. Moreover

$$\partial_\alpha \log w_\alpha(y) = \frac{e^{-y}}{e^{\alpha e^{-y}} - 1}$$

is strictly increasing in y . Markov's parameter theorem for orthogonal polynomials therefore gives strict spectral flow: every zero of every fixed $p_N(\cdot; \alpha)$ is strictly increasing in α .

As $\alpha \downarrow 0$, Y_α converges with all moments to $\text{Exp}(1)$; for fixed n ,

$$a_{\alpha,n} \rightarrow 2n + 1, \quad b_{\alpha,n} \rightarrow n^2,$$

and finite spectra tend to Laguerre zeros.

As $\alpha \rightarrow \infty$, with $L = \log \alpha$,

$$Y_\alpha/L \implies \text{Unif}[0, 1]$$

with convergence of every moment. Hence, for fixed n ,

$$\frac{a_{\alpha,n}}{L} \rightarrow \frac{1}{2}, \quad \frac{b_{\alpha,n}}{L^2} \rightarrow \frac{n^2}{4(4n^2 - 1)},$$

and the finite spectra divided by L tend to shifted-Legendre zeros.

The Hankel endpoints are

$$\frac{H_{\alpha,d}}{\alpha^d} \rightarrow \prod_{j=0}^{d-1} (j!)^2 \quad (\alpha \downarrow 0),$$

and

$$\frac{H_{\alpha,d}}{(\log \alpha)^{d^2}} \rightarrow \det \left[\frac{1}{i+j+1} \right]_{0 \leq i,j < d} = \prod_{j=0}^{d-1} \frac{(j!)^4}{(2j)!(2j+1)!} \quad (\alpha \rightarrow \infty).$$

30.7. 6. Large-degree law and determinant free energy

For fixed $\alpha > 0$, concavity of $1 - e^{-\alpha x}$ gives

$$(1 - e^{-\alpha})e^{-y} \leq w_\alpha(y) \leq \alpha e^{-y}. \quad (\text{TP-8})$$

Thus the weight is a bounded positive perturbation of the Laguerre weight. Its Gram determinant obeys

$$(1 - e^{-\alpha})^d \prod_{j=0}^{d-1} (j!)^2 \leq H_{\alpha,d} \leq \alpha^d \prod_{j=0}^{d-1} (j!)^2. \quad (\text{TP-9})$$

In particular

$$\lim_{d \rightarrow \infty} \frac{H_{\alpha,d}^{1/d^2}}{d} = e^{-3/2}. \quad (\text{TP-10})$$

Standard norm-comparison/weighted-potential theory for Laguerre weights then gives the scaled zero law

$$\frac{1}{N} \sum_{j=1}^N \delta_{\lambda_{N,j}/N} \implies \frac{1}{2\pi} \sqrt{\frac{4-x}{x}} \mathbf{1}_{(0,4)}(x) dx,$$

the Marchenko–Pastur law of parameter one.

30.8. 7. Related literature and exact scope

The exact polyexponential kernel $\Gamma(q) \text{Ein}_q(\alpha)$, the associated order-axis Jacobi measure, and the arithmetic-generic positive Grassmannian/oscillatory-matrix consequences are not supplied by the sources cited below. The analytic ingredients are classical: exponential-kernel total positivity and Karlin's basic composition formula. Close precedents are:

- D. St. P. Richards, *Totally positive kernels, Polya frequency functions, and generalized hypergeometric series*, LAA 137/138 (1990), 467–478, DOI 10.1016/0024-3795(90)90139-4.
- V. Buchstaber and A. Glutsyuk, *Total positivity, Grassmannian and modified Bessel functions*, arXiv:1708.02154 (integer-index Bessel TP matrices).
- D. S. P. Salazar, *Strict Total Positivity from Spectral Darboux and Toeplitz Smoothing Mechanisms*, arXiv:2607.02778 (2026), which solves the real-order Bessel question but contains no Ein/polyexponential result.
- A. Postnikov, *Total positivity, Grassmannians, and networks*, arXiv:math/0609764.

The resulting package is an explicit **arithmetic-generic STP kernel** simultaneously realizing generic positive Grassmannians, oscillatory matrices with generic Galois groups, and a determinate Jacobi spectral flow. Its arithmetic conclusions are relative to [001], while the positivity and spectral deductions use classical composition and moment machinery.

31. Order-axis Hankel double scaling: exact ramp equilibrium, free energy, and arithmetic phase transition

31.1. 0. Scope and normalization

For $\alpha > 0$, put

$$w_\alpha(y) = 1 - e^{-\alpha e^{-y}}, \quad \mu_{\alpha,k} = \int_0^\infty y^k w_\alpha(y) dy = k! \text{Ein}_{k+1}(\alpha),$$

and define the size- N order-axis Hankel determinant

$$H_N(\alpha) = \det[\mu_{\alpha,i+j}]_{i,j=0}^{N-1}.$$

The purpose of this note is to close the simultaneous limit

$$\frac{\log \alpha_N}{N} \longrightarrow c \in \mathbf{R}$$

at the level of the equilibrium measure, zero distribution, determinant free energy, and the transition from Marchenko–Pastur to arcsine behavior. The final section records the arithmetic strengthening supplied by [001].

Throughout,

$$\mathcal{I}_c(\mu) = \int_0^\infty (x - c)_+ d\mu(x) - \iint \log |x - y| d\mu(x) d\mu(y),$$

and $\mathcal{F}(c) = \inf_{\mu \in \mathcal{P}([0,\infty))} \mathcal{I}_c(\mu)$.

31.2. 1. Master theorem

31.2.1 Theorem 1 (exact order-axis double-scaling law)

Let $\alpha_N > 0$ and assume

$$c_N := \frac{\log \alpha_N}{N} \longrightarrow c \in \mathbf{R}.$$

Then:

1. The effective fields after $y = Nx$,

$$Q_N(x) = -\frac{1}{N} \log w_{\alpha_N}(Nx),$$

converge uniformly on $[0, \infty)$ to

$$V_c(x) = (x - c)_+.$$

2. The normalized zero-counting measures of the degree- N monic orthogonal polynomials for $w_{\alpha_N}(y)dy$, after division of their zeros by N , converge weakly to the unique minimizer μ_c of $\mathcal{I}_c$.
3. The Hankel determinant satisfies

$$\log H_N(\alpha_N) = N^2 \log N - N^2 \mathcal{F}(c) + o(N^2).$$

4. For $c \leq 0$,

$$d\mu_c(x) = \frac{1}{2\pi} \sqrt{\frac{4-x}{x}} \mathbf{1}_{(0,4)}(x) dx, \quad \mathcal{F}(c) = \frac{3}{2} - c.$$

5. For $c > 0$, let $\phi = \phi(c) \in (0, \pi/2)$ be the unique solution

$$c = \frac{2\pi \cos^2 \phi}{\phi + \sin \phi \cos \phi}. \quad (\text{DS-1.1})$$

Set

$$D(\phi) = \phi + \sin \phi \cos \phi, \quad b = b(c) = \frac{2\pi}{D(\phi)}. \quad (\text{DS-1.2})$$

Thus $c = b \cos^2 \phi$ and $b - c = b \sin^2 \phi$. The support is exactly $[0, b]$, and on $0 < x < b$, $x \neq c$, its density is

$$\begin{aligned} \rho_c(x) = & \frac{\phi}{\pi^2} \sqrt{\frac{b-x}{x}} \\ & + \frac{1}{2\pi^2} \log \left| \frac{\sqrt{c(b-x)} + \sqrt{(b-c)x}}{\sqrt{c(b-x)} - \sqrt{(b-c)x}} \right|. \end{aligned} \quad (\text{DS-1.3})$$

Equivalently,

$$\rho_c(x) = \frac{1}{2\pi^2} \sqrt{\frac{b-x}{x}} \left[2\phi + \sqrt{\frac{x}{b-x}} \log \left| \frac{\cot \phi + \sqrt{x/(b-x)}}{\cot \phi - \sqrt{x/(b-x)}} \right| \right]. \quad (\text{DS-1.4})$$

6. The mass lying in the linearly confined part of the potential is

$$p(c) := \mu_c([c, b]) = \frac{b\phi^2}{\pi^2} = \frac{2\phi^2}{\pi D(\phi)}. \quad (\text{DS-1.5})$$

7. The equilibrium free energy is

$$\mathcal{F}(c) = \frac{3}{2} + \log \frac{2D(\phi)}{\pi} - \frac{2\phi \cos^2 \phi (2\phi + \sin \phi \cos \phi)}{D(\phi)^2}, \quad c > 0. \quad (\text{DS-1.6})$$

It obeys the envelope identity

$$\mathcal{F}'(c) = -p(c). \quad (\text{DS-1.7})$$

31.3. 2. Uniform reduction to the ramp potential

For every $t > 0$,

$$(1 - e^{-1}) \min(1, t) \leq 1 - e^{-t} \leq \min(1, t).$$

Taking $t = \alpha_N e^{-Nx}$ gives the uniform bounds

$$(x - c_N)_+ \leq Q_N(x) \leq (x - c_N)_+ + \frac{-\log(1 - e^{-1})}{N}. \quad (\text{DS-2.1})$$

Since

$$\sup_{x \geq 0} |(x - c_N)_+ - (x - c)_+| \leq |c_N - c|,$$

the convergence $Q_N \rightarrow V_c$ is uniform on the whole half-line, not merely locally away from the kink.

Heine's identity and $y_i = Nx_i$ give

$$H_N(\alpha_N) = \frac{N^{N^2}}{N!} \int_{[0, \infty)^N} \Delta(x)^2 \prod_{j=1}^N e^{-NQ_N(x_j)} dx_j. \quad (\text{DS-2.2})$$

The logarithmic-gas Laplace principle, together with (2.1), yields

$$\frac{1}{N^2} \log \frac{H_N(\alpha_N)}{N^{N^2}} \longrightarrow -\mathcal{F}(c). \quad (\text{DS-2.3})$$

Here is the short triangular-array reduction implicit in this step. Put

$$\delta_N = \|Q_N - V_c\|_{L^\infty([0,\infty))} = o(1)$$

and let $Z_N[Q]$ denote the N -particle integral in (2.2), with Q_N replaced by Q . Direct comparison of the integrands gives

$$e^{-N^2\delta_N} Z_N[V_c] \leq Z_N[Q_N] \leq e^{N^2\delta_N} Z_N[V_c]. \quad (\text{DS-2.4})$$

Thus the partition-function limit for the triangular array follows from the classical fixed-field limit for V_c . The same reduction applies to zeros. Write $\|p\|_{Q,N}^2 = \int_0^\infty |p(x)|^2 e^{-NQ(x)} dx$. If $p_N^{(N)}$ is the monic degree- N extremal polynomial for Q_N , and $p_N^{(c)}$ is the corresponding extremal polynomial for V_c , then

$$\|p_N^{(N)}\|_{V_c,N}^2 \leq e^{2N\delta_N} \|p_N^{(c)}\|_{V_c,N}^2. \quad (\text{DS-2.5})$$

Consequently $p_N^{(N)}$ is asymptotically extremal for the fixed admissible field V_c , and the standard weighted zero-distribution theorem gives the same equilibrium limit. Equations (2.4)–(2.5) isolate all dependence on the varying sequence Q_N .

For precision, the standard varying-weight result used here is the following Gonchar–Rakhmanov/Mhaskar–Saff form. If $Q_N \rightarrow Q$ uniformly on a closed real set, Q is continuous and admissible ($Q(x) - 2\log(1 + |x|) \rightarrow +\infty$), and the reference measure has positive density almost everywhere, then the normalized zeros of the monic degree- N $L^2(e^{-NQ_N} dx)$ -extremal polynomials converge to the weighted equilibrium measure for Q . The corresponding log-gas partition functions obey the energy Laplace principle (2.3). On an unbounded set this is reduced to the compact theorem by the classical restricted-range inequality.

All hypotheses are explicit here. Equation (2.1) gives uniform convergence, and, because c_N is eventually bounded,

$$Q_N(x) \geq x - C \quad (x \geq C + 1)$$

with one constant C . Thus admissibility, exponential tightness, and the restricted-range reduction are uniform in N . Lebesgue measure has positive density everywhere. This proves both the zero-counting statement and (2.3). This is the only place where general weighted-potential theory is used; all remaining formulas are obtained explicitly below.

For $c \leq 0$, $V_c(x) = x - c$ on $[0, \infty)$. Adding the constant $-c$ does not alter the minimizer, so it is the Laguerre/Marchenko–Pastur equilibrium measure. Its first moment is one and its logarithmic energy gives $\mathcal{F}(0) = 3/2$: explicitly, the MP law has $\int x d\mu_0 = 1$, $\int \log x d\mu_0 = -1$, and evaluating the Euler–Lagrange constant at the hard edge gives $\ell = -2$, so $\mathcal{F}(0) = (1 - \ell)/2 = 3/2$. Hence $\mathcal{F}(c) = 3/2 - c$.

31.4. 3. Solution of the ramp equilibrium problem for $c > 0$

31.4.1 3.1 Resolvent and endpoint

Suppose the support is $[0, b]$, with $b > c$. Let

$$M_c(z) = \int_0^b \frac{d\mu_c(x)}{z-x}, \quad R(z) = \sqrt{\frac{z-b}{z}}, \quad R(z) \rightarrow 1 \quad (z \rightarrow \infty).$$

The differentiated Euler–Lagrange condition is

$$M_{c,+}(x) + M_{c,-}(x) = V'_c(x) = \mathbf{1}_{(c,b)}(x). \quad (\text{DS-3.1})$$

The scalar Riemann–Hilbert solution is

$$M_c(z) = \frac{R(z)}{2\pi} \int_c^b \frac{1}{z-s} \sqrt{\frac{s}{b-s}} ds. \quad (\text{DS-3.2})$$

The condition $M_c(z) = z^{-1} + O(z^{-2})$ is precisely

$$\int_c^b \sqrt{\frac{s}{b-s}} ds = 2\pi. \quad (\text{DS-3.3})$$

Put $c = b \cos^2 \phi$, $b - c = b \sin^2 \phi$. Evaluating (3.3) gives

$$b(\phi + \sin \phi \cos \phi) = 2\pi,$$

which is (1.1)–(1.2). The map

$$c(\phi) = \frac{2\pi \cos^2 \phi}{D(\phi)}$$

is a decreasing bijection from $(0, \pi/2)$ onto $(0, \infty)$, with $c(\phi) \rightarrow \infty$ as $\phi \downarrow 0$ and $c(\phi) \rightarrow 0$ as $\phi \uparrow \pi/2$, since

$$c'(\phi) = -\frac{4\pi \cos \phi (\cos \phi + \phi \sin \phi)}{D(\phi)^2} < 0. \quad (\text{DS-3.4})$$

Thus the endpoint is unique.

The integral in (3.2) is elementary. With consistent branches one may also write

$$M_c(z) = \frac{1}{\pi} \left[\arctan \left(\tan \phi \sqrt{\frac{z}{z-b}} \right) - \phi \sqrt{\frac{z-b}{z}} \right]. \quad (\text{DS-3.5})$$

31.4.2 3.2 Density and Frostman verification

Taking the jump of (3.2) gives

$$\rho_c(x) = \frac{1}{2\pi^2} \sqrt{\frac{b-x}{x}} \text{PV} \int_c^b \frac{1}{s-x} \sqrt{\frac{s}{b-s}} ds. \quad (\text{DS-3.6})$$

If

$$a = \sqrt{\frac{x}{b-x}}, \quad u_c = \sqrt{\frac{c}{b-c}} = \cot \phi,$$

then direct integration yields

$$\text{PV} \int_c^b \frac{1}{s-x} \sqrt{\frac{s}{b-s}} ds = 2\phi + a \log \left| \frac{u_c + a}{u_c - a} \right|. \quad (\text{DS-3.7})$$

The right side is strictly positive both for $a < u_c$ and $a > u_c$. Hence (1.3) defines a positive probability density on $(0, b)$.

Equation (3.1) makes the Frostman expression constant on $[0, b]$. For $x > b$, the derivative of

$$V_c(x) - 2 \int \log |x-y| d\mu_c(y)$$

is $1 - 2M_c(x) > 0$: the Stieltjes transform decreases from $M_c(b+) = 1/2$. Thus the required Frostman inequality holds off the support. This proves that (1.3) is the unique equilibrium measure, rather than only a formal solution of the singular-integral equation.

The three distinguished local behaviors are

$$\rho_c(x) \sim \frac{\phi \sqrt{b}}{\pi^2 \sqrt{x}} \quad (x \downarrow 0), \quad (\text{DS-3.8})$$

$$\rho_c(x) = \frac{1}{2\pi^2} \log \frac{4c(b-c)}{b|x-c|} + O(1) \quad (x \rightarrow c), \quad (\text{DS-3.9})$$

and

$$\rho_c(x) \sim \frac{\phi + \cot \phi}{\pi^2 \sqrt{b}} \sqrt{b-x} \quad (x \uparrow b). \quad (\text{DS-3.10})$$

Thus the hard edge survives, the ramp kink produces an integrable logarithmic bulk singularity, and the right endpoint is soft.

31.5. 4. Tail mass and exact free energy

Under $x = b \sin^2 \theta$, the kink corresponds to $\theta = \pi/2 - \phi$. Integrating (1.4) over $[c, b]$ gives

$$p(c) = \frac{b}{\pi^2} \int_0^{\tan \phi} \left[\frac{2\phi t^2}{(1+t^2)^2} + \frac{t}{(1+t^2)^2} \log \frac{\tan \phi + t}{\tan \phi - t} \right] dt. \quad (\text{DS-4.1})$$

The two integrals are

$$\phi(\phi - \sin \phi \cos \phi), \quad \phi \sin \phi \cos \phi,$$

respectively. Hence (1.5) follows.

Since μ_c has no atom at c , the envelope theorem gives

$$\mathcal{F}'(c) = \int \partial_c(x - c)_+ d\mu_c(x) = -\mu_c((c, \infty)) = -p(c). \quad (\text{DS-4.2})$$

Using $c = 2\pi \cos^2 \phi / D(\phi)$, one checks

$$\frac{d}{d\phi} \left[\log D(\phi) - \frac{2\phi \cos^2 \phi (2\phi + \sin \phi \cos \phi)}{D(\phi)^2} \right] = -p(c(\phi))c'(\phi). \quad (\text{DS-4.3})$$

At $\phi = \pi/2$, $c = 0$, $D = \pi/2$, and $\mathcal{F}(0) = 3/2$. Integrating (4.3) proves (1.6).

31.6. 5. The MP-to-arcsine interpolation and its phase transition

31.6.1 5.1 Onset at $c = 0$

Write $\phi = \pi/2 - \varepsilon$. Then

$$c = 4\varepsilon^2 + O(\varepsilon^4), \quad b = 4 + \frac{2}{3\pi}c^{3/2} + O(c^2), \quad (\text{DS-5.1})$$

$$p(c) = 1 - \frac{2}{\pi}\sqrt{c} + O(c), \quad (\text{DS-5.2})$$

and

$$\boxed{\mathcal{F}(c) = \frac{3}{2} - c + \frac{4}{3\pi}c^{3/2} + O(c^2) \quad (c \downarrow 0).} \quad (\text{DS-5.3})$$

For $c < 0$, $\mathcal{F}(c) = 3/2 - c$. Therefore $\mathcal{F}$ is C^1 but not C^2 at $c = 0$; more precisely,

$$\mathcal{F}''(c) \sim \frac{1}{\pi\sqrt{c}} \quad (c \downarrow 0).$$

This is a genuine continuous hard-edge onset transition with singular exponent $3/2$. Calling it a third-order transition would be inaccurate under the usual derivative-based convention.

31.6.2 5.2 Arcsine limit

As $c \rightarrow \infty$,

$$\phi = \frac{\pi}{c} - \frac{2\pi^3}{3c^3} + O(c^{-5}), \quad b = c + \frac{\pi^2}{c} + O(c^{-3}), \quad (\text{DS-5.4})$$

$$p(c) = \frac{1}{c} - \frac{\pi^2}{3c^3} + O(c^{-5}), \quad (\text{DS-5.5})$$

and

$$\mathcal{F}(c) = \log \frac{4}{c} - \frac{\pi^2}{6c^2} + O(c^{-4}). \quad (\text{DS-5.6})$$

Let ν_c be the pushforward of μ_c by $x \mapsto x/c$. Then

$$\boxed{\nu_c \Longrightarrow \frac{\mathbf{1}_{(0,1)}(u)}{\pi\sqrt{u(1-u)}} du \quad (c \rightarrow \infty).} \quad (\text{DS-5.7})$$

Indeed, for fixed $0 < u < 1$, (1.3) gives

$$c\rho_c(cu) \longrightarrow \frac{1}{\pi\sqrt{u(1-u)}},$$

while $b/c \rightarrow 1$ and the tail mass above the kink is $p(c) \rightarrow 0$. Thus this single explicit equilibrium family starts at the parameter-one Marchenko–Pastur law and ends, after its natural macroscopic rescaling, at the arcsine equilibrium law of a compact interval.

31.7. 6. Arithmetic strengthening from [001]

Let

$$K = \overline{\mathbf{Q}}(e^\beta : \beta \in \overline{\mathbf{Q}}).$$

The arithmetic input [001, Corollary 8.4] says that all $\text{Ein}_q(\alpha)$, jointly for $q \geq 1$ and nonzero algebraic α , are algebraically independent over K .

31.7.1 Theorem 2 (an algebraically independent phase-transition sequence)

For every sequence $\alpha_N \in \overline{\mathbf{Q}}_{>0}$, the countable family

$$\{H_N(\alpha_N) : N \geq 1\}$$

is algebraically independent over K , in the finite-subset sense.

Proof For fixed N , the coordinate

$$\mu_{\alpha_N, 2N-2} = (2N-2)! \text{Ein}_{2N-1}(\alpha_N)$$

occurs only in the bottom-right entry of the N -by- N Hankel matrix. Expansion in this coordinate gives

$$H_N(\alpha_N) = H_{N-1}(\alpha_N)\mu_{\alpha_N, 2N-2} + B_N, \quad (\text{DS-6.1})$$

where B_N does not involve $\mu_{\alpha_N, 2N-2}$, and $H_{N-1}(\alpha_N) \neq 0$ by strict positivity. No earlier determinant $H_j(\alpha_j)$, $j < N$, uses the coordinate $(\alpha_N, 2N-1)$. Since all underlying polyexponential coordinates are jointly algebraically independent, (6.1) proves the assertion inductively. For completeness, there is no hidden cancellation in this pivot argument. If a nonzero relation $R(H_1, \dots, H_N) = 0$ had degree d in its last variable, induction would make its leading coefficient $R_d(H_1, \dots, H_{N-1})$ nonzero. After substituting (6.1) and viewing the result as a polynomial in the new coordinate $\mu_{\alpha_N, 2N-2}$, its leading coefficient would be

$$R_d(H_1, \dots, H_{N-1})H_{N-1}(\alpha_N)^d \neq 0.$$

This contradicts the algebraic independence of that coordinate from all coordinates occurring earlier. The reasoning also covers repeated values among the α_N , because the order $2N - 1$ coordinate cannot occur in a determinant of smaller size.

For every real c , one can choose positive rational α_N with $\log \alpha_N / N \rightarrow c$. Combining Theorems 1 and 2 therefore produces, for every $c \in \mathbf{R}$, an explicit sequence of positive, jointly algebraically independent transcendental Hankel determinants satisfying

$$\log H_N(\alpha_N) = N^2 \log N - N^2 \mathcal{F}(c) + o(N^2),$$

with the nonanalytic 3/2-onset at $c = 0$ and the MP-to-arcsine spectral interpolation. This is the cleanest direct bridge between the arithmetic theorem of [001] and a macroscopic random-matrix phase diagram.

31.8. 7. Literature boundary and honest level assessment

The general mechanisms used here are classical:

- logarithmic equilibrium and varying-weight zero asymptotics go back to Gonchar–Rakhmanov and weighted-potential theory;
- the Laguerre endpoint $c \leq 0$ is the parameter-one Marchenko–Pastur law;
- the $c \rightarrow \infty$ endpoint is the classical equilibrium/arcsine law on an interval;
- large- N asymptotics for analytic one-cut varying weights were developed by Deift–Kriecherbauer–McLaughlin–Venakides–Zhou.

The cited results do not directly treat the exact weight $1 - e^{-\alpha e^{-y}}$, the exact ramp field $(x - c)_+$ with formulas (1.1)–(1.7), or its coupling to the polyexponential Hankel tower. The remaining local problem is the analysis of the logarithmic kink (3.9): the mean spacing there is of order $(N \log N)^{-1}$ in the scaled x -coordinate, while the original Gumbel smoothing of the weight occurs on the larger N^{-1} scale. A rigorous local kernel, recurrence-coefficient asymptotics, and a uniform critical expansion across $c = 0$ would constitute a substantially deeper Riemann–Hilbert problem.

31.8.1 Primary background located

- A. A. Gonchar and E. A. Rakhmanov, *Equilibrium measure and the distribution of zeros of extremal polynomials*, Math. USSR-Sb. 53 (1986), 119–130.
- P. Deift, T. Kriecherbauer, K. T.-R. McLaughlin, S. Venakides and X. Zhou, *Uniform asymptotics for polynomials orthogonal with respect to varying exponential weights and applications to universality questions in random matrix theory*, Comm. Pure Appl. Math. 52 (1999), 1335–1425.
- A. B. J. Kuijlaars and P. M. J. Tibboel, *The asymptotic behaviour of recurrence coefficients for orthogonal polynomials with varying exponential weights*, [arXiv:0708.3956](https://arxiv.org/abs/0708.3956).

32. Fixed-cut incomplete Gamma tail: differential transcendence, zero density, and the Euler-motive barrier

32.1. Status and scope

This section combines the all-order algebraic-point theorem of companion paper [001] with the universal inverse-trace identities developed above. Accordingly, every arithmetic-independence or transcendence statement in this section has status **R-PASS**. Analytic identities are unconditional. No claim about the individual irrationality or transcendence of γ or δ , and no comparison-injectivity claim, is made or used.

Put

$$K := \overline{\mathbf{Q}}(e^\alpha : \alpha \in \overline{\mathbf{Q}}), \quad E := E_1(1) = \frac{\delta}{e}, \quad \eta := \text{Ein}(1) = \gamma + E.$$

Write the Laurent expansion and the upper-tail moments as

$$\Gamma(s) = \frac{1}{s} + \sum_{k \geq 0} g_k s^k, \quad T_k := \frac{1}{k!} \int_1^\infty e^{-u} \frac{(\log u)^k}{u} du.$$

The split at one proved in [001] is

$$c_k := g_k - T_k = (-1)^{k+1} \text{Ein}_{k+1}(1), \quad (\text{GT-1})$$

and the family $(c_k)_{k \geq 0}$ is jointly algebraically independent over K .

32.2. 1. Exact finite gamma-jet field and the tail transcendence-degree bound

32.2.1 Lemma 1 (exact gamma-jet field)

For $N = 0$,

$$K(g_0) = K(\gamma).$$

For every $N \geq 1$,

$$\boxed{K(g_0, \dots, g_N) = K\left(\gamma, \pi^2, \zeta(3), \zeta(5), \dots, \zeta\left(2 \left\lfloor \frac{N}{2} \right\rfloor + 1\right)\right)}. \quad (\text{GT-2})$$

Consequently,

$$\text{trdeg}_K K(g_0, \dots, g_N) \leq 2 + \left\lfloor \frac{N}{2} \right\rfloor \quad (N \geq 1). \quad (\text{GT-3})$$

Proof The standard identity

$$\Gamma(1+s) = \exp\left(-\gamma s + \sum_{j \geq 2} \frac{(-1)^j \zeta(j)}{j} s^j\right)$$

shows triangularly that $g_0, \dots, g_N$ and $\gamma, \zeta(2), \dots, \zeta(N+1)$ generate the same field: taking the formal logarithm gives the inverse triangular transformation. The even zeta values all lie in $\mathbf{Q}(\pi^2)$, while the odd indices in $[3, N+1]$ are precisely

$$3, 5, \dots, 2 \left\lfloor \frac{N}{2} \right\rfloor + 1,$$

of which there are $\lfloor N/2 \rfloor$. This proves (2) and (3). Notice that (3) is only an upper bound: no algebraic independence of γ, π^2 , or odd zeta values over K is being assumed. $\square$

32.2.2 Theorem 2 (half-rank lower bound for the fixed-cut tail)

For every $N \geq 1$,

$$\boxed{\text{trdeg}_K K(T_0, \dots, T_N) \geq \left\lfloor \frac{N}{2} \right\rfloor - 1.} \quad (\text{GT-4})$$

For $N = 0$, the corresponding lower bound is 0. In particular,

$$\text{trdeg}_K K(T_0, T_1, \dots) = \infty. \quad (\text{GT-5})$$

Proof By (1),

$$K(c_0, \dots, c_N) \subset K(g_0, \dots, g_N, T_0, \dots, T_N).$$

The left side has transcendence degree $N+1$ over K . Subadditivity and (3) therefore give

$$\begin{aligned} N+1 &\leq \text{trdeg}_K K(g_0, \dots, g_N, T_0, \dots, T_N) \\ &\leq 2 + \left\lfloor \frac{N}{2} \right\rfloor + \text{trdeg}_K K(T_0, \dots, T_N), \end{aligned}$$

which is (4). $\square$

The bound is collective. It implies that infinitely many tail moments are transcendental over K , with lower density at least $1/2$ among the indices in the weak counting sense supplied by (4), but it does **not** name $T_0 = \delta/e$ as one of them.

32.3. 2. Mahler's coefficient-field theorem and a fixed-cut DT theorem

32.3.1 The exact Mahler input

The needed result is the formal-power-series theorem in K. Mahler, “A remark on algebraic differential equations,” *Rend. Accad. Naz. Lincei*, Ser. VIII **50** (1971), 402–412, especially §§3–5 ([scan](#)).

Mahler’s theorem (form used here). Let L be any field of characteristic zero and let $f(z) = \sum_{n \geq 0} f_n z^n \in L[[z]]$. If f satisfies a nonzero algebraic differential equation

$$F(z, f, f', \dots, f^{(m)}) = 0, \quad F \in L[z, Y_0, \dots, Y_m],$$

then the coefficient field $\mathbf{Q}(f_0, f_1, \dots)$ is a finitely generated field extension of $\mathbf{Q}$; in particular, it has finite transcendence degree over $\mathbf{Q}$.

Rational functions in z are allowed after clearing denominators. The statement is formal and includes singular formal solutions. Mahler first constructs a defining equation over the coefficient field. The Hurwitz–Popken recurrence in §4 has the eventual form

$$A(n)f_{n+1} = B_n(f_0, \dots, f_n), \tag{GT-6}$$

where A is a nonzero polynomial and the coefficients used in all B_n ’s belong to the finite set of coefficients of the defining equation. Since a nonzero polynomial has only finitely many integer zeros, all sufficiently late coefficients are rational functions of finitely many early coefficients and finitely many equation coefficients. Section 5 records the result as

$$\mathbf{Q}(f_0, f_1, \dots) = \mathbf{Q}(s_1, \dots, s_r, t),$$

with $s_1, \dots, s_r$ algebraically independent and t algebraic over $\mathbf{Q}(s_1, \dots, s_r)$.

32.3.2 Theorem 3 (fixed-cut upper incomplete Gamma is differentially transcendental)

Let

$$U(s) := \Gamma(s, 1) = \int_1^\infty e^{-u} u^{s-1} du.$$

Then

$$\boxed{U(s) \text{ is differentially transcendental over } \mathbf{C}(s).} \tag{GT-7}$$

Equivalently, no nonzero polynomial in $s, U, U', \dots, U^{(m)}$, with coefficients in $\mathbf{C}$, vanishes identically.

Proof Uniform domination on compact subsets of the s -plane permits differentiation under the integral, so U is entire and

$$U(s) = \sum_{k \geq 0} T_k s^k. \tag{GT-8}$$

If U were differentially algebraic over $\mathbf{C}(s)$, clearing denominators and applying Mahler’s theorem with $L = \mathbf{C}$ would make $\mathbf{Q}(T_0, T_1, \dots)$ finite-transcendence-degree over $\mathbf{Q}$. But (4) tends to infinity. Algebraic independence over K implies algebraic independence over $\mathbf{Q}$, so $\text{trdeg}_{\mathbf{Q}} \mathbf{Q}(T_0, T_1, \dots) = \infty$, a contradiction. $\square$

This strictly strengthens the safe consequences “not D-finite” and “the coefficient sequence is not P-recursive.” It also implies that U belongs to no finite-type ordinary Picard–Vessiot

extension of $\mathbf{C}(s)$, because every element of such an extension is differentially algebraic over the base differential field.

32.3.3 Spectral formal-series corollary

Let $m_k \geq 2$ be arbitrary and put

$$\sigma_k := P_{m_k}(T_k^{-1}),$$

where P_m is the universal inverse-trace polynomial of v2.3. For every finite index set I ,

$$\text{trdeg}_K K(\sigma_k : k \in I) = \text{trdeg}_K K(T_k : k \in I), \quad (\text{GT-9})$$

because $\sigma_k \in K(T_k)$ and T_k is algebraic over $K(\sigma_k)$. Hence the formal series $\sum_{k \geq 0} \sigma_k z^k$ is also differentially transcendental over $\mathbf{C}(z)$. Convergence is not needed for this statement: Mahler's theorem is formal.

32.3.4 Exact overlap with Arreche (2013)

C. E. Arreche, “A Galois-theoretic proof of the differential transcendence of the incomplete Gamma function,” *J. Algebra* **389** (2013), 119–127 ([arXiv:1304.1917](https://arxiv.org/abs/1304.1917)), proves

$$\gamma(t, x) = \int_0^x u^{t-1} e^{-u} du \quad \text{is } \partial_t\text{-transcendental over } \mathbf{C}(t, x). \quad (\text{GT-10})$$

Its setting is parameterized Picard–Vessiot theory: x is the main differential variable, t is the parameter, and

$$\partial_x^2 Y - \frac{t-1-x}{x} \partial_x Y = 0. \quad (\text{GT-11})$$

Arreche's criterion in fact applies to any nonconstant solution of (11), so it also covers the bivariate upper incomplete Gamma after placing that solution in the ambient differential extension.

The overlap and non-overlap are therefore precise:

- Both results assert differential transcendence in the **parameter/order** direction.
- Arreche's theorem is unconditional and bivariate over $\mathbf{C}(t, x)$, obtained from a parameterized Galois group.
- Theorem 3 above first fixes the cut at $x = 1$ and then proves that the one-variable entire function $s \mapsto \Gamma(s, 1)$ is DT over $\mathbf{C}(s)$, using the arithmetic independence theorem of [001] and Mahler's coefficient theorem.
- The bivariate assertion (10) does **not formally specialize** to the fixed-cut assertion (7): differential transcendence need not survive specialization of a second variable. Conversely, (7) says nothing about the full PPV group in the x -direction.

Thus (7) is not a corollary of the cited Arreche theorem as stated. A final priority claim would still require a dedicated search of the broader shift-difference/hypertranscendence literature, because U also satisfies $U(s+1) = sU(s) + e^{-1}$.

32.4. 3. A half-density theorem for wholly transcendental tail divisors

For $k \geq 0$, set

$$F_k(z) := \text{Ein}(z) - T_k.$$

32.4.1 Lemma 4 (unique real tail root)

There is a unique $r_k \in (0, 1)$ with

$$\text{Ein}(r_k) = T_k, \tag{GT-12}$$

and the sequence (r_k) is strictly decreasing.

Proof With $u = e^y$, put

$$I_k := k!T_k = \int_0^\infty e^{-e^y} y^k dy.$$

Integration by parts gives

$$(k+1)I_k = \int_0^\infty e^y e^{-e^y} y^{k+1} dy > I_{k+1},$$

so $T_{k+1} < T_k$. Also $0 < T_k \leq T_0 = E < \eta = \text{Ein}(1)$. The real function Ein is strictly increasing because $(1 - e^{-x})/x > 0$ for every real $x \neq 0$. This proves existence, uniqueness, and strict decrease. $\square$

32.4.2 Lemma 5 (only possible algebraic zero)

The only possible algebraic zero of F_k is r_k .

Proof There is only one real zero by strict monotonicity. If a nonreal algebraic zero z existed, then $\bar{z} \neq z$ would also be algebraic and

$$\text{Ein}(z) = T_k = \text{Ein}(\bar{z}).$$

This is a polynomial relation between two distinct coordinates of the jointly algebraically independent family in [001, Cor. 8.4], a contradiction. $\square$

32.4.3 Theorem 6 (half-density all-zero theorem)

Let

$$A_N := \#\{0 \leq k \leq N : r_k \in \overline{\mathbf{Q}}\}.$$

Then for every $N \geq 1$,

$$\boxed{A_N \leq 2 + \left\lfloor \frac{N}{2} \right\rfloor.} \quad (\text{GT-13})$$

Consequently at least

$$\boxed{\left\lceil \frac{N}{2} \right\rceil - 1} \quad (\text{GT-14})$$

of the divisors $F_0, \dots, F_N$ have **all** their zeros transcendental. In density notation,

$$\liminf_{N \rightarrow \infty} \frac{\#\{0 \leq k \leq N : \text{every zero of } F_k \text{ is transcendental}\}}{N+1} \geq \frac{1}{2}. \quad (\text{GT-15})$$

Proof Let $I \subset \{0, \dots, N\}$ be the set of indices with algebraic r_k . The $2|I|$ resonant coordinates

$$\{\text{Ein}_{k+1}(1), \text{Ein}_1(r_k) : k \in I\}$$

are pairwise distinct and jointly algebraically independent over K by [001, Cor. 8.4]. Via (1) and $g_k = c_k + T_k$, the invertible blockwise linear transformation

$$(c_k, T_k) \longleftrightarrow (c_k, g_k)$$

shows that the selected g_k 's are algebraically independent over K . Hence

$$|I| \leq \text{trdeg}_K K(g_0, \dots, g_N) \leq 2 + \left\lfloor \frac{N}{2} \right\rfloor,$$

which proves (13). Lemma 5 then gives (14) and (15). $\square$

32.4.4 A sharp level-zero fork

The unique real zero at level zero is

$$r_0 = \text{Ein}^{-1}(E_1(1)) \approx 0.2321964241289549.$$

Exactly one of the following holds:

1. every zero of $\text{Ein}(z) - E_1(1)$ is transcendental;
2. r_0 is algebraic, and γ and δ are algebraically independent over K .

Indeed, in the second case $\text{Ein}(1) = \eta$ and $\text{Ein}(r_0) = E$ are jointly algebraically independent. The invertible linear/scalar change

$$(\eta, E) \longleftrightarrow (\gamma = \eta - E, \delta = eE)$$

proves the assertion. This is a genuine zero criterion, but not an unconditional irrationality result: the arithmetic nature of r_0 is unknown.

32.4.5 An Euler–Gompertz pencil strengthening

For real algebraic $t \in [0, 1]$, put

$$c_t := (1 - t)\gamma + tE, \quad \text{Ein}(q_t) = c_t, \quad q_t \in (0, 1). \quad (\text{GT-16})$$

The elementary bounds $E < e^{-1} < 1/2 < \gamma$ show that the c_t 's, and hence the q_t 's, are distinct as t varies. Each divisor $\text{Ein} - c_t$ has at most one algebraic zero, namely q_t , and among the entire algebraic pencil there is at most one algebraic q_t . For if q_s, q_t were algebraic with $s \neq t$, the three distinct resonant values η, c_s, c_t would be jointly algebraically independent, whereas

$$(t - s)\eta + (1 - 2t)c_s - (1 - 2s)c_t = 0. \quad (\text{GT-17})$$

Moreover $q_{1/2}$ is unconditionally transcendental because $2 \text{Ein}(q_{1/2}) - \text{Ein}(1) = 0$. If the one possible exceptional q_t is algebraic, then $t \neq 1/2$, and the matrix taking (γ, E) to (η, c_t) has determinant $2t - 1 \neq 0$. Therefore that exception would again imply algebraic independence of γ and δ over K .

32.5. 4. What the inverse traces do and do not add to the Euler motive

Let $P_m \in \mathbf{Q}[X]$ be the monic universal trace polynomial of v2.3,

$$\mathcal{T}_m(c) = P_m(c^{-1}), \quad m \geq 2,$$

and recall the exact recovery identity

$$c^{-1} = 144\mathcal{T}_4(c) - 144\mathcal{T}_2(c)^2 - 14\mathcal{T}_2(c). \quad (\text{GT-18})$$

32.5.1 Exact spectral reformulations

For distinct algebraic $s, t \in [0, 1]$ and arbitrary $m, n \geq 2$,

$$\boxed{\text{trdeg}_K K(\mathcal{T}_m(c_s), \mathcal{T}_n(c_t)) = \text{trdeg}_K K(\gamma, \delta) \in \{1, 2\}.} \quad (\text{GT-19})$$

Each level is algebraic over any one of its nonconstant trace coordinates, and the two distinct linear pencil levels recover γ, E . Thus γ, δ are algebraically independent over K if and only if any such pair of distinct pencil traces is algebraically independent. This is an exact algebraic-matroid/spectral criterion, not a proof of either side.

For the Euler exponential motive with period matrix

$$P_\gamma = \begin{pmatrix} 1 & \gamma \\ 0 & 2\pi i \end{pmatrix},$$

full comparison injectivity of

$$\overline{\mathbf{Q}}[a, a^{-1}, b] \longrightarrow \mathbf{C}, \quad a \mapsto 2\pi i, \quad b \mapsto \gamma, \quad (\text{GT-20})$$

is equivalent to

$$\mathrm{trdeg}_{\overline{\mathbf{Q}}} \overline{\mathbf{Q}}(2\pi i, \gamma) = 2. \quad (\text{GT-21})$$

If $S_m = P_m(\gamma^{-1})$, (18) gives

$$\boxed{(20) \text{ injective} \iff \mathrm{trdeg}_{\overline{\mathbf{Q}}} \overline{\mathbf{Q}}(2\pi i, S_2, S_4) = 2.} \quad (\text{GT-22})$$

Again, (22) is a birational re-encoding of the same unknown period statement.

32.5.2 The exact barriers

1. **Global DT does not isolate the constant term.** Equation (7) only says that infinitely much arithmetic complexity occurs throughout the tail. It is compatible with $T_0 = \delta/e$ being algebraic. Likewise (4) can be carried entirely by higher T_k 's.
2. **The gamma-jet upper field has enough unknown capacity.** The odd zeta values and π^2 in (2) can absorb roughly half of the finite transcendence rank. No argument above forces γ to contribute a transcendence degree.
3. **Inverse traces are not automatically motivic periods.** The period algebra of a Tannakian category is not generally closed under inversion of an arbitrary numerical period coordinate. Formula (18) analytically recovers γ , but it does not construct S_2, S_4 as matrix coefficients of a tensor construction on M_γ . Consequently it supplies no new comparison injectivity by itself.
4. **Degree-one fullness is already the hard γ problem.** The Euler motive is known to be a non-split extension $0 \rightarrow \mathbf{Q}(0) \rightarrow M_\gamma \rightarrow \mathbf{Q}(-1) \rightarrow 0$. In this fixed rank-two setting, the assertion “every numerical relation $\lambda\gamma + \beta = 0$ is motivic and forces splitting” is logically equivalent to transcendence of γ : non-splitting makes fullness imply transcendence, while transcendence leaves no such relation to lift.
5. **Bare Frobenius does not retain the extension.** The unfiltered rank-two Frobenius object splits automatically in the regime proved in [001]. Any successful arithmetic comparison must retain filtered, syntomic, or coherent adelic norm data and prove a numerical-to-motivic lifting theorem. The new inverse-root or Hankel identities do not supply that missing functorial comparison.
6. **Function-level differential Galois rigidity is a different axis.** The conclusion that $U(s)$ lies in no finite Picard–Vessiot extension concerns variation in the parameter s . It does not turn a numerical relation at $s = 0$ into a morphism of the Euler motive and therefore does not imply irrationality of δ , transcendence of γ , or period-map injectivity.

32.5.3 One genuine motivic-size consequence

No single fixed finite-dimensional Tannakian object with finite-type motivic Galois group can contain the entire numerical tail tower $(T_k)_{k \geq 0}$ in the field generated by its period-matrix entries: that field has finite transcendence degree, whereas (5) has infinite transcendence degree. Any motivic realization of all tail jets must therefore use an increasing tower or an infinite/pro-object. This is a structural consequence, not a new statement about M_γ 's rank-two comparison map.

32.6. 5. Consolidated classification

Claim	Status from [001] + v2.3	Arithmetic force on γ, δ
$\text{trdeg}_K K(T_0, \dots, T_N) \geq \lceil N/2 \rceil - 1$	proved	none individually
$\Gamma(s, 1)$ DT over $\mathbf{C}(s)$	proved via Mahler	none individually
no finite ordinary PV extension contains $\Gamma(s, 1)$	proved	none individually
lower density $\geq 1/2$ of tail divisors with all zeros transcendental	proved	none individually
algebraic $r_0 \Rightarrow \gamma, \delta$ jointly AI over K	proved conditional fork	very strong if trigger is settled
at most one algebraic root in the Euler–Gompertz pencil	proved	an exception forces joint AI
two trace coordinates recover an Euler/pencil level	proved	exact criterion only
full Euler period-map injectivity	open	equivalent to π, γ algebraic independence
γ transcendental from motivic non-splitting alone	not proved	needs numerical-to-motivic fullness
δ irrational/transcendental	not proved	tail rank does not select T_0

The strongest transfer package is therefore the combination of (i) a **fixed-cut differential-transcendence theorem**, (ii) a **half-density all-zero theorem**, and (iii) sharp **conditional zero/trace criteria**. None closes Euler’s or Gompertz’s individual arithmetic problem without one further trigger: an algebraic exceptional inverse root, or a numerical-to-motivic comparison theorem.

33. A finite-rank Gamma-jet spectral sieve across all polyexponential orders

Status and scope. The arithmetic statements in this section have status **R-PASS** and use the joint algebraic-independence theorem of [001]. The analytic fiber statements use the all-order inverse-surface theorem proved above. No unconditional irrationality or transcendence statement about Euler’s constant is asserted.

Put

$$K = \overline{\mathbf{Q}}(e^\xi : \xi \in \overline{\mathbf{Q}})$$

and assume the arithmetic input

$$(\text{Ein}_q(\beta))_{(q,\beta) \in \mathbf{N}_{\geq 1} \times \overline{\mathbf{Q}}^\times} \text{ is jointly algebraically independent over } K \quad (\text{GS-0.1})$$

in the finite-subset sense; the coordinate labels are the ordered pairs (q, β) .

For positive algebraic α and $k \geq 0$, define

$$U_{\alpha,k} = [s^k] \Gamma(s, \alpha) = \frac{1}{k!} \int_{\alpha}^{\infty} e^{-u} \frac{(\log u)^k}{u} du. \quad (\text{GS-0.2})$$

Writing

$$\Gamma(s) = s^{-1} + \sum_{k \geq 0} g_k s^k,$$

the lower/upper incomplete-Gamma splitting gives

$$U_{\alpha,k} = g_k + R_{\alpha,k}, \quad (\text{GS-0.3})$$

where, with $L_{\alpha} = \log \alpha$,

$$R_{\alpha,k} = -\frac{L_{\alpha}^{k+1}}{(k+1)!} + \sum_{j=0}^k \frac{(-1)^j L_{\alpha}^{k-j}}{(k-j)!} \text{Ein}_{j+1}(\alpha). \quad (\text{GS-0.4})$$

33.1. 1. The all-order spectral sieve

Let $\Lambda \subset \overline{\mathbf{Q}}_{\geq 1}$ consist of distinct positive algebraic numbers whose multiplicative span has rank $r < \infty$. For $\alpha \in \Lambda$, $q \geq 1$, and fixed $k \geq 0$, set

$$\mathcal{D}_{\alpha,q,k} = \{z \in \mathbf{C} : \text{Ein}_q(z) = U_{\alpha,k}\}. \quad (\text{GS-1.1})$$

Theorem 1 (finite-rank all-order spectral sieve). Among all pairs $(\alpha, q) \in \Lambda \times \mathbf{N}$, at most $r+1$ divisors $\mathcal{D}_{\alpha,q,k}$ contain an algebraic point. More precisely:

1. if a divisor contains an algebraic point, that point is its unique real point β in $(0, 1)$; in particular it contains no nonreal algebraic point;
2. for any fixed α , at most one outer order q can give such an algebraic point;
3. every divisor outside the at most $r+1$ exceptional pairs is infinite and consists wholly of transcendental numbers.

33.1.1 Proof

For $\alpha \geq 1$, every integrand in (0.2) is nonnegative and

$$\sum_{k \geq 0} U_{\alpha,k} = \int_{\alpha}^{\infty} e^{-u} du = e^{-\alpha}.$$

Consequently

$$0 < U_{\alpha,k} < e^{-\alpha} \leq e^{-1}. \quad (\text{GS-1.2})$$

On the other hand, the alternating series gives, for every $q \geq 1$,

$$\text{Ein}_q(1) > 1 - \frac{1}{2^{q+1}} \geq \frac{3}{4} > e^{-1}. \quad (\text{GS-1.3})$$

The real restriction of Ein_q is strictly increasing, vanishes at zero, and is onto. Therefore the equation $\text{Ein}_q(x) = U_{\alpha,k}$ has exactly one real solution, and (1.2)–(1.3) put it in $(0, 1)$.

If a nonreal algebraic β belonged to the divisor, then its distinct conjugate would also belong to it because Ein_q has real coefficients. This would give

$$\text{Ein}_q(\beta) = \text{Ein}_q(\bar{\beta}),$$

contradicting (0.1). Thus any algebraic point is the unique real point just described.

For fixed k , differentiation under the integral sign gives

$$\frac{\partial}{\partial \alpha} U_{\alpha,k} = -\frac{e^{-\alpha}(\log \alpha)^k}{k! \alpha} < 0 \quad (\alpha > 1), \quad (\text{GS-1.4})$$

with strict decrease from $\alpha = 1$ as well. Hence distinct cuts give distinct levels. Also, if the same cut α had algebraic inverse points β_1, β_2 at two distinct outer orders q_1, q_2 , then

$$\text{Ein}_{q_1}(\beta_1) = U_{\alpha,k} = \text{Ein}_{q_2}(\beta_2),$$

again contradicting (0.1). Thus bad pairs use distinct cuts.

Suppose now that s distinct bad pairs (α_i, q_i) exist, with algebraic inverse points $\beta_i \in (0, 1)$. Let X be the collection of all source coordinates

$$\text{Ein}_j(\alpha_i), \quad 1 \leq i \leq s, \quad 1 \leq j \leq k+1,$$

and put $M = |X|$. The new coordinates

$$Y_i = \text{Ein}_{q_i}(\beta_i) = U_{\alpha_i,k}$$

have mutually distinct coordinate labels, and their labels are disjoint from those of X , because $\beta_i \in (0, 1)$ while $\alpha_i \geq 1$. Assumption (0.1) therefore gives

$$\text{trdeg}_K K(X, Y_1, \dots, Y_s) = M + s. \quad (\text{GS-1.5})$$

Choose logarithms $\lambda_1, \dots, \lambda_{r'}$, with $r' \leq r$, spanning the logarithms of the selected cuts over $\mathbf{Q}$. Equations (0.3)–(0.4) show that

$$K(X, Y_1, \dots, Y_s) \subset K(X, \lambda_1, \dots, \lambda_{r'}, g_k).$$

The right side has transcendence degree at most $M + r' + 1$. Comparison with (1.5) yields

$$s \leq r' + 1 \leq r + 1. \quad (\text{GS-1.6})$$

Finally, the all-order inverse-surface theorem shows that every fiber of Ein_q is infinite. This proves the theorem. $\square$

33.2. 2. The rank-one cut tower

Fix a positive algebraic $A > 1$ and put $\alpha_n = A^n$, $n \geq 1$. Its multiplicative rank is one.

Corollary 2 (two-exception theorem across the whole (n, q) -grid). For each fixed $k \geq 0$, among all pairs $(n, q) \in \mathbf{N}^2$, at most two divisors

$$\text{Ein}_q(z) - U_{A^n, k}$$

contain any algebraic zero. Every other divisor is infinite and all of its zeros are transcendental.

This is stronger than a density-one statement in each row: the total number of exceptional entries in the entire countable two-dimensional grid is at most two.

At outer order $q = 1$, every positive real level is regular, because the critical values of Ein are nonreal. Hence, for fixed k , all but at most two of

$$\text{Ein}(z) - U_{A^n, k}$$

are infinite, simple, pairwise disjoint divisors consisting wholly of transcendental points. Their inverse monodromy is the full finitary symmetric group.

The most classical row is $k = 0$:

$$U_{A^n, 0} = E_1(A^n).$$

Thus, for example, among every divisor

$$\text{Ein}_q(z) - E_1(2^n), \quad n, q \geq 1,$$

at most two can contain an algebraic point.

33.3. 3. A cofinite algebraically independent spectral forest

The fixed-order upper-jet defect theorem from the Moving-Level manuscript says that, after deleting at most $r + 1$ cuts, the levels $U_{\alpha, k}$ are algebraically independent over K . Delete also the at most $r + 1$ cuts occurring in Theorem 1. We obtain:

Corollary 3 (cofinite all-transcendental trace forest). There is a subset $E \subset \Lambda$, $|E| \leq 2r + 2$, such that:

- the levels $\{U_{\alpha, k} : \alpha \in \Lambda \setminus E\}$ are jointly algebraically independent over K ;
- for every $\alpha \notin E$ and every $q \geq 1$, the divisor $\mathcal{D}_{\alpha, q, k}$ consists wholly of transcendental points;
- if one chooses arbitrary orders $q_\alpha \geq 1$ and $m_\alpha \geq 2$, then the reciprocal traces

$$T_{q_\alpha, m_\alpha}(U_{\alpha, k}) = P_{q_\alpha, m_\alpha}(U_{\alpha, k}^{-1})$$

are jointly algebraically independent over K .

For the tower A^n , one may take $|E| \leq 4$. At $q = 1$ this gives a cofinite family of pairwise disjoint, simple, wholly transcendental spectral divisors whose arbitrary one-trace transversal is jointly algebraically independent.

The last assertion follows because every $P_{q,m}(X) \in \mathbf{Q}[X]$ is a nonconstant monic polynomial: adjoining $P_{q,m}(U^{-1})$ makes U algebraic, so separate independent level coordinates remain independent after the coordinatewise polynomial maps.

33.4. 4. Extremal rigidity

The defect bound also records exactly what two exceptional algebraic roots would cost.

Corollary 4 (two exceptions force Euler–logarithm independence). Let $A > 1$ be algebraic. If, at one fixed k , two distinct pairs $(n_1, q_1), (n_2, q_2)$ have algebraic inverse points, then

$$\log A \quad \text{and} \quad g_k$$

are algebraically independent over K . For $k = 0$, since $g_0 = -\gamma$, this forces

$$\boxed{\gamma \text{ and } \log A \text{ jointly algebraically independent over } K.}$$

Indeed, with two bad pairs, equality holds at both ends of the transcendence-degree comparison:

$$M + 2 = \text{trdeg}_K K(X, Y_1, Y_2) \leq \text{trdeg}_K K(X, \log A, g_k) \leq M + 2.$$

Thus $\log A, g_k$ are algebraically independent even over $K(X)$. This is a conditional fork, not a proof of the transcendence of γ .

33.5. 5. Significance and boundary

The result is a Diophantine finiteness theorem for algebraic inverse images of explicit upper incomplete-Gamma jets. Its distinctive point is the quantifier order: the $r + 1$ bound holds **simultaneously over every outer polyexponential order q** . In rank one, a two-parameter infinite grid has at most two algebraically contaminated divisors.

The bound is the strongest conclusion available from (0.1) without new information on the r logarithmic coordinates and the single Gamma coefficient g_k . Removing the last $r + 1$ possibilities would require additional arithmetic input; the extremal rigidity corollary explains why Euler’s constant already appears at $k = 0$.

34. Arithmetic Transfer Closures from the Companion Independence Theorem

Put

$$K = \overline{\mathbf{Q}}(e^\xi : \xi \in \overline{\mathbf{Q}}), \quad \text{Ein}_q(z) = \sum_{n \geq 1} \frac{(-1)^{n-1} z^n}{n^q n!} \quad (q \geq 1).$$

Thus $\text{Ein}_1 = \text{Ein}$. This section combines two inputs already isolated elsewhere in the monograph:

1. the companion theorem [001] that the coordinates

$$\{\text{Ein}_q(\alpha) : q \geq 1, \alpha \in \overline{\mathbf{Q}}^\times\}$$

are countably algebraically independent over K ; and

2. the universal trace identity

$$T_m(c) = P_m(c^{-1}) \quad (m \geq 2), \quad c^{-1} = 144T_4(c) - 144T_2(c)^2 - 14T_2(c).$$

Every conclusion in this section that uses the first input is marked **R-PASS** and is relative to [001]. Formal trace identities, positivity, analytic non-holonomy, and the general p -adic criteria are marked **U-PASS**. Finite residue-field certificates are marked **X-PASS**. None of the results below determines the arithmetic nature of an individual classical connection constant.

34.1. Algebraic inputs in signed polyexponential levels

Theorem 34.1 (Signed algebraic-input classification (**R-PASS**)). *Let $\alpha, \beta \in \overline{\mathbb{Q}}^\times$, $q \geq 1$, and $a \in \overline{\mathbb{Q}}^\times$. Then*

$$\text{Ein}(\beta) = a \text{Ein}_q(\alpha) \iff (q, \beta, a) = (1, \alpha, 1).$$

Consequently, every solution of $\text{Ein}(z) = a \text{Ein}_q(\alpha)$ is transcendental except for the possible algebraic solution $z = \alpha$ in the tautological case $q = 1$, $a = 1$.

Proof. If $(1, \beta) \neq (q, \alpha)$, the displayed equality is a nonzero K -linear relation between two distinct coordinates in the [001] family. If the coordinates coincide, it becomes $(1-a) \text{Ein}(\alpha) = 0$. Algebraic independence implies $\text{Ein}(\alpha) \neq 0$, and hence $a = 1$. Finally, $z = 0$ cannot solve the equation because $a \text{Ein}_q(\alpha) \neq 0$; applying the classification to any nonzero algebraic solution proves the last assertion. $\square$

For example, if $\alpha > 0$ is algebraic and $q \geq 2$, strict monotonicity and unboundedness of Ein on the positive real axis give a unique $x_{q,\alpha} > 0$ such that

$$\text{Ein}(x_{q,\alpha}) = \text{Ein}_q(\alpha).$$

The theorem says that $x_{q,\alpha}$ is transcendental over $\overline{\mathbb{Q}}$.

34.2. Gamma-jet levels and their inverse spectra

Write

$$\Gamma(s) = \frac{1}{s} + \sum_{k \geq 0} g_k s^k, \quad U_k = \frac{1}{k!} \int_1^\infty e^{-u} (\log u)^k \frac{du}{u},$$

and set

$$c_k = g_k - U_k = (-1)^{k+1} \text{Ein}_{k+1}(1).$$

For the zero multiset $Z_k = \{\rho \in \mathbb{C} : \text{Ein}(\rho) = c_k\}$, put

$$\tau_{m,k} = \sum_{\rho \in Z_k} \rho^{-m} = P_m(c_k^{-1}) \quad (m \geq 2).$$

Theorem 34.2 (Gamma-jet spectral divisors (**R-PASS**)). *The following statements hold.*

1. *Every point of every divisor Z_k , $k \geq 0$, is transcendental.*
2. *Each $\tau_{m,k}$ is transcendental over K .*
3. *For an arbitrary selection of orders $m_k \geq 2$, the family*

$$\{\tau_{m_k,k} : k \geq 0\}$$

is countably algebraically independent over K .

Proof. The levels c_k are nonzero algebraic multiples of distinct members of the [001] family, and hence are jointly algebraically independent. For $k = 0$ the signed level is $-\text{Ein}_1(1)$; for $k \geq 1$ its order is $k + 1 \geq 2$. In either case [theorem 34.1](#) excludes an algebraic zero.

If $P_m(c_k^{-1})$ were algebraic over K , then c_k^{-1} would be algebraic over K , because it is a root of a nonconstant polynomial over the algebraic closure of K . This is impossible. For a finite set of indices, each c_k is algebraic over $K(\tau_{m_k, k})$. Equality of transcendence degrees therefore transfers the algebraic independence of the levels to the selected traces. $\square$

34.3. Positive-integer incomplete-Gamma derivatives

Let

$$\Gamma_{<}(s, 1) = \int_0^1 e^{-t} t^{s-1} dt, \quad D_{r,k} = \left. \frac{\partial^k}{\partial s^k} \Gamma_{<}(s, 1) \right|_{s=r} \quad (r \geq 1, k \geq 1).$$

Theorem 34.3 (Fixed-parameter Gamma derivatives (R-PASS)). *For every fixed positive integer r , the countable family*

$$\{D_{r,k} : k \geq 1\}$$

is algebraically independent over K . In particular every one of its members is transcendental over K . At $r = 1$,

$$D_{1,k} = (-1)^k k! \text{Ein}_k(1).$$

No assertion of independence is made after combining different values of r , since the incomplete-Gamma recurrence creates relations between those rows.

Proof. Expand

$$\mathcal{R}(u) = \frac{1}{u} - \Gamma_{<}(u, 1) = \sum_{j \geq 0} R_j u^j, \quad R_j = (-1)^j \text{Ein}_{j+1}(1).$$

Iterating $\Gamma_{<}(s+1, 1) = s\Gamma_{<}(s, 1) - e^{-1}$ gives a polynomial $Q_r \in \mathbb{Z}[u]$ such that

$$\Gamma_{<}(u+r, 1) = \frac{(u)_r}{u} - (u)_r \mathcal{R}(u) - e^{-1} Q_r(u), \quad (u)_r = u(u+1) \cdots (u+r-1).$$

Comparison of the coefficient of u^k , for $k \geq 1$, yields

$$\frac{D_{r,k}}{k!} = -(r-1)! R_{k-1} + A_{r,k}(R_0, \dots, R_{k-2}),$$

where $A_{r,k}$ is affine over K . This is an invertible affine lower-triangular transformation on every finite initial segment. The [001] independence of $R_0, R_1, \dots$ proves the claim. The formula at $r = 1$ follows directly by differentiation under the integral sign. $\square$

34.4. The order-axis Stieltjes–Hankel tower

For $\alpha \in \overline{\mathbb{Q}}_{>0}$ and $k \geq 0$, define

$$\mu_{\alpha,k} = k! \text{Ein}_{k+1}(\alpha).$$

Mellin inversion of n^{-k-1} gives the two positive representations

$$\mu_{\alpha,k} = \int_0^1 \frac{1 - e^{-\alpha t}}{t} (-\log t)^k dt = \int_0^\infty y^k (1 - e^{-\alpha e^{-y}}) dy. \quad (264)$$

Thus $(\mu_{\alpha,k})_{k \geq 0}$ is a strict Stieltjes moment sequence. Put

$$H_{\alpha,d} = \det(\mu_{\alpha,i+j})_{0 \leq i,j < d} \quad (d \geq 1).$$

Theorem 34.4 (Order-axis Hankel tower (U-PASS positivity; R-PASS independence)). *One has*

$$H_{\alpha,d} > 0 \quad (\alpha \in \overline{\mathbb{Q}}_{>0}, d \geq 1),$$

and the complete family

$$\{H_{\alpha,d} : \alpha \in \overline{\mathbb{Q}}_{>0}, d \geq 1\}$$

is algebraically independent over K . Consequently, if one order $m_{\alpha,d} \geq 2$ is selected for every pair (α, d) , then

$$\{T_{m_{\alpha,d}}(H_{\alpha,d}) : \alpha \in \overline{\mathbb{Q}}_{>0}, d \geq 1\}$$

is algebraically independent over K .

Proof. The second integral in (264) is the moment functional of a positive density on an interval with infinite support. Its Gram determinants are therefore strictly positive.

The [001] theorem makes all variables $\mu_{\alpha,k}$, over all α and k , algebraically independent over K . Within one α -block,

$$\frac{\partial H_{\alpha,d}}{\partial \mu_{\alpha,2d-2}} = H_{\alpha,d-1}, \quad H_{\alpha,0} = 1.$$

The Jacobian of $(H_{\alpha,1}, \dots, H_{\alpha,D})$ with respect to $(\mu_{\alpha,0}, \mu_{\alpha,2}, \dots, \mu_{\alpha,2D-2})$ is triangular with nonzero diagonal. Distinct α -blocks use disjoint variables. Hence every finite family of the Hankel determinants is algebraically independent. The final assertion follows because each nonconstant trace polynomial leaves its level algebraic over the generated one-coordinate trace field. $\square$

34.5. Explicit relative transcendental combinations

Let

$$S_m = T_m(\gamma) = P_m(\gamma^{-1}) \quad (m \geq 2)$$

and let

$$X \in \{\delta, E_1(1), \text{Ei}(1), \text{Ci}(1), \text{Chi}(1)\}.$$

All constants in this display are real.

Theorem 34.5 (Complex separation of relative classes (R-PASS)). *For every $a \in \overline{\mathbb{Q}} \setminus \mathbb{R}$,*

$$S_m + aX$$

is transcendental over K . Moreover e and $S_m + aX$ are algebraically independent over $\mathbb{Q}$. In particular, each number $S_m + iX$ is an explicit member of this family.

Proof. The field K is stable under complex conjugation. If $z = S_m + aX$ were algebraic over K , then so would be $\bar{z} = S_m + \bar{a}X$, and hence

$$X = \frac{z - \bar{z}}{a - \bar{a}} \quad \text{and} \quad S_m = z - aX$$

would both be algebraic over K . Since $S_m = P_m(\gamma^{-1})$ with P_m nonconstant, this would also make γ algebraic over K . The five-relative-class theorem, which is itself a consequence of [001], excludes the simultaneous algebraicity of the class of γ and the class containing X . Thus z is transcendental over K .

If e and z satisfied a nonzero polynomial over $\mathbb{Q}$, the transcendence of e would make z algebraic over $\mathbb{Q}(e)$, hence over K , a contradiction. $\square$

Remark 34.6. The theorem is a collective separation statement relative to [001]. It does not imply an irrationality or transcendence result for either summand separately.

34.6. Defect at most one after adjoining two connection constants

Set

$$F = K(\gamma, \delta), \quad d = \text{trdeg}_K F.$$

Because $\text{Ein}(1) = \gamma + \delta/e$ is transcendental over K under [001], one has $d \in \{1, 2\}$. Write $Y_{\alpha,q} = \text{Ein}_q(\alpha)$ and exclude the anchor $(\alpha, q) = (1, 1)$.

Theorem 34.7 (Two-constant defect bound (R-PASS)). *For every finite set I of non-anchor indices,*

$$\text{trdeg}_F F(Y_{\alpha,q} : (\alpha, q) \in I) \geq |I| + 1 - d.$$

Thus the loss of rank over F is at most one. More precisely:

1. *if $d = 1$, all non-anchor coordinates remain algebraically independent over F ;*
2. *if one non-anchor coordinate is algebraic over F , then $d = 2$ and all the remaining non-anchor coordinates are algebraically independent over F .*

The same statements hold after replacing each selected coordinate by one nonconstant inverse trace $P_m(Y_{\alpha,q}^{-1})$.

Proof. The anchor belongs to F , while [001] gives

$$|I| + 1 = \text{trdeg}_K K(Y_{1,1}, Y_{\alpha,q} : (\alpha, q) \in I) \leq d + \text{trdeg}_F F(Y_{\alpha,q} : (\alpha, q) \in I).$$

This proves the inequality and the first refinement. If a coordinate Y_0 is algebraic over F , applying the inequality to $I \cup \{0\}$, with $0 \notin I$, gives

$$\text{trdeg}_F F(Y_{\alpha,q} : (\alpha, q) \in I) \geq |I| + 2 - d.$$

The case $I = \emptyset$ forces $d = 2$, and the general case then gives full rank $|I|$. Finally, a nonconstant one-variable trace coordinate and its level generate fields differing only by an algebraic extension, so all transcendence degrees are preserved. $\square$

34.7. The complete trace relation ideal and spectral rigidity

The next result is formal and does not use [001]. Let L be a field of characteristic zero, let $M \subset \{2, 3, \dots\}$ be finite with $2, 4 \in M$, and introduce variables $X_{i,m}$ for $1 \leq i \leq n$, $m \in M$. Put

$$R_i = 144X_{i,4} - 144X_{i,2}^2 - 14X_{i,2}$$

and

$$J_M = \langle X_{i,2} - P_2(R_i), \quad X_{i,m} - P_m(R_i) : m \in M \setminus \{2, 4\}, \quad 1 \leq i \leq n \rangle.$$

Proposition 34.8 (Complete universal relation ideal (U-PASS)). *For the homomorphism*

$$\phi : L[X_{i,m} : 1 \leq i \leq n, \quad m \in M] \longrightarrow L[Z_1, \dots, Z_n], \quad X_{i,m} \longmapsto P_m(Z_i),$$

one has $\ker \phi = J_M$, and ϕ is surjective. Consequently

$$L[X_{i,m}]/J_M \simeq L[Z_1, \dots, Z_n].$$

In particular, J_M is a prime complete-intersection ideal and its variety is smooth, normal, factorial, and isomorphic to $\mathbb{A}_L^n$.

Proof. The recovery identity gives

$$144P_4(Z_i) - 144P_2(Z_i)^2 - 14P_2(Z_i) = Z_i,$$

so every Z_i lies in the image. In the quotient by J_M , all coordinates except $X_{i,4}$ are polynomials in R_i , while the definition of R_i expresses $X_{i,4}$ as a polynomial in R_i and $X_{i,2} = P_2(R_i)$. Hence the quotient is generated freely by $R_1, \dots, R_n$, and the induced inverse sends Z_i to R_i . This proves the asserted isomorphism. The geometric properties follow from the polynomial coordinate ring; the displayed generators have the expected codimension and form a regular sequence. $\square$

Corollary 34.9 (No hidden cross-level relations (U-PASS formally; R-PASS for [001]. levels))

If the levels $c_1, \dots, c_n$ are algebraically independent over L , then J_M is the complete ideal of algebraic relations among the numbers $T_m(c_i)$. In particular this applies to every finite family of distinct [001] coordinates and to the derived independent levels above.

Proof. The elements c_i^{-1} are algebraically independent, so evaluation $L[Z_1, \dots, Z_n] \rightarrow L[c_1^{-1}, \dots, c_n^{-1}]$ is injective. Compose this map with ϕ . $\square$

Proposition 34.10 (Minimal two-moment rigidity (U-PASS)). *The pair $(T_2(c), T_4(c))$ recovers c and therefore the complete divisor of $\text{Ein} - c$. No single trace T_m is globally injective as a function of $c \in \mathbb{C}^\times$, since P_m has degree $m \geq 2$. Moreover the only nontrivial collision of (T_2, T_3) is*

$$\{c, d\} = \{6 - 2\sqrt{3}, 6 + 2\sqrt{3}\}, \quad (T_2, T_3) = \left(-\frac{1}{24}, \frac{1}{96}\right).$$

Thus trace order four is the smallest maximal order that gives a globally rigid trace pair in this family.

Proof. Recovery from (T_2, T_4) is the displayed universal identity. A nonconstant degree- m polynomial cannot be injective on $\mathbb{C}$, and the point $c = 0$, which is outside the level domain, does not remove all generic fibers. The collision calculation follows by eliminating $x = c^{-1}$ and $y = d^{-1}$ from $P_2(x) = P_2(y)$ and $P_3(x) = P_3(y)$; the non-diagonal branch has $x + y = 1/2$, $xy = 1/24$, giving the stated reciprocal levels. $\square$

34.8. Analytic non-holonomy of the moving tower

Recall the dyadic levels

$$Y_N = (N + 1) \text{Ein}(2^N) - N \text{Ein}(2^{N+1}) = \gamma + \varepsilon_N, \quad \varepsilon_N \sim \frac{N + 1}{2^N} e^{-2^N},$$

and put $\tau_{m,N} = P_m(Y_N^{-1})$.

Theorem 34.11 (All-order moving-index non-holonomy (U-PASS)). *For every fixed $m \geq 2$, neither $(\tau_{m,N})_{N \geq 1}$ nor its signed increment sequence $(\tau_{m,N+1} - \tau_{m,N})_{N \geq 1}$ is P -recursive over $\mathbb{C}$. Equivalently, neither ordinary generating function is D -finite.*

Proof. Let $g_m(x) = P_m(x^{-1})$. This is nonconstant and analytic at $x = \gamma$. If $r \geq 1$ is the first nonzero Taylor order of $g_m(\gamma + u) - g_m(\gamma)$, then

$$\tau_{m,N} - T_m(\gamma) = b_m \varepsilon_N^r (1 + o(1)), \quad b_m \neq 0.$$

For each fixed $j > 0$,

$$\frac{\varepsilon_{N+j}}{\varepsilon_N} = \exp(-(2^j - 1)2^N + O(N)),$$

which tends to zero faster than every rational function of N . In a hypothetical polynomial-coefficient recurrence for the centered sequence, divide by the term having the least shift with nonzero coefficient. Every later-shift ratio vanishes, leaving $1 = 0$. The increment has the same leading scale, with coefficient $-b_m$, so the same argument applies. P-recursive coefficient sequences are equivalent to D-finite ordinary generating functions. $\square$

There is a second, independent analytic obstruction. The meromorphic function

$$\mathcal{R}(u) = u^{-1} - \Gamma_{<}(u, 1)$$

has poles at every negative integer. A D-finite meromorphic function can have finite-plane singularities only among the finitely many zeros of the leading polynomial of its differential equation. Therefore $\mathcal{R}$ is not D-finite over $\mathbb{C}(u)$. This analytic statement is **U-PASS** and does not depend on any arithmetic independence assertion.

34.9. Exact p -adic closures and their remaining barriers

For the Euler factorial Padé system, recall

$$Q_n(t) = \sum_{h=0}^n h! \binom{n}{h}^2 (-t)^h.$$

For a prime p , $t \in \mathbb{Z}_p^\times$, and $0 \leq r < p$, set

$$q_{r,p}(X) = \sum_{h=0}^r h! \binom{r}{h}^2 (-X)^h \in \mathbb{F}_p[X].$$

Theorem 34.12 (Unit-boundary residue closure (**U-PASS** with **X-PASS** certificates)). *If $q_{r,p}(\bar{t}) \neq 0$ for every $0 \leq r < p$, then every $Q_n(t)$ is a p -adic unit and the full Padé sequence converges to*

$$E_p(t) = \sum_{n \geq 0} n! t^n.$$

Exact finite residue-field calculation gives the sample classes

p	$t \pmod{p}$
3	2
5	2, 3, 4
7	4, 5
11	2, 3, 5, 7
13	6, 9, 11, 12.

Together with the separate modulo-four calculation at $p = 2$, this proves full convergence at $t = -1$ for $p = 2, 3, 5, 13$. In particular, at $p = 13$,

$$v_{13}\left(E_{13}(-1) - \frac{P_n(-1)}{Q_n(-1)}\right) \geq 2v_{13}(n!) = \frac{n - s_{13}(n)}{6}.$$

Proof. Modulo p , terms with $h \geq p$ vanish and the remaining binomial coefficients depend only on $n \bmod p$. Hence $Q_n(t) \equiv q_{r,p}(\bar{t}) \pmod{p}$ when $n \equiv r \pmod{p}$. The unit conclusion and the standard Padé remainder bound then give convergence. The listed classes are exact finite certificates. For $p = 2, t = -1$, one has $v_2(Q_n(-1)) = 0$ for even n and 1 for odd n . Legendre's formula gives the last equality. $\square$

Theorem 34.13 (One rational subsequence at all fixed places (**U-PASS**)). *Let $N_j = \text{lcm}(1, \dots, j)$. For every fixed negative integer t , the rational numbers*

$$\frac{P_{N_j}(t)}{Q_{N_j}(t)}$$

converge at the real place to the Stieltjes value $\int_0^\infty e^{-x}(1-tx)^{-1} dx$, and converge in $\mathbb{Q}_p$ to $E_p(t)$ for every fixed prime p .

Proof. The Carlitz congruence $Q_{n+k}(T) \equiv Q_n(T) \pmod{k\mathbb{Z}[T]}$ gives $Q_{N_j}(T) \equiv 1 \pmod{N_j\mathbb{Z}[T]}$. Thus the denominators are eventually units at each fixed prime, and the Padé remainder tends to zero. Real Stieltjes–Padé convergence holds for every $t < 0$, hence also along this subsequence. $\square$

Finally, the uniformly convergent interpolation

$$q_p(x) = \sum_{h \geq 0} h! \binom{x}{h}^2 \quad (x \in \mathbb{Z}_p)$$

is 1-Lipschitz, satisfies $q_p(n) = Q_n(-1)$, and gives the exact **U-PASS** equivalence

$$\sup_{n \geq 0} v_p(Q_n(-1)) = \infty \iff q_p \text{ has a zero in } \mathbb{Z}_p.$$

Open p -adic gates (OPEN**).** Full convergence at $t = -1$ is not known for every prime. A zero of q_p records unbounded denominator valuation; by itself it does not imply failure of convergence, which requires comparison with $v_p(n!)$. The fixed-prime valuation gain supplied here is only linear in n , whereas the relevant global denominator height has an $n \log n$ term, so these estimates do not furnish an irrationality criterion for $E_p(-1)$. The common subsequence theorem is placewise and does not assert convergence in a restricted adelic product topology.

34.10. Transfer boundary

The results of this section are exact transports: [001] supplies the arithmetic rank, and the trace, Gamma, Hankel, or conjugation maps preserve enough of that rank to yield the displayed **R-PASS** conclusions. The formal geometry and analytic statements marked **U-PASS** require no such input. None of these mechanisms isolates an individual boundary constant from an infinite limit, improves the fixed-prime height budget to the required global scale, or converts a trace coordinate into a new comparison theorem. Those are the remaining gates rather than suppressed hypotheses.

35. Multigenerator Jossen-type reductions for the resonant polyexponential algebra

Scope. This section proves a rigorous reduction and two complete non-cyclic subclasses. It does **not** claim the full multigenerator Jossen problem for the algebra below. Statements using the algebraic-point independence theorem of [001] have status **R-PASS**.

35.1. 1. The algebra and the arithmetic generic-point theorem

Put

$$E_{q,\lambda}(z) = \text{Ein}_q(\lambda z), \quad \mathcal{A} = \overline{\mathbf{Q}}[E_{q,\lambda} : q \geq 1, \lambda \in \overline{\mathbf{Q}}^\times],$$

and let

$$K_{\text{exp}} = \overline{\mathbf{Q}}(e^\alpha : \alpha \in \overline{\mathbf{Q}}).$$

Only finitely many generators occur in any element of $\mathcal{A}$.

35.1.1 Theorem 1.1 (polynomiality and algebraic-time genericity)

After identifying repeated pairs (q, λ) , the functions $E_{q,\lambda}$ are algebraically independent over $\overline{\mathbf{Q}}$. Hence

$$\mathcal{A} \simeq \overline{\mathbf{Q}}[X_{q,\lambda}]$$

is a polynomial UFD in the finite-subset sense.

For every $\beta \in \overline{\mathbf{Q}}^\times$, evaluation at $z = \beta$ is injective on $\mathcal{A}$, and the point

$$\Phi(\beta) = (\text{Ein}_q(\lambda\beta))_{q,\lambda}$$

is a generic point over K_{exp} : every finite set of its distinct coordinates is algebraically independent over K_{exp} .

Consequently, if $F \in \mathcal{A}$ is nonconstant, then

$$F(\beta) \notin K_{\text{exp}} \quad (\beta \in \overline{\mathbf{Q}}^\times). \quad (\text{J-1.1})$$

More generally, $\Phi(\beta)$ misses every proper algebraic subvariety defined over K_{exp} .

Proof Apply Theorem 8.3 and Corollary 8.4 of 001 to the finitely many pairs occurring in a proposed relation, with the arguments $\lambda\beta$. They are distinct exactly when the corresponding λ 's are distinct, and the theorem simultaneously includes all selected orders q . A functional polynomial relation would specialize at any fixed nonzero algebraic β , contradicting the value theorem. The remaining claims are immediate.

This is stronger than ordinary Zariski density: **every** nonzero algebraic time is generic. It does not say anything comparable at a transcendental time.

35.1.2 Corollary 1.2 (complete algebraic-zero exclusion)

For nonconstant $F \in \mathcal{A}$ and $a \in K_{\text{exp}}$, every zero of $F(z) - a$, other than a possible zero at $z = 0$, is transcendental. In particular, no nonzero element of $\mathcal{A}$ has a nonzero algebraic zero.

35.1.3 Proposition 1.3 (every fiber is infinite)

For every nonconstant $F \in \mathcal{A}$ and every $a \in \mathbf{C}$, the fiber

$$\{z \in \mathbf{C} : F(z) = a\}$$

is infinite.

Proof Every element of $\mathcal{A}$ is an E-function and is of exponential type, hence has order at most one. If $F - a$ had only finitely many zeros, Hadamard factorization would give

$$F(z) - a = e^{Az+B}R(z) \quad (\text{J-1.2})$$

with $R \in \mathbf{C}[z]$.

The functional proof of 001, Lemmas 8.1–8.2, is unchanged after adjoining one extra exponential e^{Az} : take a $\mathbf{Z}$ -basis of the finitely generated additive group spanned by A and all slopes λ occurring in F , and repeat the Laurent-character/Darboux-polynomial argument. Thus the selected $E_{q,\lambda}$'s remain algebraically independent over

$$\mathbf{C}(z, e^{Az}, e^{\lambda z} : \lambda \text{ selected}).$$

Equation (1.2) is a nontrivial polynomial relation over this field, a contradiction.

35.2. 2. Exact reduction of both parts of Jossen's conjecture

Let $F, G \in \mathcal{A} \setminus \{0\}$. Since only finitely many variables occur, their gcd $D = \gcd_{\mathcal{A}}(F, G)$ is well-defined. Write

$$F = DF_0, \quad G = DG_0, \quad \gcd_{\mathcal{A}}(F_0, G_0) = 1. \quad (\text{J-2.1})$$

35.2.1 Theorem 2.1 (the exact remaining common-zero obstruction)

1. Neither F_0 nor G_0 can vanish at a nonzero algebraic point.
2. If $F_0(0) = G_0(0) = 0$, the common zero at the origin is explained by the nonunit E-function $h(z) = z$: both F_0/z and G_0/z are E-functions.
3. Therefore the only unresolved case of Jossen part (ii) for $\mathcal{A}$ is a **nonzero transcendental** point ξ satisfying

$$F_0(\xi) = G_0(\xi) = 0. \quad (\text{J-2.2})$$

Equivalently, for a finite coordinate map $\Phi : \mathbf{C} \rightarrow \mathbf{A}^n$, the hard locus is the transcendental-time pullback

$$\Phi^{-1}(V(P, Q)), \quad \text{codim } V(P, Q) \geq 2, \quad (\text{J-2.3})$$

for coprime $P, Q \in \overline{\mathbf{Q}}[X_1, \dots, X_n]$.

The factor z in item 2 is standard E-function zero removal at the algebraic point zero. The contribution of $[001]$ is to remove all other algebraic times simultaneously.

35.2.2 Theorem 2.2 (an entire-quotient counterexample would be infinite)

Assume F/G is entire. With (2.1), exactly one of the following occurs.

- G_0 is constant, in which case $F/G = F_0/G_0 \in \mathcal{A}$, so Jossen part (i) holds inside $\mathcal{A}$.
- G_0 is nonconstant. Then G_0 has infinitely many zeros by Proposition 1.3, every one of them is a zero of F_0 , and all but a possible zero at the origin are transcendental. Thus a failure of divisibility in $\mathcal{A}$ would force an **infinite transcendental codimension-two intersection carrying the complete divisor of G_0** .

This is a useful sharpening of the gate: one isolated mysterious root is the obstruction for part (ii), whereas part (i) requires an infinite divisor-sized exceptional intersection.

35.2.3 Origin contact and transversality

For a finite list $E_i = E_{q_i, \lambda_i}$,

$$E_i(z) = \lambda_i z + O(z^2).$$

Let P_d be the first nonzero homogeneous part of $P \in \overline{\mathbf{Q}}[X_1, \dots, X_n]$. If

$$P_d(\lambda_1, \dots, \lambda_n) \neq 0,$$

then

$$\text{ord}_{z=0} P(E_1(z), \dots, E_n(z)) = d. \quad (\text{J-2.4})$$

In particular, the origin is a simple zero precisely under the familiar first-order transversality condition

$$\sum_i \lambda_i \partial_i P(0) \neq 0. \quad (\text{J-2.5})$$

If the expression in (2.5) vanishes, higher jets give an effective valuation algorithm. This completely settles the algebraic-time Jacobian question but does not control (2.2).

35.3. 2A. A general affine E-surface theorem

There is a substantially broader two-dimensional closure that does not use any special identity of Ein.

35.3.1 Theorem 2A.1 (factorial E-surface criterion)

Let B be a finitely generated factorial $\overline{\mathbf{Q}}$ -domain of Krull dimension two, embedded as a subring of the ring of E-functions. Assume that B contains at least one nonconstant polynomial coordinate

$$\tau(z) \in \overline{\mathbf{Q}}[z].$$

Then both parts of Jossen's conjecture hold for every nonzero $f, g \in B$ (so in particular the quotient below has $g \neq 0$):

1. if f/g is entire, then f/g is an E-function;
2. if f and g have a common zero, they have a common nonunit E-function divisor.

Proof of the common-zero assertion Take a gcd in the UFD B :

$$f = df_0, \quad g = dg_0, \quad \gcd_B(f_0, g_0) = 1. \quad (\text{J-2A.1})$$

Let ξ be a common zero. If $d(\xi) = 0$, then d itself is a nonunit E-function divisor of f and g . Assume $d(\xi) \neq 0$. Then $f_0(\xi) = g_0(\xi) = 0$. Since f_0, g_0 are nonzero and B is a factorial domain, neither the height-zero prime nor any height-one prime contains both of them. Hence every prime over (f_0, g_0) has height two. Because $\dim B = 2$,

$$B/(f_0, g_0)$$

is a zero-dimensional affine $\overline{\mathbf{Q}}$ -algebra. The kernel of evaluation at ξ is a prime over (f_0, g_0) , hence is maximal. By Zariski's lemma its residue field is algebraic over $\overline{\mathbf{Q}}$, hence equals $\overline{\mathbf{Q}}$. In particular $\tau(\xi) \in \overline{\mathbf{Q}}$. Since ξ is a root of the nonzero polynomial $\tau(T) - \tau(\xi)$, this proves $\xi \in \overline{\mathbf{Q}}$.

If $\xi \neq 0$, Beukers's algebraic-point zero-removal proposition gives

$$\frac{f_0(z)}{z - \xi}, \quad \frac{g_0(z)}{z - \xi}$$

as E-functions; for $\xi = 0$, the same conclusion follows directly by shifting the Taylor coefficients of an E-function that vanishes at the origin. Thus $z - \xi$ is the required common nonunit E-function divisor (after multiplying the displayed quotients by d).

Proof of the entire-quotient assertion Cancel d as in (2A.1); the cancelled quotient extends to the same entire function across the zeros of d . If f_0/g_0 is entire, every zero of g_0 is a common zero of f_0, g_0 . The zero-dimensional scheme above has only finitely many closed points. Each such point fixes an algebraic value of τ , and a nonconstant polynomial has finite fibers. Thus g_0 has finitely many zeros, all algebraic. Let

$$r(z) = \prod_{\xi} (z - \xi)^{\text{ord}_{\xi} g_0} \in \overline{\mathbf{Q}}[z]. \quad (\text{J-2A.2})$$

Repeated algebraic-point zero removal shows that g_0/r is a zero-free E-function. A zero-free entire function of order at most one is Ce^{az} by Hadamard factorization. Here

$$C = (g_0/r)(0) \in \overline{\mathbf{Q}}^{\times}, \quad a = (g_0/r)'(0)/(g_0/r)(0) \in \overline{\mathbf{Q}},$$

so $g_0/r = Ce^{az}$ is a unit E-function. Entirety also gives $\text{ord}_{\xi} f_0 \geq \text{ord}_{\xi} g_0$, hence f_0/r is an E-function. Finally,

$$\frac{f}{g} = \frac{f_0/r}{Ce^{az}}$$

is an E-function.

35.3.2 Corollary 2A.2 (one arbitrary E-function plus z)

Let F be any nonpolynomial E-function. Then z and $F(z)$ are algebraically independent: an entire function algebraic over $\overline{\mathbf{Q}}(z)$ is a polynomial. Consequently

$$B_F = \overline{\mathbf{Q}}[z, F(z)] \simeq \overline{\mathbf{Q}}[X, Y]$$

is a factorial affine E-surface, and **both parts of Jossen's conjecture hold unconditionally throughout** B_F .

In direct plane language, if coprime $P, Q \in \overline{\mathbf{Q}}[X, Y]$ have a common pullback zero ξ , then $(\xi, F(\xi)) \in V(P, Q)$. The plane intersection is finite and defined over $\overline{\mathbf{Q}}$, so ξ is algebraic and $z - \xi$ is the common E-factor. For an entire quotient, the same finite intersection traps all denominator zeros and the preceding Hadamard argument completes the proof.

Equivalently, for arbitrary P, Q and $D = \gcd(P, Q)$, one has the exact set-theoretic decomposition

$$\begin{aligned} Z(P(z, F)) \cap Z(Q(z, F)) &= Z(D(z, F)) \\ &\cup S_{P, Q, F}, \end{aligned} \tag{J-2A.3}$$

where $S_{P, Q, F}$ is a finite set of algebraic numbers. Thus every transcendental common zero is automatically carried by the explicit common E-factor $D(z, F)$; only finitely many algebraic residual points require the linear factors $z - \xi$.

The polynomial case reduces to the univariate algebra $\overline{\mathbf{Q}}[z]$ and is elementary. Thus the corollary holds for every E-function F , with the nonpolynomial case being the genuinely two-dimensional one.

More generally, if $\tau \in \overline{\mathbf{Q}}[z]$ is any nonconstant polynomial and F is a nonpolynomial E-function, then $\tau(z), F(z)$ are algebraically independent (otherwise F would be algebraic over $\overline{\mathbf{Q}}(z)$). Therefore

$$\overline{\mathbf{Q}}[\tau(z), F(z)]$$

also satisfies both parts of Jossen.

For $F = e^{\beta z}$, this recovers the familiar one-support exponential- polynomial situation. The point of Corollary 2A.2 is that no exponential, hypergeometric, differential-order, or zero-geometry assumption on F is needed beyond the E-function axioms.

In particular, every nonconstant $F \in \mathcal{A}$ is nonpolynomial (by the functional independence in 001), and hence

$$\boxed{\overline{\mathbf{Q}}[z, F] \text{ satisfies both parts of Jossen}} \tag{J-2A.4}$$

for **every single polyexponential polynomial observable** F . This is a strict extension of the cyclic result $\overline{\mathbf{Q}}[F]$: the numerator and denominator may depend independently on the physical variable z and on $F(z)$.

This surface theorem explains precisely why adding a second independent polyexponential beyond z, F changes the problem: in Krull dimension three, two coprime equations can leave a positive-dimensional intersection, and the argument forcing ξ algebraic disappears.

35.3.3 Corollary 2A.3 (holonomic transfer)

The proof is not intrinsically restricted to Siegel E-functions. Let $\mathcal{H}_{\overline{\mathbf{Q}}}^{\text{fin}}$ be the ring of finite-order entire functions that are D-finite over $\overline{\mathbf{Q}}(z)$ and have algebraic Taylor coefficients. It is closed under sums and products and under division by $z - \xi$ at an algebraic zero, with finite order preserved. A zero-free finite-order entire function h is $Ce^{A(z)}$ with A a polynomial. Normalize this representation by taking $A(0) = 0$ and $C = h(0)$. Then $C \in \overline{\mathbf{Q}}^\times$, and the formal logarithm of h/C shows that $A \in \overline{\mathbf{Q}}[z]$. Hence $h^{-1} = C^{-1}e^{-A(z)}$ is again D-finite of finite order with algebraic Taylor coefficients.

Therefore Theorem 2A.1, with “E-function” replaced by “finite-order entire D-finite function with algebraic Taylor coefficients,” remains valid. In particular, every factorial affine holonomic surface containing a nonconstant polynomial coordinate is closed under its entire quotients and has the same common-zero factor property. This is an elementary algebraic–analytic transfer to the holonomic setting.

35.4. 3. A complete genuinely bivariate blow-up subclass inside $\mathcal{A}$

The next result is not merely cyclic in one generator. It exhibits the geometric mechanism by which a codimension-two point becomes an E-divisor.

Choose

$$H(z) = \text{Ein}(\lambda z), \quad R(z) = \text{Ein}_p(\mu z),$$

with $(p, \mu) \neq (1, \lambda)$, and fix $a \in \overline{\mathbf{Q}}$, $m \geq 1$. Theorem 1.1 makes H, R algebraically independent. Put

$$U = a + H^m R. \tag{J-3.1}$$

Then H, U are also algebraically independent, so $\mathcal{B} = \overline{\mathbf{Q}}[H, U]$ is a genuine polynomial two-generator subalgebra of $\mathcal{A}$.

For $0 \neq P \in \overline{\mathbf{Q}}[X, Y]$, expand

$$P(X, a + Y) = \sum_{i,j \geq 0} c_{ij} X^i Y^j$$

and define the weighted exceptional valuation

$$\nu_m(P) = \min\{i + mj : c_{ij} \neq 0\}. \tag{J-3.2}$$

35.4.1 Theorem 3.1 (exact pullback gcd)

If $P, Q \in \overline{\mathbf{Q}}[X, Y]$ are coprime, then, up to a nonzero algebraic constant,

$$\gcd_{\mathcal{A}}(P(H, U), Q(H, U)) = H^{\min(\nu_m(P), \nu_m(Q))}. \tag{J-3.3}$$

Proof Substitution gives

$$P(H, U) = \sum c_{ij} H^{i+mj} R^j = H^{\nu_m(P)} P^\sharp(H, R),$$

where $P^\sharp$ is not divisible by H . After inverting H , the substitution is the isomorphism

$$\overline{\mathbf{Q}}[X, Y, X^{-1}] \xrightarrow{\sim} \overline{\mathbf{Q}}[H, R, H^{-1}], \quad X \mapsto H, \quad Y \mapsto a + H^m R,$$

whose inverse sends R to $(Y - a)/X^m$. Coprimality therefore forbids every common irreducible factor other than H , and (3.3) follows.

35.4.2 Corollary 3.2 (complete Jossen common-zero theorem at the contracted point)

Assume

$$\sqrt{(P, Q)} = (X, Y - a), \tag{J-3.4}$$

so the affine complete intersection of P and Q is exactly the point $(0, a)$. Then

$$\{P(H, U) = Q(H, U) = 0\} = \{H = 0\}. \tag{J-3.5}$$

The right side is infinite; zero is its only algebraic point, and all its other points are transcendental. Every zero is simple. All of these common zeros, including the transcendental ones, are explained by the explicit nonunit E-function factor in (3.3). Thus Jossen part (ii) holds completely for this bivariate class.

Indeed,

$$H'(z) = \frac{1 - e^{-\lambda z}}{z}.$$

A multiple nonzero zero would have $\lambda z = 2\pi i k$. But

$$\Re \operatorname{Ein}(2\pi i k) = \int_0^1 \frac{1 - \cos(2\pi k t)}{t} dt > 0 \quad (k \neq 0), \tag{J-3.6}$$

so this is impossible. The zero at zero is simple because $H'(0) = \lambda$. There are infinitely many zeros since otherwise Hadamard factorization would make $H = e^{Az+B} S(z)$ and hence H' would have only finitely many zeros, whereas $H'(2\pi i k/\lambda) = 0$ for every $k \neq 0$. Their transcendence follows from Theorem 1.1.

A basic example is

$$P = X, \quad Q = Y - a,$$

for which the coprime bivariate pair pulls back to

$$H, \quad H^m R.$$

It has infinitely many transcendental common zeros, but the common factor is exactly H . Geometrically, (3.1) contracts the divisor $H = 0$ to the codimension-two point $(0, a)$; Jossen's factor is the exceptional divisor.

35.4.3 Theorem 3.3 (an exact non-cyclic entire-quotient window)

Let $1 \leq \nu \leq m$ and $P \in \overline{\mathbf{Q}}[X, Y] \setminus \{0\}$. Then

$$\frac{P(H, U)}{H^\nu} \text{ is entire} \iff \partial_X^j P(0, a) = 0 \quad (0 \leq j < \nu) \quad (\text{J-3.7})$$

and, whenever these equivalent conditions hold,

$$\frac{P(H, U)}{H^\nu} \in \mathcal{A}. \quad (\text{J-3.8})$$

Hence Jossen part (i) holds for this whole bivariate numerator class and the denominators H^ν , $1 \leq \nu \leq m$.

Proof For weights strictly below $\nu \leq m$, only the pure terms $c_{i0}X^i$ can occur in (3.2); every term involving $Y - a$ acquires a factor H^m . Thus the derivative conditions in (3.7) are equivalent to $\nu_m(P) \geq \nu$, which makes the quotient in (3.8) a polynomial in H, R . If one derivative does not vanish, the first offending pure H^i -term produces a pole at every simple zero of H . This proves the reverse implication.

For example,

$$\frac{U - a}{H^m} = R$$

recovers an independent polyexponential coordinate as an entire E-function quotient; this is not contained in the cyclic algebra $\overline{\mathbf{Q}}[H]$.

35.5. 4. A natural differential two-generator subclass

Let

$$H = \text{Ein}(\lambda z), \quad \Theta H = zH' = 1 - e^{-\lambda z}.$$

The pair $(H, \Theta H)$ is algebraically independent over $\overline{\mathbf{Q}}$, by the functional exponential/polyexponential theorem in 001. Equation (3.6) proves

$$\{H = 0\} \cap \{\Theta H = 0\} = \{0\}. \quad (\text{J-4.1})$$

Consequently, for any $P, Q \in \overline{\mathbf{Q}}[X, Y]$ satisfying

$$\sqrt{(P, Q)} = (X, Y), \quad (\text{J-4.2})$$

the common-zero set of $P(H, \Theta H)$ and $Q(H, \Theta H)$ is exactly $\{0\}$, and the common zero is explained by the E-function factor z . This is a complete non-cyclic common-zero theorem in the first differential closure of $\mathcal{A}$.

There is also a broad imported quotient theorem. Fischler–Rivoal prove that if g^r is an E-function and $L \in \overline{\mathbf{Q}}(z)[d/dz]$ is such that $L(g)/g$ is entire, then $L(g)/g$ is a polynomial. Thus Jossen part (i) holds for every pair $(L(G), G)$ lying in $\mathcal{A}^2$. For example, Θ maps

$$\overline{\mathbf{Q}}[\text{Ein}_q(\lambda z) : q \geq 2] \quad \text{into} \quad \mathcal{A}.$$

For a nonconstant G in this subalgebra, $\Theta G/G$ is in fact never entire: Proposition 1.3 supplies a nonzero zero of G , and the logarithmic derivative has a pole there.

35.6. 5. Why the full multigenerator statement is still open

35.6.1 5.1 Algebraic independence is not codimension-two avoidance

An algebraically nondegenerate finite-order entire curve can meet a codimension-two target infinitely often. The elementary map

$$z \longmapsto (\sin z, z \sin z)$$

is algebraically nondegenerate and meets $(0, 0)$ at every $z \in \pi\mathbf{Z}$. Here the hidden divisor $\sin z$ explains the intersection. This is exactly the sort of divisor that Jossen asks one to recover inside the E-function ring. Therefore Theorem 1.1 alone cannot settle (2.3).

35.6.2 5.2 Why standard Nevanlinna gcd estimates do not immediately close it

Nevanlinna gcd theorems on algebraic tori give $o(T)$ -type bounds for codimension-two intersections under zero-free or small-zero hypotheses on the coordinate functions. The polyexponential coordinates have infinite divisors of their own, so those hypotheses are not automatic. Even an $o(T)$ upper bound would need a matching lower bound for the zero counting function of every possible denominator $Q(\Phi)$ before it could prove Jossen part (i).

Thus a usable Nevanlinna route requires two new ingredients:

1. a gcd estimate adapted to the additive/iterated-integral differential group containing the Ein_q 's, rather than a torus; and
2. a uniform lower zero-density theorem for every nonconstant element of $\mathcal{A}$, stronger than Proposition 1.3.

35.6.3 5.3 The exact remaining structural gate

Any one of the following would close a substantial part of the remaining problem.

- **Transcendental codimension-two saturation:** if coprime $P, Q \in \mathcal{A}$ have a nonzero common root, prove that their pullbacks share a nonunit E-function factor (necessarily outside $\mathcal{A}$).
- **Strong avoidance:** prove that a full- $\mathcal{A}$ -coprime pair has no nonzero common root at all. This is stronger than Jossen and may be false; it should not be asserted without a new zero theorem.
- **Divisor-sized gcd estimate:** for coprime P, Q , show that the common-zero counting function is too small to contain the complete divisor of $Q(\Phi)$. Together with Theorem 2.2 this proves the entire-quotient part.

- **Bourget-type seed:** already the assertion that two distinct primitive coordinates $\text{Ein}_q(\lambda z)$ and $\text{Ein}_p(\mu z)$ have no common nonzero zero is not supplied by 001. Proving it uniformly would be a natural first step.

35.7. 6. Literature boundary and exact limitations

Fischler–Rivoal state Jossen’s conjecture as follows:

1. if E-functions f, g have an entire quotient f/g , then that quotient is an E-function;
2. if two E-functions have a common root, then they have a common nonunit E-function divisor.

They record that part (ii) is known when the common root is algebraic, part (i) is known for polynomial denominators, and prove the differential-operator case used above. Under Schanuel they obtain much stronger common-zero results for exponential polynomials. The transcendental-root case for general E-functions remains the hard boundary.

Primary sources:

- S. Fischler and T. Rivoal, *Zeros of E-functions and of exponential polynomials defined over $\overline{\mathbb{Q}}$* , *Rend. Circ. Mat. Palermo (2)* **75** (2026), Art. 158; see also the [2025 preprint](#), especially Conjecture 1.1, Theorem 1.2, and Sections 5–6.
- F. Beukers, *A refined version of the Siegel–Shidlovskii theorem*, *Ann. of Math.* 163 (2006), 369–379, especially the algebraic-point zero-removal proposition used in Theorem 2A.1.
- A. Levin and J. T.-Y. Wang, *Greatest common divisors of analytic functions and Nevanlinna theory on algebraic tori*.
- 001, Theorem 8.3 and Corollary 8.4, for the resonant value and functional independence used throughout.

The precise remaining obstruction is the transcendental codimension-two saturation gate of Section 5.3. None of the preceding subclasses is used to claim the full multigenerator conjecture.

Part VI

Arithmetic Barriers and Remaining Gates

36. The critical moving-point barrier

We first separate the coefficient of γ from the gamma-free tail. Fix $q > 1$, put

$$f_q(t) = E_1(q^t),$$

and consider a finitely supported coefficient vector (c_n) . The two linear statistics

$$\omega(c) = \sum_n c_n, \quad \mu(c) = \sum_n n c_n$$

give the exact decomposition

$$\sum_n c_n \text{Ein}(q^n) = \omega(c)\gamma + \mu(c) \log q + \sum_n c_n f_q(n). \quad (265)$$

We call $\omega = 1, \mu = 0$ the augmentation-one, log-balanced stratum. The dyadic level belongs to this stratum. The next theorem completely solves its natural cumulative-positive cone.

Direct differentiation gives

$$f'_q(t) = -(\log q)e^{-q^t} < 0, \quad f''_q(t) = (\log q)^2 q^t e^{-q^t} > 0. \quad (266)$$

Theorem 36.1 (Sharp cumulative-positive extremality). *Let $M \geq 2$, and suppose that real coefficients $c_0, \dots, c_M$ satisfy*

$$\sum_{n=0}^M c_n = 1, \quad \sum_{n=0}^M n c_n = 0.$$

Put $C_k = \sum_{n=0}^k c_n$, and assume $C_k \geq 0$ for $0 \leq k < M$. Then the complete sharp range is

$$\begin{aligned} M f_q(M-1) - (M-1) f_q(M) &\leq \sum_{n=0}^M c_n f_q(n) \\ &\leq f_q(M) + M \{f_q(0) - f_q(1)\}. \end{aligned} \quad (267)$$

The lower endpoint is attained uniquely by

$$c_{M-1} = M, \quad c_M = -(M-1), \quad c_0 = \dots = c_{M-2} = 0. \quad (268)$$

The upper endpoint is attained uniquely by $c_0 = M, c_1 = -M, c_M = 1$, with the other coefficients zero. Consequently the moving level $Y_{M-1}(q) - \gamma$ is the unique minimum tail in this cone, and

$$\min \sum_n c_n f_q(n) \sim \frac{M}{q^{M-1}} e^{-q^{M-1}}.$$

Proof. Abel summation and the moment normalization give

$$\begin{aligned} \sum_{n=0}^M c_n f_q(n) &= f_q(M) + \sum_{k=0}^{M-1} C_k \{f_q(k) - f_q(k+1)\}, \\ \sum_{k=0}^{M-1} C_k &= M. \end{aligned}$$

By strict convexity, the positive differences $f_q(k) - f_q(k+1)$ decrease strictly with k . The displayed sum is therefore minimized by putting all mass M at $k = M-1$, and maximized by putting it at $k = 0$. Unwinding the partial sums gives the two unique coefficient vectors. The feasible C -vectors form the simplex $\{C_k \geq 0 : \sum C_k = M\}$; its continuous linear image is the whole interval between these two endpoints, which proves the asserted complete range. The asymptotic is (51). $\square$

The same doubly exponential scale survives when the two levels use multiplicatively independent bases.

Proposition 36.2 (Two-base moving-level obstruction). *For $q > 1$, write*

$$Y_N(q) = (N+1) \operatorname{Ein}(q^N) - N \operatorname{Ein}(q^{N+1}) = \gamma + \varepsilon_N(q).$$

Let

$$M_N = \left\lfloor \frac{N \log 2}{\log 3} \right\rfloor + 1, \quad W_N = Y_N(2) - Y_{M_N}(3).$$

Then $W_N > 0$ for all sufficiently large N , and

$$\boxed{-\log W_N = 2^N + N \log 2 - \log(N+1) + O(1).} \quad (269)$$

In particular $-\log W_N = 2^N + O(N)$. Thus changing from one multiplicative orbit to two independent ones does not remove the moving-point scale of the fixed-form obstruction.

Proof. Put $x = 2^N$ and $y = 3^{M_N}$. Since $N \log 2 / \log 3$ is not an integer, $y > x$; both are integers, so $y \geq x + 1$. The standard bounds

$$\frac{e^{-u}}{u+1} < E_1(u) < \frac{e^{-u}}{u}$$

show, uniformly here, that

$$\varepsilon_N(2) = \frac{N+1}{x} e^{-x} \{1 + O(x^{-1})\}.$$

Moreover $M_N/N \rightarrow \log 2 / \log 3 < 1$, and the same bounds give

$$\frac{\varepsilon_{M_N}(3)}{\varepsilon_N(2)} \leq \left(\frac{\log 2}{\log 3} + o(1) \right) e^{-1} < 1.$$

Hence W_N is positive and differs from $\varepsilon_N(2)$ by a factor bounded above and below away from zero. Taking logarithms yields (269). $\square$

Outside this cone, augmentation one alone has no sign content.

Proposition 36.3 (Three-point flexibility). *Let $0 < a < b < c$ be positive integers. Put*

$$c^{(0)} = \left(\frac{b}{b-a}, -\frac{a}{b-a}, 0 \right), \quad v = (c-b, a-c, b-a).$$

For $t \in \mathbb{R}$, the vector $c(t) = c^{(0)} + tv$, supported at a, b, c , is augmentation one and log balanced. Its tail is $A + tD$, where

$$A > 0, \quad D = (b-a)(c-b)(c-a)f_q[a, b, c] > 0.$$

Hence rational admissible coefficients give a dense set of tail values in $\mathbb{R}$; in particular both signs and arbitrarily small nonzero values occur. For example, at $q = 2$ and exponents $1, 2, 3$,

$$-E_1(2) + 5E_1(4) - 3E_1(8) < 0.$$

Proof. The two normalizations are immediate. The slope D is a positive second divided difference by (266). Since $A + D\mathbb{Q}$ is dense in $\mathbb{R}$, all qualitative assertions follow. For the explicit example use

$$E_1(x) > \frac{e^{-x}}{x+1}, \quad E_1(x) < \frac{e^{-x}}{x} :$$

the displayed value is less than $-e^{-2}/3 + 5e^{-4}/4 < 0$. $\square$

The next estimate identifies the coefficient height at which such cancellation can begin. Write a rational coefficient vector primitively as $c_j = a_j/D$, with $D \geq 1$, and put $H(c) = \max(D, |a_1|, \dots, |a_r|)$.

Proposition 36.4 (First-tail dominance). *Let $n_1 < \dots < n_r$ be integers with $a_1 \neq 0$. If*

$$(r-1)H(c) \frac{1+q^{-n_1}}{q} e^{-(q-1)q^{n_1}} \leq \frac{1}{2}, \quad (270)$$

then the tail has the sign of a_1 and

$$\left| \sum_{j=1}^r c_j E_1(q^{n_j}) \right| \geq \frac{e^{-q^{n_1}}}{2H(c)(q^{n_1}+1)}. \quad (271)$$

Thus, for fixed q, r , cancellation of the first active tail requires

$$\log H(c) \geq (q-1)q^{n_1} + O(1).$$

Proof. The reverse triangle inequality gives

$$\left| \sum_j c_j E_1(q^{n_j}) \right| \geq \frac{E_1(q^{n_1}) - (r-1)H(c)E_1(q^{n_1+1})}{H(c)}.$$

The elementary bounds used in the preceding proof imply

$$\frac{E_1(q^{n+1})}{E_1(q^n)} \leq \frac{1+q^{-n}}{q} e^{-(q-1)q^n}.$$

Now apply (270). □

For consecutive triples the transition is exact. Let

$$F_{j,N} = E_1(q^{N+j}), \quad A_N = (N+2)F_{1,N} - (N+1)F_{2,N}, \quad B_N = F_{0,N} - 2F_{1,N} + F_{2,N},$$

and $\rho_N = B_N/A_N$. Both A_N, B_N are positive.

Theorem 36.5 (Exact denominator phase transition). *Every rational augmentation-one, log-balanced coefficient vector on $N, N+1, N+2$ has the form*

$$c(s) = (s, N+2-2s, s-N-1).$$

For $s = a/D$ with $D \geq 1$, its tail $\varepsilon_N(s)$ satisfies

$$\left| \varepsilon_N(a/D) \right| = \frac{B_N}{D} \left| a + \frac{D}{\rho_N} \right|, \quad (272)$$

and hence

$$\min_{a \in \mathbb{Z}} |\varepsilon_N(a/D)| = \frac{B_N}{D} \left\| \frac{D}{\rho_N} \right\|.$$

In particular, $D \leq \rho_N/2$ implies $|\varepsilon_N(a/D)| \geq A_N$ for every integer a . Conversely, whenever $\rho_N \geq \frac{1}{2}$ (hence for all sufficiently large N), choosing a positive integer D nearest to ρ_N and taking $a = -1$ gives either an exact zero or

$$0 < |\varepsilon_N(-1/D)| \leq \frac{A_N}{2D}. \quad (273)$$

Furthermore, with $X = q^N$,

$$\rho_N \sim \frac{q}{N+2} e^{(q-1)X}. \quad (274)$$

Thus the denominator threshold is exactly $e^{(q-1)q^N}/N$ up to a constant factor.

Proof. Substitution gives $\varepsilon_N(s) = A_N + B_N s$, proving (272). If $D/\rho_N \leq 1/2$, its nearest integer is zero. If $|D - \rho_N| \leq 1/2$, then $|1 - D/\rho_N| \leq 1/(2\rho_N)$, which gives (273). Finally (51) yields

$$B_N \sim E_1(X), \quad A_N \sim (N+2)E_1(qX),$$

and hence (274). $\square$

When q is algebraic, exact zero can occur for at most one N ; by theorem 9.5, any such zero would already force the algebraic independence of γ, δ over K . Discarding that possible one index, the nearest-denominator construction gives normalized coefficient height

$$H_N = qe^{(q-1)X}(1 + o(1))$$

and

$$0 < |\varepsilon_N| \leq \frac{(N+2)^2}{2q^2 X} e^{-(2q-1)X}(1 + o(1)). \quad (275)$$

Consequently a uniform lower bound, on normalized nonzero rational augmentation-one triples, of the form

$$|\varepsilon| \geq C e^{-\beta X} H^{-\kappa} X^{-\lambda}$$

requires

$$\boxed{\beta + \kappa(q-1) \geq 2q-1.} \quad (276)$$

At equality it also requires $\lambda \geq 1$. In particular, at the natural point cost $\beta = 1$, one must have $\kappa \geq 2$. This exponent refers to the affine normalization $\sum c_j = 1$; clearing denominators multiplies the value and changes the corresponding homogeneous height exponent.

The preceding sections construct exact finite spectral data converging very rapidly to Euler-level traces. To extract the arithmetic nature of the limit, one would need a lower bound uniform in a moving algebraic sample point. We first show that sublinear point cost is impossible.

Let

$$F_1(z) = \text{Ein}(z), \quad F_2(z) = \text{Ein}(2z), \quad F_3(z) = \text{Ein}(4z),$$

and let $P_0 = X_1 - 2X_2 + X_3$.

Theorem 36.6 (A fixed-form critical-scale no-go). *For $x \geq 1$,*

$$R(x) := P_0(F_1(x), F_2(x), F_3(x)) = E_1(x) - 2E_1(2x) + E_1(4x) > 0, \quad (277)$$

and

$$R(x) = \frac{e^{-x}}{x} \left(1 + O(x^{-1}) + O(e^{-x}) \right). \quad (278)$$

Consequently, for every $A > 0$ and every $\vartheta < 1$,

$$0 < R(n) < e^{-An^\vartheta} \quad (279)$$

for all sufficiently large integers n .

More sharply, for every $C_0 > 0$, a lower bound

$$R(n) \geq C_0 e^{-\sigma n} n^{-\lambda} \quad (280)$$

cannot hold for all large n if either $\sigma < 1$, or if $\sigma = 1$ and $\lambda < 1$.

In particular, no lower bound

$$|P(F(\xi))| \geq \exp \left[-C(1 + |\xi|)^\vartheta \Phi(\deg P, \log H(P)) \right] \quad (281)$$

can hold uniformly for all positive integral ξ , all nonzero values and all fixed E -function systems if $\vartheta < 1$, when C and Φ are independent of ξ .

Proof. The connection formula (1) cancels γ , $\log x$, and the scale logarithms because

$$1 - 2 + 1 = 0, \quad \log x - 2 \log(2x) + \log(4x) = 0.$$

This proves the identity in (277). Using $E_1(x) = \int_1^\infty e^{-xt} dt/t$, its integrand numerator is, with $u = e^{-xt}$,

$$u - 2u^2 + u^4 = u(1 - u)(1 - u - u^2) > 0,$$

because $0 < u \leq e^{-1} < (\sqrt{5} - 1)/2$. This proves positivity. Asymptotic formula (278) follows from (51); the terms at $2x$ and $4x$ are exponentially smaller. Hence

$$-\log R(n) = n + \log n + o(1).$$

Since $n^\vartheta = o(n)$, (279) follows. In (281), P_0 has fixed degree one and fixed height two, so its right-hand exponent is $O(n^\vartheta)$, contradicting (279). Finally, (278) gives

$$\frac{R(n)}{C_0 e^{-\sigma n} n^{-\lambda}} = C_0^{-1} e^{-(1-\sigma)n} n^{\lambda-1} (1 + o(1)),$$

which tends to zero in each of the two stated cases and contradicts (280). $\square$

Remark 36.7 (Exact scope of the no-go). The theorem gives the necessary condition $\vartheta \geq 1$ under the normalization (281), and at the critical exponential scale it forces at least the polynomial loss n^{-1} . It does not prove the existence or sufficiency of such a bound. If arbitrary point-dependent constants are allowed, or if the function system itself is changed with the sample point, the exponent ϑ alone is not an invariant.

Remark 36.8 (Augmentation zero versus one). The fixed form $P_0 = (1, -2, 1)$ has $\omega(P_0) = 0$ and zero logarithmic moment, so it lies in the gamma-free stratum of (265). It disproves bounds that treat all coefficient directions uniformly; by itself it does not disprove a theorem restricted to $\omega = 1$. Theorems 36.1 and 36.5 describe what survives, and what fails, after that restriction.

The same fixed second difference is built into the dyadic tower.

Proposition 36.9 (Increment identity). *For every $N \geq 1$,*

$$\boxed{Y_N - Y_{N+1} = (N + 1)R(2^N)}. \quad (282)$$

Consequently

$$\varepsilon_N = \sum_{k \geq N} (k + 1)R(2^k). \quad (283)$$

Proof. Expanding (48) gives

$$Y_N - Y_{N+1} = (N + 1)\{\text{Ein}(2^N) - 2 \text{Ein}(2^{N+1}) + \text{Ein}(2^{N+2})\},$$

which is (282). Telescoping and $Y_N \rightarrow \gamma$ give (283). $\square$

Thus the obstruction to a sublinear measure is not merely analogous to the tower: it is its exact level increment.

We now formulate the remaining arithmetic question with a coefficient that is fixed across every level. For $s \in \mathbb{C}$, define

$$L_N(s) = 2sY_N^2 + Y_N - 2. \quad (284)$$

Theorem 36.10 (Common-coefficient critical-slope criterion). *For $s \in \mathbb{C}$, the limit*

$$\lambda(s) = \lim_{N \rightarrow \infty} \frac{-\log |L_N(s)|}{2^N}$$

exists whenever the logarithm is defined for all sufficiently large N , and

$$\lambda(s) = \begin{cases} 1, & s = S_2, \\ 0, & s \neq S_2. \end{cases} \quad (285)$$

For $s = S_2$, more precisely,

$$L_N(S_2) = \frac{4-\gamma}{\gamma} \varepsilon_N + 2S_2 \varepsilon_N^2 > 0, \quad (286)$$

$$-\log L_N(S_2) = 2^N + N \log 2 - \log(N+1) + O(1). \quad (287)$$

Hence

$$\boxed{S_2 \in \overline{\mathbb{Q}} \iff \exists s \in \overline{\mathbb{Q}} \text{ with } \lambda(s) = 1.} \quad (288)$$

If $s \in \overline{\mathbb{Q}}$, the family $\{L_N(s) : N \geq 1\}$ is countably algebraically independent over K .

Proof. Since $2S_2\gamma^2 + \gamma - 2 = 0$, substituting $Y_N = \gamma + \varepsilon_N$ gives (286); combine this with (50) to get (287). If $s \neq S_2$, then

$$L_N(s) \longrightarrow 2\gamma^2(s - S_2) \neq 0,$$

so the normalized logarithm tends to zero. This proves (285) and (288).

Finally, for algebraic s , one has $s \in K$ and

$$2sY_N^2 + Y_N - (L_N(s) + 2) = 0.$$

Thus $K(Y_N : N \in I)$ is algebraic over $K(L_N(s) : N \in I)$ for every finite set I ; when $s = 0$, the relation is linear. Algebraic independence of the Y_N therefore transfers to the $L_N(s)$. $\square$

There is an equivalent bounded-distortion form that uses only finite-stage gamma-free data.

Corollary 36.11 (Bounded-distortion criterion). *One has*

$$\frac{L_N(S_2)}{Y_N - Y_{N+1}} \longrightarrow \frac{4-\gamma}{\gamma}, \quad (289)$$

whereas for $s \neq S_2$,

$$\left| \frac{L_N(s)}{Y_N - Y_{N+1}} \right| \longrightarrow \infty.$$

Consequently

$$S_2 \in \overline{\mathbb{Q}} \iff \exists s \in \overline{\mathbb{Q}} : \left\{ \frac{L_N(s)}{Y_N - Y_{N+1}} \right\}_{N \gg 1} \text{ is bounded.} \quad (290)$$

Proof. By proposition 36.9 and (278),

$$Y_N - Y_{N+1} \sim \frac{N+1}{2^N} e^{-2^N} \sim \varepsilon_N.$$

Now use (286); if $s \neq S_2$, the numerator tends to a nonzero constant while the denominator tends to zero. $\square$

Remark 36.12 (Why a generic critical measure is not enough). The fixed-form obstruction has size $R(x) \sim e^{-x}/x$. Under the algebraicity hypothesis $s = S_2$, the target at $x = 2^N$ has size

$$L_N(S_2) \sim C_\gamma(\log x) \frac{e^{-x}}{x}.$$

It is therefore larger, by a logarithmic factor, than an unconditional fixed-form example already permits. A lower bound constrained only by this worst-case pointwise scale cannot distinguish the two. A surviving route must use additional structure, for example the repetition of the *same* algebraic coefficient s over all levels.

Existing quantitative E -function theorems cited here are fixed-point theorems: the constants in the algebraic-independence measure depend on the chosen nonzero algebraic evaluation point; see Adamczewski–Faverjon [1] and Fischler–Rivoal [5]. They do not directly provide the moving uniformity required by [theorem 36.10](#). This statement is only about the scope of those theorems, not an impossibility claim for all future methods.

37. An exact asymmetric Sondow integral

One of the discarded barrier heuristics attempted to assign a universal entropy floor to an asymmetric two-variable Sondow integral. The correct replacement is an exact finite formula. For integers $A \geq 0$ and $B \geq 1$, put

$$\mathcal{I}_{A,B} = \int_0^1 \int_0^1 \frac{x^A(1-x)^B y^A(1-y)^B}{(1-xy)(-\log(xy))} dx dy. \quad (291)$$

Theorem 37.1 (Exact asymmetric evaluation). *One has*

$$\begin{aligned} \mathcal{I}_{A,B} = & \binom{2B}{B} \gamma \\ & + 2 \sum_{j=0}^B \binom{B}{j}^2 (H_j - H_{B-j}) \log((A+j)!) \\ & - \sum_{j=0}^B \binom{B}{j}^2 H_{A+j}. \end{aligned} \quad (292)$$

In particular,

$$\mathcal{I}_{1,2} = 6\gamma + 3 \log 6 - \frac{53}{6}. \quad (293)$$

Proof. Use Tonelli's theorem,

$$-\frac{1}{\log(xy)} = \int_0^\infty (xy)^u du, \quad \frac{1}{1-xy} = \sum_{k \geq 0} (xy)^k,$$

and the beta integral. With

$$R_B(t) = \frac{B!}{t(t+1) \cdots (t+B)},$$

this gives

$$\mathcal{I}_{A,B} = \sum_{m=A+1}^{\infty} \int_m^{\infty} R_B(t)^2 dt.$$

The rational kernel has the partial fraction identity

$$\left\{ \frac{B!}{t(t+1)\cdots(t+B)} \right\}^2 = \sum_{j=0}^B \binom{B}{j}^2 \left\{ \frac{1}{(t+j)^2} + \frac{2(H_j - H_{B-j})}{t+j} \right\}. \quad (294)$$

Put $a_j = \binom{B}{j}^2$ and $h_j = H_j - H_{B-j}$. Integrating and summing only up to $m = M$ gives the finite identity

$$\begin{aligned} \mathcal{I}_{A,B}^{(M)} &= \sum_{j=0}^B a_j (H_{M+j} - H_{A+j}) \\ &\quad - 2 \sum_{j=0}^B a_j h_j \log \frac{(M+j)!}{(A+j)!}. \end{aligned}$$

The potentially divergent pieces cancel by

$$\sum_j a_j = \binom{2B}{B}, \quad \sum_j a_j h_j = 0,$$

together with the derivative of Vandermonde's identity

$$2 \sum_j j a_j h_j = \binom{2B}{B}.$$

Indeed $H_{M+j} = \log M + \gamma + o(1)$, while Stirling's formula gives

$$\log(M+j)! = M(\log M - 1) + (j + \tfrac{1}{2}) \log M + \tfrac{1}{2} \log(2\pi) + o(1).$$

Letting $M \rightarrow \infty$ in the finite identity gives

$$\binom{2B}{B} \gamma - \sum_j a_j H_{A+j} + 2 \sum_j a_j h_j \log((A+j)!),$$

which is (292). Direct substitution gives (293). $\square$

Formula (292) replaces, rather than supports, any fixed numerical entropy floor for this asymmetric family. The remaining Diophantine bottleneck is the transverse height of the prime-exponent vector in the logarithmic kernel. We make no novelty claim beyond this exact evaluation without a separate priority comparison with the broader Sondow literature [34].

38. The Prévost grid and residual-prime saturation

For $0 \leq m \leq n$, Prévost's family [15] can be written

$$J_{n,m} = \gamma + L_{n,m} - 2H_n, \quad (295)$$

$$L_{n,m} = \sum_{k=0}^n (-1)^{n+k} \binom{n}{k} \binom{n+k}{k} \log(n-m+k+1). \quad (296)$$

Let $d_n = \text{lcm}(1, \dots, n)$ and $j_{n,m} = d_n J_{n,m}$. If $m/n \rightarrow \alpha \in (0, 1)$, its saddle rate is

$$|J_{n,m}| = \exp\{(\log r(\alpha) + o(1))n\}, \quad r(\alpha) = \alpha^\alpha (1-\alpha)^{2-2\alpha} (2-\alpha)^{\alpha-2}, \quad (297)$$

up to a polynomial factor. The unique optimum is

$$\alpha_* = 1 - \frac{1}{\sqrt{2}}, \quad r(\alpha_*) = 3 - 2\sqrt{2}, \quad -\log r(\alpha_*) = \log(3 + 2\sqrt{2}). \quad (298)$$

We first isolate an exact linear-algebra obstruction that is independent of the saddle estimate. Fix a finite set I of cells and collect the normalized forms into

$$\mathbf{j} = \gamma \mathbf{B} - \mathbf{A} + \sum_{\ell \in \mathcal{P}_I} (\log \ell) \mathbf{V}_\ell \in \mathbb{R}^I, \quad (299)$$

where $\mathcal{P}_I$ is the finite set of primes occurring in the logarithmic coordinates. The vectors $\mathbf{B}, \mathbf{A}, \mathbf{V}_\ell$ are integral. We write $B(x) = \langle \mathbf{B}, x \rangle$, and similarly for the other coordinate functionals.

Theorem 38.1 (Residual-prime Hilbert saturation). *Let $\mathcal{S} \subset \mathcal{P}_I$, let $a/b \in \mathbb{Q}$ be reduced with $b > 0$, and put*

$$E_{\mathcal{S}, a/b} = \bigcap_{\ell \notin \mathcal{S}} \ker V_\ell \cap \ker(aB - bA).$$

If P is orthogonal projection onto this subspace, then

$$\boxed{bP\mathbf{j} = (b\gamma - a)P\mathbf{B} + b \sum_{p \in \mathcal{S}} (\log p) P\mathbf{V}_p.} \quad (300)$$

Let $c = (c_p)_{p \in \mathcal{S}} \in \mathbb{Z}^{\mathcal{S}}$ be primitive and nonzero. Choose $h_p \in \mathbb{Z}$ with $\sum h_p c_p = 1$, set

$$\mathbf{T} = \sum_{p \in \mathcal{S}} h_p \mathbf{V}_p, \quad \alpha_c = \sum_{p \in \mathcal{S}} c_p \log p,$$

write $T(x) = \langle \mathbf{T}, x \rangle$, and define

$$E_{c, a/b} = E_{\mathcal{S}, a/b} \cap \bigcap_{p \in \mathcal{S}} \ker(V_p - c_p T).$$

Denote its orthogonal projection by P_c . If $\gamma = a/b$, then

$$\boxed{P_c \mathbf{j} = \alpha_c P_c \mathbf{T}.} \quad (301)$$

If $\Lambda = E_{c, a/b} \cap \mathbb{Z}^I$ and $T(\Lambda) = g\mathbb{Z}$, $g > 0$, then the real quotient minimum

$$\lambda_{\mathbb{R}} = \inf \{ \|x\| : x \in E_{c, a/b}, T(x) = g \}$$

satisfies

$$\boxed{\lambda_{\mathbb{R}} = \frac{g}{\|P_c \mathbf{T}\|}, \quad \lambda_{\mathbb{R}} \|P_c \mathbf{j}\| = g |\alpha_c|.} \quad (302)$$

Proof. Project (299). All primes outside $\mathcal{S}$ vanish and $aP\mathbf{B} = bP\mathbf{A}$, proving (300). The fixed-direction equations give $P_c \mathbf{V}_p = c_p P_c \mathbf{T}$, and (301) follows under $\gamma = a/b$. Finally, the restriction of T to $E_{c, a/b}$ is represented by $P_c \mathbf{T}$. Cauchy–Schwarz is sharp in its direction and gives the first identity in (302); the second follows from (301). $\square$

Corollary 38.2 (Phase-blind quotient barrier). *Under the hypotheses of the second part of theorem 38.1, and under $\gamma = a/b$,*

$$\lambda_{\mathbb{R}} \|\mathbf{j}\| \geq g |\alpha_c|.$$

Thus a strict inequality in the opposite direction cannot follow from this real Euclidean quotient identity and Cauchy–Schwarz alone. Integral divisibility or genuinely lattice-theoretic information is not excluded. Any strict reverse inequality would already imply $\gamma \neq a/b$.

Remark 38.3 (The one-prime shear). Let P now denote orthogonal projection onto $\bigcap_{\ell \neq 2} \ker V_\ell$, with no candidate rationality hyperplane imposed, and suppose $x = P\mathbf{V}_2 \neq 0$. Put

$$z = P(\gamma\mathbf{B} - \mathbf{A}), \quad \varepsilon = \|P\mathbf{j}\|.$$

Then $P\mathbf{j} = z + (\log 2)x$, and hence

$$\frac{\det \text{Gram}(x, z)}{\|x\|^2} = \|P_{x^\perp} P\mathbf{j}\|^2 \leq \varepsilon^2, \quad \frac{\langle x, z \rangle}{\|x\|^2} = -\log 2 + O\left(\frac{\varepsilon}{\|x\|}\right). \quad (303)$$

Thus $-\log 2$ is the longitudinal shear coefficient, while $P_{x^\perp} P\mathbf{j}$ is the narrow transverse residual. For an external proxy $\tilde{\gamma} = a/b \neq \gamma$,

$$P(\tilde{\gamma}\mathbf{B} - \mathbf{A}) = z + (\tilde{\gamma} - \gamma)P\mathbf{B}.$$

More explicitly, if an integral coefficient vector c kills the odd-prime labels and has residual label $k = V_2(c)$, then

$$(aB(c) - bA(c)) + bk \log 2 = b\langle \mathbf{j}, c \rangle + (a - b\gamma)B(c).$$

Thus a proxy form is not controlled by the analytic norm of $P\mathbf{j}$ unless $a/b = \gamma$; finite-window diagnostics based on an external proxy cannot be used as irrationality evidence.

38.1. The fixed-place Ridout deficit

For a fixed finite set S of rational primes and $0 \neq r \in \mathbb{Z}$, write

$$r_S = \prod_{p \in S} p^{v_p(r)}, \quad r_{S^c} = |r|/r_S.$$

Proposition 38.4 (Fixed- S Ridout budget). *Let ξ be a real algebraic irrational number. For every $\epsilon > 0$, all but finitely many reduced fractions a/b with $b > 0$, $a \neq 0$, and $|\xi - a/b| < 1$ satisfy*

$$|b\xi - a| \geq \frac{a_S}{b_{S^c} b^\epsilon}. \quad (304)$$

Consequently, suppose along a sequence that

$$\log b_N = hN + o(N), \quad \log(a_N)_S + \log(b_N)_S = \sigma N + o(N), \quad |b_N \xi - a_N| = e^{-(\beta + o(1))N}.$$

Assume here that $h > 0$, that $b_N \rightarrow \infty$, and that the reduced fractions a_N/b_N take infinitely many distinct values. If $\beta > h - \sigma$, then ξ is not algebraic irrational. If $\beta \leq h - \sigma$, these rates alone do not enter the range excluded by Ridout's theorem.

Proof. Ridout's theorem [17] says that

$$\prod_{p \in S} |ab|_p \min\{1, |\xi - a/b|\} < b^{-2-\epsilon}$$

has only finitely many reduced solutions. For a good approximant the left side is $|\xi - a/b|/(a_S b_S)$. Multiplying the complementary inequality by b gives (304); in the homogeneous-height form of the theorem, $|\xi - a/b| < 1$ makes $\max(|a|, b) \ll_\xi b$, and applying the theorem with $\epsilon/2$ absorbs the fixed comparison constant for large b . Taking logarithms and then letting $\epsilon \downarrow 0$ proves the rate assertion. $\square$

Theorem 38.5 (The Prévost saddle-lcm ledger). *For any integer sequence m_n with $m_n/n \rightarrow \alpha_*$,*

$$|d_n J_{n,m_n}| = \exp\{-(\log(3 + 2\sqrt{2}) - 1 + o(1))n\}.$$

For every fixed finite set S ,

$$\log(d_n)_S = |S| \log n + O_S(1) = o(n), \quad \log d_n = n + o(n).$$

Thus, in the favorable height-one ledger in which the lcm is the only denominator cost and a fixed S contributes no additional exponential divisibility, the analytic exponent falls short of the Ridout threshold by

$$\boxed{\Delta_R = 2 - \log(3 + 2\sqrt{2}) = 0.2372528259 \dots} \quad (305)$$

This is an exact saddle-denominator deficit, not yet a rational approximation to γ , because logarithmic coordinates remain in J_{n,m_n} .

Proof. Combine (297) with the prime-number theorem form $\log d_n = \psi(n) = n + o(n)$. Differentiating $\log r$ gives

$$(\log r)'(\alpha) = \log \frac{\alpha(2 - \alpha)}{(1 - \alpha)^2},$$

which proves the unique optimum (298). Finally $v_p(d_n) = \lfloor \log n / \log p \rfloor$, so every fixed S contributes only $o(n)$. Comparing the resulting exponent $\log(3 + 2\sqrt{2}) - 1$ with the height-one threshold (1) gives (305). $\square$

Corollary 38.6 (Conditional rational-extraction budget). *Suppose an elimination of the logarithmic coordinates produces infinitely many distinct reduced fractions a_n/b_n such that*

$$\log b_n = n + o(n), \quad \log(a_n)_S + \log(b_n)_S = o(n),$$

and preserves the saddle rate

$$|b_n \gamma - a_n| = \exp\{-(\log(3 + 2\sqrt{2}) - 1 + o(1))n\}.$$

Then proposition 38.4 does not exclude algebraic irrationality of γ . Within this height-one template, crossing the Ridout threshold requires an additional fixed- S valuation density greater than Δ_R , an archimedean gain of the same size, or a corresponding reduction in height.

Remark 38.7 (Exact scope). This is a budget theorem, not an exclusion of every linear combination of Prévost forms. Further cancellation could increase the archimedean exponent, and a new automatic divisibility theorem could increase the S -part. Neither input is presently available. Enlarging a fixed finite set S alone cannot help because $\log(d_n)_S = o(n)$.

Proposition 38.8 (Bounded-degree logarithmic transfer). *After cancelling all odd-prime coordinates, write a resulting form as*

$$J = B\gamma - A + k \log 2, \quad A, B, k \in \mathbb{Z},$$

and assume $k \neq 0$. If γ is algebraic of degree d , then

$$\beta = \frac{A - B\gamma}{k}$$

has degree at most d , naïve height at most $C_\gamma \max(1, |A|, |B|, |k|)^d$, and

$$|\log 2 - \beta| = \frac{|J|}{|k|}.$$

Thus any such attack is bounded by the degree- d algebraic approximation exponent of $\log 2$, with an additional factor d in the height exponent.

Proof. If f is the minimal polynomial of γ and $B \neq 0$, then β is a zero of

$$B^d f\left(\frac{A - kX}{B}\right),$$

and the affine transformation shows that its degree is exactly d . The displayed polynomial has coefficient height $O_\gamma(\max(1, |A|, |B|, |k|)^d)$; the standard factor-height bound gives the same estimate, up to a constant depending on γ , for the minimal polynomial of β . The case $B = 0$ is rational, and the approximation identity is immediate. $\square$

Remark 38.9. The available bounds are

$$\mu(\log 2) \leq 3.574553902525\dots, \quad \mu_2(\log 2) \leq 12.841618132152\dots;$$

see [12, 13]. Thus a rational branch must beat the first exponent, while the degree-two height transfer makes $2\mu_2(\log 2) = 25.683236264304\dots$ the corresponding coefficient-height target. These are target exponents once the growth of $\max(1, |A|, |B|, |k|)$ is known; no comparison with a projected Prévost family is claimed here without a separate height lemma.

39. Fixed-prime Rodrigues twists and universal saddle cancellation

The deficit (305) suggests inserting a factor p^{cn} , for one fixed prime p , directly into the Prévost kernel. The following deformation preserves integrality, degree, and a linear number of moments, and produces this factor in the endpoint coefficient. Even if the whole endpoint factor is favorably credited as nonarchimedean gain, the real saddle cancels it in the precise sense of theorem 39.4.

Fix a prime p . For $n \geq 1$, $1 \leq \ell \leq n$, put

$$\mathcal{P}_{n,\ell,p}(x) = \frac{(-1)^\ell}{\ell!} \frac{d^\ell}{dx^\ell} \left[x^\ell (1-x)^\ell (1+(p-1)x)^{n-\ell} \right]. \quad (306)$$

Write $\mathcal{P}_{n,\ell,p}(x) = \sum_{k=0}^n c_k x^k$.

Proposition 39.1 (Arithmetic and orthogonality). *One has $\mathcal{P}_{n,\ell,p} \in \mathbb{Z}[x]$, $\deg \mathcal{P}_{n,\ell,p} = n$,*

$$\int_0^1 x^j \mathcal{P}_{n,\ell,p}(x) dx = 0 \quad (0 \leq j < \ell),$$

and

$$\boxed{\mathcal{P}_{n,\ell,p}(1) = p^{n-\ell}}. \quad (307)$$

Proof. Expanding before differentiation gives

$$c_k = (-1)^\ell \binom{\ell+k}{\ell} \sum_{a+b=k} (-1)^a \binom{\ell}{a} \binom{n-\ell}{b} (p-1)^b,$$

which proves integrality and the degree assertion. Integration by parts ℓ times proves the moment identities; all boundary terms vanish. At $x = 1$, only the term in which all derivatives fall on $(1-x)^\ell$ survives, giving (307). $\square$

Put

$$W(x) = \frac{1}{\log x} + \frac{1}{1-x}$$

and, for $0 \leq m \leq \ell$, define

$$\mathcal{J}_{n,\ell,m,p} = \int_0^1 x^{\ell-m} \mathcal{P}_{n,\ell,p}(x) W(x) dx. \quad (308)$$

Proposition 39.2 (Exact logarithmic linear form). *Let*

$$Q_{n,\ell,p} = p^{n-\ell}, \quad A_{n,\ell,p} = \sum_{k=0}^n c_k H_k,$$

and

$$L_{n,\ell,m,p} = \sum_{k=0}^n c_k \log(\ell - m + k + 1).$$

Then

$$\boxed{\mathcal{J}_{n,\ell,m,p} = Q_{n,\ell,p} \gamma - A_{n,\ell,p} + L_{n,\ell,m,p}}, \quad (309)$$

and $d_n A_{n,\ell,p} \in \mathbb{Z}$.

Proof. Use Euler's integral $\gamma = \int_0^1 W(x) dx$ and Frullani's identity

$$\int_0^1 \frac{x^r - 1}{\log x} dx = \log(r+1).$$

The rational part is

$$\begin{aligned} & \int_0^1 \frac{\mathcal{P}(1) - x^{\ell-m} \mathcal{P}(x)}{1-x} dx \\ &= \int_0^1 \frac{\mathcal{P}(1) - \mathcal{P}(x)}{1-x} dx + \int_0^1 \mathcal{P}(x) \frac{1 - x^{\ell-m}}{1-x} dx. \end{aligned}$$

The second integral vanishes by the moment conditions, and the first is $\sum c_k H_k$. This proves (309); its denominator divides d_n . $\square$

Prévost's positive Markov–Stieltjes representation is

$$W(u) = \int_0^1 \frac{\omega(t)}{1 - (1-u)t} dt, \quad \omega(t) = \frac{1}{t\{\log^2(t^{-1} - 1) + \pi^2\}} > 0. \quad (310)$$

Proposition 39.3 (Positive double integral). *One has*

$$(-1)^m \mathcal{J}_{n,\ell,m,p} = \int_0^1 \int_0^1 \frac{u^\ell (1-u)^\ell (1+(p-1)u)^{n-\ell} (1-t)^{\ell-m} t^m}{(1-(1-u)t)^{\ell+1}} \omega(t) dt du > 0. \quad (311)$$

Proof. Insert (310), use Fubini, and integrate by parts ℓ times. The polynomial part is annihilated, while

$$\frac{d^\ell}{du^\ell} \frac{u^{\ell-m}}{1 - (1-u)t} = (-1)^m \ell! \frac{(1-t)^{\ell-m} t^m}{(1 - (1-u)t)^{\ell+1}}.$$

This gives (311) and positivity. $\square$

Let $\theta = \ell/n$, $\alpha = m/\ell$, and set

$$\phi_\alpha(u, t) = \frac{u(1-u)(1-t)^{1-\alpha}t^\alpha}{1-(1-u)t}, \quad (312)$$

$$\Phi_{\theta, \alpha, p}(u, t) = \phi_\alpha(u, t)^\theta (1 + (p-1)u)^{1-\theta}, \quad (313)$$

$$R_{\theta, \alpha, p} = -\log \max_{0 \leq u, t \leq 1} \Phi_{\theta, \alpha, p}(u, t). \quad (314)$$

We extend ϕ_α , and hence $\Phi_{\theta, \alpha, p}$, continuously by zero at the boundary corner $(u, t) = (0, 1)$. Indeed the quotient in (312) tends to zero there for $0 < \alpha < 1$.

For the finite parameters $\theta_n = \ell/n$, $\alpha_n = m/\ell$, the positive double integral is

$$(-1)^m \mathcal{J}_{n, \ell, m, p} = \int_0^1 \int_0^1 \Phi_{\theta_n, \alpha_n, p}(u, t)^n \frac{\omega(t)}{1-(1-u)t} dt, du.$$

The remaining measure has finite mass

$$\int_0^1 \int_0^1 \frac{\omega(t)}{1-(1-u)t} dt, du = \int_0^1 W(u) du = \gamma.$$

This gives the upper Laplace bound, while a small rectangle around an interior maximizer gives the matching lower bound. Consequently, whenever $\ell/n \rightarrow \theta \in (0, 1]$ and $m/\ell \rightarrow \alpha \in (0, 1)$,

$$\lim_{n \rightarrow \infty} |\mathcal{J}_{n, \ell, m, p}|^{1/n} = e^{-R_{\theta, \alpha, p}}. \quad (315)$$

Theorem 39.4 (Universal fixed-prime saddle cancellation). *For every prime p , $0 < \theta \leq 1$, and $0 < \alpha < 1$,*

$$\boxed{R_{\theta, \alpha, p} + (1-\theta) \log p \leq \log(3 + 2\sqrt{2})}. \quad (316)$$

Thus no member of (306) improves the original Prévost saddle after the complete endpoint factor $p^{n-\ell}$ is favorably credited as fixed- p gain.

Proof. Since

$$e^{-\{R_{\theta, \alpha, p} + (1-\theta) \log p\}} = \max_{u, t} \left[\phi_\alpha(u, t)^\theta \left(\frac{1 + (p-1)u}{p} \right)^{1-\theta} \right],$$

it is enough to bound this normalized maximum from below. Now For $0 \leq u \leq 1$,

$$\frac{1 + (p-1)u}{p} = u + \frac{1-u}{p} \geq u.$$

At $t = 1/2$, the dependence on α disappears:

$$\phi_\alpha\left(u, \frac{1}{2}\right) = \frac{u(1-u)}{1+u}.$$

Consequently the normalized maximum is at least

$$\max_{0 \leq u \leq 1} u \left(\frac{1-u}{1+u} \right)^\theta \geq \max_{0 \leq u \leq 1} \frac{u(1-u)}{1+u} = 3 - 2\sqrt{2}.$$

Taking negative logarithms proves (316). □

Corollary 39.5 (The fixed-place deficit survives). *Even under the favorable assumptions that logarithmic coordinates are removed at no exponential cost, $d_n = e^{n+o(n)}$ is the only denominator cost, and the full $p^{n-\ell}$ factor survives primitive reduction, the integerized exponent satisfies*

$$(R_{\theta,\alpha,p} - 1) + (1 - \theta) \log p \leq \log(3 + 2\sqrt{2}) - 1 < 1.$$

Thus, even after the entire endpoint factor is credited, this favorable ledger does not cross the fixed- S Ridout threshold; its remaining gap is at least Δ_R from (305).

Proposition 39.6 (Balanced factorial ratios have zero fixed-prime slope). *Let*

$$\mathcal{R}(n, k) = \frac{\prod_{i=1}^r L_i(n, k)!}{\prod_{j=1}^s M_j(n, k)!},$$

where the integral linear forms are nonnegative and $O(n)$ on the index range, and suppose $\sum_i L_i = \sum_j M_j$. Then for every fixed prime p ,

$$|v_p(\mathcal{R}(n, k))| = O(\log n)$$

uniformly wherever the ratio is defined. In particular its fixed-prime valuation density is zero.

Proof. Legendre's formula gives a sum over p^ν . At each level the linear parts cancel by balance, leaving a bounded floor error; only $O(\log n)$ levels are nonzero. $\square$

Remark 39.7. This covers the standard balanced Apéry, Franel, Domb, and Christol factorial ratios. The valuation exponents supplied directly by the digit-count and p -Lucas mechanisms of [3] involve at most $1 + \lfloor \log_p n \rfloor$ base- p digits and therefore provide only an $O(\log n)$ guaranteed gain. This does not exclude additional problem-specific cancellation inside a multisum. If a factorial-balance defect is $\delta n + O(1)$, $\delta \neq 0$, Stirling's formula produces a $\delta n \log n$ term in the archimedean ledger. A successful kernel must therefore use information not already encoded by the balanced factorial ratio or the positive endpoint weight twist analyzed here.

40. The Stokes-framing barrier

The preceding barriers distinguish gamma-free directions from the Euler extension. The analogous distinction for an elementary differential operator annihilating the Euler E -function is exact.

Put

$$\mathcal{E}(z) = \sum_{n \geq 1} \frac{z^n}{n n!}, \quad F(x) = \mathcal{E}(-x) = -\text{Ein}(x),$$

and let $\partial = \partial_x$. Consider

$$\mathcal{L} = (\partial + 1) \circ \partial \circ x \circ \partial, \quad \mathcal{L}y = xy''' + (x + 2)y'' + y'. \quad (317)$$

Theorem 40.1 (Euler Stokes-framing barrier). *On $\Omega = \mathbb{C} \setminus (-\infty, 0]$, with principal branches,*

$$\mathcal{B}_\infty = (1, \log x, E_1(x)), \quad \mathcal{B}_0 = (1, F(x), \log x)$$

are solution bases of $\mathcal{L}$, and

$$\begin{pmatrix} 1 \\ F(x) \\ \log x \end{pmatrix} = C_\gamma \begin{pmatrix} 1 \\ \log x \\ E_1(x) \end{pmatrix}, \quad C_\gamma = \begin{pmatrix} 1 & 0 & 0 \\ -\gamma & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (318)$$

The complete upper/lower lateral transition across the negative real axis is

$$\begin{pmatrix} 1 \\ \log_+ x \\ E_{1,+}(x) \end{pmatrix} = S \begin{pmatrix} 1 \\ \log_- x \\ E_{1,-}(x) \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 & 0 \\ 2\pi i & 1 & 0 \\ -2\pi i & 0 & 1 \end{pmatrix}. \quad (319)$$

In particular S is independent of γ , and the coefficient row $c_\gamma = (-\gamma, -1, -1)$ satisfies

$$\boxed{c_\gamma S = c_\gamma} \quad (320)$$

for every complex value of γ .

For every $t \in \overline{\mathbb{Q}}$, $F + t$ is an E -function solution of the same operator. With $\mathcal{B}_0^{(t)} = (1, F + t, \log x)$, its connection matrix is

$$C_t = U_t C_\gamma, \quad U_t = \begin{pmatrix} 1 & 0 & 0 \\ t & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (321)$$

Consequently, under $\gamma \in \overline{\mathbb{Q}}$, the choice $t = \gamma$ removes the Euler entry by an algebraic change of Frobenius framing without changing the operator or any unframed Stokes conjugacy class.

Proof. Since $xF'(x) = e^{-x} - 1$,

$$(\partial + 1)\partial(xF') = (\partial + 1)(-e^{-x}) = 0.$$

Moreover,

$$W(1, \log x, E_1(x)) = \frac{e^{-x}}{x^2} \neq 0,$$

and $\det C_\gamma = 1$, so both displayed triples are indeed bases. The connection identity

$$F(x) = -\gamma - \log x - E_1(x)$$

proves (318). For $r > 0$,

$$\log_+(-r) - \log_-(-r) = 2\pi i, \quad E_{1,+}(-r) - E_{1,-}(-r) = -2\pi i,$$

which gives (319). Matrix multiplication proves (320). Adding the constant solution t adds t times the first row of the connection matrix to its second row, proving (321). $\square$

Corollary 40.2 (Exterior blindness). *One has*

$$\det C_\gamma = 1, \quad \det S = 1, \quad \text{tr } S = 3.$$

The displayed lateral transition and every invariant of its unframed conjugacy class are independent of γ . Their determinant, trace, and exterior-power characteristic data lose the off-diagonal Euler framing.

Remark 40.3 (Strict factors and framed Stokes constants). Fischler–Rivoal use “Stokes constants” for sector-dependent coefficients of a distinguished E -function in a formal basis at infinity; in that broader sense $-\gamma$ is such a coefficient. After the logarithmic formal monodromy is separated, the strict factor is $E_1 \mapsto E_1 - 2\pi i$ and is gamma-free. Their arithmetic theorem is a containment theorem for possible coefficients, not a numerical-to-motivic separation theorem; see [4].

Remark 40.4 (Class-group averaging does not erase γ). Suppose a Kronecker–Chowla–Selberg family is written in class-group components as $A_C = \kappa\gamma + B_C$, with the same Euler coefficient in each class. Character projection gives

$$\sum_C \overline{\chi(C)} A_C = \kappa\gamma \sum_C \overline{\chi(C)} + \sum_C \overline{\chi(C)} B_C.$$

Thus the trivial character, equivalently the class-group average, preserves the Euler term. It disappears only in nontrivial character components or other coefficient-sum-zero directions. In representation-theoretic language, γ occupies the trivial isotypic component. This corrects the opposite averaging heuristic; it is not a new Chowla–Selberg theorem. See [24].

41. The mixed-Gevrey aperture

Fischler–Rivoal call a formal Laurent series mixed arithmetic-Gevrey when its nonnegative part is an E -function and its negative part is an arithmetic Gevrey series of order one [6]. The Euler connection formula produces a particularly transparent family in this class.

For $\lambda \in \overline{\mathbb{Q}}_{>0}$, set

$$\hat{D}_\lambda(z) = \sum_{n \geq 0} (-1)^n n! \lambda^{-n-1} z^{-n-1}, \quad U_\lambda(z) = e^{\lambda z}, \quad W_\lambda(z) = e^{\lambda z} \text{Ein}(\lambda z) - \hat{D}_\lambda(z). \quad (322)$$

Proposition 41.1 (Dilation system and functional rank). *Direction zero is not anti-Stokes for $\hat{D}_\lambda$, and its Borel sum is*

$$(\hat{D}_\lambda)_0(z) = e^{\lambda z} E_1(\lambda z). \quad (323)$$

Consequently

$$(W_\lambda)_0(z) = e^{\lambda z} (\gamma + \log \lambda + \log z), \quad (324)$$

on the principal positive sector, and

$$\begin{pmatrix} U_\lambda \\ W_\lambda \end{pmatrix}' = \begin{pmatrix} \lambda & 0 \\ z^{-1} & \lambda \end{pmatrix} \begin{pmatrix} U_\lambda \\ W_\lambda \end{pmatrix}. \quad (325)$$

For distinct $\lambda_1, \dots, \lambda_m \in \overline{\mathbb{Q}}_{>0}$, the $2m$ mixed functions $U_{\lambda_i}, W_{\lambda_i}$ are linearly independent over $\mathbb{C}(z)$.

Proof. The Laplace integral of the formal negative part is

$$\int_0^\infty \frac{e^{-\lambda z t}}{1+t} dt = e^{\lambda z} E_1(\lambda z),$$

which proves (323); its Borel singularity lies on the negative direction. Subtracting this identity from $e^{\lambda z} \text{Ein}(\lambda z)$ and using (1) proves (324). Direct differentiation proves (325).

Suppose a rational-function linear relation among all the displayed functions exists, and sum it in direction zero. Analytic continuation once around zero adds $2\pi i e^{\lambda_i z}$ to $(W_{\lambda_i})_0$ and fixes U_{λ_i} . Subtraction gives a rational-function relation among exponentials with distinct exponents, so all coefficients of the W -terms vanish. The remaining exponential relation then forces all U -coefficients to vanish. $\square$

Proposition 41.2 (Unconditional non-divisibility). *For every $\lambda \in \overline{\mathbb{Q}}_{>0}$ and $a \in \overline{\mathbb{Q}}$, the formal quotient*

$$\frac{W_\lambda - aU_\lambda}{z-1} \quad (326)$$

is not a mixed arithmetic-Gevrey function.

Proof. Write the normalized positive part of the numerator as

$$F(z) = e^{\lambda z} \{\text{Ein}(\lambda z) - a\}$$

and its negative part as $-\widehat{D}_\lambda(z)$, which has zero constant term. If the quotient were mixed, comparison of nonnegative and nonpositive Laurent parts after multiplication by $z - 1$ would force $F(1) \in \overline{\mathbb{Q}}$. But

$$F(1) = e^\lambda \{\text{Ein}(\lambda) - a\},$$

which is transcendental because $\text{Ein}(\lambda)$ is algebraically independent over $K \supset \overline{\mathbb{Q}}(e^\lambda)$ by [theorem 2.1](#). This is a contradiction. $\square$

We can now state the exact conditional payoff. Write $\mathbf{E}, \mathbf{D}$ for the rings of algebraic-point values of E -functions and of directionally summed arithmetic Gevrey-one series, respectively, in the notation of Fischler–Rivoal.

Theorem 41.3 (Conditional simultaneous dilation theorem). *Assume the intersection conjecture*

$$\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}. \quad (327)$$

For any distinct $\lambda_1, \dots, \lambda_m \in \overline{\mathbb{Q}}_{>0}$, the $2m$ numbers

$$\boxed{e^{\lambda_i}, \quad e^{\lambda_i(\gamma + \log \lambda_i)} \quad (1 \leq i \leq m)} \quad (328)$$

are linearly independent over $\overline{\mathbb{Q}}$. In particular every $\gamma + \log \lambda_i$ is transcendental; the case $\lambda = 1$ gives the conditional transcendence of γ .

Proof. Assume that the values in (328) satisfy a linear relation over $\overline{\mathbb{Q}}$. Using $W_\lambda = e^{\lambda z} \text{Ein}(\lambda z) - \widehat{D}_\lambda$, move all summed Gevrey terms to the right. The left side is in $\mathbf{E}$, the right side is in $\mathbf{D}$, so their common value is algebraic by (327). The left side has the form

$$\sum_i a_i e^{\lambda_i} + \sum_i b_i e^{\lambda_i} \text{Ein}(\lambda_i).$$

The $\text{Ein}(\lambda_i)$ are algebraically independent over the field K containing all e^{λ_i} , so algebraicity forces every $b_i = 0$. The Lindemann–Weierstrass theorem applied to $0, \lambda_1, \dots, \lambda_m$ then forces every $a_i = 0$. This proves linear independence. The final transcendence assertion follows because an algebraic ratio would make the two values in its block linearly dependent. $\square$

The dilation construction has an all-order resonant extension. Write $\begin{bmatrix} n \\ r \end{bmatrix}$ for the unsigned Stirling number of the first kind, and for $r \geq 1$ put

$$\widehat{D}_{\lambda,r}(z) = \sum_{n \geq 0} (-1)^n \begin{bmatrix} n+1 \\ r \end{bmatrix} \lambda^{-n-1} z^{-n-1}, \quad (329)$$

and

$$\mathcal{W}_{\lambda,0}(z) = e^{\lambda z}, \quad \mathcal{W}_{\lambda,r}(z) = e^{\lambda z} \text{Ein}_r(\lambda z) - \widehat{D}_{\lambda,r}(z) \quad (r \geq 1). \quad (330)$$

Proposition 41.4 (All-order mixed jet). *Every $\mathcal{W}_{\lambda,r}$ is mixed arithmetic-Gevrey, direction zero is non-anti-Stokes, and*

$$\boxed{\sum_{r \geq 0} (\mathcal{W}_{\lambda,r})_0(z) t^r = e^{\lambda z} \Gamma(1-t) (\lambda z)^t.} \quad (331)$$

They satisfy the triangular differential system

$$\mathcal{W}'_{\lambda,0} = \lambda \mathcal{W}_{\lambda,0}, \quad \mathcal{W}'_{\lambda,r} = \lambda \mathcal{W}_{\lambda,r} + z^{-1} \mathcal{W}_{\lambda,r-1} \quad (r \geq 1). \quad (332)$$

For distinct positive algebraic λ_i , every finite family of the functions $\mathcal{W}_{\lambda_i,r}$ is linearly independent over $\mathbb{C}(z)$.

Proof. The normalized Borel transform of (329) is

$$\frac{(-1)^{r-1}}{\lambda(r-1)!} \frac{\log^{r-1}(1 + \xi/\lambda)}{1 + \xi/\lambda},$$

a G -function whose singularities lie on the negative ray. Hence the formal series is arithmetic Gevrey of order one and direction zero is admissible. The elementary split

$$\Gamma(1-t)x^t = 1 + t \sum_{r \geq 1} \text{Ein}_r(x) t^{r-1} - tx^t \Gamma(-t, x)$$

together with the large- x expansion

$$e^x x^t \Gamma(-t, x) \sim \sum_{n \geq 0} (-1)^n (t+1)_n x^{-n-1}$$

proves (331), since $[t^{r-1}](t+1)_n = \begin{bmatrix} n+1 \\ r \end{bmatrix}$. Differentiating the generating identity and comparing coefficients proves (332).

The sectorial sum on the right of (331) is $e^{\lambda z}$ times a polynomial in $\log z$ of degree r , with nonzero leading coefficient $1/r!$. Repeated monodromy differences about zero strip off the highest logarithmic degree. At each step the linear independence of exponentials with distinct exponents forces the corresponding rational coefficients to vanish. Descending induction on r proves the last assertion. $\square$

Theorem 41.5 (Conditional all-order Gamma-jet independence). *Assume (327). Let $\lambda_1, \dots, \lambda_m \in \overline{\mathbb{Q}}_{>0}$ be distinct and choose integers $R_i \geq 0$. Then all numbers*

$$e^{\lambda_i [t^r]} \{\Gamma(1-t)\lambda_i^t\}, \quad 1 \leq i \leq m, \quad 0 \leq r \leq R_i, \quad (333)$$

are linearly independent over $\overline{\mathbb{Q}}$. In particular

$$1, \Gamma'(1), \Gamma''(1), \dots \quad (334)$$

are linearly independent over $\overline{\mathbb{Q}}$.

Proof. In a putative relation, split every $r \geq 1$ value as the value of the E -function $e^{\lambda z} \text{Ein}_r(\lambda z)$ minus the summed Gevrey value $(\widehat{D}_{\lambda,r})_0$, and move all Gevrey terms to one side. By (327), the common value is algebraic. The other side is a linear form in the jointly algebraically independent coordinates $\text{Ein}_r(\lambda_i)$ over K , by (2); therefore every coefficient with $r \geq 1$ vanishes. Lindemann–Weierstrass then removes the remaining $r = 0$ exponential terms. Formula (331) at $z = 1$ proves (333). Taking $\lambda = 1$ and using $[t^r]\Gamma(1-t) = (-1)^r \Gamma^{(r)}(1)/r!$ proves (334). $\square$

Remark 41.6 (Priority and scope). Fischler–Rivoal already prove from (327) the transcendence of Gamma derivatives at the relevant rational points; in particular the one-point transcendence of γ is not claimed as new here. Their stronger mixed zero-removal Conjecture 3 implies (327); alternatively, their conditional linear specialization theorem and proposition 41.1 give a second proof of the displayed rank. The contribution of theorems 41.3 and 41.5 is the explicit

simultaneous dilation family, its resonant all-order lift, the use of the companion multipoint theorem, and the full linear rank. It does not assert algebraic independence: products of mixed functions need not be mixed, so the cited specialization theorem is deliberately linear. The value $(W_1)_0(1) = e\gamma$ is the same Euler extension coordinate that reappears in the framed motive below, but no implication between the two conjectural routes is known.

42. The Euler–Tate matched-line target

The exponential-motive construction supplies the canonical framing absent from the scalar Stokes conjugacy class.

Theorem 42.1 (Euler–Mascheroni motive; Fresán–Jossen). *There is a rank-two exponential motive $M(\gamma)$ with adapted de Rham and Betti bases (d_0, d_1) and (b_0, b_1) for which, with columns indexed by the de Rham basis and rows by the Betti basis, the comparison matrix is*

$$P_\gamma = \begin{pmatrix} 1 & \gamma \\ 0 & 2\pi i \end{pmatrix}. \quad (335)$$

Equivalently,

$$\text{comp}(d_0) = b_0, \quad \text{comp}(d_1) = \gamma b_0 + 2\pi i b_1.$$

It lies in a non-split exact sequence

$$0 \longrightarrow \mathbb{Q}(0) \longrightarrow M(\gamma) \longrightarrow \mathbb{Q}(-1) \longrightarrow 0, \quad (336)$$

satisfies $\text{Hom}(M(\gamma), \mathbb{Q}(0)) = 0$, and has motivic Galois group

$$G_{M(\gamma)} = \left\{ \begin{pmatrix} 1 & b \\ 0 & a \end{pmatrix} : a \in \mathbb{G}_m, b \in \mathbb{G}_a \right\}.$$

Proof. This is the Euler–Mascheroni construction and Galois-group computation of Fresán–Jossen [7, §12.8]. $\square$

Its determinant is $2\pi i$, the same as that of the semisimplification, and is again blind to the extension coordinate. The missing input can be stated for this one object.

Proposition 42.2 (Complete matched-line classification). *Every de Rham line mapping isomorphically to the quotient is uniquely of the form*

$$W_{\text{dR}}(x) = \overline{\mathbb{Q}}(xd_0 + d_1), \quad x \in \overline{\mathbb{Q}},$$

and every Betti line with the same property is uniquely of the form

$$W_{\text{B}}(u) = \overline{\mathbb{Q}}(ub_0 + b_1), \quad u \in \overline{\mathbb{Q}}.$$

The two lines are matched if and only if

$$\boxed{x + \gamma = 2\pi i u.} \quad (337)$$

Consequently a matched pair exists if and only if

$$\boxed{\gamma \in \overline{\mathbb{Q}} + 2\pi i \overline{\mathbb{Q}} = \overline{\mathbb{Q}} + \pi \overline{\mathbb{Q}}.} \quad (338)$$

Proof. The displayed parameterizations are the affine charts of lines transverse to the respective subobjects. Comparison sends

$$xd_0 + d_1 \mapsto (x + \gamma)b_0 + 2\pi i b_1.$$

This spans $ub_0 + b_1$ exactly when (337) holds. Eliminating $x, u \in \overline{\mathbb{Q}}$ gives (338); the two fields coincide because $2i \in \overline{\mathbb{Q}}^\times$. $\square$

Problem 42.3 (Euler–Tate matched-line lifting). After extension to $\overline{\mathbb{Q}}$, suppose one-dimensional de Rham and Betti subspaces both map isomorphically onto the corresponding realization of $\mathbb{Q}(-1)$, and comparison maps one line to the other. Prove that the matched pair is induced by a subobject

$$\mathbb{Q}(-1)_{\overline{\mathbb{Q}}} \hookrightarrow M(\gamma)_{\overline{\mathbb{Q}}}.$$

Theorem 42.4 (Strengthened payoff of matched-line lifting). *A positive solution of problem 42.3 implies that*

$$\boxed{\gamma \notin \overline{\mathbb{Q}} + \pi\overline{\mathbb{Q}}.} \tag{339}$$

Equivalently, $1, \pi, \gamma$ are linearly independent over $\overline{\mathbb{Q}}$. In particular $\gamma, \gamma/\pi$, and S_2 are transcendental.

Proof. If $\gamma \in \overline{\mathbb{Q}} + \pi\overline{\mathbb{Q}}$, proposition 42.2 supplies a matched pair. The proposed lifting would give a copy of $\mathbb{Q}(-1)$ mapping nontrivially to the quotient in (336). The composite is a nonzero scalar, so rescaling gives a section and hence a splitting. Moreover $\text{Hom}(-, -)$ commutes here with extension from $\mathbb{Q}$ to $\overline{\mathbb{Q}}$, so the Hom-vanishing in theorem 42.1 remains zero. This contradiction proves (339). Membership in $\overline{\mathbb{Q}} + \pi\overline{\mathbb{Q}}$ is exactly a nontrivial $\overline{\mathbb{Q}}$ -linear relation among $1, \pi, \gamma$. Algebraicity of either γ or γ/π would give such a relation. Finally, if S_2 were algebraic, $2S_2\gamma^2 + \gamma - 2 = 0$ would make γ algebraic. $\square$

Remark 42.5 (Rational and algebraic versions). A matched-line lifting statement restricted to $\mathbb{Q}$ -defined lines would already exclude the corresponding rational matched locus. The $\overline{\mathbb{Q}}$ -coefficient formulation in problem 42.3 is the version that yields the full linear-independence conclusion (339).

Remark 42.6 (Exact scope of current rapid-decay structures). Snodgrass constructs cyclotomic rational structures on rapid-decay cohomology under a hypothesis chosen so that Stokes constants do not enter [18, §1.2]. This supplies a realization and period pairing, but not the numerical-to-motivic fullness asserted in problem 42.3. The problem is a single-object rank-one target, far more narrowly scoped than the full exponential period conjecture but not presently proved.

Remark 42.7 (Three independent apertures). The common-coefficient problem problem 43.1 is quantitative and uses one algebraic coefficient across infinitely many moving levels. The matched-line problem is categorical and uses one canonical framed extension. The mixed-Gevrey route uses the intersection $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$. Each would prove the transcendence of γ ; matched-line lifting would additionally prove the transcendence of γ/π , while the mixed route gives an all-order linear-independence family. No implication among the three inputs is known.

43. Conclusions and remaining problems

The completed results fall into the following thematic groups.

- (1) Every divisor $\text{Ein} - c$ has a universal polynomial inverse-trace calculus. Its full trace ring is exactly $F[c^{-1}]$, generated by the two even traces T_2, T_4 , while every transversal moving selection preserves algebraic independence. The only nontrivial collision of (T_2, T_3) is the pair $\{6 - 2\sqrt{3}, 6 + 2\sqrt{3}\}$; at every order there are also algebraic levels with zero inverse trace and wholly transcendental spectral divisors. Finite-rank multiplicative groups have defect at most their rank and admit cofinite jointly independent subfamilies; the complete gamma-free, log-free cancellation space is identified. The tails $E_1(\alpha)$, and jointly the four classical families $E_1, \text{Ei}, \text{Ci}, \text{Si}$, have defect at most one more than the multiplicative rank. Regularized lower incomplete-Gamma jets have defect at most the rank, uniformly across all jet orders, while fixed upper-jet rows have one further possible defect. Sine-integral graph ideals, Dickman log-Laplace relations, and characteristic-zero algebraic matroids give further realizations of this engine. At every finite set of levels, the complete trace field is exactly the rational quotient by the one-dimensional gauge shift.
- (2) The dyadic and fixed-power balanced levels approach γ through explicit algebraically independent finite-stage data. Their order-two and order-four traces give positive algebraically independent spectral decompositions of S_2 and S_4 ; the two positive carrier traces recover γ . The order-two carrier also gives an order-zero Laguerre–Pólya determinant. Neither the dyadic level sequence nor its second trace is P-recursive.
- (3) The gamma-free map $\mathcal{R}$ has transcendental inverse images at every algebraic level in its range, giving the explicit family r_m . Normalized finite Ein-certificates are unique. This turns all cross-index real Euler witnesses into one countable exclusion theorem and yields the relative five-class theorem. The critical cancellation coefficients $u_N(q)$ have algebraic defect at most one.
- (4) For the 2026 Challenge recurrence, the terminating ${}_2F_2$ state is explicitly gauge-equivalent to the contragredient normalized official Meijer– G state. The finite bilinear identity fixes the conserved pairing; a flat dual-number SL_3 , equivalently SL_6 , includes the numerator. The Beta-log transform connects the Challenge to the cubic family and every fixed offset. A single rational carrier yields the denominator, numerator, full zeta log-jets, and a monic contact polynomial. The exact maxima mixture, Meijer stop-loss interpretation, Poisson–binomial law, zero-density limit, regularized Gamma determinant, strengthened divisor, factorial-six Casoratian, all-initial-value limit classification, and integer solution lattice complete the finite structure.
- (5) The Euler–Padé convergents have a finite residue criterion on the p -adic unit boundary and converge as a full sequence at $t = -1$ for $p = 2, 3, 5, 13$. One lcm-indexed rational subsequence converges placewise at the real place and at every fixed prime. The denominator interpolation q_p identifies unbounded valuations with a p -adic zero, without confusing that event with failure of convergence.
- (6) The dyadic augmentation tail is the unique minimum in the cumulative-positive cone, while outside that cone three points already produce dense tail values. For fixed q , the exact denominator transition for consecutive augmentation-one triples occurs at $e^{(q-1)q^N}/N$. A moving-point lower bound covering the displayed fixed system cannot have sublinear point cost; at linear cost the example forces the stated height loss. The two-base form $Y_N(2) - Y_{M_N}(3)$ retains the same leading 2^N logarithmic scale. The exact asymmetric Sondow formula replaces the invalid universal entropy-floor heuristic.
- (7) In each nondegenerate fixed residual direction of the Prévost grid, the real Hilbert quotient is exactly saturated. For the uncombined optimally saddled form, and absent additional exponential divisibility or cancellation, the baseline fixed-place ledger leaves the deficit

$2 - \log(3 + 2\sqrt{2})$. The bounded-degree reduction transfers a surviving one-prime problem to quantitative approximation of $\log 2$, but does not close it. The fixed-prime Rodrigues deformation produces integral polynomials, exact logarithmic linear forms, positivity, and the endpoint factor $p^{(1-\theta)^n}$. Even after the whole factor is favorably credited as fixed-place gain, the universal saddle-cancellation inequality shows that this ledger does not improve the Prévost baseline. Balanced factorial ratios themselves have only $O(\log n)$ fixed-prime valuation.

- (8) The displayed unframed lateral transition of the elementary Euler differential operator is gamma-free. The missing information is the canonical framing. In the mixed arithmetic-Gevrey route, the intersection conjecture $\mathbf{E} \cap \mathbf{D} = \overline{\mathbb{Q}}$ yields simultaneous linear independence of the dilation family and of all derivatives $1, \Gamma'(1), \Gamma''(1), \dots$. These conclusions remain conditional. In the Euler–Mascheroni motive the same extension coordinate is the off-diagonal entry of the period matrix. The explicit matched-line lifting problem [problem 42.3](#) would force the $\overline{\mathbb{Q}}$ -linear independence of $1, \pi, \gamma$, not merely the transcendence of γ .

The principal quantitative unresolved problem can now be stated without spectral language.

Problem 43.1 (Common-coefficient separation). Prove that for every $s \in \overline{\mathbb{Q}}$,

$$\limsup_{N \rightarrow \infty} \frac{-\log |2sY_N^2 + Y_N - 2|}{2^N} < 1,$$

or establish any equivalent separation that exploits the same coefficient s across all N .

By [theorem 36.10](#), this problem is equivalent to the transcendence of S_2 , hence to the transcendence of γ . It is therefore a precise quantitative target, not a smaller lemma already within current fixed-point E -function theory. Independently, [problem 42.3](#) is a categorical target for the canonical rank-two Euler extension; its $\overline{\mathbb{Q}}$ -version would additionally prove the transcendence of γ/π . The mixed-Gevrey intersection conjecture gives a logically independent conditional route and the stronger all-order linear-independence package, but it is not presently known.

We emphasize the corresponding negative conclusions. The algebraic independence of the finite stages does not determine the arithmetic nature of their limit. The Laguerre–Pólya determinant is an exact spectral carrier but is not proved arithmetic or D -finite; the regularized Gamma determinant does not prove the irrationality of any odd zeta value. Certificate uniqueness shows that producing more linear witnesses of the same normalized type creates mutually exclusive alternatives rather than independent chances to prove transcendence. The specific Prévost and Rodrigues families analyzed here do not pay their complete height ledger, even under the favorable fixed-place accounting stated above. The p -adic convergence theorems prove neither the irrationality of $E_p(-1)$, nor Kurepa’s conjecture, nor full convergence for every prime. The external relation classifications do not close a famous transcendence conjecture. Finally, unframed Stokes data forget the Euler extension coordinate, and the mixed-Gevrey theorems that recover it are explicitly conditional. Accordingly, no unconditional theorem in this paper proves the irrationality or transcendence of γ , δ , S_2 , or any odd zeta value; the exact Euler apertures above state what is still missing.

44. Consolidated theorem-status and dependency ledger

This ledger governs every statement in the monograph when a theorem heading does not itself carry a badge. Analytic identities and monodromy results listed as **U-PASS** are independent of the companion arithmetic-independence paper. Every transcendence or algebraic-independence

transfer whose input is the algebraic-point theorem of [001] is marked **R-PASS** here, even when the subsequent deduction is elementary.

Result family		Status	Exact scope
Universal inverse traces		U-PASS	$T_m(c) = P_m(c^{-1})$, exact two-trace recovery, the fixed-level trace ring, and the unique (T_2, T_3) collision.
Moving-level arithmetic		R-PASS	Finite-rank/cofinite independence, the dyadic algebraically independent tower, inverse-level transcendence, algebraic matroid realizations, and transcendental algebraic-level spectral divisors.
Polyexponential structure	Faber	U-PASS	Absolute indecomposability for all $q \geq 1, m \geq 2$, and $G_{q,m}^{\text{geom}} \in \{A_m, S_m\}$ for $m \geq 3$.
Polyexponential symmetric monodromy		U-PASS	Every even degree; every prime degree; $m = p + 2$ for prime $p \geq 5$; and, for each fixed m , all sufficiently large integer orders q .
Polyexponential monodromy		X-PASS	Every $q \geq 1$ for $m \leq 1001$, and the original $q = 1$ tower for $m \leq 10^6$. These are group statements, not all-degree Morse statements.
Fixed-order polylogarithmic window		U-PASS/X-PASS	For $\tau = 1, 2, 3, 4$, every odd $7 \leq m \leq 1001$ has geometric and arithmetic group S_m . The row $(\tau, m) = (4, 17)$ is covered by the unconditional prime-degree theorem; all other rows use the exact archived window audit.
Derivative separability		U-PASS/X-PASS	At least half of the roots of $P'_{q,2s}$ are simple and $P'_{q,p+1}$ is square-free; exact residual computations cover the displayed finite ranges. The global DS assertion remains open.
Polylogarithmic tower		U-PASS	For every integer $\tau \geq 1$, all degrees are absolutely indecomposable; every prime degree is Morse with group S_m ; and the degrees $m = p + 2$ have group S_m . The corresponding off-diagonal collision polynomials are absolutely irreducible in those two symmetric families.
Factorially weighted Faber transfer		U-PASS	For $f_{\sigma,\tau}$, the single-spike hypotheses give an isolated transposition. Full S_m additionally requires a separate indecomposability input, so only this local conclusion is asserted.
Log-Stirling/supernecklace		U-PASS	All-degree indecomposability, the $m = p + 2$ symmetric family, the supernecklace EGF, and off-diagonal absolute irreducibility in that family.
Inverse surfaces		U-PASS	No finite asymptotic values, maximal finitary symmetric inverse monodromy, algebraic freedom of finite distinct branches and their first jets, and the local-freedom/global-trace-collapse synthesis.

Result family	Status	Exact scope
Ramanujan–Gamma package	U-PASS	Recurrence closed forms, flat fields, Meijer– G /hypergeometric transmutation, Beta–log transform, probability carrier, contact polynomials, and the regularized Gamma determinant.
Euler Padé/continued fraction	U-PASS	The finite-residue unit criterion; full $t = -1$ convergence for $p = 2, 3, 5, 13$; the common real/all-fixed-prime lcm subsequence; and the exact $\mathbb{Z}_p$ -zero reformulation.
Total positivity and Jacobi tower	U-PASS/R-PASS	The STP kernel, moment determinacy, Jacobi operator, interlacing, spectral flow and asymptotics are unconditional. Arithmetic independence, generic Grassmannian/Galois, eigenvalue, node, and weight conclusions are relative to [001].
Double scaling	U-PASS/R-PASS	The ramp equilibrium, free energy, and Marchenko–Pastur-to-arcsine transition are unconditional; the arithmetic independent phase sequence is relative to [001].
Gamma-tail and spectral sieve	R-PASS	Fixed-cut half-rank, differential transcendence, wholly transcendental divisor density, all-order spectral sieves, and the order-axis arithmetic consequences use [001].
Jossen reductions	U-PASS/R-PASS	The factorial affine E-surface statement is unconditional (with priority caveat); polyexponential polynomiality, reductions, and noncyclic subclasses use [001]. The full multigenerator problem is not claimed.
Density-one odd monodromy	C-PASS	For $q = 1$, the exceptional count is $O_\varepsilon(X^{15/16+\varepsilon})$ under the displayed derivative-separability hypothesis DS.

44.1. Statements deliberately left open

The following assertions are not consequences of this volume and are never used as hidden inputs:

- full geometric S_m in every odd composite degree, or a uniform odd-square arithmetic theorem;
- Morse behavior for every trace polynomial, global DS, or global critical-value noncollision for every $q \geq 2$;
- general indecomposability, full S_m , or a fixed- m large- τ Morse theorem for the factorially weighted $f_{\sigma,\tau}$ family;
- higher inverse-jet freedom and a minimal differential-order theorem in every polyexponential order;
- irrationality or transcendence of any individual $E_p(-1)$, full continued-fraction convergence at every prime, Kurepa’s conjecture, or convergence in the restricted adelic-product topology;

- irrationality or transcendence of γ , δ , or an odd zeta value; comparison-map injectivity for the proposed Euler motive;
- the full multigenerator Jossen problem, local kink universality for the double-scaling model, or unproved quantitative Hilbert/DLS assertions.

44.2. Corrections incorporated in Version 3.0

Version 3.0 incorporates the sharp threshold $m \geq 5$ for the inverse-level corollary; the explicit row-space step in the linear-image rank bound; the scoped moving-level statement; the cancellation check for $v_p(b) = q - 1$; the endpoint/interior Newton-polygon wording; the exact `candidate-count=10` search policy; the missing $p = 11$ Euler–Padé row; and a direct link to the mean-square prime-gap input. The values (7) and (15) in the $m = 27$ table are retained because they are discriminants; the alternative values (22) and (14) arise from a different resultant calculation. The invalid general criterion formerly labelled barriers4 Criterion 7.1 and the hidden-C1 versions of its Gamma claims are excluded.

44.3. Additions incorporated in Version 3.1

Version 3.1 supplies the complete Maynard bibliography for the mean-square prime-gap theorem, proves strict negativity of every positive-degree coefficient of $z/\text{Li}_\tau(z)$, and uses the resulting Faber obstructions to establish all-degree absolute indecomposability for every integer polylogarithmic order. It also adds the polylogarithmic prime-degree Morse theorem, the unconditional $m = p + 2$ symmetric family, the associated off-diagonal absolute irreducibility, and exact fixed-order window certificates through odd degree 1001. These conclusions do not extend the unresolved factorially weighted $f_{\sigma,\tau}$ family.

Part VII

Proof Certificates and Reproducibility

A. An Ein-linear realization of characteristic-zero algebraic matroids

This appendix is the only place where the paper uses the linearization theorem of Rosen–Sidman–Theran. It is kept separate so that the moving trace and Ramanujan results do not appear to depend on it.

Theorem A.1 (Ein-realization of algebraic matroids). *Let M be a finite characteristic-zero algebraic matroid of rank r on a ground set E . There are a matrix $B = (b_{je}) \in \overline{\mathbb{Q}}^{r \times E}$ representing M , distinct $\beta_1, \dots, \beta_r \in \overline{\mathbb{Q}}^\times$, and numbers*

$$\xi_j = \text{Ein}(\beta_j), \quad W_e = \sum_{j=1}^r b_{je} \xi_j, \quad (340)$$

such that, for every $S \subseteq E$,

$$\boxed{\text{trdeg}_K K(W_e : e \in S) = r_M(S).} \quad (341)$$

Moreover the complete relation ideal is

$$\boxed{\ker(K[X_e : e \in E] \longrightarrow K[W_e : e \in E]) = \left\langle \sum_{e \in E} c_e X_e : c \in \overline{\mathbb{Q}}^E, Bc = 0 \right\rangle.} \quad (342)$$

Proof. Extend a characteristic-zero field of algebraic realization to its algebraic closure; the algebraic-dependence matroid is unchanged. By [33, Thm. 1.10], the resulting matroid is linearly representable over that algebraically closed field. Linear representability of a fixed finite matroid is described by the vanishing of its nonbasis minors and the nonvanishing of its basis minors. Introducing one auxiliary inverse for the product of the basis minors converts these conditions into polynomial equations over $\mathbb{Q}$. Since they have a solution in an algebraically closed characteristic-zero field, the Nullstellensatz gives a solution in $\overline{\mathbb{Q}}$. Thus a representing matrix B may be chosen over $\overline{\mathbb{Q}}$.

Choose distinct nonzero algebraic β_j . By theorem 2.1, the ξ_j are algebraically independent over K . If the columns of B indexed by S have rank q , choose q independent columns. The corresponding W_e are algebraically independent by lemma 2.2, giving the lower bound q ; every remaining column is a $\overline{\mathbb{Q}}$ -linear combination of those columns, giving the upper bound. This proves (341). Finally the symmetric algebra of the column span is the polynomial ring on the X_e modulo the linear forms coming from $\ker B$, which proves (342). $\square$

Remark A.2 (Exact scope). The construction realizes the rank function of M and the linearized relation ideal (342). It does not reproduce the nonlinear circuit polynomials of a chosen algebraic presentation of M , and it is not a claim that such a presentation becomes linear as an algebraic variety.

B. Creative-telescoping certificates for the Euler recurrence

For $a = n + 3$, $b = n + 4$, define meromorphically

$$T_{n,k} = b \frac{(a!b!)^2}{\Gamma(k+1)^3 \Gamma(a-k+1) \Gamma(b-k+1)}, \quad \mathcal{L}_{n,k} = 3H_k - H_{a-k} - H_{b-k},$$

where $H_z = \psi(z+1) + \gamma$. The regularized endpoint values are

$$T_{n,b} \mathcal{L}_{n,b} = a!, \quad T_{n,k} \mathcal{L}_{n,k} = 0 \quad (k > b).$$

Indeed, at $k = b$ the simple zero of $1/\Gamma(a-k+1)$ cancels the simple pole of $-H_{a-k}$, with limiting product $a!$. For $k > b$, the two reciprocal-gamma zeros dominate the at-most-simple digamma pole, so the regularized product vanishes. Thus (138) and (139) are respectively $\sum_{k \geq 0} T_{n,k}$ and $\sum_{k \geq 0} T_{n,k} \mathcal{L}_{n,k}$.

Let

$$c_0 = -A_n, \quad c_1 = B_n, \quad c_2 = -C_n, \quad c_3 = D_n,$$

and $\Delta_k u(k) = u(k+1) - u(k)$. The certificates are

$$\sum_{j=0}^3 c_j T_{n-j,k} = \Delta_k \{T_{n,k} h_{n,k}\}, \quad (343)$$

$$\sum_{j=0}^3 c_j T_{n-j,k} \mathcal{L}_{n-j,k} = \Delta_k \{T_{n,k} (h_{n,k} \mathcal{L}_{n,k} + g_{n,k})\}. \quad (344)$$

Here

$$h_{n,k} = -\frac{k^3}{(n+3)^3(n+4)^3} \mathcal{H}(n,k), \quad g_{n,k} = \frac{k^2}{(n+3)^3(n+4)^3} \mathcal{G}(n,k),$$

where

$$\begin{aligned} \mathcal{H}(n,k) = & (8n^3 + 75n^2 + 231n + 232)k^3 \\ & - (48n^4 + 562n^3 + 2436n^2 + 4626n + 3248)k^2 \\ & + (96n^5 + 1388n^4 + 7924n^3 + 22278n^2 + 30772n + 16668)k \\ & - (64n^6 + 1120n^5 + 8071n^4 + 30590n^3 + 64148n^2 + 70347n + 31388), \end{aligned}$$

and

$$\begin{aligned} \mathcal{G}(n,k) = & (48n^3 + 450n^2 + 1386n + 1392)k^3 \\ & - (240n^4 + 2810n^3 + 12180n^2 + 23130n + 16240)k^2 \\ & + (384n^5 + 5552n^4 + 31696n^3 + 89112n^2 + 123088n + 66672)k \\ & - (192n^6 + 3360n^5 + 24213n^4 + 91770n^3 + 192444n^2 + 211041n + 94164). \end{aligned}$$

To verify (343), divide by $T_{n,k}$, use

$$\frac{T_{n,k+1}}{T_{n,k}} = \frac{(n+3-k)(n+4-k)}{(k+1)^3},$$

and, for $0 \leq j \leq 3$,

$$\frac{T_{n-j,k}}{T_{n,k}} = \frac{n-j+4}{n+4} \frac{(n+3-k)^j (n+4-k)^j}{\{(n+3)^j (n+4)^j\}^2},$$

where $x^j = x(x-1)\cdots(x-j+1)$, and clear denominators. For (344), also use

$$\mathcal{L}_{n,k+1} - \mathcal{L}_{n,k} = \frac{3}{k+1} + \frac{1}{n+3-k} + \frac{1}{n+4-k}.$$

The remaining shifts are reduced by

$$\mathcal{L}_{n-j,k} - \mathcal{L}_{n,k} = \sum_{r=0}^{j-1} \left(\frac{1}{n+3-k-r} + \frac{1}{n+4-k-r} \right).$$

Both become polynomial identities in n, k . The lower boundary vanishes because of the factors k^3, k^2 , and the reciprocal gamma factors give the upper boundary. Summing proves (136) for both closed forms for every $n \geq 0$. The accompanying source package contains an exact symbolic verifier for these identities and for (158)–(159).

C. Finite certificate for the integer initial lattice

The official dual transition and cyclic gauge may be taken as

$$R(n) = \begin{pmatrix} 6n^2 + 21n + 18 & -8n - 11 & 3 \\ 3n^3 + 12n^2 + 16n + 7 & -3n^2 - 7n - 4 & n + 1 \\ (n+1)^4 & 0 & 0 \end{pmatrix} \quad (345)$$

and

$$U(n) = \begin{pmatrix} 1 & 6n^2 + 21n + 18 & 15n^4 + 147n^3 + 520n^2 + 792n + 440 \\ 0 & 3n^3 + 12n^2 + 16n + 7 & 10n^5 + 96n^4 + 353n^3 + 625n^2 + 537n + 179 \\ 0 & (n+1)^4 & 6n^6 + 57n^5 + 213n^4 + 402n^3 + 408n^2 + 213n + 45 \end{pmatrix}. \quad (346)$$

Let the row-companion transfer of (136) be

$$C_n^{\text{comp}} = \begin{pmatrix} 0 & 0 & D_n/A_n \\ 1 & 0 & -C_n/A_n \\ 0 & 1 & B_n/A_n \end{pmatrix}. \quad (347)$$

Direct multiplication gives the exact coboundary

$$R(n)U(n+1) = U(n)C_n^{\text{comp}}. \quad (348)$$

For a raw row c , the integer recurrence state at level N is

$$cR(0) \cdots R(N-1)U(N). \quad (349)$$

Indeed (348) proves by induction that (349) equals $(u_{N-3}, u_{N-2}, u_{N-1})$ when $cU(0)$ is the initial row; the empty product is used at $N = 0$. Let

$$d_1 = (8, -22, 10), \quad d_2 = (0, 2, -6), \quad d_3 = (0, -1, 7).$$

Direct multiplication gives

$$\frac{d_1}{8}U(0) = (1, 0, 4), \quad \frac{d_2}{8}U(0) = (0, 1, 11), \quad \frac{d_3}{8}U(0) = (0, 0, 17), \quad (350)$$

and the finite congruence certificate

$$R(0) \cdots R(7) \equiv 0 \pmod{8}, \quad (351)$$

$$d_i R(0) \cdots R(N-1)U(N) \equiv 0 \pmod{8} \quad (1 \leq i \leq 3, 0 \leq N < 8). \quad (352)$$

For $N \geq 8$, (351) supplies the factor eight; for $N < 8$, (352) does. Hence the three rows in (350) generate integral solutions at all levels, completing the sufficiency argument in theorem 22.4. The source verifier checks every displayed matrix identity over $\mathbb{Z}$.

D. A numerical ledger

The following values provide reproducibility checks for the indexing in (59). They are not used in any proof.

Table 5: Positive increments and positive zeros of Δ_2 .

N	d_N	$r_N = \sqrt{2/d_N}$
0	1.474569389580200	1.1646150303
1	$5.633945512124626 \times 10^{-1}$	1.8841205074
2	$9.586804440833174 \times 10^{-2}$	4.5674948216
3	$1.339700176289053 \times 10^{-3}$	38.637693279
4	$2.954643417466919 \times 10^{-7}$	2601.7314103
5	$2.050814843583590 \times 10^{-14}$	9.8753335×10^6

The indexing is tied to the convention Y_N for $N \geq 1$:

$$d_0 = \tau_1, \quad d_N = \tau_{N+1} - \tau_N \quad (N \geq 1).$$

Starting a level sequence at $N = 0$ would give a different, though equally valid, spectral decomposition.

E. Exact computational certificates

The computer-assisted statements in this volume use exact integer or finite-field arithmetic. Floating-point computations are never used as proof of a Galois-group, separability, or valuation assertion. The complete scripts and captured outputs are shipped in the source archive under `computation/`.

E.1. All-order rectangle

The command

```
python3 computation/trace_allq_modp_audit.py -bound 1001 -candidate-count 10
```

examines, for every eligible odd degree, at most the ten largest strict-window primes in descending order. It checks every relevant order residue class, records the prime-degree fallback rows, and emits the combined certificates at $m = 27$ and $m = 989$. The seven $m = 27$ discriminant rows use the values displayed in the main text; the search does not replace them by resultants.

E.2. Fixed-order polylogarithmic window

The command

```
python3 computation/polylog_window_prime_audit.py -bound 1001
```

uses exact arithmetic in $\mathbb{F}_p$ to test every strict-window prime $(m+1)/2 < p < m$ for each $\tau \in \{1, 2, 3, 4\}$ and odd $7 \leq m \leq 1001$. It constructs the residual polynomial R_r from (116) and checks $\gcd(R_r, R'_r) = 1$. There are no window exceptions for $\tau = 1, 2, 3$; for $\tau = 4$ the

sole window exception is $m = 17$, where the unconditional prime-degree part of [Theorem 16.3](#) proves S_{17} . Thus the exception concerns only the window certificate, not the conclusion of [Corollary 16.4](#).

The frozen files have SHA-256 hashes

- `polylog_window_prime_audit.py`:
dd42b574f3eb5144cd49316530911f4cee0e3722ed03dda7382bd10ec8a31cd2;
- `polylog_window_prime_audit_1001.txt`:
8f952488d2011c1391ae15a4f607715c7fc7de73a00ce8cae00fa973f938cbdf.

E.3. Original tower through one million

The command

```
python3 computation/window_prime_selection_audit.py -bound 1000000
```

checks the original $q = 1$ tower. For each odd composite degree it first tests the largest strict-window prime and then the next candidate if needed. The output records the two failures of the largest-prime choice and the successful second-prime certificates.

E.4. Morse and derivative certificates

The lower-degree exact Morse certificates are generated by

```
python3 computation/trace_morse_certificates.py -max-m 55 -prime-bound 3000.
```

The residual even-derivative computations are implemented independently in the supplied C++ source and their captured outputs are included for bounds 1000 and 5000. These finite certificates support exactly the ranges stated in the theorem ledger; they are not extrapolated to a universal DS theorem.

E.5. Reproducibility convention

Each source-archive file is covered by the top-level SHA-256 manifest. A successful independent reproduction consists of:

1. verifying the manifest;
2. compiling the monograph with the documented pdfLaTeX command;
3. running the quick certificate suite;
4. optionally running the extended 10^6 -degree suite and comparing its terminal summary with the captured output.

The source archive records the interpreter/compiler versions used for the frozen build. No network access is required to compile the PDF or rerun the certificates.

References

- [1] B. Adamczewski and C. Faverjon, Algebraic independence measures for values of E -functions and M -functions, arXiv:2502.09999 (2025). <https://arxiv.org/abs/2502.09999>.
- [2] NIST Digital Library of Mathematical Functions, §6.12, asymptotic expansions of exponential and logarithmic integrals, release 1.2.7 (2026). <https://dlmf.nist.gov/6.12>.
- [3] E. Delaygue, Arithmetic properties of Apéry-like numbers, *Compos. Math.* **154** (2018), 249–274. <https://doi.org/10.1112/S0010437X17007552>.
- [4] S. Fischler and T. Rivoal, Arithmetic theory of E -operators, *J. Éc. polytech. Math.* **3** (2016), 31–65. <https://doi.org/10.5802/jep.28>.
- [5] S. Fischler and T. Rivoal, Values of E -functions are not Liouville numbers, *J. Éc. polytech. Math.* **11** (2024), 1–18. <https://doi.org/10.5802/jep.249>.
- [6] S. Fischler and T. Rivoal, Relations between values of arithmetic Gevrey series, and applications to values of the Gamma function, *J. Number Theory* **261** (2024), 36–54. <https://arxiv.org/abs/2301.13518>.
- [7] J. Fresán and P. Jossen, *Exponential Motives*, monograph manuscript, version of April 2024. <http://javier.fresan.perso.math.cnrs.fr/expmot.pdf>.
- [8] KEIO, Exponential-integral periods across arithmetic, Padé, Hankel, reciprocal geometry, and Frobenius framing, Version 6.3, research manuscript, 108 pages (2026).
- [9] Th. Kaluza, *Über die Koeffizienten reziproker Potenzreihen*, *Math. Z.* **28** (1928), 161–170. <https://doi.org/10.1007/BF01181155>.
- [10] J. C. Lagarias, Euler’s constant: Euler’s work and modern developments, *Bull. Amer. Math. Soc.* **50** (2013), 527–628. <https://doi.org/10.1090/S0273-0979-2013-01423-X>.
- [11] B. Ya. Levin, *Lectures on Entire Functions*, Translations of Mathematical Monographs 150, American Mathematical Society, Providence, RI, 1996.
- [12] R. Marcovecchio, The Rhin–Viola method for $\log 2$, *Acta Arith.* **139** (2009), 147–184. <https://doi.org/10.4064/aa139-2-5>.
- [13] R. Marcovecchio, Multiple Legendre polynomials in Diophantine approximation, arXiv:2512.12890 (2025). <https://arxiv.org/abs/2512.12890>.
- [14] J. Maynard, *On the difference between consecutive primes*, arXiv:1201.1787v3 [math.NT] (2012), Theorem 3.1. <https://arxiv.org/abs/1201.1787>.
- [15] M. Prévost, A family of criteria for irrationality of Euler’s constant, *J. Comput. Appl. Math.* **219** (2008), no. 2, 484–492. <https://arxiv.org/abs/math/0507231>.
- [16] M. Shalyt, R. Kalisch, C. Schneider, H. Barkan, E. Leibtag, J. Campbell, S. Weinbaum, T. Monderer, A. Narayanan, and I. Kaminer, The Ramanujan Challenge for AI, arXiv:2607.09721 (2026). <https://arxiv.org/abs/2607.09721>.
- [17] D. Ridout, The p -adic generalization of the Thue–Siegel–Roth theorem, *Mathematika* **5** (1958), 40–48. <https://doi.org/10.1112/S0025579300001339>.
- [18] B. Snodgrass, Periods of E -operators, arXiv:2608.06005v2 (2026). <https://arxiv.org/abs/2608.06005>.
- [19] W. Van Assche and T. Wolfs, Rational approximation of Euler’s constant using multiple orthogonal polynomials, *Combinatorics and Number Theory* **14** (2025), no. 2, 141–162. <https://doi.org/10.2140/cnt.2025.14.141>.

- [20] N. I. Akhiezer, *The Classical Moment Problem and Some Related Questions in Analysis*, Oliver & Boyd, Edinburgh, 1965.
- [21] É. Delaygue, A Lindemann–Weierstrass theorem for E -functions, *J. Reine Angew. Math.* **820** (2025), 75–85. <https://doi.org/10.1515/crelle-2024-0090>.
- [22] A.-M. Ernvall-Hytönen, T. Matala-Aho, and L. Seppälä, Euler’s factorial series, Hardy integral, and continued fractions, *J. Number Theory* **244** (2023), 224–250. <https://doi.org/10.1016/j.jnt.2022.09.007>.
- [23] D. Grahovac, A. Kovtun, N. N. Leonenko, and A. Pepelyshev, Dickman type stochastic processes with short- and long-range dependence, *Stochastics* **97** (2025), no. 6, 722–743. <https://doi.org/10.1080/17442508.2025.2522789>.
- [24] B. H. Gross, On the periods of abelian integrals and a formula of Chowla and Selberg, *Invent. Math.* **45** (1978), no. 2, 193–211. <https://doi.org/10.1007/BF01390273>.
- [25] V. V. Petrov, *Sums of Independent Random Variables*, Ergebnisse der Mathematik und ihrer Grenzgebiete 82, Springer, Berlin, 1975.
- [26] M. R. Murty and E. Saha, Transcendental values of the incomplete gamma function and related questions, *Arch. Math.* **105** (2015), 271–283. <https://doi.org/10.1007/s00013-015-0800-3>.
- [27] Kh. Hessami Pilehrood and T. Hessami Pilehrood, On a continued fraction expansion for Euler’s constant, *J. Number Theory* **133** (2013), no. 2, 769–786. <https://doi.org/10.1016/j.jnt.2012.08.016>.
- [28] Kh. Hessami Pilehrood and T. Hessami Pilehrood, Rational approximations to values of Bell polynomials at points involving Euler’s constant and zeta values, *J. Aust. Math. Soc.* **92** (2012), no. 1, 71–98. <https://doi.org/10.1017/S1446788712000134>.
- [29] G. Pólya and I. Schur, Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen, *J. Reine Angew. Math.* **144** (1914), 89–113.
- [30] M. R. Powers, A sequence of nested exponential random variables with connections to two constants of Euler, arXiv:2302.04605v4 (2023). <https://arxiv.org/abs/2302.04605>.
- [31] M. R. Powers, Transcendence results for $\Gamma^{(n)}(1)$ and related sequences of generalized constants, arXiv:2511.01849v7 (2026). <https://arxiv.org/abs/2511.01849>.
- [32] Ramanujan Machine Group, *Ramanujan Challenge Proofs*, official solution manuscript, 45 pages (2026). https://ramanujanmachine.com/wp-content/uploads/2026/08/Ramanujan_Challenge_Proofs_02_08.pdf. Supplementary code: <https://github.com/RamanujanMachine/RamanujanChallengeSolutions>.
- [33] Z. Rosen, J. Sidman, and L. Theran, Linearizing algebraic matroids, arXiv:2507.07220v3 (2026). <https://arxiv.org/abs/2507.07220>.
- [34] J. Sondow, Criteria for irrationality of Euler’s constant, *Proc. Amer. Math. Soc.* **131** (2003), no. 11, 3335–3344. <https://doi.org/10.1090/S0002-9939-03-07081-3>.
- [35] W. Van Assche and T. Wolfs, Multiple orthogonal polynomials associated with the exponential integral, *Stud. Appl. Math.* **151** (2023), no. 2, 411–449. <https://doi.org/10.1111/sapm.12608>.